

26. General PWM Timer (GPTW)

This MCU has a general PWM timer (GPTW) consisting of a four-channel 32-bit timer.

26.1 Overview

Table 26.1 lists the specifications for the GPTW, and Table 26.2 shows the functions of the GPTW. Figure 26.1 shows a block diagram of the GPTW.

Table 26.1 GPTW Specifications

Item	Description
Functions	• 32 bits × 4 channels • Up-counting or down-counting (sawtooth waves) or up/down-counting (triangle waves) for each counter. • Clock sources independently selectable for each channel • Two I/O pins per channel • Two output compare/input capture registers per channel • For the two output compare/input capture registers of each channel, four registers are provided as buffer registers and are capable of operating as comparison registers when buffering is not in use. • In output compare operation, buffer switching can be at crests or troughs, enabling the generation of laterally asymmetric PWM waveforms. • Registers for setting up frame cycles in each channel (with capability for generating interrupts at overflow or underflow) • Generation of dead times in PWM operation • Simultaneous start/stop/clearing of desired channel counters • Operation of count start/count stop/counter clearing/up-counting/down-counting/input capture by maximum of eight ELC events based on the ELC setting • Operation of count start/count stop/counter clearing/up-counting/down-counting/input capture by detecting two input signal conditions • Operation of count start/count stop/counter clearing/up-counting/down-counting/input capture by maximum of four external triggers • Function to control output negation by requests for disabling of output from the POEG • A/D conversion start trigger generation function • Event signals for compare match A to F and for overflow/underflow can be output to the ELC • Input capture input can select noise filter function • Bus clock: PCLKA, GPTW count reference clock: PCLKA

Table 26.2 GPTW Functions

Item		GPTW0 to GPTW3
Timer counter		32-bit
Count clocks		PCLKA, PCLKA/2, PCLKA/4, PCLKA/8, PCLKA/16, PCLKA/32, PCLKA/64, PCLKA/256, PCLKA/1024, GTETRGA, GTETRGB, GTETRGC, GTETRGD
Output compare/input capture registers (GTCCR)		GTCCRA, GTCCRB
Compare/buffer registers		GTCCRC, GTCCRD, GTCCRE, GTCCRF
PWM period setting register		GTPR
PWM period setting buffer registers		GTPBR, GTPDBR
Input capture input/ compare match output/ PWM output pin		GTIOCnA, GTIOCnB
External trigger input pin (via the POEG)		GTETRGA, GTETRGB, GTETRGC, GTETRGD
Counter clear sources		Compare match for the GTPR register, input capture, ELC input, the state of input pins, and external trigger input
Compare match output	Low output	✓
	High output	✓
	Toggle output	✓
Input capture function		✓
Automatic addition of dead time		✓
PWM mode	Sawtooth-wave PWM mode 1	✓
	Sawtooth-wave one-shot pulse mode	✓
	Triangle-wave PWM mode 1/2/3	✓
Phase counting function		✓
Buffer operation	Double buffer	✓
	Simultaneous operation disable control for multiple channels	✓
One-shot operation		✓
DMAC/DTC activation		All interrupt sources
A/D conversion start trigger		Compare match of GTADTRA or GTADTRB register
Interrupt sources		Nine sources • GTCCRA register compare match/input capture (GTCIA_n) • GTCCRB register compare match/input capture (GTCIB_n) • GTCCRC register compare match (GTCIC_n) • GTCCRD register compare match (GTCID_n) • Dead time error (GDTEn) • GTCCRE register compare match (GTCIE_n) • GTCCRF register compare match (GTCIF_n) • GTCNT counter overflow (GTPR register compare match) (GTCIV_n) • GTCNT counter underflow (GTCIU_n)
Interrupt skipping function		• Skipping of interrupts of GTCNT counter overflow (GTPR register compare match) (GTCIV_n) and GTCNT counter underflow (GTCIU_n) (interlocked with other interrupts and A/D conversion start requests) • Buffer transfer skipping function
Event operation by the ELC		✓
Noise filter function		✓

✓: Possible

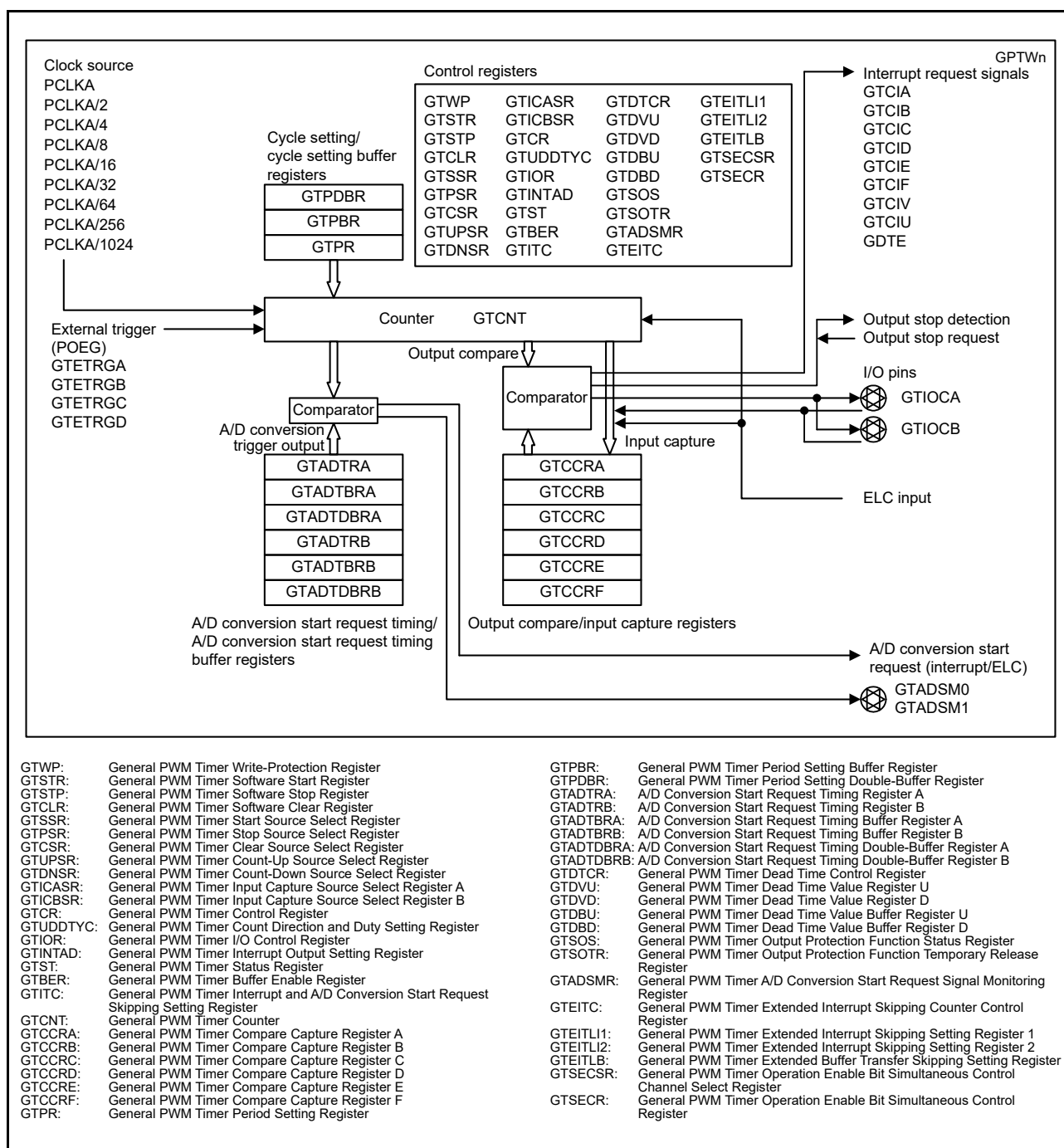


Figure 26.1 GPTW Block Diagram (n = 0 to 3)

Table 26.3 lists the I/O pins used in the GPTW.

Table 26.3 GPTW I/O Pins (n = 0 to 3)

Channel	Pin Name	I/O	Function
GPTW	GTETRGA	Input	External trigger input pin A (input via the POEG)
	GTETRGB	Input	External trigger input pin B (input via the POEG)
	GTETRG	Input	External trigger input pin C (input via the POEG)
	GTETRGD	Input	External trigger input pin D (input via the POEG)
	GTADSM0	Output	A/D conversion start request monitor 0 output pin
	GTADSM1	Output	A/D conversion start request monitor 1 output pin
GPTWn	GTIOCnA	I/O	GTCCRA register input capture input/output compare output/PWM output pin
	GTIOCnB	I/O	GTCCRB register input capture input/output compare output/PWM output pin

26.2 Register Descriptions

26.2.1 General PWM Timer Write-Protection Register (GTWP)

Address(es): GPTW0.GTWP 000C 2000h, GPTW1.GTWP 000C 2100h, GPTW2.GTWP 000C 2200h,
GPTW3.GTWP 000C 2300h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
PRKEY[7:0]								—	—	—	CMNWP	CLRWP	STPWP	STRWP	WP
Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															

Bit	Symbol	Bit Name	Description	R/W
b0	WP	Register Write Disabled	0: Write to the register is enabled 1: Write to the register is disabled	R/W
b1	STRWP	GTSTR.CSTRT Bit Write Disabled	0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
b2	STPWP	GTSTP.CSTOP Bit Write Disabled	0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
b3	CLRWP	GTCLR.CCLR Bit Write Disabled	0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
b4	CMNWP	Common Register Write Disabled	0: Write to the register is enabled 1: Write to the register is disabled	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b8	PRKEY[7:0]	GTWP Key Code	These bits are read as 0. To modify the WP, STRWP, STPWP, CLRWP, and CMNWP bits, write A5h.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

GTWP enables or disables writing to registers to prevent accidental modification.

Protection by the GTWP register only covers the CPU write operation.

Such protection does not cover a register updating generated in coordination with the CPU write.

For registers that are write enabled or disabled depending on the setting of the GTWP register, see section 26.8.1, Write-Protection for Registers.

WP Bit (Register Write Disabled)

Write enable/disable to the GPTW can be selected with this bit.

Registers which are targets of write enable/disable are as follows:

GTSSR, GTPSR, GTCSSR, GTUPSR, GTDNSR, GTICASR, GTICBSR, GTCR, GTUDDTYC, GTIOR, GTINTAD, GTST, GTBER, GTITC, GTCNT, GTCCRA, GTCCRB, GTCCRC, GTCCRD, GTCCRE, GTCCRF, GTPR, GTPBR, GTPDBR, GTADTRA, GTADTBRA, GTADTDBRA, GTADTRB, GTADTBRB, GTADTDBRB, GTDTCR, GTDVU, GTDVD, GTDBU, GTDBD, GTSOS, GTSOTR, GTADSMR, GTEITC, GTEITLI1, GTEITLI2, GTEITLB

STRWP Bit (GTSTR.CSTRT Bit Write Disabled)

The STRWP bits enable or disable starting the updating of counter values by writing to the CSTRTn bit corresponding to a channel number in the GTSTR register.

The bit position of each CSTRTn bit in the GTSTR registers is allocated to the channel with the corresponding number, and writing to the GTSTR register for any channel results in writing to the registers of all channels. The STRWP bit for

each channel does not control writing but only controls updating of the CSTRT bit for the corresponding channel in the case of simultaneous writing to all channels.

Thus, in the case of writing to the CSTRT bits of a channel for which the setting of the STRWP bit is 1 (disabling writing), the CSTRT bit for the given channel will not be updated, but the CSTRT bits corresponding to channel for which the setting of the STRWP bit is 0 (enabling writing) will be updated. For example, when the setting of the GPTW0.GTWP.STRWP bit is 0 (enabling writing), writing 1 to the GPTW1.GTSTR.CSTRT0 bit when its current setting is 0 will lead to updating of the value, and the GPTW0.GTCNT counter starts running. When the setting of the GPTW0.GTWP.STRWP bit is 1 (disabling writing), writing 1 to the GPTW1.GTSTR.CSTRT0 bit when its current setting is 0 will leave the bit with the value 0, and the GPTW0.GTCNT counter will not start running.

If you want to protect all bits in the GTSTR register from being updated, set the STRWP bits of all channels to 1.

STPWP Bit (GTSTP.CSTOP Bit Write Disabled)

The STPWP bits enable or disable starting the updating of counter values by writing to the CSTOPn bit corresponding to a channel number in the GTSTP register.

The bit position of each CSTOPn bit in the GTSTP registers is allocated to the channel with the corresponding number, and writing to the GTSTP register for any channel results in writing to the registers of all channels. The STPWP bit for each channel does not control writing but only controls updating of the CSTOP bit for the corresponding channel in the case of simultaneous writing to all channels.

Thus, in the case of writing to the CSTOP bits of a channel for which the setting of the STPWP bit is 1 (disabling writing), the CSTOP bit for the given channel will not be updated, but the CSTOP bits corresponding to channel for which the setting of the STPWP bit is 0 (enabling writing) will be updated. For example, when the setting of the GPTW0.GTWP.STPWP bit is 0 (enabling writing), writing 1 to the GPTW1.GTSTP.CSTOP0 bit when its current setting is 0 will lead to updating of the value, and the GPTW0.GTCNT counter is stopped. When the setting of the GPTW0.GTWP.STPWP bit is 1 (disabling writing), writing 1 to the GPTW1.GTSTP.CSTOP0 bit when its current setting is 0 will leave the bit with the value 0, and the GPTW0.GTCNT counter will not be stopped.

If you want to protect all bits in the GTSTP register from being updated, set the STPWP bits of all channels to 1.

CLRWP Bit (GTCLR.CCLR Bit Write Disabled)

The CLRWP bits enable or disable starting the updating of counter values by writing to the CCLRn bit corresponding to a channel number in the GTCLR register.

The bit position of each CCLRn bit in the GTCLR registers is allocated to the channel with the corresponding number, and writing to the GTCLR register for any channel results in writing to the registers of all channels. The CLRWP bit for each channel does not control writing but only controls updating of the CCLR bit for the corresponding channel in the case of simultaneous writing to all channels.

Thus, in the case of writing to the CCLR bits of a channel for which the setting of the CLRWP bit is 1 (disabling writing), the CCLR bit for the given channel will not be updated, but the CCLR bits corresponding to channel for which the setting of the CLRWP bit is 0 (enabling writing) will be updated. For example, when the setting of the GPTW0.GTWP.CLRWP bit is 0 (enabling writing), writing 1 to the GPTW1.GTCLR.CCLR0 bit when its current setting is 0 will lead to updating of the value, and the GPTW0.GTCNT counter is cleared.

If you want to protect all bits in the GTCLR register from being updated, set the CLRWP bits of all channels to 1.

CMNWP Bit (Common Register Write Disabled)

The CMNWP bit enables or disables starting the updating of counter values by writing to the SECSELn bit (n = 0 to 3) corresponding to a channel number in the GTSECSR register or to the GTSECR register.

The bit position of each SECSEL bit in the GTSECSR registers is allocated to the channel with the corresponding number, and writing to the GTSECSR register for any channel results in writing to the registers of all channels. Writing to the GTSECR register of any channel leads to writing to the registers of all channels. The CMNWP bit for each channel does not control writing but only controls updating of the SECSEL bit and the GTSECR register value for the corresponding channel in the case of simultaneous writing to all channels.

Thus, in the case of writing to the SECSEL bit and the GTSECR register value of a channel for which the setting of the CMNWP bit is 1 (disabling writing), the SECSEL bit and the GTSECR register value for the given channel will not be updated, but the SECSEL bit and the GTSECR register value corresponding to channel for which the setting of the CMNWP bit is 0 (enabling writing) will be updated.

For example, when the setting of the GPTW0.GTWP.CMNWP bit is 0 (enabling writing), writing to the GPTW1.GTSECSR.SECSEL0 bit will lead to updating of the value of the GPTW0.GTSECSR.SECSEL0 bit. In the same way, writing to the GPTW1.GTSECR register will update the value of the GPTW0.GTSECR register. When the setting of the GPTW0.GTWP.CMNWP bit is 1 (disabling writing), writing to the GPTW1.GTSECSR.SECSEL0 bit will not lead to updating of the value of the GPTW0.GTSECSR.SECSEL0 bit. In the same way, writing to the GPTW1.GTSECR register will not lead to updating of the value of the GPTW0.GTSECR register.

If you want to protect all bits in the GTSECSR and GTSECR registers from being updated, set the CMNWP bits of all channels to 1.

PRKEY[7:0] Bits (GTWP Key Code)

This bit controls whether the WP, STRWP, STPWP, CLRWP, and CMNWP bits can be overwritten.

26.2.2 General PWM Timer Software Start Register (GTSTR)

Address(es): GPTW0.GTSTR 000C 2004h, GPTW1.GTSTR 000C 2104h, GPTW2.GTSTR 000C 2204h, GPTW3.GTSTR 000C 2304h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	CSTRT3	CSTRT2	CSTRT1	CSTRT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CSTRT0	Channel 0 Count Start	When reading 0: The counter is stopped. 1: The counter is running. When writing 0: Ignored. 1: The counter starts running.	R/W
b1	CSTRT1	Channel 1 Count Start		R/W
b2	CSTRT2	Channel 2 Count Start		R/W
b3	CSTRT3	Channel 3 Count Start		R/W
b31 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTSTR register starts the GTCNT counter operation.

A bit position of the GTSTR register indicates a channel number. The GTSTR register of each channel is a common register, and writing 1 to the GTSTR register in any channel and updating it can start operation of the GTCNT counter in all the channels related to the position of the bit written with 1. A change in counter operation and the register value is not generated by writing 0 to the bit.

CSTRTn Bit (Channel n Count Start) (n = 0 to 3)

This bit starts the GTCNT counter for channel n.

A read value indicates the state of the counter operation in a corresponding channel (GTCR.CST bit). The bit with 0 indicates that the counter is stopped, while the bit with 1 indicates that the counter is in operation.

26.2.3 General PWM Timer Software Stop Register (GTSTP)

Address(es): GPTW0.GTSTP 000C 2008h, GPTW1.GTSTP 000C 2108h, GPTW2.GTSTP 000C 2208h, GPTW3.GTSTP 000C 2308h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	CSTOP3	CSTOP2	CSTOP1	CSTOP0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	CSTOP0	Channel 0 Count Stop	When reading 0: The counter is operating. 1: The counter is stopped. When writing 0: Ignored. 1: The counter is stopped.	R/W
b1	CSTOP1	Channel 1 Count Stop		R/W
b2	CSTOP2	Channel 2 Count Stop		R/W
b3	CSTOP3	Channel 3 Count Stop		R/W
b31 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTSTP register stops the GTCNT counter operation.

A bit position of the GTSTP register indicates a channel number. The GTSTP register of each channel is a common register, and writing 1 to the GTSTP register in any channel and updating it can stop operation of the GTCNT counter in all the channels related to the position of the bit written with 1. A change in counter operation and the register value is not generated by writing 0 to the bit.

CSTOPn Bit (Channel n Count Stop) (n = 0 to 3)

This bit stops the GTCNT counter for channel n.

A read value indicates the state of the counter operation in a corresponding channel (inverted GTCR.CST bit). The bit with 0 indicates that the counter is in operation, while the bit with 1 indicates that the counter is stopped.

26.2.4 General PWM Timer Software Clear Register (GTCLR)

Address(es): GPTW0.GTCLR 000C 200Ch, GPTW1.GTCLR 000C 210Ch, GPTW2.GTCLR 000C 220Ch, GPTW3.GTCLR 000C 230Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	CCLR3	CCLR2	CCLR1	CCLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CCLR0	Channel 0 Counter Clear	0: Ignored. 1: The counter is to be cleared.	W
b1	CCLR1	Channel 1 Counter Clear		W
b2	CCLR2	Channel 2 Counter Clear		W
b3	CCLR3	Channel 3 Counter Clear		W
b31 to b4	—	Reserved	The write value should be 0.	W

The GTCLR register is a write-only register which sets clearing of the GTCNT counter.

A bit position of the GTCLR register indicates a channel number. The GTCLR register of each channel is a common register, and writing 1 to the GTCLR register in any channel and updating it changes to 0 of the GTCNT counter in all the channels related to the position of the bit written with 1. A change in counter operation and the register value is not generated by writing 0 to the bit.

CCLRn Bit (Channel n Counter Clear) (n = 0 to 3)

When the counting direction flag is set for decrementation (GTST.TUCF flag = 0) with sawtooth-wave mode selected in the GTCR.MD[2:0] bits, the value of the GTCNT counter becomes that of the corresponding GTPR register in response to writing 1 to the CCLRn bit. The value of the counter becomes 0000 0000h with other settings.

26.2.5 General PWM Timer Start Source Select Register (GTSSR)

Address(es): GPTW0.GTSSR 000C 2010h, GPTW1.GTSSR 000C 2110h, GPTW2.GTSSR 000C 2210h, GPTW3.GTSSR 000C 2310h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	CSTRT	—	—	—	—	—	—	—	SSELC H	SSELC G	SSELC F	SSELC E	SSELC D	SSELC C	SSELC B	SSELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SSCBF AH	SSCBF AL	SSCBR AH	SSCBR AL	SSCAF BH	SSCAF BL	SSCAR BH	SSCAR BL	SSGTR GDF	SSGTR GDR	SSGTR GCF	SSGTR GCR	SSGTR GBF	SSGTR GBR	SSGTR GAF	SSGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SSGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTETRGA signal. 1 0: The counter starts at a falling edge of the GTETRGA signal. 1 1: The counter starts at both edges of the GTETRGA signal.	R/W
b1	SSGTRGAF			R/W
b2	SSGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTETRGB signal. 1 0: The counter starts at a falling edge of the GTETRGB signal. 1 1: The counter starts at both edges of the GTETRGB signal.	R/W
b3	SSGTRGBF			R/W
b4	SSGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTETRGC signal. 1 0: The counter starts at a falling edge of the GTETRGC signal. 1 1: The counter starts at both edges of the GTETRGC signal.	R/W
b5	SSGTRGCF			R/W
b6	SSGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTETRGD signal. 1 0: The counter starts at a falling edge of the GTETRGD signal. 1 1: The counter starts at both edges of the GTETRGD signal.	R/W
b7	SSGTRGDF			R/W
b8	SSCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter starts at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter starts at a rising edge of the GTIOCnA signal.	R/W
b9	SSCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	SSCAFBL	GTIOCnA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOCnA signal is not used as a trigger to start counting. 0 1: The counter starts at a falling edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter starts at a falling edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter starts at a falling edge of the GTIOCnA signal.	R/W
b11	SSCAFBH			R/W
b12	SSCBRAL	GTIOCnB Signal Rising Edge Applying Condition Select	b13 b12 0 0: Rising edge of the GTIOCnB signal is not used as a trigger to start counting. 0 1: The counter starts at a rising edge of the GTIOCnB signal while the GTIOCnA pin is driven low. 1 0: The counter starts at a rising edge of the GTIOCnB signal while the GTIOCnA pin is driven high. 1 1: The counter starts at a rising edge of the GTIOCnB signal.	R/W
b13	SSCBRAH			R/W
b14	SSCBFAL	GTIOCnB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOCnB signal is not used as a trigger to start counting. 0 1: The counter starts at a falling edge of the GTIOCnB signal while the GTIOCnA pin is driven low. 1 0: The counter starts at a falling edge of the GTIOCnB signal while the GTIOCnA pin is driven high. 1 1: The counter starts at a falling edge of the GTIOCnB signal.	R/W
b15	SSCBFAH			R/W
b16	SSELCA	ELCA Event Source Count Start Enable	0: Disables count start by the ELCA event input 1: Enables count start by the ELCA event input	R/W
b17	SSELCB	ELCB Event Source Count Start Enable	0: Disables count start by the ELCB event input 1: Enables count start by the ELCB event input	R/W
b18	SSELCC	ELCC Event Source Count Start Enable	0: Disables count start by the ELCC event input 1: Enables count start by the ELCC event input	R/W
b19	SSELCD	ELCD Event Source Count Start Enable	0: Disables count start by the ELCD event input 1: Enables count start by the ELCD event input	R/W
b20	SSELCE	ELCE Event Source Count Start Enable	0: Disables count start by the ELCE event input 1: Enables count start by the ELCE event input	R/W
b21	SSELCF	ELCF Event Source Count Start Enable	0: Disables count start by the ELCF event input 1: Enables count start by the ELCF event input	R/W
b22	SSELCG	ELCG Event Source Count Start Enable	0: Disables count start by the ELCG event input 1: Enables count start by the ELCG event input	R/W
b23	SSELCH	ELCH Event Source Count Start Enable	0: Disables count start by the ELCH event input 1: Enables count start by the ELCH event input	R/W
b30 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	CSTRT	Software Source Count Start Enable	0: Disables count start by the GTSTR register 1: Enables count start by the GTSTR register	R/W

n = 0 to 3

The GTSSR register sets a count start source for the GTCNT counter.

Inputs from the GTETRGA, GTETRGB, GTETRG, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.6 General PWM Timer Stop Source Select Register (GTPSR)

Address(es): GPTW0.GTPSR 000C 2014h, GPTW1.GTPSR 000C 2114h, GPTW2.GTPSR 000C 2214h,
GPTW3.GTPSR 000C 2314h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	CSTOP	—	—	—	—	—	—	—	PSELC H	PSELC G	PSELC F	PSELC E	PSELC D	PSELC C	PSELC B	PSELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	PSCBF AH	PSCBF AL	PSCBR AH	PSCBR AL	PSCAF BH	PSCAF BL	PSCAR BH	PSCAR BL	PSGTR GDF	PSGTR GDR	PSGTR GCF	PSGTR GCR	PSGTR GBF	PSGTR GBR	PSGTR GAF	PSGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PSGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTETRGA signal. 1 0: The counter stops at a falling edge of the GTETRGA signal. 1 1: The counter stops at both edges of the GTETRGA signal.	R/W
b1	PSGTRGAF			R/W
b2	PSGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTETRGB signal. 1 0: The counter stops at a falling edge of the GTETRGB signal. 1 1: The counter stops at both edges of the GTETRGB signal.	R/W
b3	PSGTRGBF			R/W
b4	PSGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTETRGC signal. 1 0: The counter stops at a falling edge of the GTETRGC signal. 1 1: The counter stops at both edges of the GTETRGC signal.	R/W
b5	PSGTRGCF			R/W
b6	PSGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTETRGD signal. 1 0: The counter stops at a falling edge of the GTETRGD signal. 1 1: The counter stops at both edges of the GTETRGD signal.	R/W
b7	PSGTRGDF			R/W
b8	PSCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter stops at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter stops at a rising edge of the GTIOCnA signal.	R/W
b9	PSCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	PSCAFBL	GTIOCnA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOCnA signal is not used as a trigger to stop counting. 0 1: The counter stops at a falling edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter stops at a falling edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter stops at a falling edge of the GTIOCnA signal.	R/W
b11	PSCAFBH			R/W
b12	PSCBRAL	GTIOCnB Signal Rising Edge Applying Condition Select	b13 b12 0 0: Rising edge of the GTIOCnB signal is not used as a trigger to stop counting. 0 1: The counter stops at a rising edge of the GTIOCnB signal while the GTIOCnA pin is driven low. 1 0: The counter stops at a rising edge of the GTIOCnB signal while the GTIOCnA pin is driven high. 1 1: The counter stops at a rising edge of the GTIOCnB signal.	R/W
b13	PSCBRAH			R/W
b14	PSCBFAL	GTIOCnB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOCnB signal is not used as a trigger to stop counting. 0 1: The counter stops at a falling edge of the GTIOCnB signal while the GTIOCnA pin is driven low. 1 0: The counter stops at a falling edge of the GTIOCnB signal while the GTIOCnA pin is driven high. 1 1: The counter stops at a falling edge of the GTIOCnB signal.	R/W
b15	PSCBFAH			R/W
b16	PSELCA	ELCA Event Source Count Stop Enable	0: Disables count stop by the ELCA event input 1: Enables count stop by the ELCA event input	R/W
b17	PSELCB	ELCB Event Source Count Stop Enable	0: Disables count stop by the ELCB event input 1: Enables count stop by the ELCB event input	R/W
b18	PSELCC	ELCC Event Source Count Stop Enable	0: Disables count stop by the ELCC event input 1: Enables count stop by the ELCC event input	R/W
b19	PSELCD	ELCD Event Source Count Stop Enable	0: Disables count stop by the ELCD event input 1: Enables count stop by the ELCD event input	R/W
b20	PSELCE	ELCE Event Source Count Stop Enable	0: Disables count stop by the ELCE event input 1: Enables count stop by the ELCE event input	R/W
b21	PSELCF	ELCF Event Source Count Stop Enable	0: Disables count stop by the ELCF event input 1: Enables count stop by the ELCF event input	R/W
b22	PSELCG	ELCG Event Source Count Stop Enable	0: Disables count stop by the ELCG event input 1: Enables count stop by the ELCG event input	R/W
b23	PSELCH	ELCH Event Source Count Stop Enable	0: Disables count stop by the ELCH event input 1: Enables count stop by the ELCH event input	R/W
b30 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	CSTOP	Software Source Count Stop Enable	0: Disables count stop by the GTSTP register 1: Enables count stop by the GTSTP register	R/W

n = 0 to 3

The GTPSR register sets a count stop source for the GTCNT counter.

Inputs from the GTETRGA, GTETRGB, GTETRG, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.7 General PWM Timer Clear Source Select Register (GTCSR)

Address(es): GPTW0.GTCSR 000C 2018h, GPTW1.GTCSR 000C 2118h, GPTW2.GTCSR 000C 2218h,
GPTW3.GTCSR 000C 2318h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
CCLR	—	—	—	—	—	—	—	CSELC H	CSELC G	CSELC F	CSELC E	CSELC D	CSELC C	CSELC B	CSELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
CSCBF AH	CSCBF AL	CSCBR AH	CSCBR AL	CSCAF BH	CSCAF BL	CSCAR BH	CSCAR BL	CSGTR GDF	CSGTR GDR	CSGTR GCF	CSGTR GCR	CSGTR GBF	CSGTR GBR	CSGTR GAF	CSGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CSGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTETRGA signal. 1 0: The counter is cleared at a falling edge of the GTETRGA signal. 1 1: The counter is cleared at both edges of the GTETRGA signal.	R/W
b1	CSGTRGAF			R/W
b2	CSGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTETRGB signal. 1 0: The counter is cleared at a falling edge of the GTETRGB signal. 1 1: The counter is cleared at both edges of the GTETRGB signal.	R/W
b3	CSGTRGBF			R/W
b4	CSGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTETRGC signal. 1 0: The counter is cleared at a falling edge of the GTETRGC signal. 1 1: The counter is cleared at both edges of the GTETRGC signal.	R/W
b5	CSGTRGCF			R/W
b6	CSGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTETRGD signal. 1 0: The counter is cleared at a falling edge of the GTETRGD signal. 1 1: The counter is cleared at both edges of the GTETRGD signal.	R/W
b7	CSGTRGDF			R/W
b8	CSCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter is cleared at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter is cleared at a rising edge of the GTIOCnA signal.	R/W
b9	CSCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	CSCAFBL	GTIOcNA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOcNA signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven low. 1 0: The counter is cleared at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven high. 1 1: The counter is cleared at a falling edge of the GTIOcNA signal.	R/W
b11	CSCAFBH			R/W
b12	CSCBRAL	GTIOcNB Signal Rising Edge Applying Condition Select	b11 b12 0 0: Rising edge of the GTIOcNB signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is cleared at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is cleared at a rising edge of the GTIOcNB signal.	R/W
b13	CSCBRAH			R/W
b14	CSCBFAL	GTIOcNB Signal Falling Edge Applying Condition Select	b13 b14 0 0: Falling edge of the GTIOcNB signal is not used as a trigger to clear the counter. 0 1: The counter is cleared at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is cleared at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is cleared at a falling edge of the GTIOcNB signal.	R/W
b15	CSCBFAH			R/W
b16	CSELCA	ELCA Event Source Counter Clear Enable	0: Disables counter clear by the ELCA event input 1: Enables counter clear by the ELCA event input	R/W
b17	CSELCB	ELCB Event Source Counter Clear Enable	0: Disables counter clear by the ELCB event input 1: Enables counter clear by the ELCB event input	R/W
b18	CSELCC	ELCC Event Source Counter Clear Enable	0: Disables counter clear by the ELCC event input 1: Enables counter clear by the ELCC event input	R/W
b19	CSELCD	ELCD Event Source Counter Clear Enable	0: Disables counter clear by the ELCD event input 1: Enables counter clear by the ELCD event input	R/W
b20	CSELCE	ELCE Event Source Counter Clear Enable	0: Disables counter clear by the ELCE event input 1: Enables counter clear by the ELCE event input	R/W
b21	CSELCF	ELCF Event Source Counter Clear Enable	0: Disables counter clear by the ELCF event input 1: Enables counter clear by the ELCF event input	R/W
b22	CSELCG	ELCG Event Source Counter Clear Enable	0: Disables counter clear by the ELCG event input 1: Enables counter clear by the ELCG event input	R/W
b23	CSELCH	ELCH Event Source Counter Clear Enable	0: Disables counter clear by the ELCH event input 1: Enables counter clear by the ELCH event input	R/W
b30 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	CCLR	Software Source Counter Clear Enable	0: Disables counter clear by the GTCLR register 1: Enables counter clear by the GTCLR register	R/W

n = 0 to 3

The GTCSR register sets a counter clear source for the GTCNT counter.

Counter clearing can be executed whether the counter is running (the GTCR.CST bit = 1) or stopped (the GTCR.CST bit = 0).

Inputs from the GTETRGA, GTETRGB, GTETRG C, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.8 General PWM Timer Count-Up Source Select Register (GTUPSR)

Address(es): GPTW0.GTUPSR 000C 201Ch, GPTW1.GTUPSR 000C 211Ch, GPTW2.GTUPSR 000C 221Ch,
GPTW3.GTUPSR 000C 231Ch

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	USELC H	USELC G	USELC F	USELC E	USELC D	USELC C	USELC B	USELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
USCBF AH	USCBF AL	USCBR AH	USCBR AL	USCAF BH	USCAF BL	USCAR BH	USCAR BL	USGTR GDF	USGTR GDR	USGTR GCF	USGTR GCR	USGTR GBF	USGTR GBR	USGTR GAF	USGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	USGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTETRGA signal. 1 0: The counter is incremented at a falling edge of the GTETRGA signal. 1 1: The counter is incremented at both edges of the GTETRGA signal.	R/W
b1	USGTRGAF			R/W
b2	USGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTETRGB signal. 1 0: The counter is incremented at a falling edge of the GTETRGB signal. 1 1: The counter is incremented at both edges of the GTETRGB signal.	R/W
b3	USGTRGBF			R/W
b4	USGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTETRGC signal. 1 0: The counter is incremented at a falling edge of the GTETRGC signal. 1 1: The counter is incremented at both edges of the GTETRGC signal.	R/W
b5	USGTRGCF			R/W
b6	USGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTETRGD signal. 1 0: The counter is incremented at a falling edge of the GTETRGD signal. 1 1: The counter is incremented at both edges of the GTETRGD signal.	R/W
b7	USGTRGDF			R/W
b8	USCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter is incremented at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter is incremented at a rising edge of the GTIOCnA signal.	R/W
b9	USCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	USCAFBL	GTIOcNA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOcNA signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven low. 1 0: The counter is incremented at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven high. 1 1: The counter is incremented at a falling edge of the GTIOcNA signal.	R/W
b11	USCAFBH			R/W
b12	USCBRAL	GTIOcNB Signal Rising Edge Applying Condition Select	b13 b12 0 0: Rising edge of the GTIOcNB signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is incremented at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is incremented at a rising edge of the GTIOcNB signal.	R/W
b13	USCBRAH			R/W
b14	USCBFAL	GTIOcNB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOcNB signal is not used as a trigger to increment the counter. 0 1: The counter is incremented at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is incremented at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is incremented at a falling edge of the GTIOcNB signal.	R/W
b15	USCBFAH			R/W
b16	USELCA	ELCA Event Source Count-Up Enable	0: Disables count-up by the ELCA event input 1: Enables count-up by the ELCA event input	R/W
b17	USELCB	ELCB Event Source Count-Up Enable	0: Disables count-up by the ELCB event input 1: Enables count-up by the ELCB event input	R/W
b18	USELCC	ELCC Event Source Count-Up Enable	0: Disables count-up by the ELCC event input 1: Enables count-up by the ELCC event input	R/W
b19	USELCD	ELCD Event Source Count-Up Enable	0: Disables count-up by the ELCD event input 1: Enables count-up by the ELCD event input	R/W
b20	USELCE	ELCE Event Source Count-Up Enable	0: Disables count-up by the ELCE event input 1: Enables count-up by the ELCE event input	R/W
b21	USELCF	ELCF Event Source Count-Up Enable	0: Disables count-up by the ELCF event input 1: Enables count-up by the ELCF event input	R/W
b22	USELCG	ELCG Event Source Count-Up Enable	0: Disables count-up by the ELCG event input 1: Enables count-up by the ELCG event input	R/W
b23	USELCH	ELCH Event Source Count-Up Enable	0: Disables count-up by the ELCH event input 1: Enables count-up by the ELCH event input	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

n = 0 to 3

The GTUPSR register sets a count-up source for the GTCNT counter.

When at least one bit among bits in the GTUPSR register is set as 1, counting of the GTCNT counter by the count clock set by the GTCR.TPCS[3:0] bits becomes invalid, and the count-up by a source set as 1 by the GTUPSR register is executed.

Number of increment in counting is one even when multiple sources are generated simultaneously.

Inputs from the GTETRGA, GTETRGB, GTETRG, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.9 General PWM Timer Count-Down Source Select Register (GTDNSR)

Address(es): GPTW0.GTDNSR 000C 2020h, GPTW1.GTDNSR 000C 2120h, GPTW2.GTDNSR 000C 2220h, GPTW3.GTDNSR 000C 2320h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	DSELC H	DSELC G	DSELC F	DSELC E	DSELC D	DSELC C	DSELC B	DSELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
DSCBF AH	DSCBF AL	DSCBR AH	DSCBR AL	DSCAF BH	DSCAF BL	DSCAR BH	DSCAR BL	DSGTR GDF	DSGTR GDR	DSGTR GCF	DSGTR GCR	DSGTR GBF	DSGTR GBR	DSGTR GAF	DSGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DSGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTETRGA signal. 1 0: The counter is decremented at a falling edge of the GTETRGA signal. 1 1: The counter is decremented at both edges of the GTETRGA signal.	R/W
b1	DSGTRGAF			R/W
b2	DSGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTETRGB signal. 1 0: The counter is decremented at a falling edge of the GTETRGB signal. 1 1: The counter is decremented at both edges of the GTETRGB signal.	R/W
b3	DSGTRGBF			R/W
b4	DSGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTETRGC signal. 1 0: The counter is decremented at a falling edge of the GTETRGC signal. 1 1: The counter is decremented at both edges of the GTETRGC signal.	R/W
b5	DSGTRGCF			R/W
b6	DSGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTETRGD signal. 1 0: The counter is decremented at a falling edge of the GTETRGD signal. 1 1: The counter is decremented at both edges of the GTETRGD signal.	R/W
b7	DSGTRGDF			R/W
b8	DSCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter is decremented at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter is decremented at a rising edge of the GTIOCnA signal.	R/W
b9	DSCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	DSCAFBL	GTIOcNA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOcNA signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven low. 1 0: The counter is decremented at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven high. 1 1: The counter is decremented at a falling edge of the GTIOcNA signal.	R/W
b11	DSCAFBH			R/W
b12	DSCBRAL	GTIOcNB Signal Rising Edge Applying Condition Select	b11 b12 0 0: Rising edge of the GTIOcNB signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is decremented at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is decremented at a rising edge of the GTIOcNB signal.	R/W
b13	DSCBRAH			R/W
b14	DSCBFAL	GTIOcNB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOcNB signal is not used as a trigger to decrement the counter. 0 1: The counter is decremented at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter is decremented at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter is decremented at a falling edge of the GTIOcNB signal.	R/W
b15	DSCBFAH			R/W
b16	DSELCA	ELCA Event Source Count-Down Enable	0: Disables count-down by the ELCA event input 1: Enables count-down by the ELCA event input	R/W
b17	DSELCB	ELCB Event Source Count-Down Enable	0: Disables count-down by the ELCB event input 1: Enables count-down by the ELCB event input	R/W
b18	DSELC	ELCC Event Source Count-Down Enable	0: Disables count-down by the ELCC event input 1: Enables count-down by the ELCC event input	R/W
b19	DSELCD	ELCD Event Source Count-Down Enable	0: Disables count-down by the ELCD event input 1: Enables count-down by the ELCD event input	R/W
b20	DSELCE	ELCE Event Source Count-Down Enable	0: Disables count-down by the ELCE event input 1: Enables count-down by the ELCE event input	R/W
b21	DSELCF	ELCF Event Source Count-Down Enable	0: Disables count-down by the ELCF event input 1: Enables count-down by the ELCF event input	R/W
b22	DSELCG	ELCG Event Source Count-Down Enable	0: Disables count-down by the ELCG event input 1: Enables count-down by the ELCG event input	R/W
b23	DSELCH	ELCH Event Source Count-Down Enable	0: Disables count-down by the ELCH event input 1: Enables count-down by the ELCH event input	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

n = 0 to 3

The GTDNSR register sets a count-down source for the GTCNT counter.

When at least one bit among bits in the GTDNSR register is set as 1, counting of the GTCNT counter by the count clock set by the GTCR.TPCS[3:0] bits becomes invalid, and the count-down by a source set as 1 by the GTDNSR register is executed.

Number of decrement in counting is one even when multiple sources are generated simultaneously.

Inputs from the GTETRGA, GTETRGA, GTETRGB, GTETRG, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.10 General PWM Timer Input Capture Source Select Register A (GTICASR)

Address(es): GPTW0.GTICASR 000C 2024h, GPTW1.GTICASR 000C 2124h, GPTW2.GTICASR 000C 2224h, GPTW3.GTICASR 000C 2324h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	ASELC H	ASELC G	ASELC F	ASELC E	ASELC D	ASELC C	ASELC B	ASELC A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
ASCBF AH	ASCBF AL	ASCBR AH	ASCBR AL	ASCAF BH	ASCAF BL	ASCAR BH	ASCAR BL	ASGTR GDF	ASGTR GDR	ASGTR GCF	ASGTR GCR	ASGTR GBF	ASGTR GBR	ASGTR GAF	ASGTR GAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ASGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTETRGA signal. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTETRGA signal. 1 1: The counter value is captured in the GTCCRA register at both edges of the GTETRGA signal.	R/W
b1	ASGTRGAF			R/W
b2	ASGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTETRGB signal. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTETRGB signal. 1 1: The counter value is captured in the GTCCRA register at both edges of the GTETRGB signal.	R/W
b3	ASGTRGBF			R/W
b4	ASGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTETRGC signal. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTETRGC signal. 1 1: The counter value is captured in the GTCCRA register at both edges of the GTETRGC signal.	R/W
b5	ASGTRGCF			R/W
b6	ASGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTETRGD signal. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTETRGD signal. 1 1: The counter value is captured in the GTCCRA register at both edges of the GTETRGD signal.	R/W
b7	ASGTRGDF			R/W
b8	ASCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter value is captured in the GTCCRA register at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter value is captured in the GTCCRA register at a rising edge of the GTIOCnA signal.	R/W
b9	ASCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	ASCAFBL	GTIOcNA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOcNA signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven low. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven high. 1 1: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNA signal.	R/W
b11	ASCAFBH			R/W
b12	ASCBRAL	GTIOcNB Signal Rising Edge Applying Condition Select	b13 b12 0 0: Rising edge of the GTIOcNB signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter value is captured in the GTCCRA register at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter value is captured in the GTCCRA register at a rising edge of the GTIOcNB signal.	R/W
b13	ASCBRAH			R/W
b14	ASCBFAL	GTIOcNB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOcNB signal is not used as a trigger to capture the counter value in the GTCCRA register. 0 1: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter value is captured in the GTCCRA register at a falling edge of the GTIOcNB signal.	R/W
b15	ASCBFAH			R/W
b16	ASELCA	ELCA Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCA event input 1: Enables GTCCRA input capture by the ELCA event input	R/W
b17	ASELCB	ELCB Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCB event input 1: Enables GTCCRA input capture by the ELCB event input	R/W
b18	ASELCC	ELCC Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCC event input 1: Enables GTCCRA input capture by the ELCC event input	R/W
b19	ASELCD	ELCD Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCD event input 1: Enables GTCCRA input capture by the ELCD event input	R/W
b20	ASELCE	ELCE Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCE event input 1: Enables GTCCRA input capture by the ELCE event input	R/W
b21	ASELCF	ELCF Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCF event input 1: Enables GTCCRA input capture by the ELCF event input	R/W
b22	ASELCG	ELCG Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCG event input 1: Enables GTCCRA input capture by the ELCG event input	R/W

Bit	Symbol	Bit Name	Description	R/W
b23	ASELCH	ELCH Event Source GTCCRA Input Capture Enable	0: Disables GTCCRA input capture by the ELCH event input 1: Enables GTCCRA input capture by the ELCH event input	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

n = 0 to 3

The GTICASR register sets an input capture source for the GTCCRA register.

When at least one bit among bits in the GTICASR register is set to 1, input capture operation making the GTCCRA register as an input capture register is performed.

Inputs from the GTETRGA, GTETRGB, GTETRGC, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.11 General PWM Timer Input Capture Source Select Register B (GTICBSR)

Address(es): GPTW0.GTICBSR 000C 2028h, GPTW1.GTICBSR 000C 2128h, GPTW2.GTICBSR 000C 2228h,
GPTW3.GTICBSR 000C 2328h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	BSELC H	BSELC G	BSELC F	BSELC E	BSELC D	BSELC C	BSELC B	BSELC A
Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
BSCBF AH	BSCBF AL	BSCBR AH	BSCBR AL	BSCAF BH	BSCAF BL	BSCAR BH	BSCAR BL	BSGTR GDF	BSGTR GDR	BSGTR GCF	BSGTR GCR	BSGTR GBF	BSGTR GBR	BSGTR GAF	BSGTR GAR
Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															

Bit	Symbol	Bit Name	Description	R/W
b0	BSGTRGAR	GTETRGA Signal Edge Select	b1 b0 0 0: The GTETRGA signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTETRGA signal. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTETRGA signal. 1 1: The counter value is captured in the GTCCRB register at both edges of the GTETRGA signal.	R/W
b1	BSGTRGAF			R/W
b2	BSGTRGBR	GTETRGB Signal Edge Select	b3 b2 0 0: The GTETRGB signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTETRGB signal. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTETRGB signal. 1 1: The counter value is captured in the GTCCRB register at both edges of the GTETRGB signal.	R/W
b3	BSGTRGBF			R/W
b4	BSGTRGCR	GTETRGC Signal Edge Select	b5 b4 0 0: The GTETRGC signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTETRGC signal. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTETRGC signal. 1 1: The counter value is captured in the GTCCRB register at both edges of the GTETRGC signal.	R/W
b5	BSGTRGCF			R/W
b6	BSGTRGDR	GTETRGD Signal Edge Select	b7 b6 0 0: The GTETRGD signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTETRGD signal. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTETRGD signal. 1 1: The counter value is captured in the GTCCRB register at both edges of the GTETRGD signal.	R/W
b7	BSGTRGDF			R/W
b8	BSCARBL	GTIOCnA Signal Rising Edge Applying Condition Select	b9 b8 0 0: Rising edge of the GTIOCnA signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven low. 1 0: The counter value is captured in the GTCCRB register at a rising edge of the GTIOCnA signal while the GTIOCnB pin is driven high. 1 1: The counter value is captured in the GTCCRB register at a rising edge of the GTIOCnA signal.	R/W
b9	BSCARBH			R/W

Bit	Symbol	Bit Name	Description	R/W
b10	BSCAFBL	GTIOcNA Signal Falling Edge Applying Condition Select	b11 b10 0 0: Falling edge of the GTIOcNA signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven low. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNA signal while the GTIOcNB pin is driven high. 1 1: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNA signal.	R/W
b11	BSCAFBH			R/W
b12	BSCBRAL	GTIOcNB Signal Rising Edge Applying Condition Select	b13 b12 0 0: Rising edge of the GTIOcNB signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter value is captured in the GTCCRB register at a rising edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter value is captured in the GTCCRB register at a rising edge of the GTIOcNB signal.	R/W
b13	BSCBRAH			R/W
b14	BSCBFAL	GTIOcNB Signal Falling Edge Applying Condition Select	b15 b14 0 0: Falling edge of the GTIOcNB signal is not used as a trigger to capture the counter value in the GTCCRB register. 0 1: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven low. 1 0: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNB signal while the GTIOcNA pin is driven high. 1 1: The counter value is captured in the GTCCRB register at a falling edge of the GTIOcNB signal.	R/W
b15	BSCBFAH			R/W
b16	BSELCA	ELCA Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCA event input 1: Enables GTCCRB input capture by the ELCA event input	R/W
b17	BSELCB	ELCB Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCB event input 1: Enables GTCCRB input capture by the ELCB event input	R/W
b18	BSELCC	ELCC Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCC event input 1: Enables GTCCRB input capture by the ELCC event input	R/W
b19	BSELCD	ELCD Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCD event input 1: Enables GTCCRB input capture by the ELCD event input	R/W
b20	BSELCE	ELCE Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCE event input 1: Enables GTCCRB input capture by the ELCE event input	R/W
b21	BSELCF	ELCF Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCF event input 1: Enables GTCCRB input capture by the ELCF event input	R/W
b22	BSELCG	ELCG Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCG event input 1: Enables GTCCRB input capture by the ELCG event input	R/W

Bit	Symbol	Bit Name	Description	R/W
b23	BSELCH	ELCH Event Source GTCCRB Input Capture Enable	0: Disables GTCCRB input capture by the ELCH event input 1: Enables GTCCRB input capture by the ELCH event input	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

n = 0 to 3

The GTICBSR register sets an input capture source for the GTCCRB register.

When at least one bit among bits in the GTICBSR register is set to 1, input capture operation making the GTCCRB register as an input capture register is performed.

Inputs from the GTETRGA, GTETRGB, GTETRGC, and GTETRGD pins are input to the GPTW via the POEG. Set the polarity of these signals with the POEG.

26.2.12 General PWM Timer Control Register (GTCR)

Address(es): GPTW0.GTCR 000C 202Ch, GPTW1.GTCR 000C 212Ch, GPTW2.GTCR 000C 222Ch,
GPTW3.GTCR 000C 232Ch

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	CST	Count Start	0: Count operation is stopped 1: Count operation is started	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	ICDS	Input Capture Operation Select at Count Stop	0: Input capture is operated at count stop 1: Input capture is not operated at count stop	R/W
b15 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b16	MD[2:0]	Mode Select	b18 b16 0 0 0: Sawtooth-wave PWM mode (single buffer or double buffer possible) 0 0 1: Sawtooth-wave one-shot pulse mode (fixed buffer operation) 0 1 0: Setting prohibited 0 1 1: Setting prohibited 1 0 0: Triangle-wave PWM mode 1 (32-bit transfer at trough) (single buffer or double buffer possible) 1 0 1: Triangle-wave PWM mode 2 (32-bit transfer at crest and trough) (single buffer or double buffer possible) 1 1 0: Triangle-wave PWM mode 3 (64-bit transfer at trough) (fixed buffer operation) 1 1 1: Setting prohibited	R/W
b22 to b19	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b26 to b23	TPCS[3:0]	Timer Prescaler Select	b26 b23 0 0 0 0: PCLKA 0 0 0 1: PCLKA/2 0 0 1 0: PCLKA/4 0 0 1 1: PCLKA/8 0 1 0 0: PCLKA/16 0 1 0 1: PCLKA/32 0 1 1 0: PCLKA/64 0 1 1 1: Setting prohibited 1 0 0 0: PCLKA/256 1 0 0 1: Setting prohibited 1 0 1 0: PCLKA/1024 1 0 1 1: Setting prohibited 1 1 0 0: GTETRGA (via the POEG) 1 1 0 1: GTETRGB (via the POEG) 1 1 1 0: GTETRGC (via the POEG) 1 1 1 1: GTETRGD (via the POEG)	R/W
b31 to b27	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTCR register controls the GTCNT counter.

CST Bit (Count Start)

This bit controls start or stop for the GTCNT counter.

[Setting conditions]

- When 1 is written to a bit related to a channel number for the GTSTR register while the GTSSR.CSTRT bit is set to 1
- When an ELC event input, an external trigger, or a condition for the GTIOCnA pin input and GTIOCnB pin input (n = 0 to 3), which is enabled as a count start source by the GTSSR register, is generated
- 1 is written by software

[Clearing conditions]

- When 1 is written to a bit related to a channel number for the GTSTP register while the GTPSR.CSTOP bit is set to 1
- When an ELC event input, an external trigger, or a condition for the GTIOCnA pin input and GTIOCnB pin input, which is enabled as a count stop source by the GTPSR register, is generated
- 0 is written by software

ICDS Bit (Input Capture Operation Select at Count Stop)

This bit selects Input capture operation at count stop when the input capture function is selected.

MD[2:0] Bits (Mode Select)

These bits select the GPTW operating mode.

Only the MD[2] bit is valid at input capture.

Counting in sawtooth-wave mode is performed with 0 for the MD[2] bit, counting in triangle-wave mode is performed with 1 for the MD[2] bit.

Set the MD[2:0] bits while the GTCNT counter operation is stopped.

During the event count operation (when at least one bit among the bits for the GTUPSR and GTDNSR registers is set to 1), setting of the MD[2:0] bits are ignored, where counting in sawtooth-wave or triangle modes is not performed. Instead, up-counting or down-counting by a source set by the GTUPSR and GTDNSR registers is performed.

TPCS[3:0] Bits (Timer Prescaler Select)

These bits select a clock for the GTCNT counter. The clock source can be selected independently for each channel.

Set the TPCS[3:0] bits while the GTCNT counter operation is stopped.

When the GTETRGA, GTETRGB, GTETRGC, or GTETRGD is selected, output for the POEG at rising becomes a clock source. Set the polarity of these signals with the POEG.

26.2.13 General PWM Timer Count Direction and Duty Setting Register (GTUDDTYC)

Address(es): GPTW0.GTUDDTYC 000C 2030h, GPTW1.GTUDDTYC 000C 2130h, GPTW2.GTUDDTYC 000C 2230h, GPTW3.GTUDDTYC 000C 2330h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	OBDTYR	OBDTYF	OBDTY[1:0]	—	—	—	—	—	OADTYR	OADTYF	OADTY[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	UDF	UD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	UD	Count Direction Setting	0: GTCNT counts down. 1: GTCNT counts up.	R/W
b1	UDF	Forcible Count Direction Setting	0: Count direction is not forcibly set. 1: Count direction is forcibly set.	R/W
b15 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b17, b16	OADTY[1:0]	GTIOcNA Pin Output Duty Setting	b17b16 0 x: Compare matches determine the duty cycle of the output on the GTIOcNA pin. 1 0: The duty cycle of the output on the GTIOcNA pin is 0%. 1 1: The duty cycle of the output on the GTIOcNA pin is 100%.	R/W
b18	OADTYF	GTIOcNA Pin Output Duty Forced Setting	0: Duty of the GTIOcNA pin output is not forcibly set. 1: Duty of the GTIOcNA pin output is forcibly set.	R/W
b19	OADTYR	Output after Release of GTIOcNA Pin Output 0%/100% Duty Cycle Settings	0: The function selected by the GTIOA[3:2] bits is applied to the output value when the duty cycle is set after release from the 0 or 100% duty-cycle setting. 1: The function selected by the GTIOA[3:2] bits is applied to the compare match output value which is masked after release from the 0 or 100% duty-cycle setting.	R/W
b23 to b20	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b25, b24	OBDTY[1:0]	GTIOcNB Pin Output Duty Setting	b25b24 0 x: Compare matches determine the duty cycle of the output on the GTIOcNB pin. 1 0: The duty cycle of the output on the GTIOcNB pin is 0%. 1 1: The duty cycle of the output on the GTIOcNB pin is 100%.	R/W
b26	OBDTYF	GTIOcNB Pin Output Duty Forced Setting	0: Duty of the GTIOcNB pin output is not forcibly set. 1: Duty of the GTIOcNB pin output is forcibly set.	R/W
b27	OBDTYR	Output after Release of GTIOcNB Pin Output 0%/100% Duty Cycle Settings	0: The function selected by the GTIOB[3:2] bits is applied to the output value when the duty cycle is set after release from the 0 or 100% duty-cycle setting. 1: The function selected by the GTIOB[3:2] bits is applied to the compare match output value which is masked after release from the 0 or 100% duty-cycle setting.	R/W
b31 to b28	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

n = 0 to 3

The GTUDDTYC register sets the count direction (up-counting/down-counting) for the GTCNT counter and configures duty setting for output from the GTIOcNA and GTIOcNB pins.

The setting is invalid during the event count operation.

(1) Setting of Count Direction

- In sawtooth-wave mode

When the UD bit value is set to 0 during up-counting, the count direction is changed at an overflow (count clock when the GTCNT counter value is equal to the GTPR register value). When the UD bit value is set to 1 during down-counting, the count direction is changed at an underflow (count clock when the GTCNT counter value is equal to 0000 0000h). If the UD bit value is changed from 1 to 0 with the UDF bit being 0 and while counting is stopped, the counter starts up-counting and the count direction is changed at an overflow (count clock when the GTCNT counter value is equal to the GTPR register value).

If the UD bit value is changed from 0 to 1 with the UDF bit being 0 and while counting is stopped, the counter starts down-counting and the count direction is changed at an underflow (count clock when the GTCNT counter value is equal to 0000 0000h).

When the UDF bit is set to 1 while counting is stopped, the UD bit value at that time is reflected in the count direction when counting starts.

- In triangle-wave mode

Even if the UD bit value is changed during counting, the change will not be reflected in the count direction.

If the UD bit value is modified while the UDF bit is 0 and counting is stopped, the change will not be reflected in the count direction when counting starts. If the UDF bit is set to 1 while counting is stopped, the UD bit value at that time is reflected in the count direction when counting starts.

UD Bit (Count Direction Setting)

This bit sets the count direction (up-counting or down-counting) for the GTCNT counter.

UDF Bit (Forcible Count Direction Setting)

This bit forcibly sets the count direction when the GTCNT counter starts operation as the UD bit value.

Only 0 should be written to this bit during count operation.

When 1 has been written to this bit while counting is stopped, this bit should be returned to 0 before counting starts.

(2) Setting of Output Duty

- In sawtooth-wave mode

When the OADTY[1:0]/OBDTY[1:0] bits are changed during up-count operation, the duty setting changed at an overflow is reflected.

When the OADTY[1:0]/OBDTY[1:0] bits are changed during down-count operation, the duty setting changed at an underflow is reflected.

When the OADTYF/OBDTYF bit is 0 while count operation is stopped and the OADTY[1:0]/OBDTY[1:0] bits are changed, the change in the duty-cycle setting is not reflected in the first count operation, but the change is reflected on an overflow in the case of counting up and on an underflow in the case of counting down.

When the OADTYF/OBDTYF bit is 1 while count operation is stopped and the OADTY[1:0]/OBDTY[1:0] bits are changed, the change in the duty-cycle setting is immediately reflected during the first count operation.

- In triangle-wave mode

When the OADTY[1:0]/OBDTY[1:0] bits are changed during count operation, the duty-cycle setting changed at an underflow is reflected.

When the OADTYF/OBDTYF bit is 0 during count operation and the OADTY[1:0]/OBDTY[1:0] bits are changed, the duty-cycle setting changed during the first count operation is not reflected, the duty-cycle setting changed at an underflow is reflected.

When the OADTYF/OBDTYF bit is 1 while count operation is stopped and the OADTY[1:0]/OBDTY[1:0] bits are changed, the change in the duty-cycle setting is immediately reflected during the first count operation.

OmDTY[1:0] Bits (GTIOCnm Pin Output Duty Setting) (n = 0 to 3; m = A, B)

These bits set the output of the duty cycle (0%, 100% or compare match control) on the GTIOCnm pin.

OmDTYF Bit (GTIOCnm Pin Output Duty Forced Setting) (n = 0 to 3; m = A, B)

This bit forcibly specifies the duty cycle at the start of the GTCNT counter in the OmDTY[1:0] bits.

Always write 0 to the bit during count operation.

When writing 1 to the bit while count operation is stopped, set the bit back to 0 by the end of the first cycle after the count operation started and specify the next cycle.

**OmDTYR Bit (Output after Release of GTIOCnm Pin Output 0%/100% Duty Cycle Settings)
(n = 0 to 3; m = A, B)**

When the setting of a GTIOCnm pin for a 0% or 100% duty cycle is forcibly changed due to a compare match, the output level on the pin is maintained after the period has elapsed if the value of the GTIOR.GTIOm[3:2] bits is 00b, but maintenance and a value to control toggling are selected after the period has elapsed if the output has become toggled and the setting is 11b.

The GPTW internally continues to perform compare match operation during duty-cycle 0% or 100% operation. When the OmDTYR bit is 1, the value after the period has elapsed due this compare match operation is target for the GTIOm[3:2] bits.

26.2.14 General PWM Timer I/O Control Register (GTIOR)

Address(es): GPTW0.GTIOR 000C 2034h, GPTW1.GTIOR 000C 2134h, GPTW2.GTIOR 000C 2234h,
GPTW3.GTIOR 000C 2334h

b31b30b29b28b27b26b25b24b23b22b21b20b19b18b17b16																
	NFC SB[1:0]		NFBEN	—	—	OBDF[1:0]		OBE	OBHLD	OBDFLT	—	GTIOB[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
b15b14b13b12b11b10b9b8b7b6b5b4b3b2b1b0																
	NFCSA[1:0]		NFAEN	—	—	OADF[1:0]		OAE	OAHL D	OADFLT	—	GTIOA[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	GTIOA[4:0]	GTIOCnA Pin Function Select	See Table 26.4.	R/W
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6	OADFLT	GTIOCnA Pin Output Value Setting at the Count Stop	0: The GTIOCnA pin outputs low when counting is stopped. 1: The GTIOCnA pin outputs high when counting is stopped.	R/W
b7	OAHL D	GTIOCnA Pin Output Retention at the Start/Stop Count	0: The GTIOCnA pin output level at start/stop of counting depends on the register setting. 1: The GTIOCnA pin output level is retained at start/stop of counting.	R/W
b8	OAE	GTIOCnA Pin Output Enable	0: Pin output is disabled 1: Pin output is enabled	R/W
b10, b9	OADF[1:0]	GTIOCnA Pin Negate Value Setting	b10 b9 0 0: None of the following sources is specified 0 1: GTIOCnA pin is placed in the Hi-Z state in response to control for output negation. 1 0: GTIOCnA pin is set to 0 in response to control for output negation. 1 1: GTIOCnA pin is set to 1 in response to control for output negation.	R/W
b12, b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	NFAEN	GTIOCnA Pin Input Noise Filter Enable	0: The noise filter for GTIOCnA pin input is disabled 1: The noise filter for GTIOCnA pin input is enabled	R/W
b15, b14	NFCSA[1:0]	GTIOCnA Pin Input Noise Filter Sampling Clock Select	b15b14 0 0: PCLKA 0 1: PCLKA/4 1 0: PCLKA/16 1 1: PCLKA/64	R/W
b20 to b16	GTIOB[4:0]	GTIOCnB Pin Function Select	See Table 26.4.	R/W
b21	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b22	OBDFLT	GTIOCnB Pin Output Value Setting at the Count Stop	0: The GTIOCnB pin outputs low when counting is stopped. 1: The GTIOCnB pin outputs high when counting is stopped.	R/W
b23	OBHLD	GTIOCnB Pin Output Retention at the Start/Stop Count	0: The GTIOCnB pin output level at start/stop of counting depends on the register setting. 1: The GTIOCnB pin output level is retained at start/stop of counting.	R/W
b24	OBE	GTIOCnB Pin Output Enable	0: Pin output is disabled 1: Pin output is enabled	R/W
b26, b25	OBDF[1:0]	GTIOCnB Pin Negate Value Setting	b26b25 0 0: None of the following sources is specified 0 1: GTIOCnB pin is placed in the Hi-Z state in response to control for output negation. 1 0: GTIOCnB pin is set to 0 in response to control for output negation. 1 1: GTIOCnB pin is set to 1 in response to control for output negation.	R/W

Bit	Symbol	Bit Name	Description	R/W
b28, b27	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b29	NFBEN	GTIOCnB Pin Input Noise Filter Enable	0: The noise filter for GTIOCnB pin input is disabled 1: The noise filter for GTIOCnB pin input is enabled	R/W
b31, b30	NFCSB[1:0]	GTIOCnB Pin Input Noise Filter Sampling Clock Select	b31b30 0 0: PCLKA 0 1: PCLKA/4 1 0: PCLKA/16 1 1: PCLKA/64	R/W

n = 0 to 3

GTIOR sets the functions of the GTIOCnA and GTIOCnB pins.

GTIOm[4:0] Bits (GTIOCnm Pin Function Select) (n = 0 to 3; m = A, B)

These bits select the GTIOCnm pin function. For details, see Table 26.4.

OmDFLT Bit (GTIOCnm Pin Output Value Setting at the Count Stop) (n = 0 to 3; m = A, B)

This bit sets whether the GTIOCnm pin outputs high or low when counting is stopped.

OmHLD Bit (GTIOCnm Pin Output Retention at the Start/Stop Count) (n = 0 to 3; m = A, B)

This bit specifies whether the GTIOCnm pin output level is retained or the level depends on the register setting when counting is started or stopped.

[Setting conditions]

- The output is retained when counting starts or stops.

[Clearing conditions]

- The value specified by the GTIOm[4] bit is output when counting starts.
- The value specified by the OmDFLT bit is output when counting stops.
- If the OmDFLT bit is modified while counting is stopped, it is immediately reflected in the output.

OmE Bit (GTIOCnm Pin Output Enable) (m = A, B)

This bit selects whether the GTIOCnm pin output is performed or not.

When the GTCCRm register is used as an input capture register (at least one bit among the bits for the GTICmSR register is set to 1), the GTIOCnm pin output is not performed regardless of the setting of the OmE bit.

OmDF[1:0] Bits (GTIOCnm Pin Negate Value Setting) (m = A, B)

Select the value for output from the GTIOCnm pin in response to a request to disable output from the POEG.

NFmEN Bit (GTIOCnm Pin Input Noise Filter Enable) (m = A, B)

This bit sets enable or disable of the noise filter function for the GTIOCnm pin input. Switch these bits while the corresponding pin function for the timer I/O control register is set as the output compare function, because unintended internal edge may be generated when these bits are switched.

NFCSm[1:0] Bits (GTIOCnm Pin Input Noise Filter Sampling Clock Select) (m = A, B)

These bits set a sampling clock for the noise filter to the GTIOCnm pin input. Set to the input capture function after waiting two cycles for the sampling clock cycle following the setting of these bits.

Table 26.4 Settings of GTIOA[4:0] Bits (GTIOB[4:0] Bits)

GTIOA[4:0] (GTIOB[4:0]) Bits					Function		
b4	b3	b2	b1	b0	b4	b3, b2	b1, b0
0	0	0	0	0	Initial output is Low.	Output retained at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
0	0	0	0	1			Low output at GTCCRA/GTCCRB compare match
0	0	0	1	0			High output at GTCCRA/GTCCRB compare match
0	0	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	0	1	0	0		Low output at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
0	0	1	0	1			Low output at GTCCRA/GTCCRB compare match
0	0	1	1	0			High output at GTCCRA/GTCCRB compare match
0	0	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	1	0	0	0		High output at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
0	1	0	0	1			Low output at GTCCRA/GTCCRB compare match
0	1	0	1	0			High output at GTCCRA/GTCCRB compare match
0	1	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	1	1	0	0		Output toggled at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
0	1	1	0	1			Low output at GTCCRA/GTCCRB compare match
0	1	1	1	0			High output at GTCCRA/GTCCRB compare match
0	1	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	0	0	0	0	Initial output is High.	Output retained at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
1	0	0	0	1			Low output at GTCCRA/GTCCRB compare match
1	0	0	1	0			High output at GTCCRA/GTCCRB compare match
1	0	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	0	1	0	0		Low output at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
1	0	1	0	1			Low output at GTCCRA/GTCCRB compare match
1	0	1	1	0			High output at GTCCRA/GTCCRB compare match
1	0	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	1	0	0	0		High output at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
1	1	0	0	1			Low output at GTCCRA/GTCCRB compare match
1	1	0	1	0			High output at GTCCRA/GTCCRB compare match
1	1	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	1	1	0	0		Output toggled at the end of a cycle	Output retained at GTCCRA/GTCCRB compare match
1	1	1	0	1			Low output at GTCCRA/GTCCRB compare match
1	1	1	1	0			High output at GTCCRA/GTCCRB compare match
1	1	1	1	1			Output toggled at GTCCRA/GTCCRB compare match

Note: In sawtooth-wave mode, "end of a cycle" refers to an overflow (the value of the GTCNT counter changing from that of the GTPR register to 0000 0000h in up-counting), an underflow (the value of the GTCNT counter changing from 0000 0000h to that of the GTPR register in down-counting), or counter clearing. It refers to a trough in triangle-wave mode (the value of the GTCNT counter changing from 0000 0000h to 0000 0001h).

Note: When the timing of an end of a cycle and the timing of a GTCCRA/GTCCRB register compare match are the same in a compare-match operation, the b3 and b2 settings are given priority in sawtooth-wave PWM mode, and the b1 and b0 settings are given priority in any other mode.

Note: During the event count operation (when at least one bit among the bits for the GTUPSR and GTDNSR registers is set to 1), the b3 and b2 settings become invalid.

26.2.15 General PWM Timer Interrupt Output Setting Register (GTINTAD)

Address(es): GPTW0.GTINTAD 000C 2038h, GPTW1.GTINTAD 000C 2138h, GPTW2.GTINTAD 000C 2238h, GPTW3.GTINTAD 000C 2338h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	GRPAB L	GRPAB H	GRPDTE	—	—	GRP[1:0]		—	—	—	—	ADTRB DEN	ADTRB UEN	ADTRA DEN	ADTRA UEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	GTINTPR[1:0]		GTINT F	GTINT E	GTINT D	GTINT C	GTINT B	GTINT A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	GTINTA	GTCCRA Register Compare Match/ Input Capture Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b1	GTINTB	GTCCRB Register Compare Match/ Input Capture Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b2	GTINTC	GTCCRC Register Compare Match Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b3	GTINTD	GTCCRD Register Compare Match Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b4	GTINTE	GTCCRE Register Compare Match Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b5	GTINTF	GTCCRF Register Compare Match Interrupt Enable	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b7, b6	GTINTPR[1:0]	GTPR Register Compare Match Interrupt Enable	b7 b6 0 0: Interrupt request is disabled. 0 1: In sawtooth-wave mode, interrupt requests are enabled at overflows. In triangle-wave mode, interrupt requests are enabled at crests. 1 0: In sawtooth-wave mode, interrupt requests are enabled at underflows. In triangle-wave mode, interrupt requests are enabled at troughs. 1 1: In sawtooth-wave mode, interrupt requests are enabled at both overflows and underflows. In triangle-wave mode, interrupt requests are enabled at both crests and troughs.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	ADTRAUEN	GTADTRA Register Compare Match (Up-Counting) A/D Conversion Start Request Enable	0: A/D conversion start request is disabled. 1: A/D conversion start request is enabled.	R/W
b17	ADTRADEN	GTADTRA Register Compare Match (Down-Counting) A/D Conversion Start Request Enable	0: A/D conversion start request is disabled. 1: A/D conversion start request is enabled.	R/W
b18	ADTRBUEN	GTADTRB Register Compare Match (Up-Counting) A/D Conversion Start Request Enable	0: A/D conversion start request is disabled. 1: A/D conversion start request is enabled.	R/W
b19	ADTRBDEN	GTADTRB Register Compare Match (Down-Counting) A/D Conversion Start Request Enable	0: A/D conversion start request is disabled. 1: A/D conversion start request is enabled.	R/W
b23 to b20	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b25, b24	GRP[1:0]	Output Stop Group Select	b25b24 0 0: Group A is selected 0 1: Group B is selected 1 0: Group C is selected 1 1: Group D is selected	R/W

Bit	Symbol	Bit Name	Description	R/W
b27, b26	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b28	GRPDTE	Dead Time Error Output Stop Detection Enable	0: Dead time error output stop detection is disabled 1: Dead time error output stop detection is enabled	R/W
b29	GRPABH	Simultaneous High Output Stop Detection Enable	0: Simultaneous high output stop detection is disabled 1: Simultaneous high output stop detection is enabled	R/W
b30	GRPABL	Simultaneous Low Output Stop Detection Enable	0: Simultaneous low output stop detection is disabled 1: Simultaneous low output stop detection is enabled	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

The GTINTAD register sets enabling or disabling for interrupt request, A/D conversion start request, and output stop detection.

GTINTA Bit (GTCCRA Register Compare Match/Input Capture Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRA register compare match/input capture (GTCIA).

GTINTB Bit (GTCCRB Register Compare Match/Input Capture Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRB register compare match/input capture (GTCIB).

GTINTC Bit (GTCCRC Register Compare Match Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRC register compare match (GTCIC).

GTINTD Bit (GTCCRD Register Compare Match Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRD register compare match (GTCID).

GTINTE Bit (GTCCRE Register Compare Match Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRE register compare match (GTCIE).

GTINTF Bit (GTCCRF Register Compare Match Interrupt Enable)

This bit enables or disables interrupt requests by the GTCCRF register compare match (GTCIF).

GTINTPR[1:0] Bits (GTPR Register Compare Match Interrupt Enable)

These bits enable or disable interrupt requests by a GTPR register compare match (GTCNT counter overflow) and those by a GTCNT counter underflow (GTCIV/GTCIU).

ADTRAUEN Bit (GTADTRA Register Compare Match (Up-Counting) A/D Conversion Start Request Enable)

This bit enables or disables A/D conversion start requests generated by the GTADTRA register compare matches during GTCNT counter up-counting.

The setting is invalid during the event count operation, and the A/D conversion start request is not generated.

ADTRADEN Bit (GTADTRA Register Compare Match (Down-Counting) A/D Conversion Start Request Enable)

This bit enables or disables A/D conversion start requests generated by the GTADTRA register compare matches during GTCNT counter down-counting.

The setting is invalid during the event count operation, and the A/D conversion start request is not generated.

ADTRBUEN Bit (GTADTRB Register Compare Match (Up-Counting) A/D Conversion Start Request Enable)

This bit enables or disables A/D conversion start requests generated by the GTADTRB register compare matches during GTCNT counter up-counting.

The setting is invalid during the event count operation, and the A/D conversion start request is not generated.

ADTRBDEN Bit (GTADTRB Register Compare Match (Down-Counting) A/D Conversion Start Request Enable)

This bit enables or disables A/D conversion start requests generated by the GTADTRB register compare matches during GTCNT counter down-counting.

The setting is invalid during the event count operation, and the A/D conversion start request is not generated.

GRP[1:0] Bits (Output Stop Group Select)

Select the group to detect disabling of output from the GPTW to the POEG and to request of disabling of output from the POEG to the GPTW.

Each signal indicating detection of disabling in response to a dead-time error or simultaneous driving of outputs to the high or low level is output to the group selected in the GRP[1:0] bits while the respective output disable detection enable bits are 1.

Requests to disabling output from the POEG for the group selected in the GRP[1:0] bits can be monitored with the GTST.ODF flag.

Set the GRP[1:0] bits while both of the OAE and OBE bits in the GTIOR register are set to 0.

GRPDTE Bit (Dead Time Error Output Stop Detection Enable)

This bit enables or disables the output stop detection by a dead time error.

The dead time error output stop detection is not generated during the event count operation.

GRPABH Bit (Simultaneous High Output Stop Detection Enable)

This bit enables or disables the output stop detection when the GTIOCnA and GTIOCnB pins have become high simultaneously.

GRPABL Bit (Simultaneous Low Output Stop Detection Enable)

This bit enables or disables the output stop detection when the GTIOCnA and GTIOCnB pins have become low simultaneously.

26.2.16 General PWM Timer Status Register (GTST)

Address(es): GPTW0.GTST 000C 203Ch, GPTW1.GTST 000C 213Ch, GPTW2.GTST 000C 223Ch,
GPTW3.GTST 000C 233Ch

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	OABLF	OABHF	DTEF	—	—	—	ODF	—	—	—	—	ADTRB DF	ADTRB UF	ADTRA DF	ADTRA UF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
TUCF	—	—	—	—	ITCNT[2:0]		—	—	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	The read value is undefined. The write value should be 1.	R/W
b10 to b8	ITCNT[2:0]	GTCIV/GTCIU Interrupt Skipping Count Counter	Counter for counting the number of times a timer interrupt has been skipped.	R
b14 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	TUCF	Count Direction Flag	0: The GTCNT counter counts downward. 1: The GTCNT counter counts upward.	R
b16	ADTRAUF	GTADTRA Register Compare Match (Up-Counting) A/D Conversion Start Request Flag	0: No GTADTRA register compare match has occurred in up-counting. 1: A GTADTRA register compare match has occurred in up-counting.	R/(W) *1
b17	ADTRADF	GTADTRA Register Compare Match (Down-Counting) A/D Conversion Start Request Flag	0: No GTADTRA register compare match has occurred in down-counting. 1: A GTADTRA register compare match has occurred in down-counting.	R/(W) *1
b18	ADTRBUF	GTADTRB Register Compare Match (Up-Counting) A/D Conversion Start Request Flag	0: No GTADTRB register compare match has occurred in up-counting. 1: A GTADTRB register compare match has occurred in up-counting.	R/(W) *1
b19	ADTRBDF	GTADTRB Register Compare Match (Down-Counting) A/D Conversion Start Request Flag	0: No GTADTRB register compare match has occurred in down-counting. 1: A GTADTRB register compare match has occurred in down-counting.	R/(W) *1
b23 to b20	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b24	ODF	Output Stop Request Flag	0: No output stop request has occurred. 1: An output stop request has occurred.	R
b27 to b25	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b28	DTEF	Dead Time Error Flag	0: No dead time error has occurred. 1: A dead time error has occurred.	R
b29	OABHF	Simultaneous High Output Flag	0: No simultaneous generation of 1 both for the GTIOCA and GTIOCB pins has occurred. 1: A simultaneous generation of 1 both for the GTIOCA and GTIOCB pins has occurred.	R
b30	OABLF	Simultaneous Low Output Flag	0: No simultaneous generation of 0 both for the GTIOCA and GTIOCB pins has occurred. 1: A simultaneous generation of 0 both for the GTIOCA and GTIOCB pins has occurred.	R
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to clear the flag. When clearing the ADTRAUF, ADTRADF, ADTRBUF, or ADTRBDF flag, be sure to write 0 only to the target flag or flags for clearing and to write 1 to the other flags not for clearing.

The GTST register indicates the status of the GPTW.

ITCNT[2:0] Bits (GTCIV/GTCIU Interrupt Skipping Count Counter)

When the GTCIV/GTCIU interrupt skipping function is used (the GTITC.IVTC[1:0] bits are set to a value other than 00b), the counter is incremented by 1 every time the GTCIV/GTCIU interrupt source is generated.

These bits are operated independently from the extended interrupt skipping by the GTEITC register.

[Clearing conditions]

- The GTCIV/GTCIU interrupt skipping function is not used (the GTITC.IVTT[2:0] bits are 000b when the IVTC[1:0] bits are 00b).
- The GTCIV/GTCIU interrupt skipping count matches the specified count (the ITCNT[2:0] bits match the skipping count specified by the IVTT[2:0] bits).
- When the count operation is stopped.

TUCF Flag (Count Direction Flag)

This flag indicates the count direction of the GTCNT counter.

This flag indicates 1 when in up-counting and indicates 0 when in down-counting during the event count operation.

ADTRAUF Flag (GTADTRA Register Compare Match (Up-Counting) A/D Conversion Start Request Flag)

This status flag indicates generation of a GTADTRA register compare match in up-counting.

[Setting condition]

- The GTCNT counter matches the GTADTRA register in up-counting

[Clearing conditions]

- 0 is written to the ADTRAUF flag

ADTRADF Flag (GTADTRA Register Compare Match (Down-Counting) A/D Conversion Start Request Flag)

This status flag indicates generation of a GTADTRA register compare match in down-counting.

[Setting condition]

- The GTCNT counter matches the GTADTRA register in down-counting

[Clearing conditions]

- 0 is written to the ADTRADF flag

ADTRBUF Flag (GTADTRB Register Compare Match (Up-Counting) A/D Conversion Start Request Flag)

This status flag indicates generation of a GTADTRB register compare match in up-counting.

[Setting condition]

- The GTCNT counter matches the GTADTRB register in up-counting

[Clearing conditions]

- 0 is written to the ADTRBUF flag

ADTRBDF Flag (GTADTRB Register Compare Match (Down-Counting) A/D Conversion Start Request Flag)

This status flag indicates generation of a GTADTRB register compare match in down-counting.

[Setting condition]

- The GTCNT counter matches the GTADTRB register in down-counting

[Clearing conditions]

- 0 is written to the ADTRBDF flag

ODF Flag (Output Stop Request Flag)

This flag monitors the output stop request for the group selected by the GTINTAD.GRP[1:0] bits.

Even if a request to disable output is de-asserted after the request is made, releasing the control of negation on the PWM pins has to wait until the end of the PWM cycle.

DTEF Flag (Dead Time Error Flag)

This flag indicates that the timer output toggle point after the automatic addition of dead time has exceeded the count period.

This flag returns to 0 when the timer output toggle point after the automatic addition of dead time is back within the period. This flag can only be read from. (Writing 0 to clear the flag is not allowed.)

When an interrupt by the DTEF flag is enabled (when the GTINTAD.GRPDTE bit is 1), the DTEF flag is output as detection of disabling of output to the POEG, and an GDTE interrupt is generated whenever the DTEF flag changes from 0 to 1.

[Setting condition]

- When a change point of the waveform after the automatic setting of dead time has exceeded the count period (in the following cases)
 - Up-counting in triangle-wave mode: GTCCRA register – GTDVU register ≤ 0
 - Down-counting in triangle-wave mode: GTCCRA register – GTDVD register < 0
 - Up-counting in sawtooth-wave one-shot pulse mode:
 - When GTCCRA register – GTDVU register < 0 , or GTCCRA register + GTDVD register $> GTPR$ register
 - Down-counting in sawtooth-wave one-shot pulse mode:
 - When GTCCRA register + GTDVU register $> GTPR$ register, or GTCCRA register – GTDVD register < 0

[Clearing condition]

- The timer output toggle point after the automatic addition of dead time is within the count period.

OABHF Flag (Simultaneous High Output Flag)

This flag indicates that the simultaneous output of 1 both from the GTIOCnA and GTIOCnB pins.

When one of the GTIOCnA and GTIOCnB pins changes to 0, the flag changes to 0.

This flag can only be read from. (Writing 0 to clear the flag is not allowed.)

When the output stop detection by the OABHF flag is enabled (the GTINTAD.GRPABH bit is 1), the OABHF flag is output as the output stop detection to the POEG. The GPTW does not have an interrupt to indicate that outputs have been simultaneous driven to the high level. Use the interrupt function in the POEG if this is necessary.

[Setting condition]

- The GTIOCnA and GTIOCnB pins output 1 simultaneously while both the OAE and OBE bits in the GTIOR register are set to 1.

[Clearing conditions]

- The GTIOCnA and GTIOCnB pins output different values while both the OAE and OBE bits are set to 1.
- The GTIOCnA and GTIOCnB pins output 0 simultaneously while both the OAE and OBE bits are set to 1.
- At least one of the OAE and OBE bits is set to 0.

OABLF Flag (Simultaneous Low Output Flag)

This flag indicates that the simultaneous output of 0 both from the GTIOCnA and GTIOCnB pins.

When one of the GTIOCnA and GTIOCnB pins changes to 1, the flag changes to 0.

This flag can only be read from. (Writing 0 to clear the flag is not allowed.)

When the output stop detection by the OABLF flag is enabled (the GTINTAD.GRPABL bit is 1), the OABLF flag is output as the output stop detection to the POEG. The GPTW does not have an interrupt to indicate that outputs have been simultaneous driven to the high level. Use the interrupt function in the POEG if this is necessary.

[Setting condition]

- The GTIOCnA and GTIOCnB pins output 0 simultaneously while both the OAE and OBE bits in the GTIOR register are set to 1.

[Clearing conditions]

- The GTIOCnA and GTIOCnB pins output different values while both the OAE and OBE bits are set to 1.
- The GTIOCnA and GTIOCnB pins output 1 simultaneously while both the OAE and OBE bits are set to 1.
- At least one of the OAE and OBE bits is set to 0.

Comparison for the output value to generate the OABHF or OABLF flag uses the value before the compare match output (PWM output) is masked by the output negate function. Even during the output negate condition, compare match operation continues internally, where the OABHF or OABLF flag is updated based on the operation results.

26.2.17 General PWM Timer Buffer Enable Register (GTBER)

Address(es): GPTW0.GTBER 000C 2040h, GPTW1.GTBER 000C 2140h, GPTW2.GTBER 000C 2240h, GPTW3.GTBER 000C 2340h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	ADTDB	ADTTB[1:0]	—	ADTDA	ADTTA[1:0]	—	CCRS WT	PR[1:0]	CCRB[1:0]	CCRA[1:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	DBRTE CB	—	DBRTE CA	—	—	—	—	BD[3]	BD[2]	BD[1]	BD[0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	BD[0]	GTCCRA/GTCCRB Registers Buffer Operation Disable	0: Buffer operation is enabled 1: Buffer operation is disabled	R/W
b1	BD[1]	GTPR Register Buffer Operation Disable		R/W
b2	BD[2]	GTADTRA/GTADTRB Registers Buffer Operation Disable		R/W
b3	BD[3]	GTDVU/GTDVD Registers Buffer Operation Disable		R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	DBRTECA	GTCCRA Register Double Buffer Repeat Operation Enable	0: GTCCRA register double buffer repeat operation is disabled 1: GTCCRA register double buffer repeat operation is enabled	R/W
b9	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10	DBRTECB	GTCCRB Register Double Buffer Repeat Operation Enable	0: GTCCRB register double buffer repeat operation is disabled 1: GTCCRB register double buffer repeat operation is enabled	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b17, b16	CCRA[1:0]	GTCCRA Register Buffer Operation	b17b16 0 0: Buffer operation is not performed 0 1: Single buffer operation (GTCCRA register ⇔ GTCCRC register) 1 x: Double buffer operation (GTCCRA register ⇔ GTCCRC register ⇔ GTCCRD register)	R/W
b19, b18	CCRB[1:0]	GTCCRB Register Buffer Operation	b19b18 0 0: Buffer operation is not performed 0 1: Single buffer operation (GTCCRB register ⇔ GTCCRE register) 1 x: Double buffer operation (GTCCRB register ⇔ GTCCRE register ⇔ GTCCRF register)	R/W
b21, b20	PR[1:0]	GTPR Register Buffer Operation	b21b20 0 0: Buffer operation is not performed 0 1: Single buffer operation (GTPBR register ⇒ GTPR register) 1 x: Double buffer operation (GTPDBR register ⇒ GTPBR register ⇒ GTPR register)	R/W
b22	CCRSWT	GTCCRA and GTCCRB Registers Forcible Buffer Operation	Writing 1 to this bit forcibly performs buffer transfer of GTCCRA and GTCCRB registers. This bit automatically returns to 0 after the writing of 1. This bit is read as 0.	R/W
b23	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b25, b24	ADTTA[1:0]	GTADTRA Register Buffer Transfer Timing Select	- Triangle waves b25b24 0 0: No transfer 0 1: Transfer at crest 1 0: Transfer at trough 1 1: Transfer at both crest and trough - Sawtooth waves b25b24 0 0: No transfer Values other than 0 0: Transfer at underflow (during down-counting), overflow (during up-counting), or counter clearing	R/W
b26	ADTDA	GTADTRA Register Double Buffer Operation	0: Single buffer operation (GTADTBRA register ⇒ GTADTRA register) 1: Double buffer operation (GTADTDBRA register ⇒ GTADTBRA register ⇒ GTADTRA register)	R/W
b27	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b29, b28	ADTTB[1:0]	GTADTRB Register Buffer Transfer Timing Select	- Triangle waves b29b28 0 0: No transfer 0 1: Transfer at crest 1 0: Transfer at trough 1 1: Transfer at both crest and trough - Sawtooth waves b29b28 0 0: No transfer Values other than 0 0: Transfer at underflow (during down-counting), overflow (during up-counting), or counter clearing	R/W
b30	ADTDB	GTADTRB Register Double Buffer Operation	0: Single buffer operation (GTADTBRB register ⇒ GTADTRB register) 1: Double buffer operation (GTADTDBRB register ⇒ GTADTBRB register ⇒ GTADTRB register)	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

The GTBER register makes settings for buffer operation.

Set the GTBER register except the BD[3:0] bits while the GTCNT counter is stopped.

BD[0] Bit (GTCCRA/GTCCRB Registers Buffer Operation Disable)

This bit disables buffer operation using the GTCCRA, GTCCRC, and GTCCRD registers together and buffer operation using the GTCCRB, GTCCRE, and GTCCRF registers together.

Even though the BD[0] bit is set to 0, buffer operation in the GTCCRB register is not performed if the GTDTCR.TDE bit is set to 1. Instead, the compare match value for negative-phase waveform with dead time is set automatically.

A value for the BD[0] bit in the channel related to the position of the bit written with 1 by the GTSECSR register can be set when 1 is written to the SBDCE or SBDCE bit in the GTSECR register.

When the DBRTEC_m (m = A, B) bit is 1, setting the BD[0] bit to 1 while the mode of operation is sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3 results in transfer from the intermediate buffer to the GTCCR_m register.

BD[1] Bit (GTPR Register Buffer Operation Disable)

This bit disables buffer operation using the GTPR, GTPBR, and GTPDBR registers together.

A value for the BD[1] bit in the channel related to the position of the bit written with 1 by the GTSECSR register can be set when 1 is written to the SBDPE or SBDPD bit in the GTSECR register.

BD[2] Bit (GTADTRA/GTADTRB Registers Buffer Operation Disable)

This bit disables buffer operation using the GTADTRA, GTADTBRA, and GTADTDBRA registers together and buffer

operation using the GTADTRB, GTADTBRB, and GTADTDBRB registers together.

The setting is invalid during the event count operation, and the buffer operation using the GTADTRA and GTADTRB registers is not performed.

A value for the BD[2] bit in the channel related to the position of the bit written with 1 by the GTSECSR register can be set when 1 is written to the SBDAE or SBDAD bit in the GTSECR register.

BD[3] Bit (GTDVU/GTDVD Registers Buffer Operation Disable)

This bit disables buffer operation using the GTDVU and GTDBU registers together and buffer operation using the GTDVD and GTDBD registers together.

Even though the BD[3] bit is set to 0, buffer operation in the GTDVD register is not performed if the GTDTCR.TDFER bit is set to 1. Instead, the value in the GTDVU register is set automatically.

The setting is invalid during the event count operation, and the buffer operation using the GTDVU and GTDVD registers is not performed.

A value for the BD[3] bit in the channel related to the position of the bit written with 1 by the GTSECSR register can be set when 1 is written to the SBDDE or SBDDB bit in the GTSECR register.

DBRTECm Bit (GTCCRm Register Double Buffer Repeat Operation Enable) (m = A, B)

This setting enables the operation to repeat a transfer to the GTCCRm register from the intermediate buffer by cycle during the buffer transfer disable period when performing the double buffer operation using the GTCCRm register. It is valid in sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3.

The disabling period of buffer transfer indicates the period that buffer transfer is stopped by setting of the BD[0] bit (CPU writing or simultaneous buffer operation control by the GTSECSR register) and the period for buffer transfer extended skipping (except the case for skipping by counting both crests and troughs) by the GTEITLB register.

When the DBRTECm bit is 1, writing by the CPU to the GTCCRm register sets the same value in temporary register x (x = C, E). The value of the GTCCR_x (x = C, E) register is also transferred to the temporary register x (x = C, E) by forcible buffer transfer.

CCRA[1:0] Bits (GTCCRA Register Buffer Operation)

These bits set buffer operation with the GTCCRA, GTCCRC, and GTCCRD registers combined. When buffer operation is restricted by the operating mode set in the GTCR register, the GTCR register setting is given priority.*¹

CCRB[1:0] Bits (GTCCRB Register Buffer Operation)

These bits set buffer operation with the GTCCRB, GTCCRE, and GTCCRF registers combined. When buffer operation is restricted by the operating mode set in the GTCR register, the GTCR register setting is given priority.*¹

PR[1:0] Bits (GTPR Register Buffer Operation)

These bits set buffer operation with the GTPR, GTPBR, and GTPDBR registers combined.

CCRSWT Bit (GTCCRA and GTCCRB Registers Forcible Buffer Operation)

Writing 1 to the CCRSWT bit forcibly performs buffer transfer of the GTCCRA and GTCCRB registers. This bit automatically returns to 0 after the writing of 1. This bit is read as 0.

This bit is valid only when counting is stopped with compare match operation specified.

ADTTA[1:0] Bits (GTADTRA Register Buffer Transfer Timing Select)

These bits set the transfer timing for buffer operation of the GTADTRA, GTADTBRA, and GTADTDBRA registers.

The setting is invalid during the event count operation.

ADTDA Bit (GTADTRA Register Double Buffer Operation)

These bits set buffer operation with the GTADTRA, GTADTBRA, and GTADTDBRA registers combined.
The setting is invalid during the event count operation.

ADTTB[1:0] Bits (GTADTRB Register Buffer Transfer Timing Select)

These bits set the transfer timing for buffer operation of the GTADTRB, GTADTBRB, and GTADTDBRB registers.
The setting is invalid during the event count operation.

ADTDB Bit (GTADTRB Register Double Buffer Operation)

These bits set buffer operation with the GTADTRA, GTADTBRA, and GTADTDBRA registers combined.
The setting is invalid during the event count operation.

Note 1. The buffer operation mode is fixed in sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3 (64-bit transfer at trough).

26.2.18 General PWM Timer Interrupt and A/D Conversion Start Request Skipping Setting Register (GTITC)

Address(es): GPTW0.GTITC 000C 2044h, GPTW1.GTITC 000C 2144h, GPTW2.GTITC 000C 2244h, GPTW3.GTITC 000C 2344h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	ADTBL	—	ADTAL	—	—	IVTT[2:0]	—	IVTC[1:0]	—	ITLF	ITLE	ITLD	ITLC	ITLB	ITLA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ITLA	GTCCRA Register Compare Match/Input Capture Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b1	ITLB	GTCCRB Register Compare Match/Input Capture Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b2	ITLC	GTCCRC Register Compare Match Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b3	ITLD	GTCCRD Register Compare Match Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b4	ITLE	GTCCRE Register Compare Match Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b5	ITLF	GTCCRF Register Compare Match Interrupt Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b7, b6	IVTC[1:0]	GTCIV/GTCIU Interrupt Skipping Function Select	b7 b6 0 0: Skipping is not performed 0 1: Both overflow and underflow for sawtooth waves and crest for triangle waves are counted and skipped. 1 0: Both overflow and underflow for sawtooth waves and trough for triangle waves are counted and skipped. 1 1: Both overflow and underflow for sawtooth waves and both crest and trough for triangle waves are counted and skipped	R/W
b10 to b8	IVTT[2:0]	GTCIV/GTCIU Interrupt Skipping Count Select	b10 b8 0 0 0: Skipping is not performed 0 0 1: Skipping count of 1 0 1 0: Skipping count of 2 0 1 1: Skipping count of 3 1 0 0: Skipping count of 4 1 0 1: Skipping count of 5 1 1 0: Skipping count of 6 1 1 1: Skipping count of 7	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b12	ADTAL	GTADTRA Register A/D Conversion Start Request Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14	ADTBL	GTADTRB Register A/D Conversion Start Request Link	0: Not linked with GTCIV/GTCIU interrupt skipping function. 1: Linked with GTCIV/GTCIU interrupt skipping function.	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

GTITC sets the skipping function for the GTCNT counter overflow (GTPR compare match) interrupt (GTCIV)/ underflow interrupt (GTCIU) and also sets whether to link the other interrupts and A/D conversion start requests with the GTCIV/GTCIU interrupt skipping function. Note that detection of disabling of output to the POEG and dead-time error interrupts cannot be linked with the GTCIV or GTCIU interrupt skipping function.

The setting is invalid during the event count operation. The setting is operated independently from the extended interrupt skipping by the GTEITC register.

ITLA Bit (GTCCRA Register Compare Match/Input Capture Interrupt Link)

This bit specifies whether to link the GTCCRA compare match/input capture interrupt (GTCIA) with the GTCIV/GTCIU interrupt skipping function.

ITLB Bit (GTCCRB Register Compare Match/Input Capture Interrupt Link)

This bit specifies whether to link the GTCCRB compare match/input capture interrupt (GTCIB) with the GTCIV/GTCIU interrupt skipping function.

ITLC Bit (GTCCRC Register Compare Match Interrupt Link)

This bit specifies whether to link the GTCCRC compare match interrupt (GTCIC) with the GTCIV/GTCIU interrupt skipping function.

ITLD Bit (GTCCRD Register Compare Match Interrupt Link)

This bit specifies whether to link the GTCCRD compare match interrupt (GTCID) with the GTCIV/GTCIU interrupt skipping function.

ITLE Bit (GTCCRE Register Compare Match Interrupt Link)

This bit specifies whether to link the GTCCRE compare match interrupt (GTCIE) with the GTCIV/GTCIU interrupt skipping function.

ITLF Bit (GTCCRF Register Compare Match Interrupt Link)

This bit specifies whether to link the GTCCRF compare match interrupt (GTCIF) with the GTCIV/GTCIU interrupt skipping function.

IVTC[1:0] Bits (GTCIV/GTCIU Interrupt Skipping Function Select)

These bits set the skipping function for the GTPR compare match (GTCNT counter overflow) interrupt (GTCIV)/ GTCNT counter underflow interrupt (GTCIU).

IVTT[2:0] Bits (GTCIV/GTCIU Interrupt Skipping Count Select)

These bits set the skipping count for the GTPR compare match (GTCNT counter overflow) interrupt (GTCIV)/GTCNT counter underflow interrupt (GTCIU).

When modifying the IVTT[2:0] bits, first set the IVTC[1:0] bits to 00b.

ADTAL Bit (GTADTRA Register A/D Conversion Start Request Link)

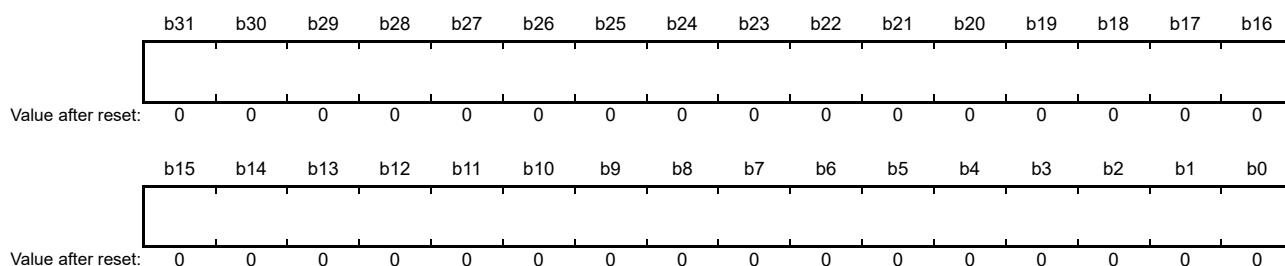
This bit specifies whether to link the A/D conversion start request, which is generated in response to a compare match with the GTCNT counter and the GTADTRA register, with the GTCIV/GTCIU interrupt skipping function.

ADTBL Bit (GTADTRB Register A/D Conversion Start Request Link)

This bit specifies whether to link the A/D conversion start request, which is generated in response to a compare match with the GTCNT counter and the GTADTRB register, with the GTCIV/GTCIU interrupt skipping function.

26.2.19 General PWM Timer Counter (GTCNT)

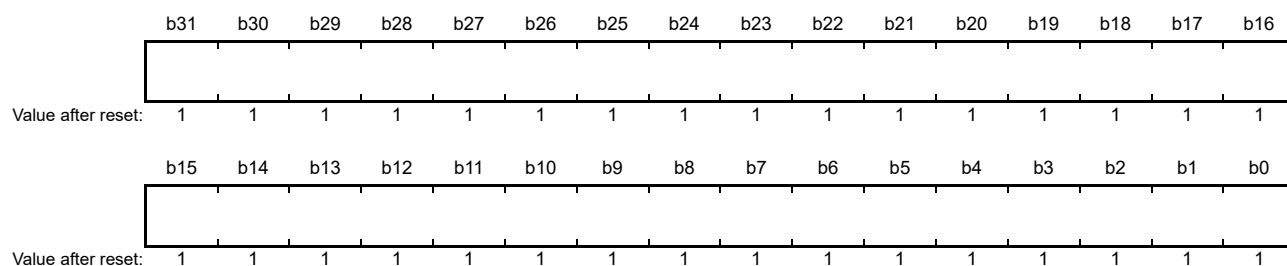
Address(es): GPTW0.GTCNT 000C 2048h, GPTW1.GTCNT 000C 2148h, GPTW2.GTCNT 000C 2248h,
GPTW3.GTCNT 000C 2348h



The GTCNT counter is a 32-bit readable/writable counter. There is one GTCNT counter for each channel. Writing is available only when counting is stopped, and thus writing is disabled while counting is in progress (when CST = 1). Access in 8-bit or 16-bit units to the GTCNT counter is prohibited, and it should be accessed in 32-bit units. Set the range of the GTCNT counter within the range of $0 \leq \text{GTCNT counter} \leq \text{GTPR register}$.

26.2.20 General PWM Timer Compare Capture Register m (GTCCRM) (m = A to F)

Address(es): GPTW0.GTCCRA 000C 204Ch, GPTW1.GTCCRA 000C 214Ch, GPTW2.GTCCRA 000C 224Ch, GPTW3.GTCCRA 000C 234Ch,
 GPTW0.GTCCRB 000C 2050h, GPTW1.GTCCRB 000C 2150h, GPTW2.GTCCRB 000C 2250h, GPTW3.GTCCRB 000C 2350h,
 GPTW0.GTCCRC 000C 2054h, GPTW1.GTCCRC 000C 2154h, GPTW2.GTCCRC 000C 2254h, GPTW3.GTCCRC 000C 2354h,
 GPTW0.GTCCRE 000C 2058h, GPTW1.GTCCRE 000C 2158h, GPTW2.GTCCRE 000C 2258h, GPTW3.GTCCRE 000C 2358h,
 GPTW0.GTCCRD 000C 205Ch, GPTW1.GTCCRD 000C 215Ch, GPTW2.GTCCRD 000C 225Ch, GPTW3.GTCCRD 000C 235Ch,
 GPTW0.GTCCRF 000C 2060h, GPTW1.GTCCRF 000C 2160h, GPTW2.GTCCRF 000C 2260h, GPTW3.GTCCRF 000C 2360h



The GTCCRM register is a 32-bit readable/writable register.

Access in 8-bit or 16-bit units to the GTCCRM register is prohibited, and it should be accessed in 32-bit units.

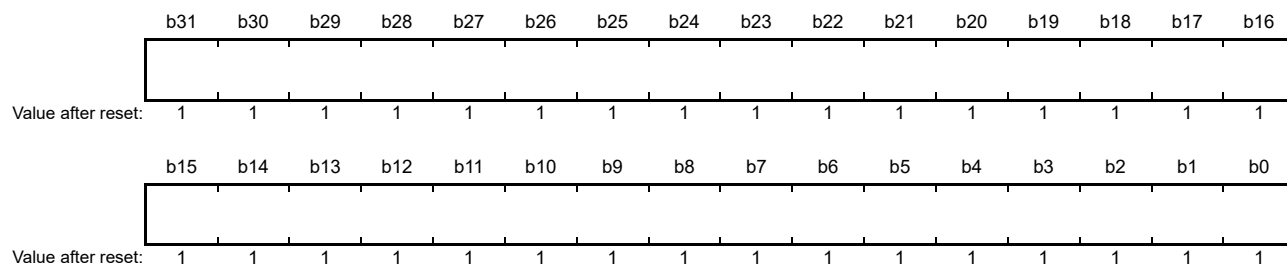
The GTCCRA and GTCCRB registers are registers used for both output compare and input capture.

The GTCCRC and GTCCRE registers are compare match registers, and can also function as buffer registers for the GTCCRA and GTCCRB registers.

The GTCCRD and GTCCRF registers are compare match registers, and can also function as buffer registers for the GTCCRC and GTCCRE registers (double buffer registers for the GTCCRA and GTCCRB registers).

26.2.21 General PWM Timer Period Setting Register (GTPR)

Address(es): GPTW0.GTPR 000C 2064h, GPTW1.GTPR 000C 2164h, GPTW2.GTPR 000C 2264h, GPTW3.GTPR 000C 2364h

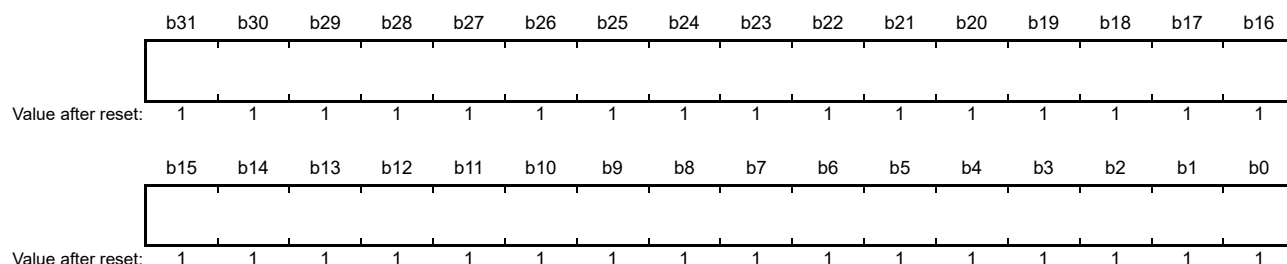


The GTPR register is a 32-bit readable/writable register that sets the maximum count value of the GTCNT counter. Access in 8-bit or 16-bit units to the GTPR register is prohibited, and it should be accessed in 32-bit units.

In sawtooth-wave mode, the value of (GTPR register + 1) is the count period. In triangle-wave mode, the value of (GTPR register value $\times$ 2) is the count period.

26.2.22 General PWM Timer Period Setting Buffer Register (GTPBR)

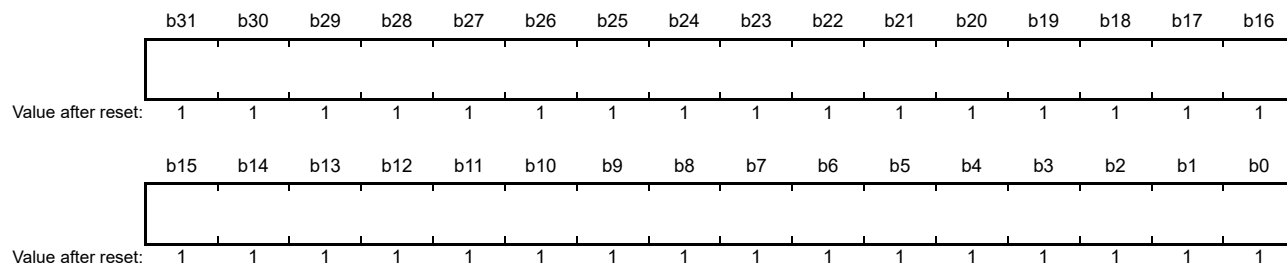
Address(es): GPTW0.GTPBR 000C 2068h, GPTW1.GTPBR 000C 2168h, GPTW2.GTPBR 000C 2268h, GPTW3.GTPBR 000C 2368h



The GTPBR register is a 32-bit readable/writable register that functions as a buffer register for the GTPR register. Access in 8-bit or 16-bit units to the GTPBR register is prohibited, and it should be accessed in 32-bit units.

26.2.23 General PWM Timer Period Setting Double-Buffer Register (GTPDBR)

Address(es): GPTW0.GTPDBR 000C 206Ch, GPTW1.GTPDBR 000C 216Ch, GPTW2.GTPDBR 000C 226Ch, GPTW3.GTPDBR 000C 236Ch

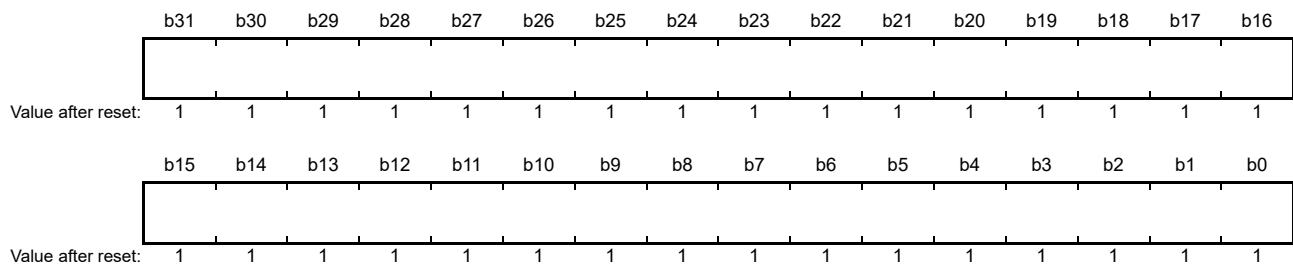


The GTPDBR register is a 32-bit readable/writable register that functions as a buffer register for the GTPBR register (a double buffer register for the GTPR register).

Access in 8-bit or 16-bit units to the GTPDBR register is prohibited, and it should be accessed in 32-bit units.

26.2.24 A/D Conversion Start Request Timing Register m (GTADTRm) (m = A, B)

Address(es): GPTW0.GTADTRA 000C 2070h, GPTW1.GTADTRA 000C 2170h, GPTW2.GTADTRA 000C 2270h,
GPTW3.GTADTRA 000C 2370h,
GPTW0.GTADTRB 000C 207Ch, GPTW1.GTADTRB 000C 217Ch, GPTW2.GTADTRB 000C 227Ch,
GPTW3.GTADTRB 000C 237Ch



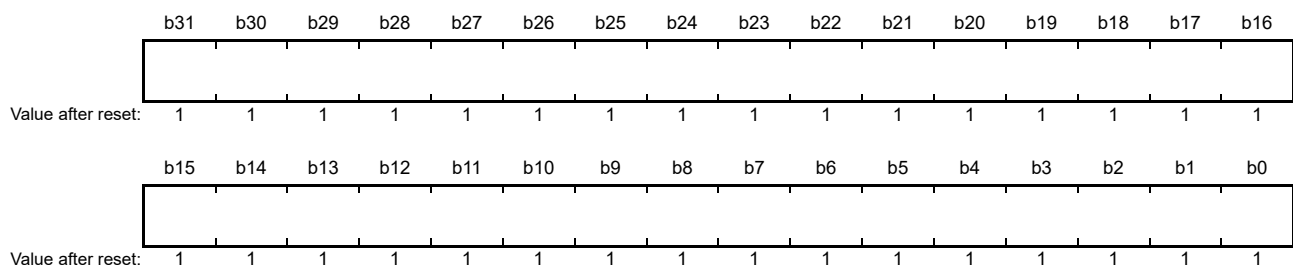
The GTADTRm register is 32-bit readable/writable register that set the timing of A/D conversion start request generation.

Access in 8-bit or 16-bit units to the GTADTRm register is prohibited, and it should be accessed in 32-bit units.

When the GTADTRm register value matches the GTCNT counter value, an A/D conversion start request is generated.

26.2.25 A/D Conversion Start Request Timing Buffer Register m (GTADTBRm) (m = A, B)

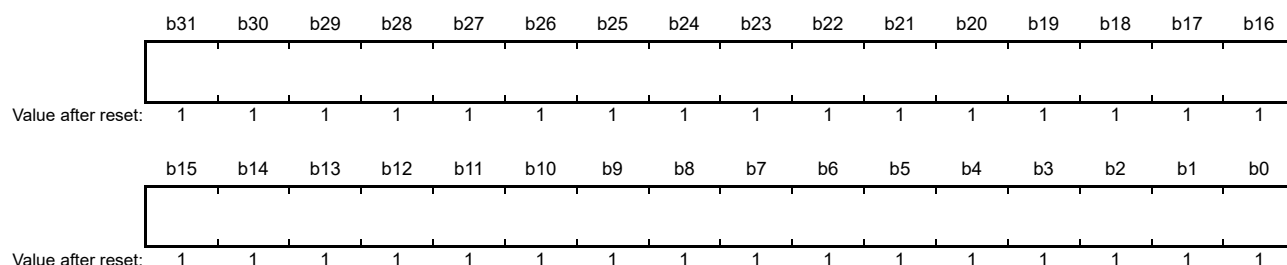
Address(es): GPTW0.GTADTBRA 000C 2074h, GPTW1.GTADTBRA 000C 2174h, GPTW2.GTADTBRA 000C 2274h,
GPTW3.GTADTBRA 000C 2374h,
GPTW0.GTADTBRB 000C 2080h, GPTW1.GTADTBRB 000C 2180h, GPTW2.GTADTBRB 000C 2280h,
GPTW3.GTADTBRB 000C 2380h



The GTADTBRm register is 32-bit readable/writable register that function as buffer register for the GTADTRm register. Access in 8-bit or 16-bit units to the GTADTBRm register is prohibited, and it should be accessed in 32-bit units.

26.2.26 A/D Conversion Start Request Timing Double-Buffer Register m (GTADTD BRm) (m = A, B)

Address(es): GPTW0.GTADTD BRA 000C 2078h, GPTW1.GTADTD BRA 000C 2178h, GPTW2.GTADTD BRA 000C 2278h, GPTW3.GTADTD BRA 000C 2378h, GPTW0.GTADTD BRB 000C 2084h, GPTW1.GTADTD BRB 000C 2184h, GPTW2.GTADTD BRB 000C 2284h, GPTW3.GTADTD BRB 000C 2384h



The GTADTD BRm register is 32-bit readable/writable register that function as buffer register for the GTADTD BRm register (double buffer register for the GTADTD BRm register).

Access in 8-bit or 16-bit units to the GTADTD BRm register is prohibited, and it should be accessed in 32-bit units.

26.2.27 General PWM Timer Dead Time Control Register (GTDTCCR)

Address(es): GPTW0.GTDTCCR 000C 2088h, GPTW1.GTDTCCR 000C 2188h, GPTW2.GTDTCCR 000C 2288h, GPTW3.GTDTCCR 000C 2388h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	TDFER	—	—	TDBDE	TDBUE	—	—	—	TDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TDE	Negative-Phase Waveform Setting	0: GTCCRB register is set without using GTDVU and GTDVD registers. 1: GTDVU and GTDVD registers are used to set the compare match value for negative-phase waveform with dead time automatically in GTCCRB register.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	TDBUE	GTDVU Register Buffer Operation Enable	0: GTDVU register buffer operation is disabled 1: GTDVU register buffer operation is enabled	R/W
b5	TDBDE	GTDVD Register Buffer Operation Enable	0: GTDVD register buffer operation is disabled 1: GTDVD register buffer operation is enabled	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	TDFER	GTDVD Register Setting	0: GTDVU and GTDVD registers are set separately. 1: The value written to GTDVU register is automatically set to GTDVD register.	R/W
b31 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTDTCCR register enables automatic setting of a compare match value for negative-phase waveform with dead time. The setting is invalid during the event count operation.

TDE Bit (Negative-Phase Waveform Setting)

This bit sets whether to use the GTDVU and GTDVD registers. When the GTDVU and GTDVD registers are used, the compare match value for a negative-phase waveform with dead time that was obtained by the compare match value of a positive-phase waveform (GTCCRA register) and the dead time value (GTDVU and GTDVD registers) is automatically set in the GTCCRB register.

The TDE bit setting is ignored in sawtooth-wave PWM mode, and the GTCCRB register is not automatic setting.

The automatically set GTCCRB register value has the following upper and lower limit values.

- Triangle wave PWM mode
 - Upper limit value: the value set in the GTPR register – 1
 - Lower limit value: 0000 0001h in up-counting, 0000 0000h in down-counting
- Sawtooth-wave one-shot pulse mode
 - Upper limit value: the value set in the GTPR register
 - Lower limit value: 0000 0000h

If the obtained GTCCRB register value is not within the range between the upper and lower limits, the upper or lower limit is set in the GTCCRB register, and the GTST.DTEF flag becomes 1. However, if the obtained GTCCRB register value exceeds the upper limit in triangle-wave PWM mode, the DTEF flag becomes 0.

TDBUE Bit (GTDVU Register Buffer Operation Enable)

This bit enables buffer operation with the GTDVU and GTDBU registers combined.

The timing of buffer transfer is at troughs in triangle-wave mode, and at overflows or underflows in sawtooth-wave mode.

TDBDE Bit (GTDVD Register Buffer Operation Enable)

This bit enables buffer operation with the GTDVD and GTDBD registers combined.

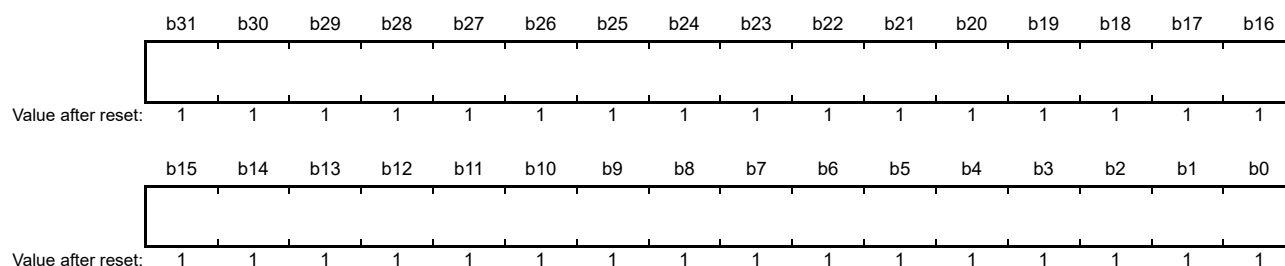
The buffer transfer timing is the trough in triangle-wave mode, and at an overflow or underflow in sawtooth-wave mode. When this bit and the TDFER bit are set to 1 simultaneously, the TDFER bit setting is given priority.

TDFER Bit (GTDVD Register Setting)

This bit sets whether or not the value written to the GTDVU register is also set to the GTDVD register automatically.

26.2.28 General PWM Timer Dead Time Value Register m (GTDVm) (m = U, D)

Address(es): GPTW0.GTDVU 000C 208Ch, GPTW1.GTDVU 000C 218Ch, GPTW2.GTDVU 000C 228Ch,
GPTW3.GTDVU 000C 238Ch,
GPTW0.GTDVD 000C 2090h, GPTW1.GTDVD 000C 2190h, GPTW2.GTDVD 000C 2290h,
GPTW3.GTDVD 000C 2390h



The GTDm register is 32-bit readable/writable register that set the dead time for generating PWM waveforms with dead time.

Access in 8-bit or 16-bit units to the GTDm register is prohibited, and it should be accessed in 32-bit units.

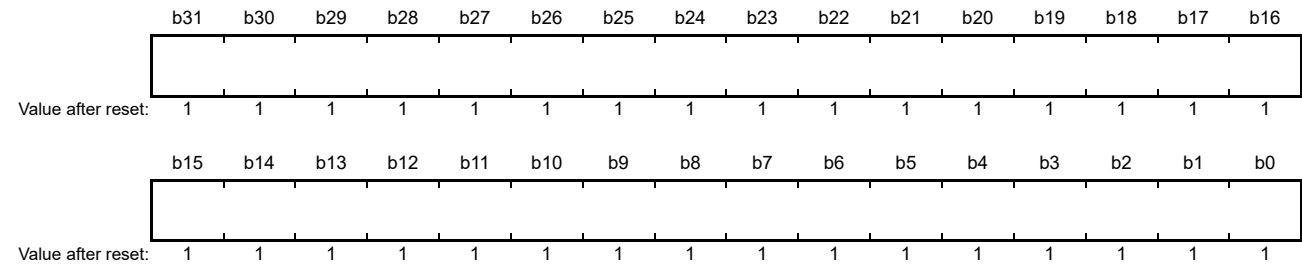
The GTDVU register is used for up-counting. The GTDVD register is used for down-counting.

The setting which the value for the GTDm register is equal to or larger than the value for the GTPR register is prohibited.

In addition, when using the automatic dead time setting function, do not set a value that makes a change point of the waveform exceeding the count period. The change point of the negative-phase waveform, which is automatically calculated, is obtained by reading the GTCCRB register. When using the GTDm register, writing to the GTCCRB register is prohibited. When setting this register to 0, waveforms without dead time are output. When the GTDTCR.TDFER bit is 1, the writing to the GTDVD register has no effect. At this time, when the GTDVD register is read, the value for the GTDVU register is read.

26.2.29 General PWM Timer Dead Time Value Buffer Register m (GTDBm) (m = U, D)

Address(es): GPTW0.GTDBU 000C 2094h, GPTW1.GTDBU 000C 2194h, GPTW2.GTDBU 000C 2294h,
GPTW3.GTDBU 000C 2394h,
GPTW0.GTDBD 000C 2098h, GPTW1.GTDBD 000C 2198h, GPTW2.GTDBD 000C 2298h,
GPTW3.GTDBD 000C 2398h



The GTDBm register is 32-bit readable/writable register that function as buffer register for the GTDVM register. Access in 8-bit or 16-bit units to the GTDBm register is prohibited, and it should be accessed in 32-bit units.

26.2.30 General PWM Timer Output Protection Function Status Register (GTSOS)

Address(es): GPTW0.GTSOS 000C 209Ch, GPTW1.GTSOS 000C 219Ch, GPTW2.GTSOS 000C 229Ch,
GPTW3.GTSOS 000C 239Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SOS[1:0]	—
Value after reset:	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b1, b0	SOS[1:0]	Output Protection Function Status	b1 b0 0 0: Normal operation 0 1: Protected state (GTCCRA = 0 is set during transfer at trough or crest) 1 0: Protected state (GTCCRA ≥ GTPR is set during transfer at trough) 1 1: Protected state (GTCCRA ≥ GTPR is set during transfer at crest)	R
b7 to b2	—	Reserved	These bits are read as 0.	R
b9, b8	—	Reserved	The read value is undefined.	R
b31 to b10	—	Reserved	These bits are read as 0.	R

The GTSOS register is a status register that indicates the status of the output protection function. The output protection function is enabled only when the dead time is automatically set (the GTDTCR.TDE bit is 1) in triangle-wave mode.

SOS[1:0] Bits (Output Protection Function Status)

This bit indicates the status of the output protection function in triangle-wave PWM mode. For details of the output protection function, see section 26.8.4, Output Protection Function for GTIOCNm Pin Output (n = 0 to 3; m = A, B).

26.2.31 General PWM Timer Output Protection Function Temporary Release Register (GTSOTR)

Address(es): GPTW0.GTSOTR 000C 20A0h, GPTW1.GTSOTR 000C 21A0h, GPTW2.GTSOTR 000C 22A0h, GPTW3.GTSOTR 000C 23A0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SOTR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SOTR	Output Protection Function Temporary Release	0: Protected state is not released 1: Protected state is released	R/W
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTSOTR register temporarily releases the protected state of GTIOCnB pin output (n = 0 to 3) when output protection has been set.

The protected state can be released only for the case of the GTSOS.SOS[1:0] bits are 10b (protected state in which GTCCRA register $\geq$ GTPR register has occurred during transfer at trough). The protected state cannot be released for any other case.

SOTR Bit (Output Protection Function Temporary Release)

This bit sets whether to temporarily release the protected state of the GTIOCnB pin output in an output protected state. After the SOTR bit has been set to 1, the output protection function is canceled from the first trough. After the SOTR bit has been set to 0, output protection is resumed from the first trough.

26.2.32 General PWM Timer A/D Conversion Start Request Signal Monitoring Register (GTADSMR)

Address(es): GPTW0.GTADSMR 000C 20A4h, GPTW1.GTADSMR 000C 21A4h, GPTW2.GTADSMR 000C 22A4h, GPTW3.GTADSMR 000C 23A4h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	ADSMEN1	—	—	—	—	—	—	ADSMS1[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	ADSMEN0	—	—	—	—	—	—	ADSMS0[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	ADSMS0[1:0]	A/D Conversion Start Request Signal Monitor 0 Selection	b1 b0 0 0: A/D conversion start request signal generated by the GTADTRA register during up-counting 0 1: A/D conversion start request signal generated by the GTADTRA register during down-counting 1 0: A/D conversion start request signal generated by the GTADTRB register during up-counting 1 1: A/D conversion start request signal generated by the GTADTRB register during down-counting	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	ADSMEN0	A/D Conversion Start Request Signal Monitor 0 Output Enabling	0: Output of A/D conversion start request signal monitor 0 is disabled. 1: Output of A/D conversion start request signal monitor 0 is enabled.	R/W
b15 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b17, b16	ADSMS1[1:0]	A/D Conversion Start Request Signal Monitor 1 Selection	b17b16 0 0: A/D conversion start request signal generated by the GTADTRA register during up-counting 0 1: A/D conversion start request signal generated by the GTADTRA register during down-counting 1 0: A/D conversion start request signal generated by the GTADTRB register during up-counting 1 1: A/D conversion start request signal generated by the GTADTRB register during down-counting	R/W
b23 to b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b24	ADSMEN1	A/D Conversion Start Request Signal Monitor 1 Output Enabling	0: Output of A/D conversion start request signal monitor 1 is disabled. 1: Output of A/D conversion start request signal monitor 1 is enabled.	R/W
b31 to b25	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to control monitors for the A/D conversion start request signal that is synchronized with a frame.

ADSMSk[1:0] Bits (A/D Conversion Start Request Signal Monitor k Selection) (k = 0, 1)

These bits are used to select A/D conversion start request signal synchronized with a frame which is monitored by the GTADSMk pin.

ADSMENk Bit (A/D Conversion Start Request Signal Monitor k Output Enabling) (k = 0, 1)

This bit enables or disables the monitor output to the GTADSMk pin.

When the output is disabled, the GTADSMk pin goes to the low level.

When the bit is 1, the signal on the GTADSMk pin goes to the high level on assertion of the signal to request to the start of A/D conversion selected by the ADSMSk[1:0] bits and returns to the low level at the end of the current cycle of the timer for the channel that generated the given signal to request the start of A/D conversion. When the counter stops, the value when the counter stopped is retained for output. Set the ADSMENk bit to 0 to output the low level.

When a signal to request the start of A/D conversion is generated at the end of a timer period, the generation of this signal has priority in terms of monitoring output and the output remains at the high level till the end of the next period.

When the output of the same A/D conversion start request signal monitoring output is enabled for multiple channels, ORed signals will be output from the GPTW.

26.2.33 General PWM Timer Extended Interrupt Skipping Counter Control Register (GTEITC)

Address(es): GPTW0.GTEITC 000C 20A8h, GPTW1.GTEITC 000C 21A8h, GPTW2.GTEITC 000C 22A8h, GPTW3.GTEITC 000C 23A8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	EITCNT2[3:0]				EITCNT2IV[3:0]				EIVTT2[3:0]				—	—	EIVTC2[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	EITCNT1[3:0]				—	—	—	—	EIVTT1[3:0]				—	—	EIVTC1[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	EIVTC1[1:0]	Extended Interrupt Skipping Counter 1 Count Source Select	b1 b0 0 0: Not counted (not skipped) 0 1: Counting both at overflow or underflow in sawtooth-wave mode, and counting crests in triangle-wave mode 1 0: Counting both at overflow or underflow in sawtooth-wave mode, and counting troughs in triangle-wave mode 1 1: Counting both at overflow or underflow in sawtooth-wave mode, and counting both crests and troughs in triangle-wave mode	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7 to b4	EIVTT1[3:0]	Extended Interrupt Skipping 1 Skipping Count Setting	Skipping count for the extended interrupt skipping 1	R/W
b11 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b12	EITCNT1[3:0]	Extended Interrupt Skipping Counter 1	The extended interrupt skipping counter 1	R
b17, b16	EIVTC2[1:0]	Extended Interrupt Skipping Counter 2 Count Source Select	b17 b16 0 0: Not counted (not skipped) 0 1: Counting both at overflow or underflow in sawtooth-wave mode, and counting crests in triangle-wave mode 1 0: Counting both at overflow or underflow in sawtooth-wave mode, and counting troughs in triangle-wave mode 1 1: Counting both at overflow or underflow in sawtooth-wave mode, and counting both crests and troughs in triangle-wave mode	R/W
b19, b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b23 to b20	EIVTT2[3:0]	Extended Interrupt Skipping 2 Skipping Count Setting	Skipping count for the extended interrupt skipping 2	R/W
b27 to b24	EITCNT2IV[3:0]	Extended Interrupt Skipping Counter 2 Initial Value	The extended interrupt skipping counter 2 initial value	R/W *1
b31 to b28	EITCNT2[3:0]	Extended Interrupt Skipping Counter 2	The extended interrupt skipping counter 2	R

Note 1. The EITCNT2IV[3:0] bits are only writable when the value other than 00b is written to the EIVTC2[1:0] bits that have been 00b.

The GTEITC register sets the extended interrupt skipping function to skip the interrupts, A/D conversion start requests, and buffer transfers independently by counting at overflow and underflow in the GTCNT counter.

The setting is operated independently from the interrupt skipping by the GTITC register.

The setting is invalid during the event count operation.

EIVTCk[1:0] Bits (Extended Interrupt Skipping k Count Source Select) (k = 1, 2)

These bits select the way of counting for the extended interrupt skipping counter k. Setting only with these bits does not skip the interrupts, A/D conversion start requests, and buffer transfers.

Skipping function for the interrupt, A/D conversion start request, and buffer transfer, all of which are a target of skipping, is set individually with the GTEITLI1, GTEITLI2, and GTEITLB registers.

EIVTTk[3:0] Bits (The Extended Interrupt Skipping k Skipping Count Setting) (k = 1, 2)

A count for the period with continuous skipping is set as a skipping count, where a period is from a generation of a count source selected by the EIVTCk[1:0] bits to the next generation of the count source. When the count source is generated while the EIVTTk[3:0] bits match the EITCNTk[3:0] bits, the EITCNTk[3:0] bits are cleared.

When these bits are 0000b, skipping is not performed.

EITCNT1[3:0] Bits (Extended Interrupt Skipping Counter 1)

The counting is incremented by 1 every time a count source (overflow/underflow/crest/trough) selected by the EIVTC1[1:0] bits is generated.

Counting is performed periodically within the range between 0 and the EIVTT1[3:0] bits.

Even if the GTCNT counter is stopped, the value is not cleared, and value at stop of the GTCNT counter is retained.

[Clearing conditions]

- 00b is written to the EIVTC1[1:0] bits
- 0000b is written to the EIVTT1[3:0] bits
- A count source (overflow/underflow/crest/trough) selected by the EIVTC1[1:0] bits is generated when the extended interrupt skipping 1 skipping count set by the EIVTT1[3:0] bits match the value for the EITCNT1[3:0]

EITCNT2IV[3:0] Bits (Extended Interrupt Skipping Counter 2 Initial Value)

These bits are the value of the initial value for the extended interrupt skipping counter 2. Writing to the EITCNT2IV[3:0] is performed only when the writing value to the EIVTC2[1:0] bits are other than 00b and when the GTEITC register is written by the access of upper 16 bits or 32 bits while the EITCNT2[3:0] bits are set not to count (EIVTC2[1:0] bits are 00b). When the EITCNT2IV[3:0] bits are written, the value written to the EITCNT2IV[3:0] bits is written to the EITCNT2[3:0] bits simultaneously.

The writing to the EITCNT2IV[3:0] bits are ignored when the EITCNT2[3:0] bits are set to count (EIVTC2[1:0] bits are other than 00b) or perform the setting of not to count (00b is written to the EIVTC2[1:0] bits).

The EITCNT2IV[3:0] bits are not reset by the writing 00b to the EIVTC2[1:0] bits.

EITCNT2[3:0] Bits (Extended Interrupt Skipping Counter 2)

The counting is incremented by 1 every time a count source (overflow/underflow/crest/trough) selected by the EIVTC2[1:0] bits is generated.

Counting is performed periodically within the range between 0 and the EIVTT2[3:0] bits.

Even if the GTCNT counter is stopped, the value is not cleared, and value at stop of the GTCNT counter is retained.

Setting of the initial value for the EITCNT2[3:0] bits are performed only when the writing value to the EIVTC2[1:0] bits are other than 00b and when the GTEITC register is written by the access of upper 16 bits or 32 bits while the extended interrupt skipping counter 2 is set as not to count (00b is written to the EIVTC2[1:0] bits).

When the initial value is set, the written value to the EITCNT2IV[3:0] bits is written to the EITCNT2[3:0] bits as the initial value.

[Clearing conditions]

- 00b is written to the EIVTC2[1:0] bits
- 0000b is written to the EIVTT2[3:0] bits
- The value other than 00b is written to the EIVTC2[1:0] bits and the 0000b is written to the EITCNT2IV[3:0] bits simultaneously while 00b is set to the EIVTC2[1:0] bits

- A count source (overflow/underflow/crest/trough) selected by the EIVTC2[1:0] bits is generated when the extended interrupt skipping 2 skipping count set by the EIVTT2[3:0] bits match the value for the EITCNT2[3:0]

26.2.34 General PWM Timer Extended Interrupt Skipping Setting Register 1 (GTEITLI1)

Address(es): GPTW0.GTEITLI1 000C 20ACh, GPTW1.GTEITLI1 000C 21ACh, GPTW2.GTEITLI1 000C 22ACh, GPTW3.GTEITLI1 000C 23ACh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	EITLU[2:0]			—	EITLV[2:0]			—	EITLF[2:0]			—	EITLE[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	EITLD[2:0]			—	EITLC[2:0]			—	EITLB[2:0]			—	EITLA[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	EITLA[2:0]	GTCCRA Register Compare Match/ Input Capture Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	EITLB[2:0]	GTCCRB Register Compare Match/ Input Capture Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10 to b8	EITLC[2:0]	GTCCRC Register Compare Match Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14 to b12	EITLD[2:0]	GTCCRD Register Compare Match Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b18 to b16	EITLE[2:0]	GTCCRE Register Compare Match Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b19	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b22 to b20	EITLF[2:0]	GTCCRF Register Compare Match Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b23	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b26 to b24	EITLV[2:0]	Overflow Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b27	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b30 to b28	EITLU[2:0]	Underflow Interrupt Extended Skipping Function Select	See Table 26.5.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

The GTEITLI1 register sets the extended skipping function for interrupts such as compare match/input capture, overflow, and underflow.

Setting only with this register does not perform skipping. The GTEITC register should be set so that a corresponding extended interrupt skipping counter operates the counting.

The setting is operated independently from the interrupt skipping by the GTITC register.

The setting is invalid during the event count operation.

EITLm[2:0] Bits (GTCCRm Register Compare Match/Input Capture Interrupt Extended Skipping Function Select) (m = A, B)

These bits select the extended interrupt skipping function to skip the compare match/input capture interrupt (GTCIm) in the GTCCRm register.

Refer to Table 26.5.

EITLx[2:0] Bits (GTCCRx Register Compare Match Interrupt Extended Skipping Function Select) (x = C, D, E, F)

These bits select the extended interrupt skipping function to skip the compare match interrupt (GTCIx) in the GTCCRx register.

Refer to Table 26.5.

EITLV[2:0] Bits (Overflow Interrupt Extended Skipping Function Select)

These bits select the extended interrupt skipping function to skip the interrupt at overflow.

Refer to Table 26.5.

EITLU[2:0] Bits (Underflow Interrupt Extended Skipping Function Select)

These bits select the extended interrupt skipping function to skip the interrupt at underflow.

Refer to Table 26.5.

Table 26.5 Setting the Function Select for the GTEITL1 Register

EITLy[2:0] Bits	Function
0 0 0	Do not perform an extended interrupt skipping
0 0 1	Skip an interrupt in the period when the value for the extended interrupt skipping counter 1 is other than 0 (An interrupt is output in the period of EITCNT1 bit = 0)
0 1 0	Skip an interrupt in the period when the value for the extended interrupt skipping counter 2 is other than 0 (An interrupt is output in the period of EITCNT2 bit = 0)
0 1 1	Skip an interrupt in the period the value for the extended interrupt skipping counter 1 or 2 is other than 0 (An interrupt is output in the period of EITCNT1 bit = 0 and EITCNT2 bit = 0)
1 0 0	Do not set this value
1 0 1	Skip an interrupt in the period when the value for the extended interrupt skipping counter 1 is other than the skipping count (An interrupt is output in the period of EITCNT1 bit = EIVTT1 bit)
1 1 0	Skip an interrupt in the period when the value for the extended interrupt skipping counter 2 is other than the skipping count (An interrupt is output in the period of EITCNT2 bit = EIVTT2 bit)
1 1 1	Skip an interrupt in the period when the value for the extended interrupt skipping counter 1 or 2 is other than the skipping count (An interrupt is output in the period of EITCNT1 bit = EIVTT1 bit and EITCNT2 bit = EIVTT2 bit)

y = A, B, C, D, E, F, V, U; k = 1, 2

Note: When the intended skipping counter is set as not to count (EIVTck[1:0]bits = 00b or EIVTTk[3:0] bits = 0000b), skipping is not performed.

Note: When the EITLy[2:0] bits are set to 011b or 111b, or when one of the skipping counters is set as not to count, skipping is not performed.

26.2.35 General PWM Timer Extended Interrupt Skipping Setting Register 2 (GTEITLI2)

Address(es): GPTW0.GTEITLI2 000C 20B0h, GPTW1.GTEITLI2 000C 21B0h, GPTW2.GTEITLI2 000C 22B0h, GPTW3.GTEITLI2 000C 23B0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	EADTBL[2:0]	—	—	—	EADTAL[2:0]	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	EADTAL[2:0]	GTADTRA A/D Conversion Start Request Extended Skipping Function Select	See Table 26.6.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	EADTBL[2:0]	GTADTRB A/D Conversion Start Request Extended Skipping Function Select	See Table 26.6.	R/W
b31 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTEITLI2 register sets the extended skipping function for A/D conversion start requests.

Setting only with this register does not perform skipping. The GTEITC register should be set so that a corresponding extended interrupt skipping counter operates the counting.

The setting is operated independently from the interrupt skipping by the GTITC register.

The setting is invalid during the event count operation.

EADTmL[2:0] Bits (GTADTRm A/D Conversion Start Request Extended Skipping Function Select)

(m = A, B)

These bits select the extended interrupt skipping function to skip A/D conversion start requests for the compare match in the GTADTRm register.

Refer to Table 26.6.

Table 26.6 Setting the Function Select for the GTEITL12 Register

EADTmL[2:0] Bits	Function
0 0 0	Do not perform an extended interrupt skipping
0 0 1	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 1 is other than 0 (An A/D conversion start request is output in the period of EITCNT1 bit = 0)
0 1 0	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 2 is other than 0 (An A/D conversion start request is output in the period of EITCNT2 bit = 0)
0 1 1	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 1 or 2 is other than 0 (An A/D conversion start request is output in the period of EITCNT1 bit = 0 and EITCNT2 bit = 0)
1 0 0	Do not set this value
1 0 1	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 1 is other than the skipping count (An A/D conversion start request is output in the period of EITCNT1 bit = EIVTT1 bit)
1 1 0	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 2 is other than the skipping count (An A/D conversion start request is output in the period of EITCNT2 bit = EIVTT2 bit)
1 1 1	Skip an A/D conversion start request in the period when the value for the extended interrupt skipping counter 1 or 2 is other than the skipping count (An A/D conversion start request is output in the period of EITCNT1 bit = EIVTT1 bit and EITCNT2 bit = EIVTT2 bit)

m = A, B; k = 1, 2

Note: When the intended skipping counter is set as not to count (EIVTCk[1:0]bits = 00b or EIVTTk[3:0] bits = 0000b), skipping is not performed.

Note: When the EADTmL[2:0] bits are set to 011b or 111b, or when one of the skipping counters is set as not to count, skipping is not performed.

26.2.36 General PWM Timer Extended Buffer Transfer Skipping Setting Register (GTEITLB)

Address(es): GPTW0.GTEITLB 000C 20B4h, GPTW1.GTEITLB 000C 21B4h, GPTW2.GTEITLB 000C 22B4h, GPTW3.GTEITLB 000C 23B4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	EBTLDVD[2:0]			—	EBTLDVU[2:0]			—	EBTLADB[2:0]			—	EBTLADA[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	EBTLPR[2:0]		—	EBTLCB[2:0]			—	EBTLCA[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	EBTLCA[2:0]	GTCCRA Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	EBTLCB[2:0]	GTCCRB Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10 to b8	EBTLPR[2:0]	GTPR Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b16	EBTLADA[2:0]	GTADTRA Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b19	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b22 to b20	EBTLADB[2:0]	GTADTRB Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b23	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b26 to b24	EBTLDVU[2:0]	GTDVU Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b27	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b30 to b28	EBTLDVD[2:0]	GTDVD Register Buffer Transfer Extended Skipping Function Select	See Table 26.7.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

The GTEITLB register sets the extended skipping function for buffer transfers.

Setting only with this register does not perform skipping. The GTEITC register should be set so that a corresponding extended interrupt skipping counter operates the counting.

The setting is operated independently from the interrupt skipping by the GTITC register.

The setting is invalid during the event count operation.

EBTLCA[2:0] Bits (GTCCRA Register Buffer Transfer Extended Skipping Function Select)

These bits select the extended buffer transfer skipping function to skip buffer transfers (transfers among the GTCCRA, GTCCRC, GTCCRD registers and a temporary register A) in the GTCCRA register.

An extended skipping of buffer transfers in the GTCCRA register is valid for forcible buffer transfers by the GTBER.CCRSWT bit while the count operation is stopped. Forcible buffer transfers in the GTCCRA register should be performed in the condition of not performing the extended buffer transfer skipping.

EBTLCB[2:0] Bits (GTCCRB Register Buffer Transfer Extended Skipping Function Select)

These bits select the extended buffer transfer skipping function to skip buffer transfers (transfers among the GTCCRB, GTCCRE, GTCCRF registers and a temporary register B) in the GTCCRB register.

An extended skipping of buffer transfers in the GTCCRB register is valid for forcible buffer transfers by the GTBER.CCRSWT bit while the count operation is stopped. Forcible buffer transfers in the GTCCRB register should be performed in the condition of not performing the extended buffer transfer skipping.

EBTLPR[2:0] Bits (GTPR Register Buffer Transfer Extended Skipping Function Select)

These bits select the extended buffer transfer skipping function to skip buffer transfers (transfers among the GTPR, GTPBR, and GTPDBR registers) in the GTPR register.

EBTLADm[2:0] Bits (GTADTRm Register Buffer Transfer Extended Skipping Function Select) (m = A, B)

These bits select the extended buffer transfer skipping function to skip buffer transfers (transfers among the GTADTRm, GTADTBm, and GTADTDBm registers) in the GTADTRm register.

EBTLDVm[2:0] Bits (GTDVm Register Buffer Transfer Extended Skipping Function Select) (m = U, D)

These bits select the extended buffer transfer skipping function to skip buffer transfers (transfers between the GTDVm and GTDBm registers) in the GTDVm register.

Table 26.7 Setting the Function Select for the GTEITLB Register

EBTLx[2:0] Bits	Function
0 0 0	Do not perform an extended interrupt skipping
0 0 1	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 1 is other than 0 (A buffer transfer is output in the period of EITCNT1 bit = 0)
0 1 0	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 2 is other than 0 (A buffer transfer is output in the period of EITCNT2 bit = 0)
0 1 1	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 1 or 2 is other than 0 (A buffer transfer is output in the period of EITCNT1 bit = 0 and EITCNT2 bit = 0)
1 0 0	Do not set this value
1 0 1	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 1 is other than the skipping count (A buffer transfer is output in the period of EITCNT1 bit = EIVTT1 bit)
1 1 0	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 2 is other than the skipping count (A buffer transfer is output in the period of EITCNT2 bit = EIVTT2 bit)
1 1 1	Skip a buffer transfer in the period when the value for the extended interrupt skipping counter 1 or 2 is other than the skipping count (A buffer transfer is output in the period of EITCNT1 bit = EIVTT1 bit and EITCNT2 bit = EIVTT2 bit)

x = CA, CB, PR, ADA, ADB, DVU, DVD; k = 1, 2

Note: When the intended skipping counter is set as not to count (EIVTCK[1:0]bits = 00b or EIVTTk[3:0] bits = 0000b), skipping is not performed.

Note: When the EBTLx[2:0] bits are set to 011b or 111b, or when one of the skipping counters is set as not to count, skipping is not performed.

26.2.37 General PWM Timer Operation Enable Bit Simultaneous Control Channel Select Register (GTSECSR)

Address(es): GPTW0.GTSECSR 000C 20D0h, GPTW1.GTSECSR 000C 21D0h, GPTW2.GTSECSR 000C 22D0h, GPTW3.GTSECSR 000C 23D0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	SECSEL3	SECSEL2	SECSEL1	SECSEL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SECSEL0	Channel 0 Operation Enable Bit Simultaneous Control Channel Select	0: Disable simultaneous control. 1: Enable simultaneous control.	R/W
b1	SECSEL1	Channel 1 Operation Enable Bit Simultaneous Control Channel Select		R/W
b2	SECSEL2	Channel 2 Operation Enable Bit Simultaneous Control Channel Select		R/W
b3	SECSEL3	Channel 3 Operation Enable Bit Simultaneous Control Channel Select		R/W
b31 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTSECSR register selects an intended channel for updating an operation enable bit by the GTSECR register.

A bit position for the GTSECSR register indicates a channel number. The GTSECSR register of each channel is a common register, and writing 1 to a bit in the GTSECSR register in any channel and updating it changes a channel, related to the position of the bit written with 1 by the GTSECSR register, to be simultaneously controlled of the operation enable bit by the GTSECR register.

Access in 8-bit or 16-bit units to the GTSECSR register is prohibited, and it should be accessed in 32-bit units.

SECSELn Bit (Operation Enable Bit Simultaneous Control Channel Select) (n = 0 to 3)

This bit enables or disables the simultaneous control of operation enable in channel n.

When the bit is set to 1, the simultaneous control is enabled, and disabled when the bit is 0.

26.2.38 General PWM Timer Operation Enable Bit Simultaneous Control Register (GTSECR)

Address(es): GPTW0.GTSECR 000C 20D4h, GPTW1.GTSECR 000C 21D4h, GPTW2.GTSECR 000C 22D4h, GPTW3.GTSECR 000C 23D4h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	SBDDD	SBDAD	SBDPD	SBDCE	—	—	—	—	SBDDE	SBDPE	SBDCE	SBDCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SBDCE	GTCCR Register Buffer Operation Simultaneous Enable	0: Disable simultaneous enabling GTCCR buffer operations 1: Enable GTCCR register buffer operations simultaneously	R/W
b1	SBDPE	GTPR Register Buffer Operation Simultaneous Enable	0: Disable simultaneous enabling GTPR buffer operations 1: Enable GTPR register buffer operations simultaneously	R/W
b2	SBDPE	GTADTR Register Buffer Operation Simultaneous Enable	0: Disable simultaneous enabling GTADTR buffer operations 1: Enable GTADTR register buffer operations simultaneously	R/W
b3	SBDDE	GTDV Register Buffer Operation Simultaneous Enable	0: Disable simultaneous enabling GTDV buffer operations 1: Enable GTDV register buffer operations simultaneously	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	SBDCE	GTCCR Register Buffer Operation Simultaneous Disable	0: Disable simultaneous disabling GTCCR buffer operations 1: Disable GTCCR register buffer operations simultaneously	R/W
b9	SBDPE	GTPR Register Buffer Operation Simultaneous Disable	0: Disable simultaneous disabling GTPR buffer operations 1: Disable GTPR register buffer operations simultaneously	R/W
b10	SBDPE	GTADTR Register Buffer Operation Simultaneous Disable	0: Disable simultaneous disabling GTADTR buffer operations 1: Disable GTADTR register buffer operations simultaneously	R/W
b11	SBDDE	GTDV Register Buffer Operation Simultaneous Disable	0: Disable simultaneous disabling GTDV buffer operations 1: Disable GTDV register buffer operations simultaneously	R/W
b31 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The GTSECR register simultaneously updates the value for operation enable bits of a channel set by the GTSECSR register.

Writing 1 to a bit in the GTSECR register in any channel and updating it updates an operation enable bit for all channels, related to the position of the bit written with 1 by the all GTSECR registers. Setting enable and disable bits for the same operation enable bit to 1 in the GTSECR is prohibited.

A bit written to 1 is automatically cleared. When the GTSECR is read, 0 is read.

Access in 8-bit or 16-bit units to the GTSECR register is prohibited, and it should be accessed in 32-bit units.

SBDCE Bit (GTCCR Register Buffer Operation Simultaneous Enable)

When 1 is written to this bit, 0 is simultaneously set to a GTBER.BD[0] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTCCRA, GTCCRC, and GTCCRD registers and using the GTCCRB, GTCCRE, and GTCCRF registers are enabled.

Simultaneous setting of SBDCE and SBDCE bits to 1 is prohibited.

SBDPE Bit (GTPR Register Buffer Operation Simultaneous Enable)

When 1 is written to this bit, 0 is simultaneously set to a GTBER.BD[1] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTPR, GTPBR, and GTPDBR registers are enabled.

Simultaneous setting of SBDPE and SBDPD bits to 1 is prohibited.

SBD AE Bit (GTADTR Register Buffer Operation Simultaneous Enable)

When 1 is written to this bit, 0 is simultaneously set to a GTBER.BD[2] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTADTRA, GTADTBRA, and GTADTDBRA registers and using the GTADTRB, GTADTBRB, and GTADTDBRB registers are enabled.

Simultaneous setting of SBD AE and SBDAD bits to 1 is prohibited.

SBDDE Bit (GTDV Register Buffer Operation Simultaneous Enable)

When 1 is written to this bit, 0 is simultaneously set to a GTBER.BD[3] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTDVU and GTDBU registers and using the GTDVD and GTDBD registers are enabled.

Simultaneous setting of SBDDE and SBDDD bits to 1 is prohibited.

SBD CD Bit (GTCCR Register Buffer Operation Simultaneous Disable)

When 1 is written to this bit, 1 is simultaneously set to a GTBER.BD[0] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTCCRA, GTCCRC, and GTCCRD registers and using the GTCCRB, GTCCRE, and GTCCRF registers are disabled.

Simultaneous setting of SBDCE and SBD CD bits to 1 is prohibited.

SBDPD Bit (GTPR Register Buffer Operation Simultaneous Disable)

When 1 is written to this bit, 1 is simultaneously set to a GTBER.BD[1] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTPR, GTPBR, and GTPDBR registers are disabled.

Simultaneous setting of SBDPE and SBDPD bits to 1 is prohibited.

SBDAD Bit (GTADTR Register Buffer Operation Simultaneous Disable)

When 1 is written to this bit, 1 is simultaneously set to a GTBER.BD[2] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTADTRA, GTADTBRA, and GTADTDBRA registers and using the GTADTRB, GTADTBRB, and GTADTDBRB registers are disabled.

Simultaneous setting of SBD AE and SBDAD bits to 1 is prohibited.

SBDDD Bit (GTDV Register Buffer Operation Simultaneous Disable)

When 1 is written to this bit, 1 is simultaneously set to a GTBER.BD[3] bit in the channels set to 1 by the GTSECSR register, and buffer operations using the GTDVU and GTDBU registers and using the GTDVD and GTDBD registers are disabled.

Simultaneous setting of SBDDE and SBDDD bits to 1 is prohibited.

26.3 Operation

26.3.1 Basic Operation

The timer in each channel performs cycle count operation by count clock or hardware source. The GTCNT counter performs up-counting, or down-counting. The timer period is controlled by the GTPR register. When the GTCNT counter value matches the values for GTCCRA or GTCCRB registers, the GTIOCnA or GTIOCnB pin outputs ($n = 0$ to 3) can be changed respectively. Also, the GTCCRA or GTCCRB registers can be used as the input capture registers by hardware source. The GTCCRC and GTCCRD registers can be performed as buffer registers for GTCCRA register, and the GTCCRE and GTCCRF registers can be performed as buffer registers for GTCCRB register.

26.3.1.1 Counter Operation

(1) Count Start/Count Stop

The counter in each channel starts counting operation when the corresponding GTCR.CST bit is set to 1, and stops counting when the bit is set to 0.

CST bit value can be changed by the following sources:

- Writing to the GTCR register
- Writing 1 to the bit corresponding to the channel number for the GTSTR register while the GTSSR.CSTRT bit is 1
- Writing 1 to the bit corresponding to the channel number for the GTSTP register while the GTPSR.CSTOP bit is 1
- Hardware source specified by the GTSSR register
- Hardware source specified by the GTPSR register

(2) Cycle Count Operation (In Up-Counting by the Count Clock)

The counter in each channel starts up-counting when the corresponding CST bit is set to 1 while the GTUPSR register and the GTDNSR register are 0000 0000h. When the value of the GTCNT counter changes from that of the GTPR register to 0000 0000h (an overflow) while the GTINTAD.GTINTPR[0] bit is 1, a GTCIV interrupt request is also issued. After GTCNT counter overflows, up-counting is resumed from 0000 0000h.

Figure 26.2 shows an example of cycle count operation in up-counting by count clock.

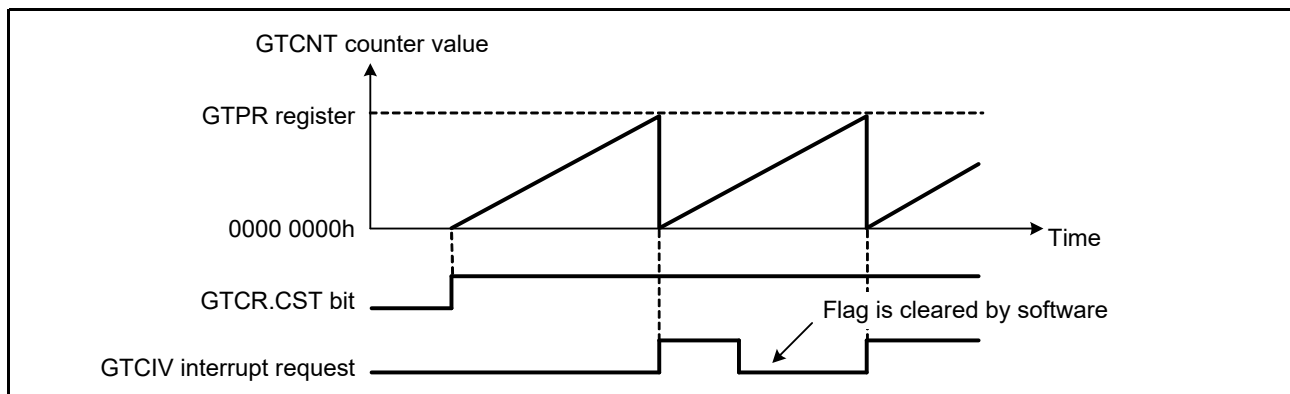


Figure 26.2 Example of Cycle Count Operation (In Up-Counting by the Count Clock)

Figure 26.3 shows an example for setting cycle count operation in up-counting.

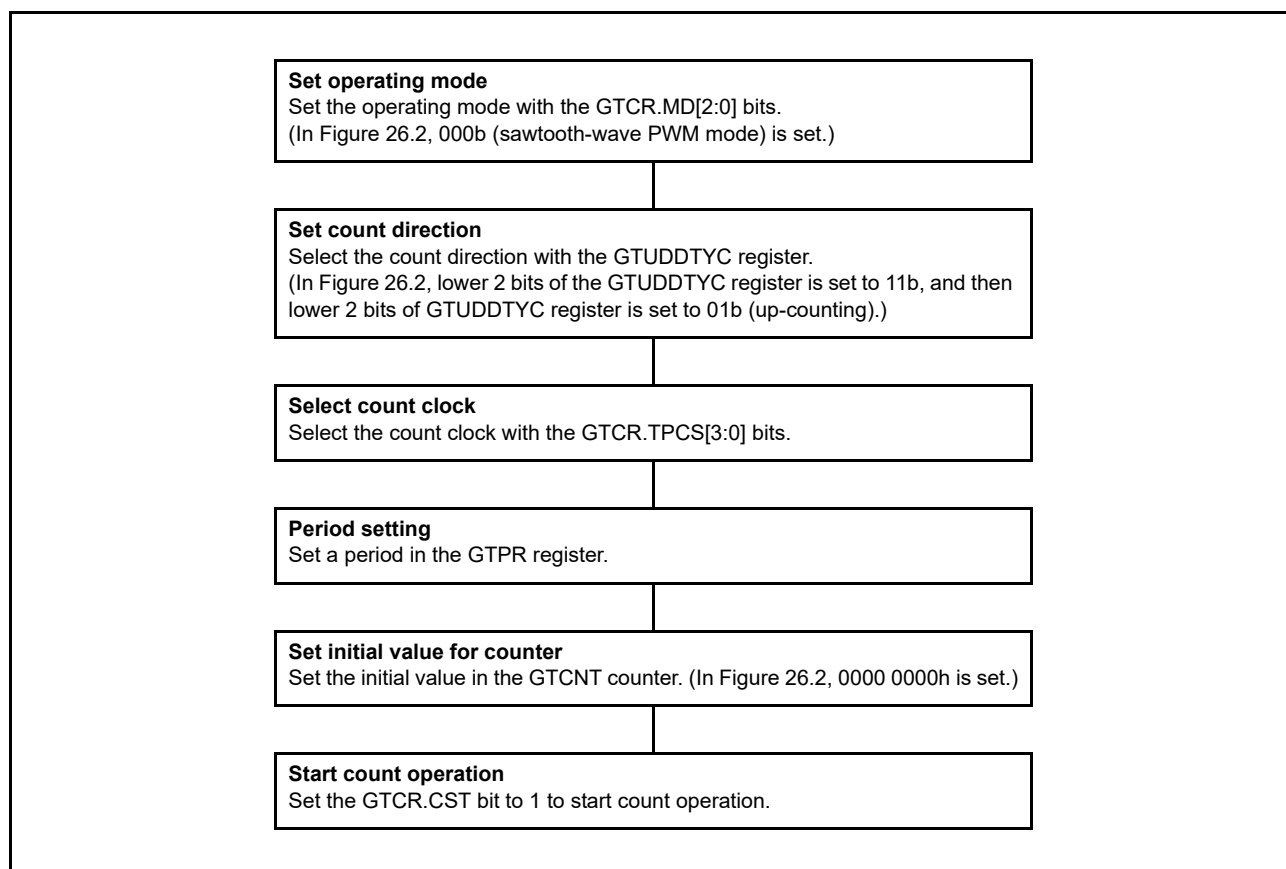


Figure 26.3 Example for Setting Cycle Count Operation (In Up-Counting by the Count Clock)

(3) Cycle Count Operation (In Down-Counting by the Count Clock)

The counter in each channel can start down-counting when the corresponding GTUDDTYC.UD bit is set in a state of the GTUPSR register and the GTDNSR register as 0000 0000h. When the value of the GTCNT counter changes from 0000 0000h to that of the GTPR register (an underflow) while the GTINTAD.GTINTPR[1] bit is 1, a GTCIU interrupt request is also issued. After the GTCNT counter underflows, down-counting is resumed from the GTPR register value.

Figure 26.4 shows an example of cycle count operation in down-counting by count clock.

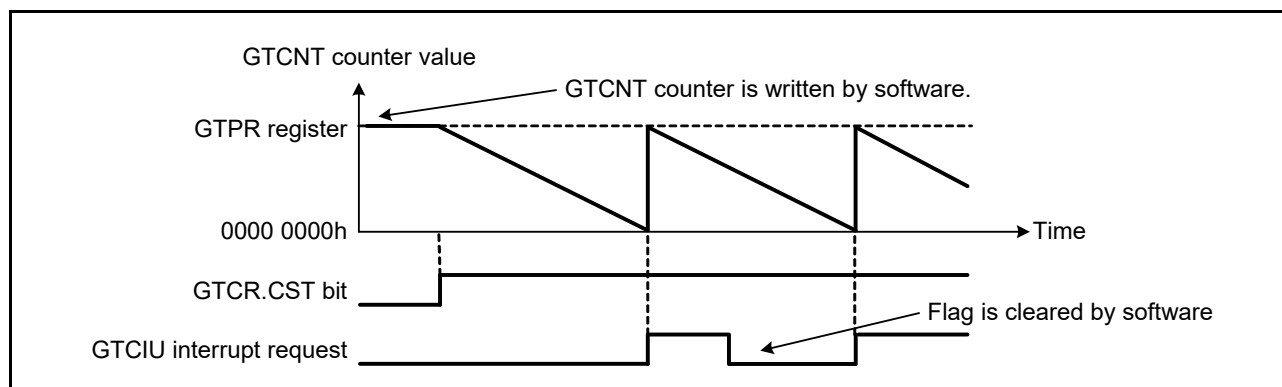


Figure 26.4 Example of Cycle Count Operation (In Down-Counting by the Count Clock)

Figure 26.5 shows an example for setting cycle count operation in down-counting by count clock.

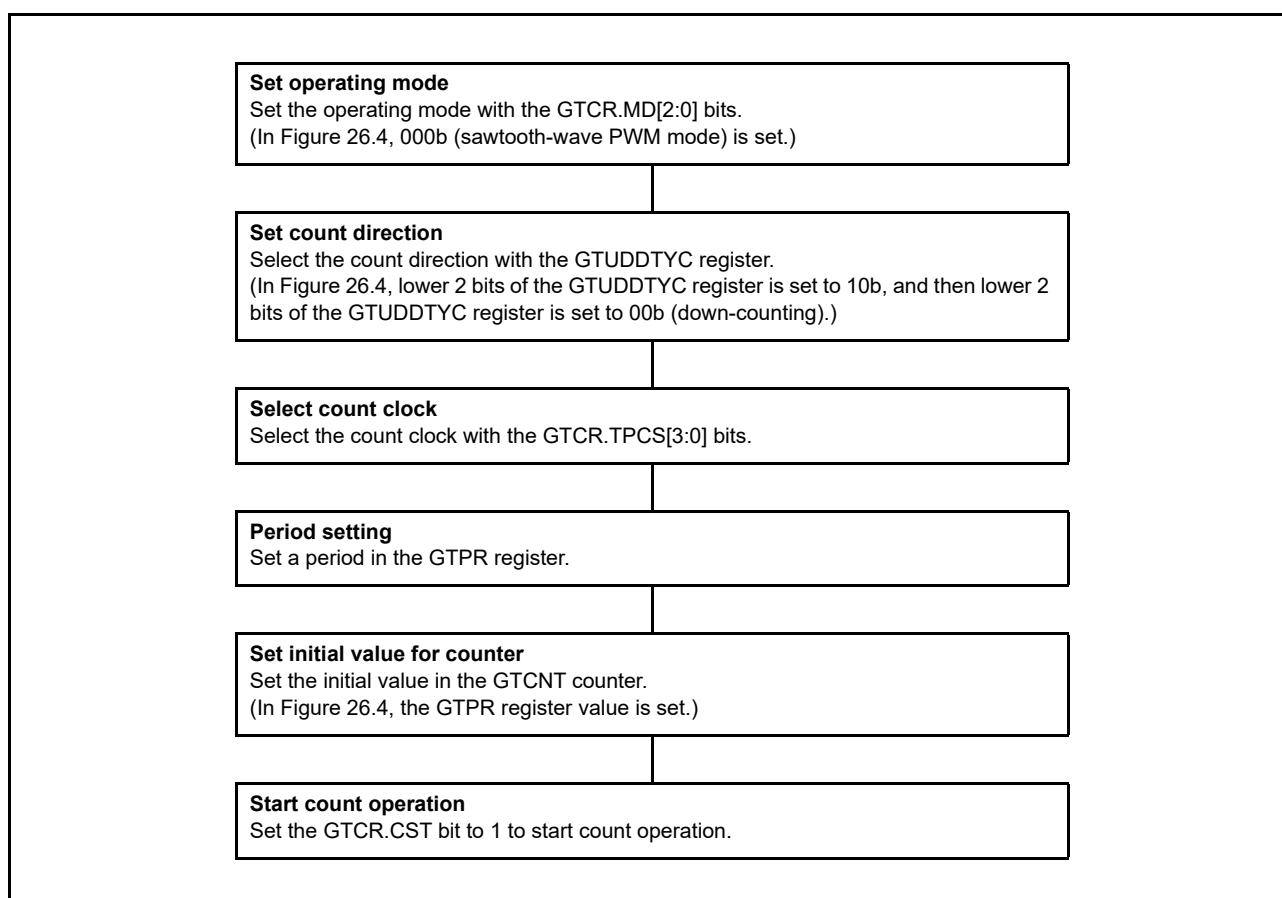


Figure 26.5 Example for Setting Cycle Count Operation (In Down-Counting by the Count Clock)

(4) Event Count Operation (In Up-Counting by Hardware Source)

The counter in each channel can start up-counting by hardware source by setting the GTUPSR register. When the GTUPSR register is set, the count clock selected by the GTCR.TPCS[3:0] bits and the count direction set by the GTUDDTYC.UD bit are invalid. If the hardware source in up-counting and the hardware source in down-counting are generated at the same time, the GTCNT counter value will not change.

Operation due to overflow in up-counting by hardware source is the same as the cycle count operation by the count clock.

To perform up-counting by hardware source, start count operation by setting the GTCR.CST bit to 1. As the starting the count operation is synchronized with the count clock selected by TPCS[3:0] bits, set the CST bit to 1, and wait until the first cycle of the count clock elapses to start up-counting operation. To start up-counting after one PCLKA from setting the CST bit to 1, set the TPCS[3:0] bits to 0000b.

Figure 26.6 to Figure 26.8 show examples of up-counting operation by hardware source .

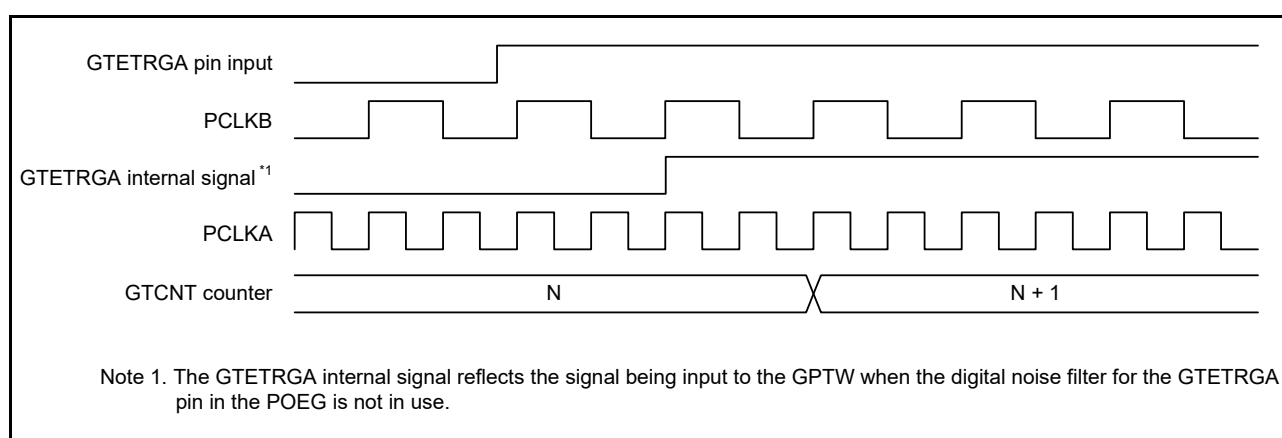


Figure 26.6 Example of Event Count Operation (Up-Counting of Rising Edges of the Input on the GTETRGA Pin)

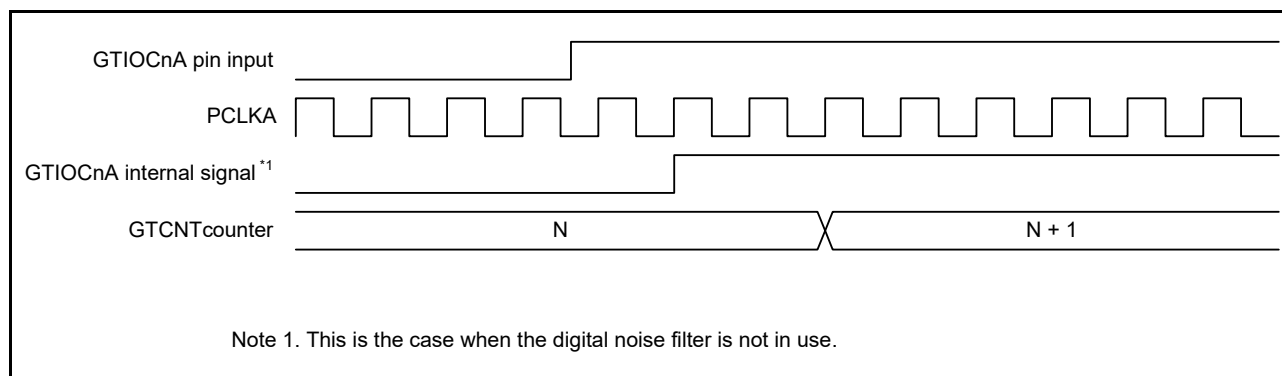


Figure 26.7 Example of Event Count Operation (Up-Counting of Rising Edges of the Input on the GTIOCnA Pin)

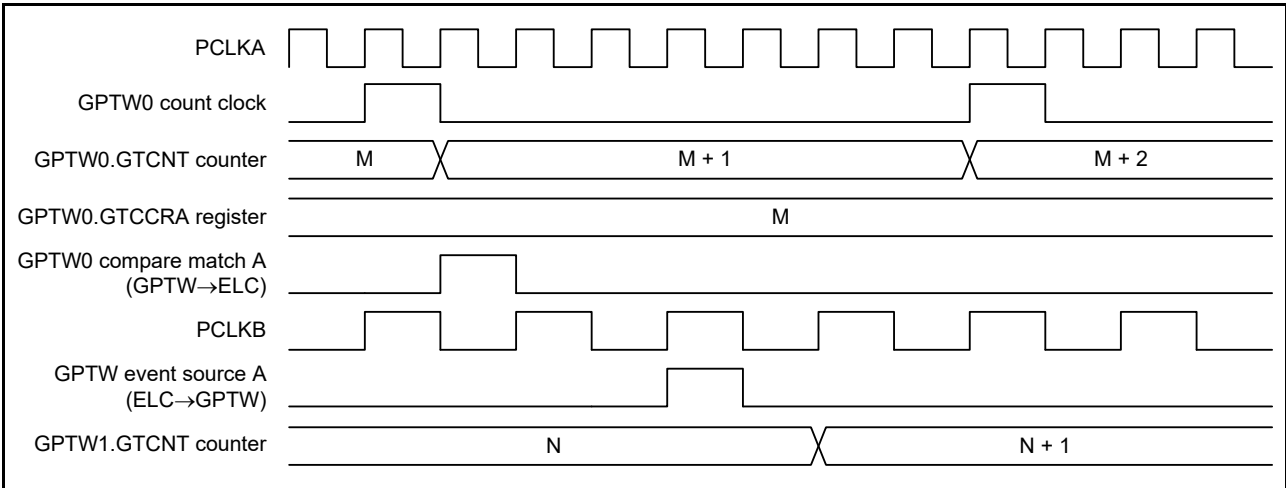


Figure 26.8 Example of Event Count Operation (Up-Counting the Number of Event Signals Input from the ELCA)

Figure 26.9 shows an example for setting cycle count operation in up-counting by hardware source.

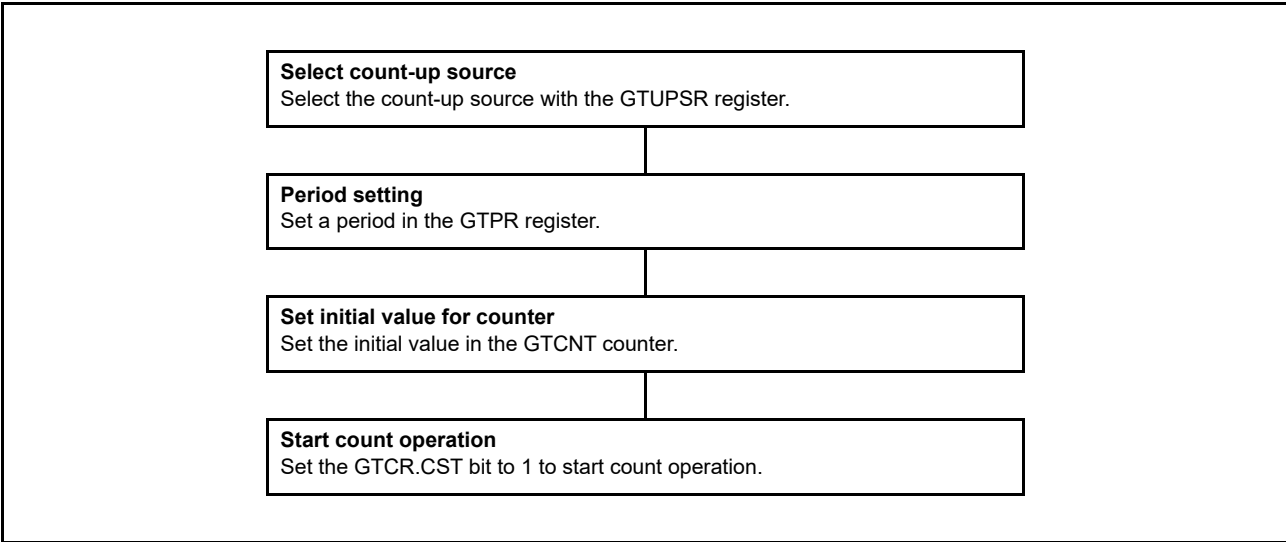


Figure 26.9 Example for Setting Event Count Operation (In Up-Counting by Hardware Source)

(5) Event Count Operation (In Down-Counting by Hardware Source)

The counter in each channel can start down-counting by hardware source by setting the GTDNSR register. When the GTDNSR register is set, the count clock selected by the GTCR.TPCS[3:0] bits and the count direction set by the GTUDDTYC.UD bit are invalid. If the hardware source in up-counting and the hardware source in down-counting are generated at the same time, the GTCNT counter value will not change.

Operation due to underflow in down-counting by hardware source is the same as the cycle count operation by the count clock.

To perform down-counting by hardware source, start count operation by setting the GTCR.CST bit to 1. As the starting the count operation is synchronized with the count clock selected by TPCS[3:0] bits, set the CST bit to 1, and wait until the first cycle of the count clock elapses to start down-counting operation. To start down-counting after one PCLKA from setting the CST bit to 1, set the TPCS[3:0] bits to 0000b.

Figure 26.10 shows an example of down-counting operation by hardware source.

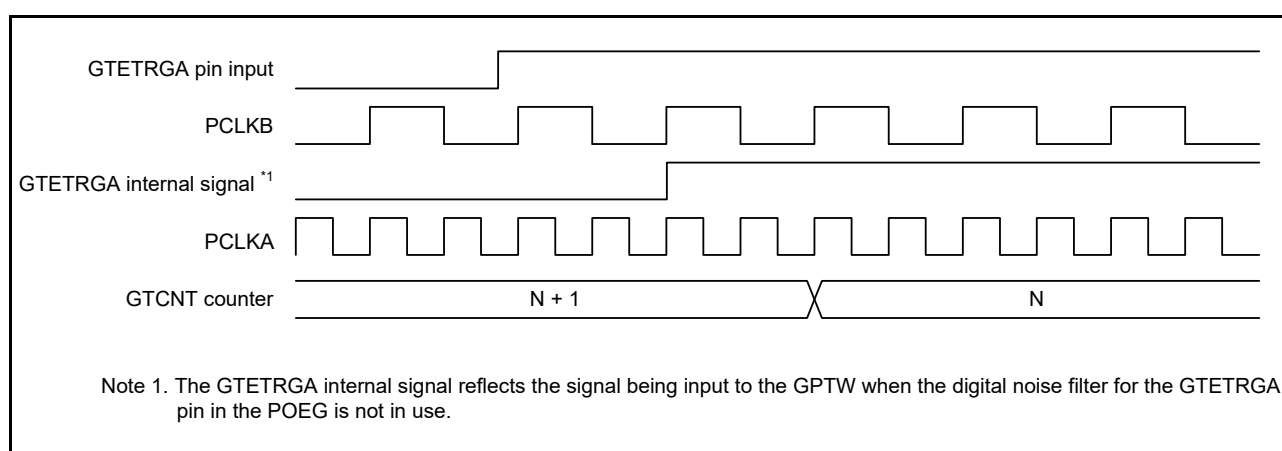


Figure 26.10 Example of Event Count Operation (Down-Counting of Rising Edges of the Input on the GTETRGA Pin)

Figure 26.11 shows an example for setting cycle count operation in down-counting by hardware source.

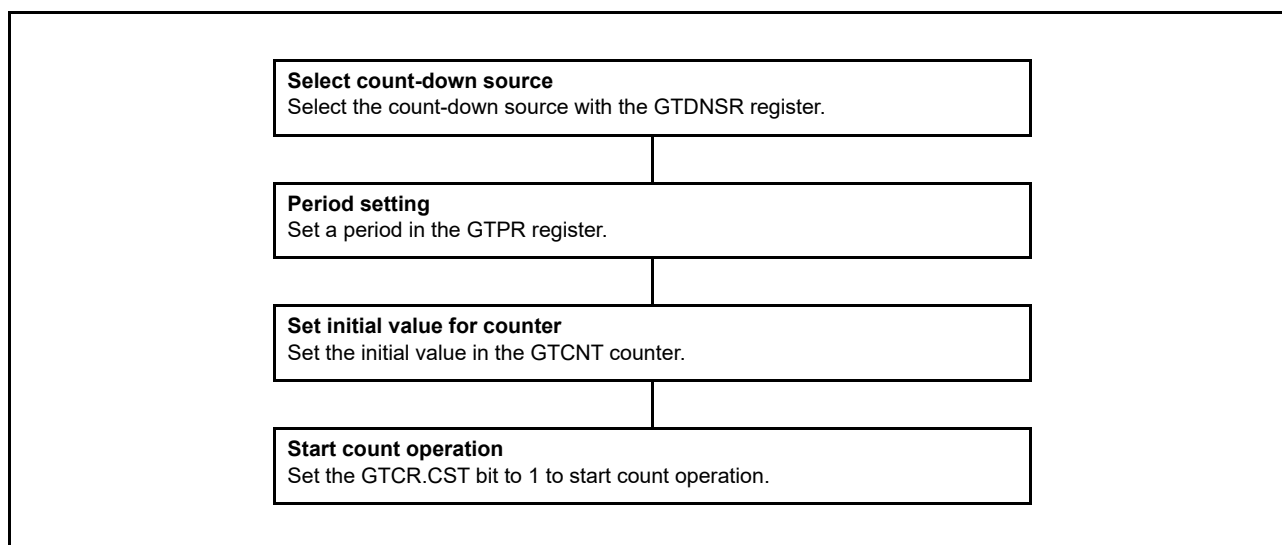


Figure 26.11 Example for Setting Event Count Operation (In Down-Counting by Hardware Source)

(6) Counter Clearing Operation

The counter in each channel can be cleared by the following sources:

- Writing to the GTCNT counter
- Writing 1 to the bit corresponding to the channel number for the GTCLR register while the GTCSR.CCLR bit is 1
- Hardware source specified by the GTCSR register

Write access during counting (when CST = 1) is disabled.

GTCLR register can be written to and cleared by hardware source whether the count operation is performed (GTCR.CST bit = 1) or is stopped (CST bit = 0). When the count direction flag is set as decrement (GTST.TUCF flag = 0) in sawtooth-wave mode selected with GTCR.MD[2:0] bits, the GTCNT counter value for writing to the GTCLR register and for clearing by hardware source becomes the GTPR register value whether the counter is in operation or is stopped. The GTCNT counter value becomes 0000 0000h for other settings.

When the event count operation is set (at least one of bits for the GTUPSR or the GTDNSR register is set as 1), writing to the GTCLR register and clearing by hardware source are performed right after the clear source is generated (the operation is based on PCLKA). For other settings, clearing operation is synchronized with the count clock selected with the GTCR.TPCS[3:0] bits.

26.3.1.2 Waveform Output by Compare Match

Compare match refers to when the GTCNT counter value matches the GTCCRA or GTCCRB register value. The output of the corresponding GTIOCnA or GTIOCnB pin (n = 0 to 3) can be set to be driven low, high, or toggled in synchronization with the counter clock (including in the case of event counting) after the match.

In addition, the GTIOCnA or GTIOCnB pin output can be driven low, high, or toggled at the “end of the cycle” which is determined by the GTPR register. The end of the cycle is as follows:

- For sawtooth waves in up-counting: When the GTCNT counter value changes from the GTPR register value to 0000 0000h (overflow)
- For sawtooth waves in down-counting: When the GTCNT counter value changes from 0000 0000h to the GTPR register value (underflow)
- For sawtooth waves with the GTCNT counter clear
- For triangle waves: When the GTCNT counter value changes from 0000 0000h to 0000 0001h (trough)

(1) Low Output and High Output

Figure 26.12 shows an example of low output and high output operation by a compare match of the GTCNT counter and GTCCRA register and of the GTCNT counter and GTCCRB register.

In this example, the GPTW0.GTCNT counter performs up-counting, and settings have been made so that high is output from the GTIOC0A pin by a GPTW0.GTCCRA register compare match, and low is output from the GTIOC0B pin by a GPTW0.GTCCRB register compare match. The pin level does not change when the specified level and pin level match.

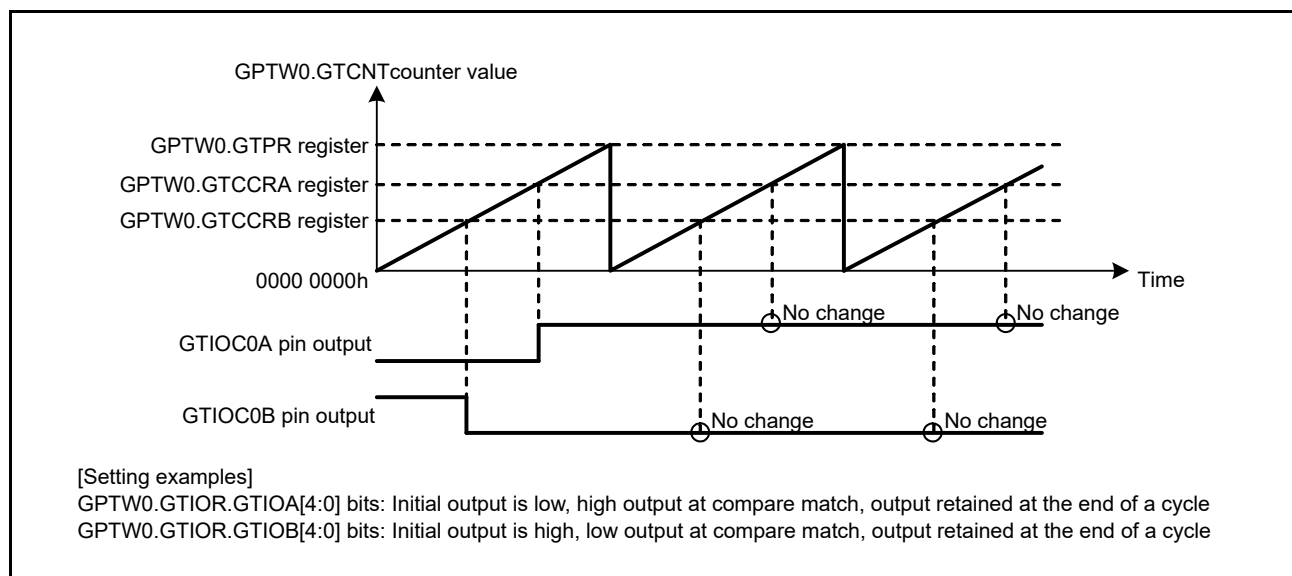


Figure 26.12 Example of Low Output and High Output Operation

Figure 26.13 shows an example for setting Low/High output operation.

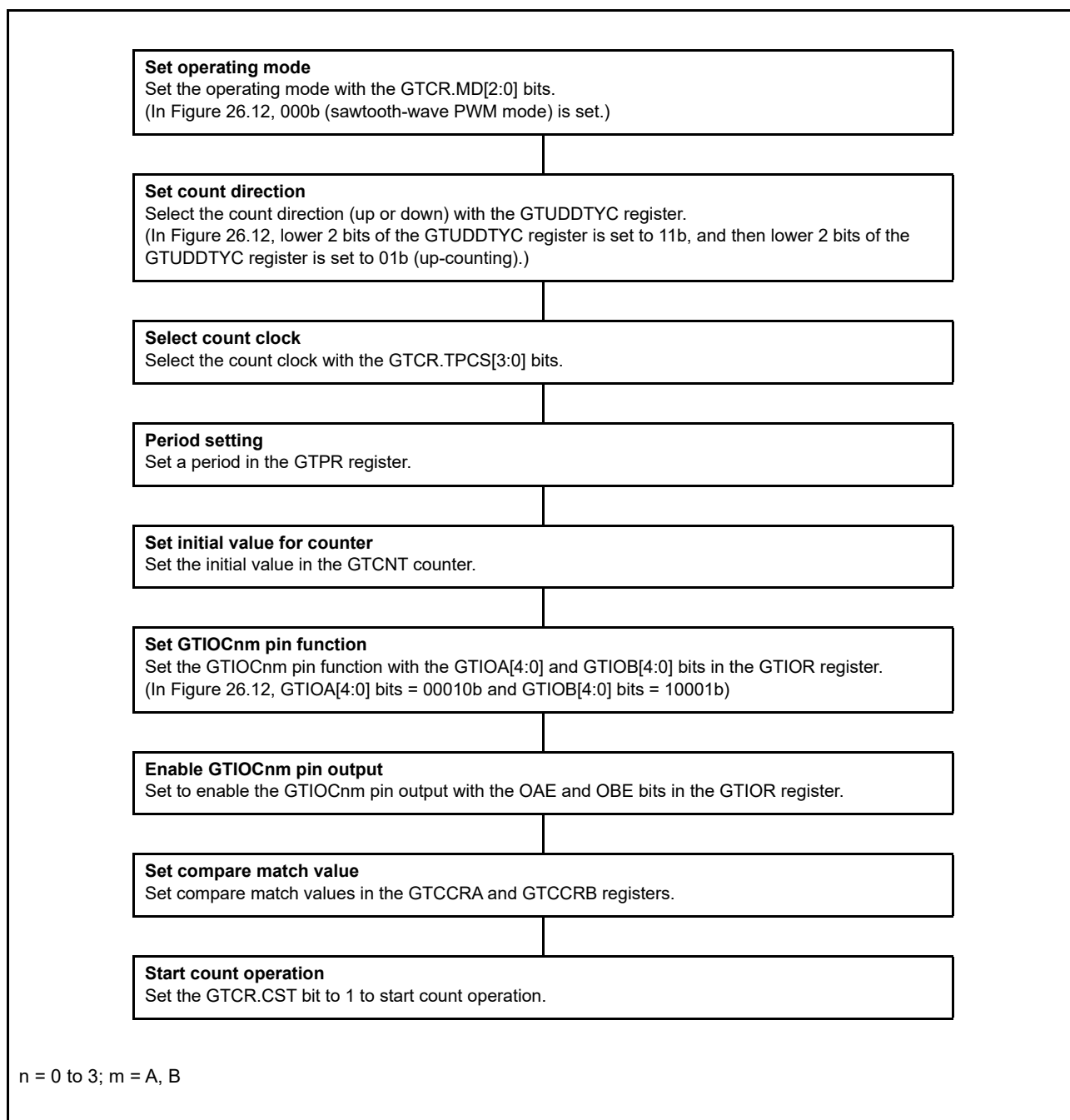


Figure 26.13 Example for Setting Low/High Output Operation

(2) Toggled Output

Figure 26.14 and Figure 26.15 show examples of toggle output operations by compare match between the GPTW0.GTCNT counter and the GTCCRA and GTCCRB registers.

In Figure 26.14, the GPTW0.GTCNT counter performs up-counting, and settings have been made so that the GTIOC0A pin output by a GPTW0.GTCCRA register compare match and GTIOC0B pin output by a GPTW0.GTCCRB register compare match are toggled.

In Figure 26.15, the GPTW0.GTCNT counter performs up-counting, and settings have been made so that the GTIOC0A pin output is toggled by a compare match of GPTW0.GTCCRA register and the GTIOC0B pin output is toggled at the end of the cycle.

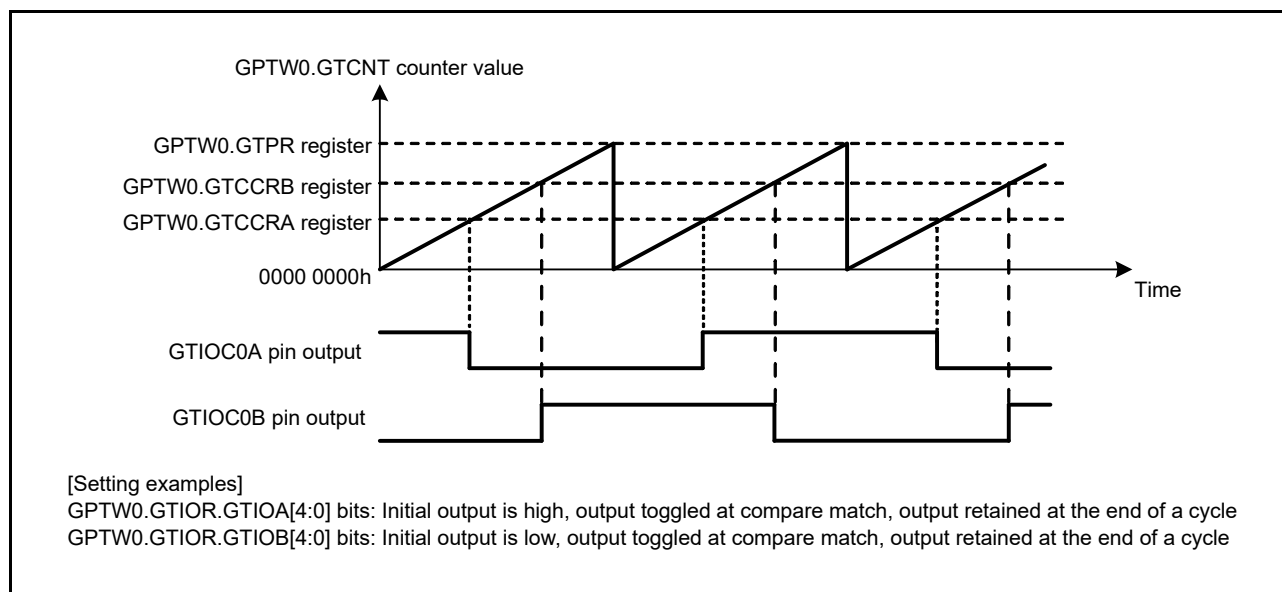


Figure 26.14 Example of Toggled Output Operation (1)

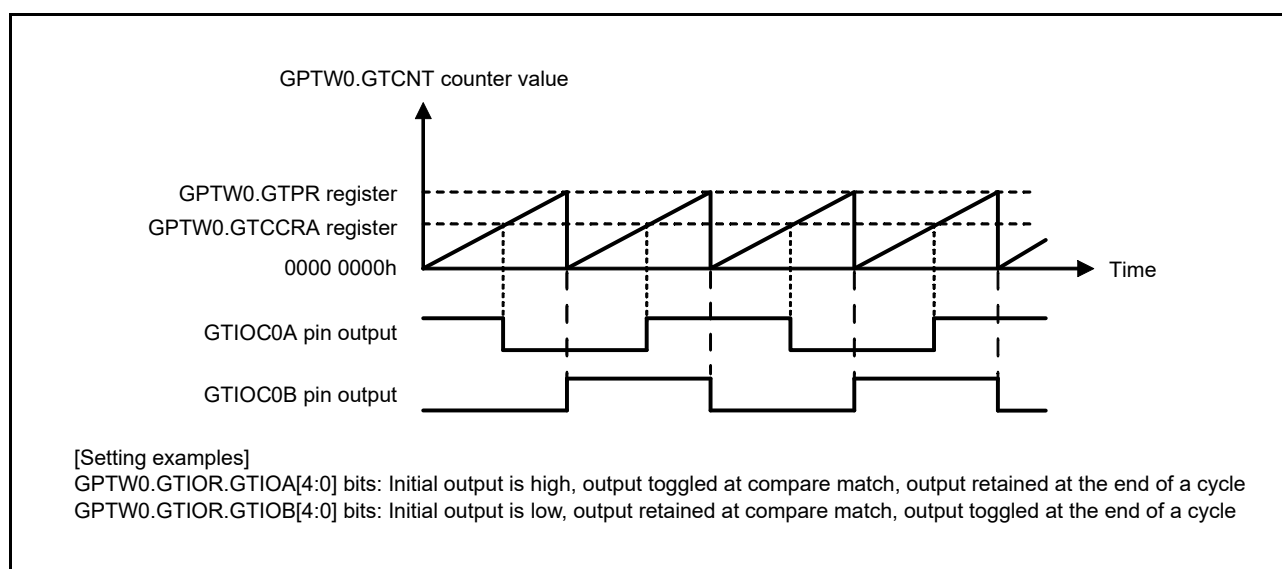


Figure 26.15 Example of Toggled Output Operation (2)

Figure 26.16 shows an example for setting toggled output operation.

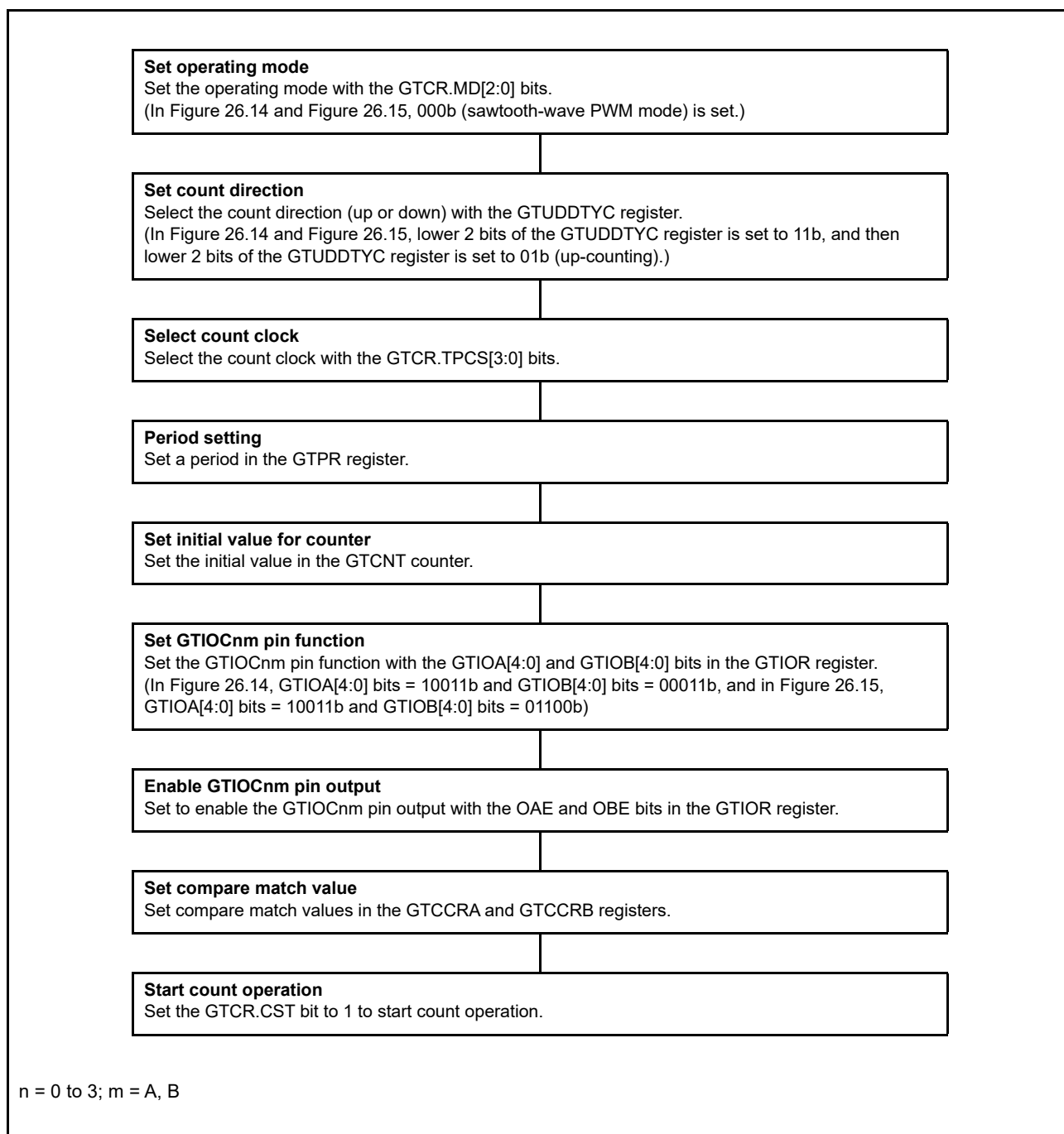


Figure 26.16 Example for Setting Toggled Output Operation

26.3.1.3 Input Capture Function

Through detecting hardware sources selected by the GTICASR and GTICBSR registers, the GTCNT counter values can be transferred to the GTCCRA and GTCCRB registers respectively.

Figure 26.17 shows an example of input capture function operation.

In this example, the GPTW0.GTCNT counter performs up-counting, and settings have been made so that an input capture is performed at both edges of the GTIOC0A pin input and at the rising edge of the GTIOC0B pin input.

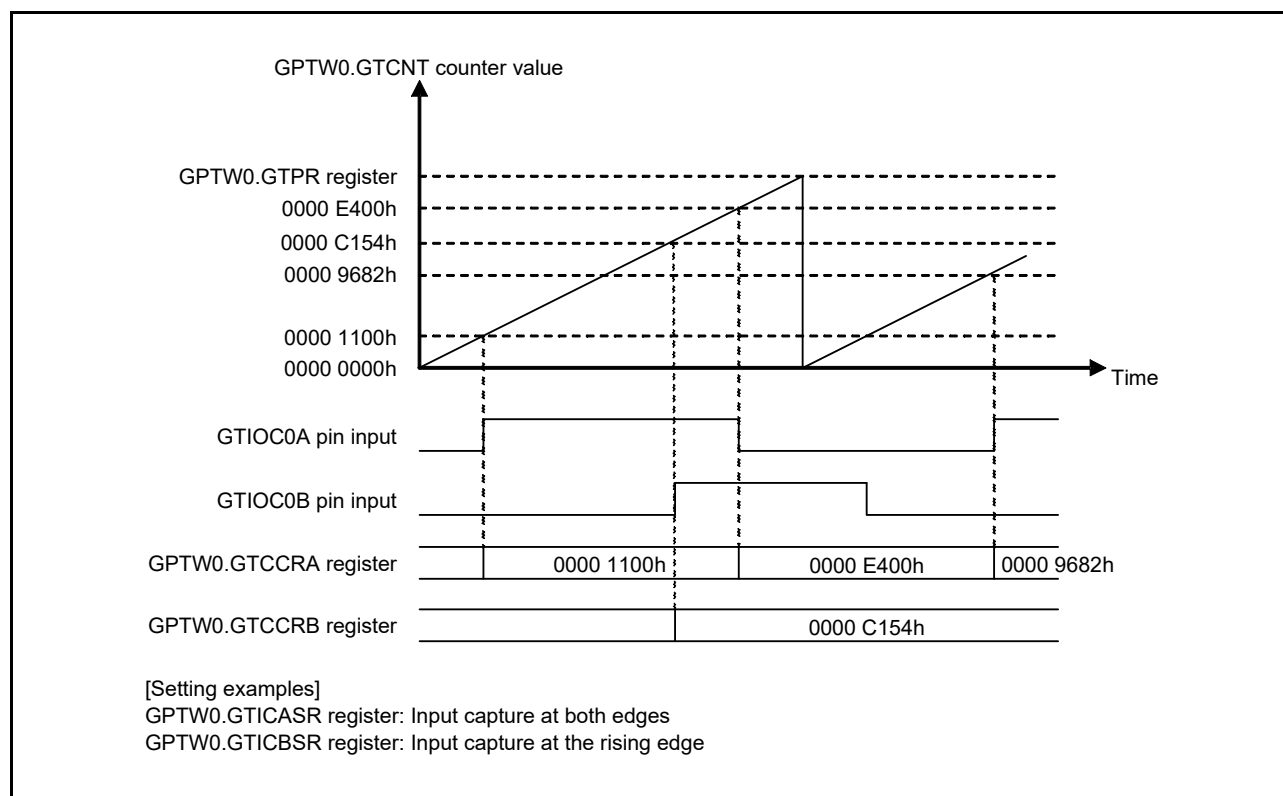


Figure 26.17 Example of Input Capture Operation

Figure 26.18 shows an example for setting input capture operation.

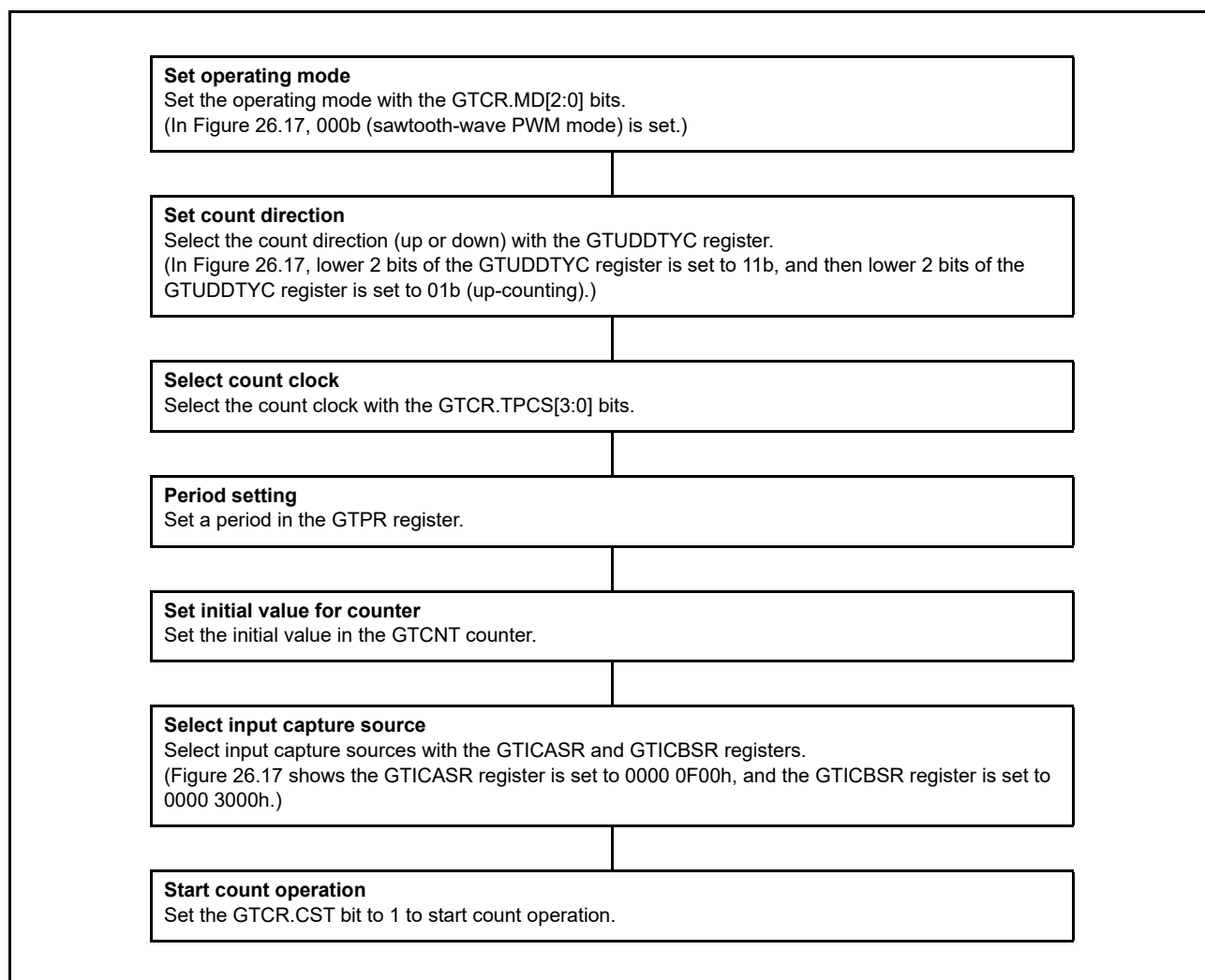


Figure 26.18 Example for Setting Input Capture Operation

Figure 26.19 shows an example of the timing of input capture operation in response to a rising edge of the input on the GTETRGA pin.

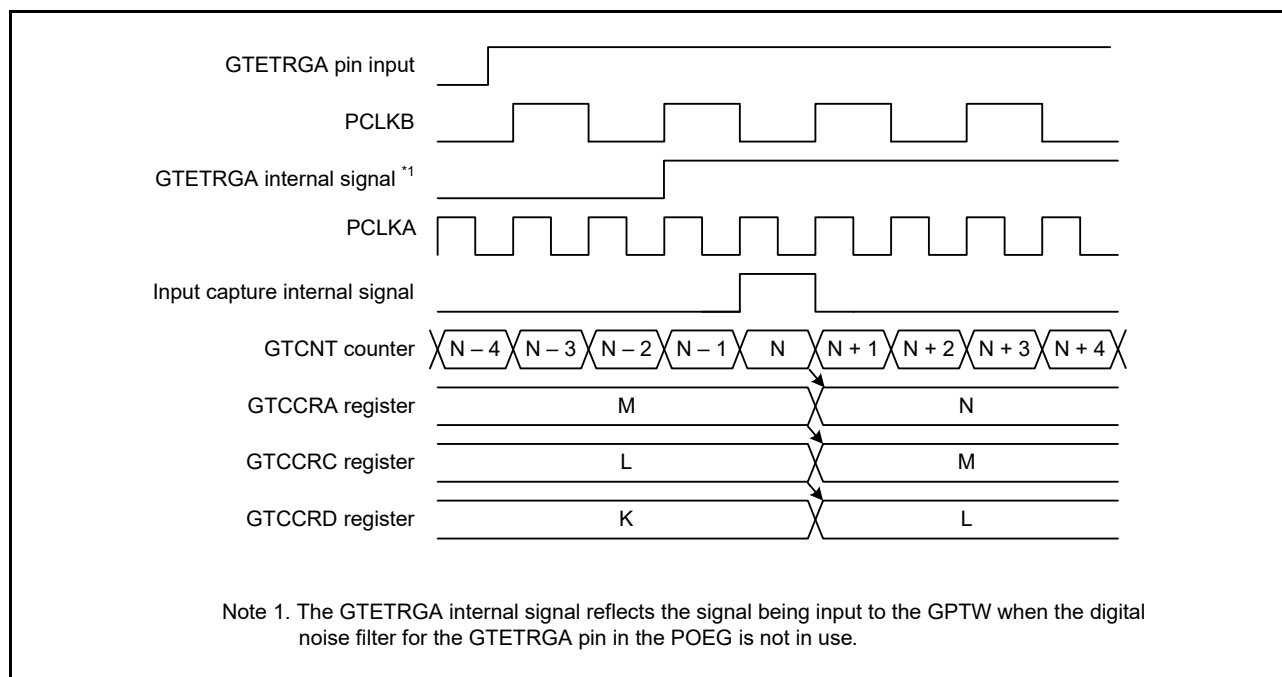


Figure 26.19 Example of the Timing of Input Capture Operation in Response to a Rising Edge of the Input on the GTETRGA Pin

Figure 26.20 shows an example of the timing of input capture operation in response to a rising edge of the input on the GTIOCnA pin.

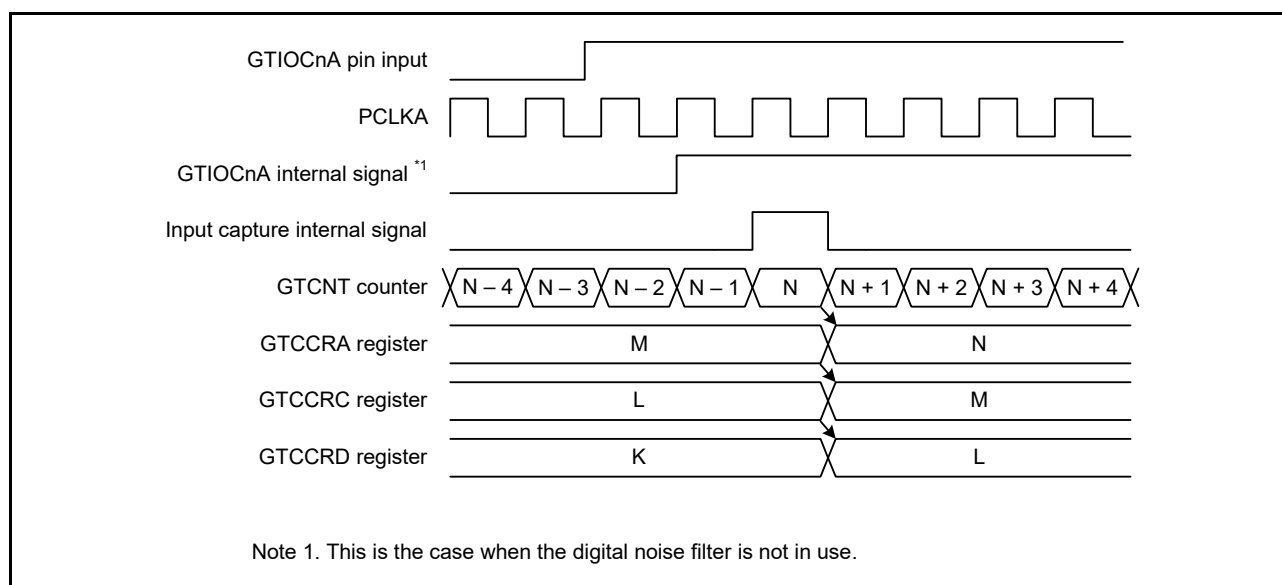


Figure 26.20 Example of the Timing of Input Capture Operation in Response to a Rising Edge of the Input on the GTIOCnA Pin

Figure 26.21 shows an example of the timing of operations for input capture in response to event input from the ELCA. This is an example of capture of the counter value by the GPTW1.GTCCRA register in response to an input signal. An event signal is output to the ELC after matches in comparison with the GPTW0.GTCCRA register. This is selected as a trigger for output to the GPTW by the ELC as event source A. The GPTW0 compare match A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and outputs the GPTW event source A signal after 1 cycle of PCLKB has elapsed.

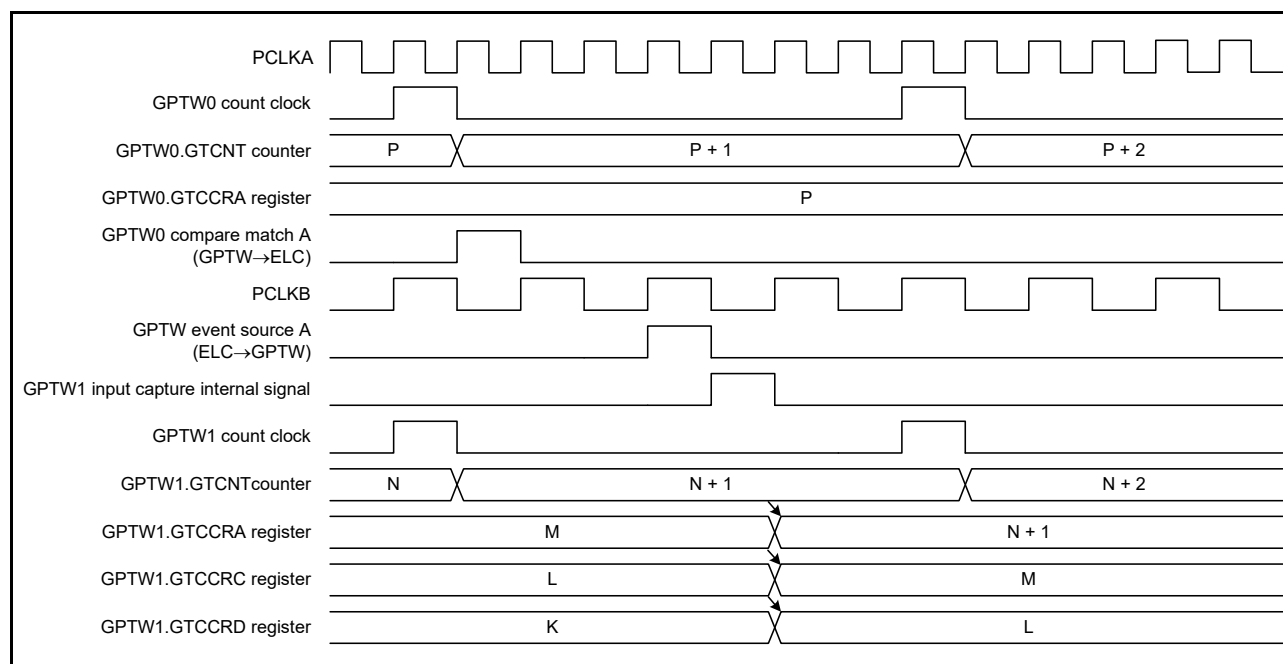


Figure 26.21 Example of the Timing of Operations for Input Capture in Response to Event Input from the ELCA

26.3.2 Buffer Operation

The following buffer operation can be set with the GTBER register.

- Buffer operation with the GTPR, GTPBR, and GTPDBR registers used together
- Buffer operation with the GTCCRA, GTCCRC, and GTCCRD registers used together
- Buffer operation with the GTCCRB, GTCCRE, and GTCCRF registers used together
- Buffer operation with the GTADTRA, GTADTBRA, and GTADTDBRA registers used together
- Buffer operation with the GTADTRB, GTADTBRB, and GTADTDBRB registers used together

The following buffer operation can be set with the GTDTCR register.

- Buffer operation with the GTDVU and GTDBU registers used together
- Buffer operation with the GTDVD and GTDBD registers used together

26.3.2.1 GTPR Register Buffer Operation

The GTPBR register can function as a buffer register for the GTPR register, and the GTPDBR register can function as a buffer register for the GTPBR register (double buffer register for the GTPR register).

The buffer transfer is performed at an overflow (in up-counting) or underflow (in down-counting) in sawtooth-wave mode or in event counting, or at a trough in triangle-wave mode.

When in sawtooth-wave mode and in event counting, the buffer transfer is performed even when the counter clearing occurs during the count is in operation.

- Hardware source clearing

(Clear source selected with GTCSR.CSGTRGAR, CSGTRGAF, CSGTRGBR, CSGTRGBF, CSGTRGCR, CSGTRGCF, CSGTRGDR, CSGTRGDF, CSCARBL, CSCARBH, CSCAFBL, CSCAFBH, CSCBRAL, CSCBRAH, CSCBFAL, CSCBFAH, CSELCA, CSELCB, CSELCC, CSELCD, CSELCE, CSELCF, CSELCH, or CSELCH bit)

- Clearing by software

(When the GTCLR.CCLRn bit is written to 1 during the GTCSR.CCLR bit is set to 1) (n = 0 to 3)

To set the GTPR register to function as double buffer, set the GTBER.PR[1:0] bits to 10b or 11b. For single buffer operation, set 01b. Not to function as buffer, set 00b.

Figure 26.22 to Figure 26.24 show examples of the GTPR register buffer operation and Figure 26.25 shows an example for setting the GTPR register buffer operation.

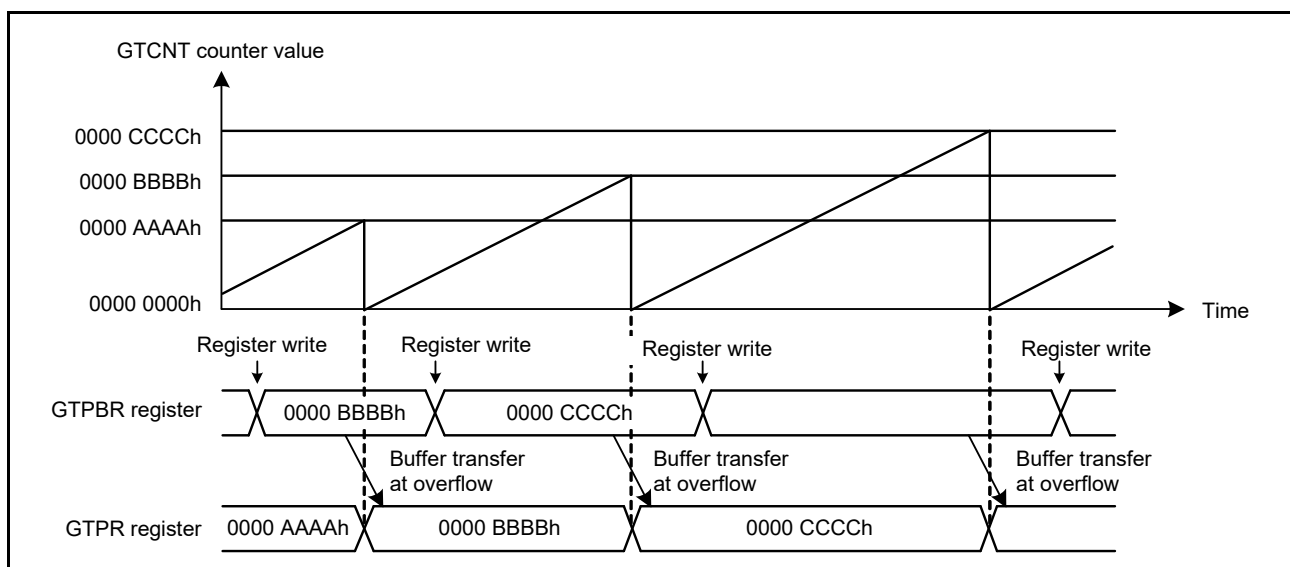


Figure 26.22 Example of the GTPR Register Buffer Operation (Sawtooth Waves in Up-Counting)

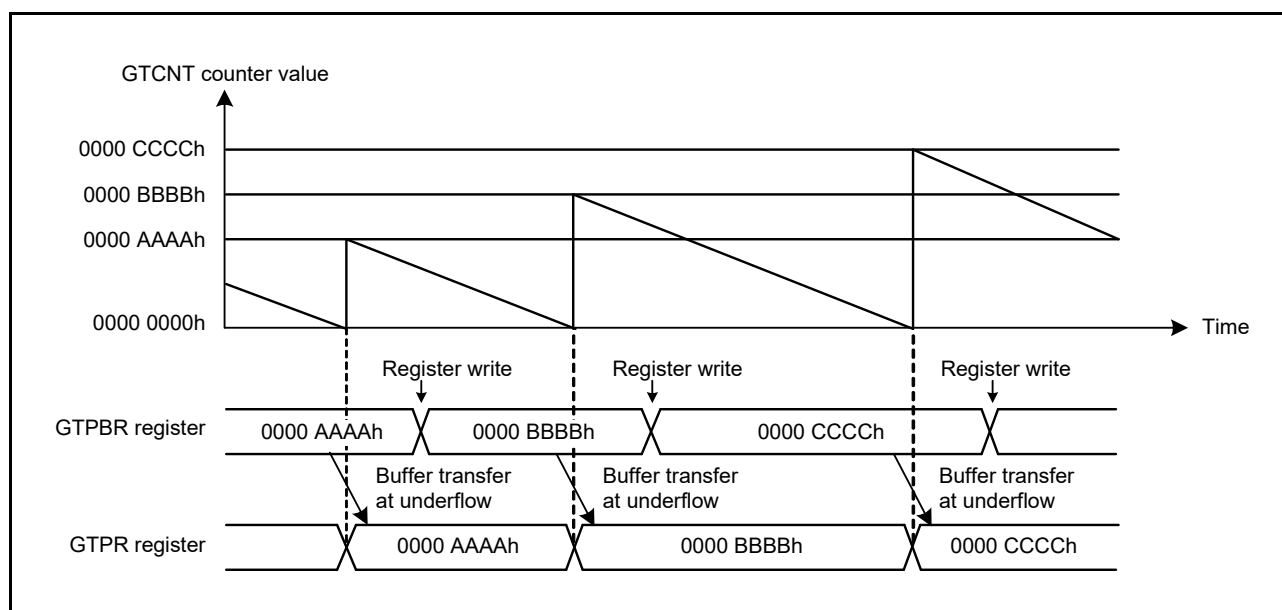


Figure 26.23 Example of the GTPR Register Buffer Operation (Sawtooth Waves in Down-Counting)

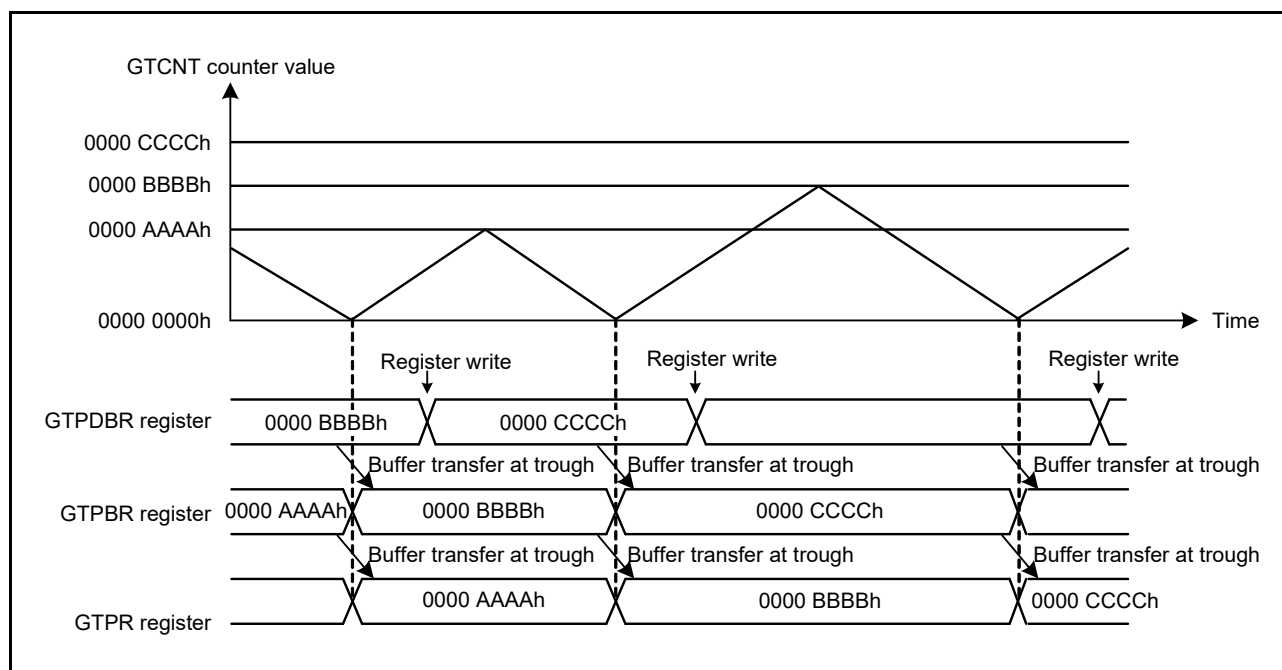


Figure 26.24 Example of the GTPR Register Double Buffer Operation (Triangle Waves)

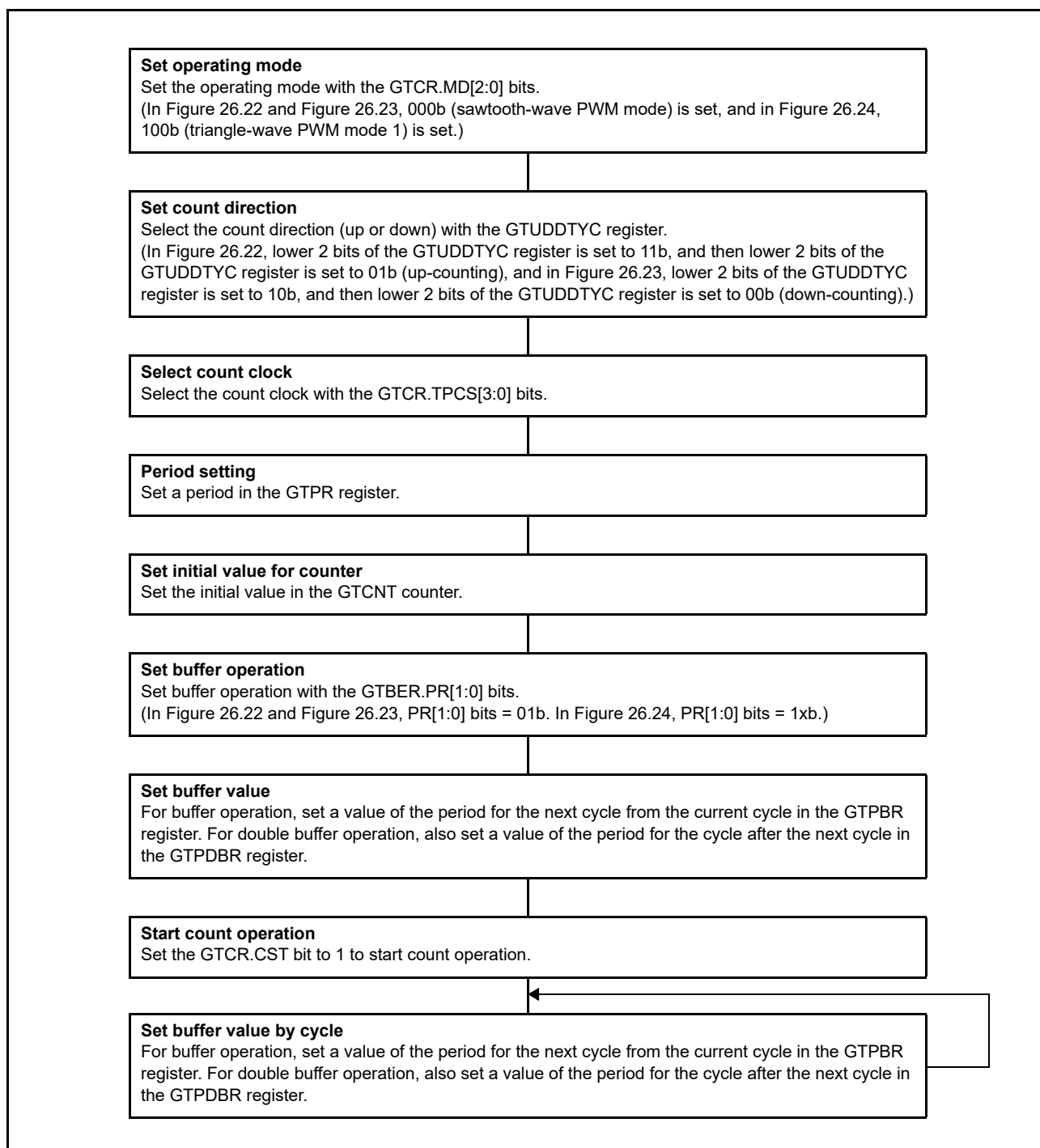


Figure 26.25 Example for Setting the GTPR Register Buffer Operation

26.3.2.2 Buffer Operation for the GTCCRA and GTCCRB Registers

The GTCCRC register can function as the GTCCRA buffer register and the GTCCRD register can function as the GTCCRC buffer register (double buffer register for the GTCCRA register). Similarly, the GTCCRE register can function as the GTCCRB buffer register and the GTCCRF register can function as the GTCCRE buffer register (double buffer register for the GTCCRB register).

To set the GTCCRA or GTCCRB register to function as a double buffer, set the GTBER.CCRA[1:0] or GTBER.CCRB[1:0] bits to 10b or 11b. For single buffer operation, set 01b. Not to function as buffer, set 00b. The following describes buffer operation during output compare and input capture operation.

(1) When the GTCCRA or GTCCRB Register Functions as Output Compare Register

Buffer transfer is performed for the following three cases:

- Buffer transfer at an overflow (in up-counting) and underflow (in down-counting)

Buffer transfer is performed at an overflow (in up-counting) or an underflow (in down-counting) in sawtooth-wave mode and in event counting, and at a trough (triangle-wave PWM mode 1) or at a crest/trough (triangle-wave PWM mode 2) in triangle-wave mode.

- Buffer transfer by counter clearing

In sawtooth-wave mode and in event counting, the buffer transfer by similar counter clearing sources in counting operation and in section 26.3.2.1, GTPR Register Buffer Operation is performed in the same way at an overflow (in up-counting) or an underflow (in down-counting).

In triangle-wave mode, buffer transfer by counter clearing is not performed.

- Forcible buffer transfer

In both sawtooth-wave mode and triangle-wave mode, buffer transfer of the GTCCRA and GTCCRB registers is forcibly performed by writing 1 to the GTBER.CCRSWT bit while the counting is stopped.

In sawtooth-wave one-shot pulse mode and triangle-wave PWM mode 3, buffer transfer from the GTCCRD register to the temporary register A and from the GTCCRF register to the temporary register B is also performed using the forcible buffer operation.

When the setting of the GTBER.DBRTCEm (m = A, B) bit is 1 in sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3, transfer from the intermediate buffers to the GTCCRm (m = A, B) registers is repeated on a cyclic basis even while buffer transfer is disabled by the setting of the GTBER.BD[0] bit or buffer transfer extended skipping function (function for repeated double-buffer operation while buffer transfer is disabled). For details, refer to section 26.8.2.2, Repeated Double-Buffered Operation When Disabling GTCCR Buffer Transfer.

Figure 26.26 to Figure 26.28 show examples of the GTCCRA and GTCCRB registers buffer operations, and Figure 26.29 shows an example for setting the GTCCRA and GTCCRB registers buffer operations.

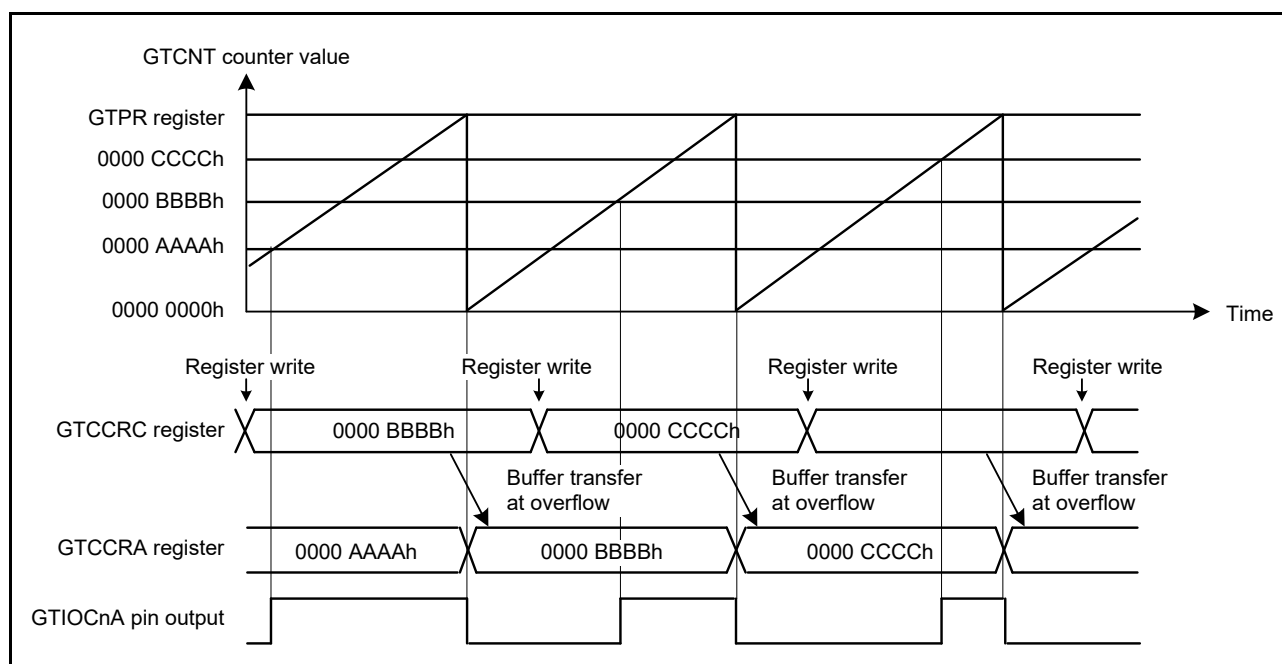


Figure 26.26 Example of GTCCRA and GTCCRB Registers Buffer Operation (Output Compare, Sawtooth-Waves in Up-Counting, High Output at GTCCRA Register Compare Match, Low Output at the End of the Cycle) (n = 0 to 3)

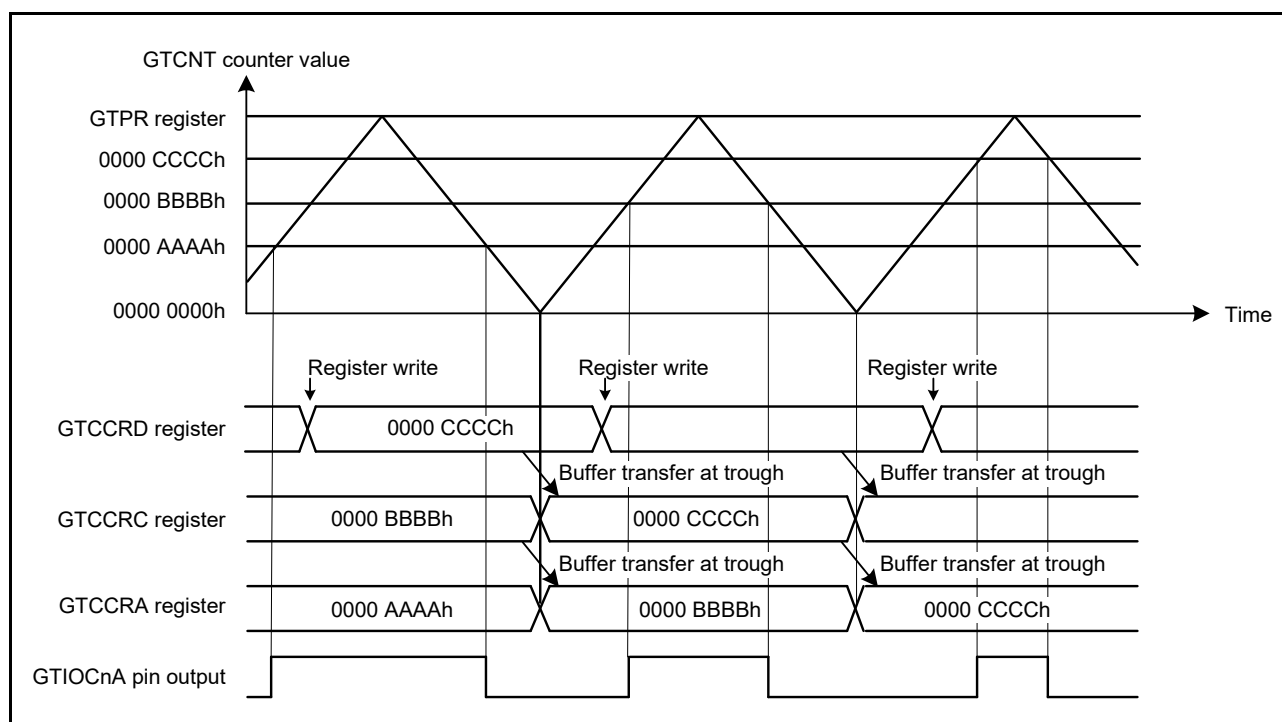


Figure 26.27 Example of GTCCRA and GTCCRB Registers Double Buffer Operation (Output Compare, Triangle Waves, Buffer Operation at Trough, Output Toggled at GTCCRA Register Compare Match, Output Retained at the End of the Cycle) (n = 0 to 3)

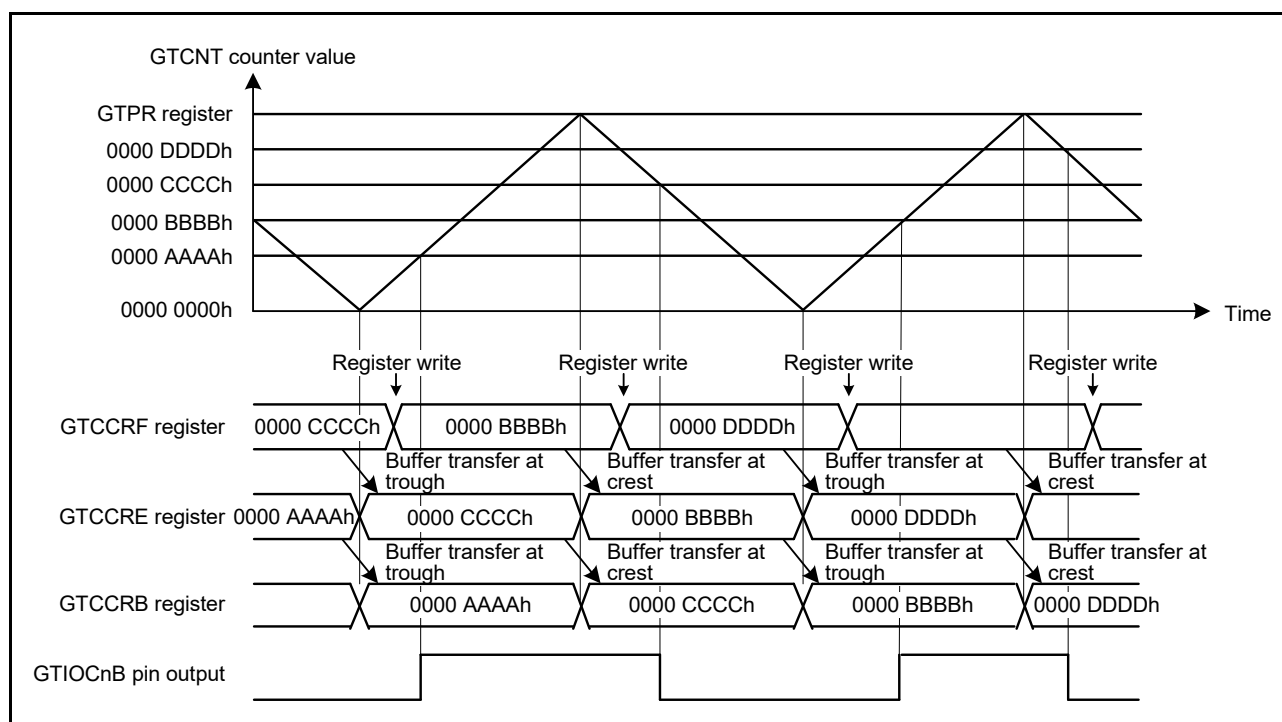


Figure 26.28 Example of GTCCRA and GTCCRB Registers Double Buffer Operation (Output Compare, Triangle Waves, Buffer Operation at Both Troughs and Crests, Output Toggled at GTCCRB Register Compare Match, Output Retained at the End of the Cycle) ($n = 0$ to 3)

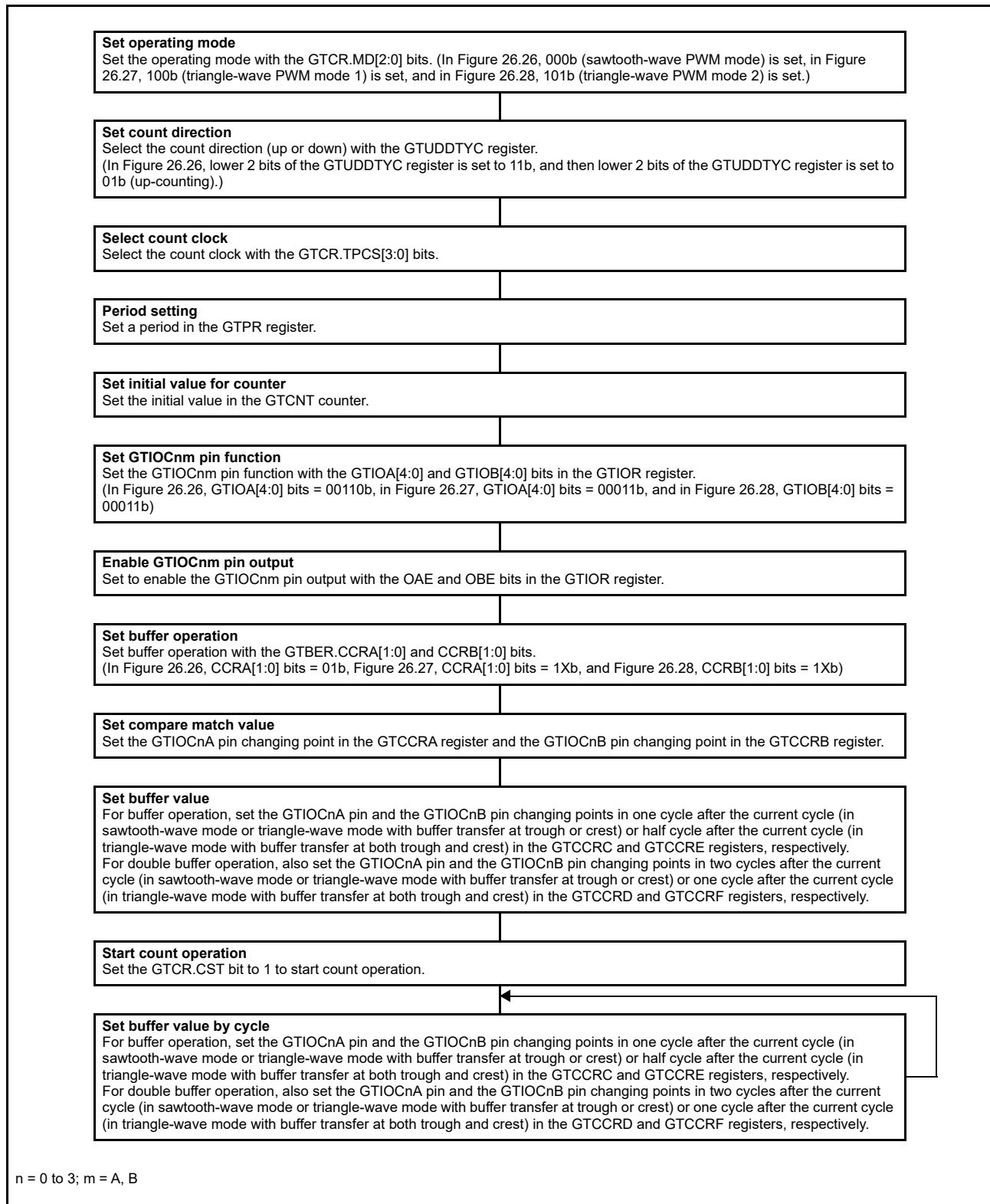


Figure 26.29 Example for Setting Buffer Operation of the GTCCRA and GTCCRB Registers (for Output Compare)

(2) When the GTCCRA or GTCCRB Register Functions as Input Capture Register

The timing of transfer to the buffers is when an input capture is generated. When an input capture is generated, the GTCNT counter value is transferred to GTCCRA and GTCCRB registers and the stored GTCCRA and GTCCRB registers values are transferred to buffer registers. Buffer transfer by counter clearing is not performed at input capture.

Figure 26.30 and Figure 26.31 show examples of buffer operation of the GTCCRA and GTCCRB registers, and Figure 26.32 shows an example for setting buffer operation of the GTCCRA and GTCCRB registers.

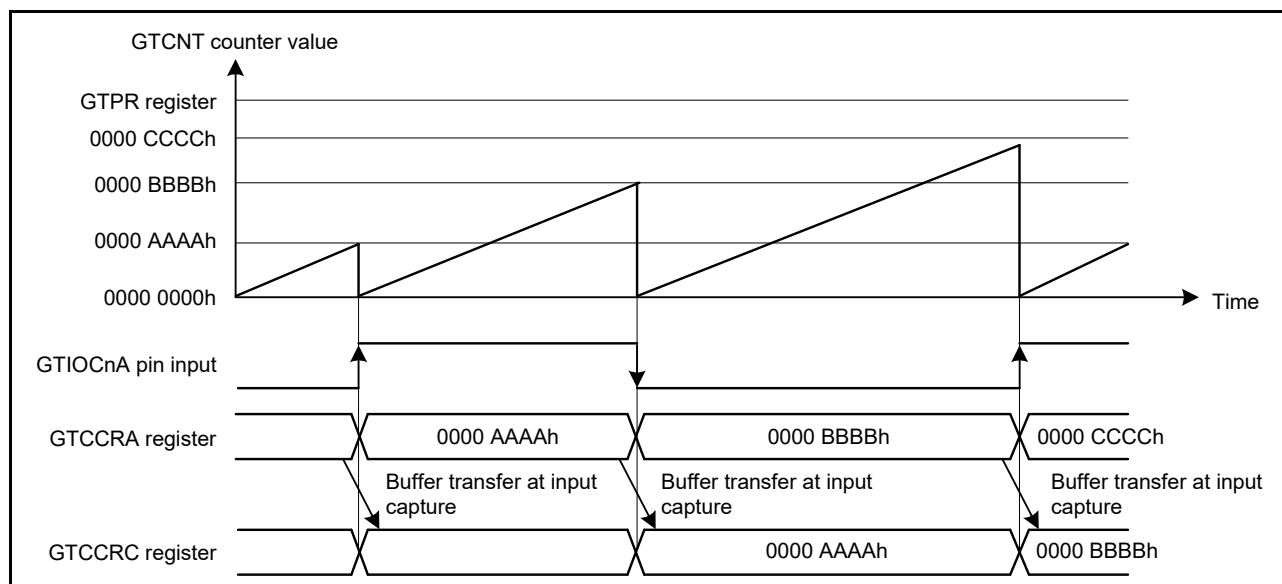


Figure 26.30 Example of Buffer Operation of the GTCCRA and GTCCRB Registers (Counting up to Produce a Sawtooth Wave, and Clearing the GTCNT Counter and Input Capture on Both Edges of the Input on the GTIOcNA Pin) (n = 0 to 3)

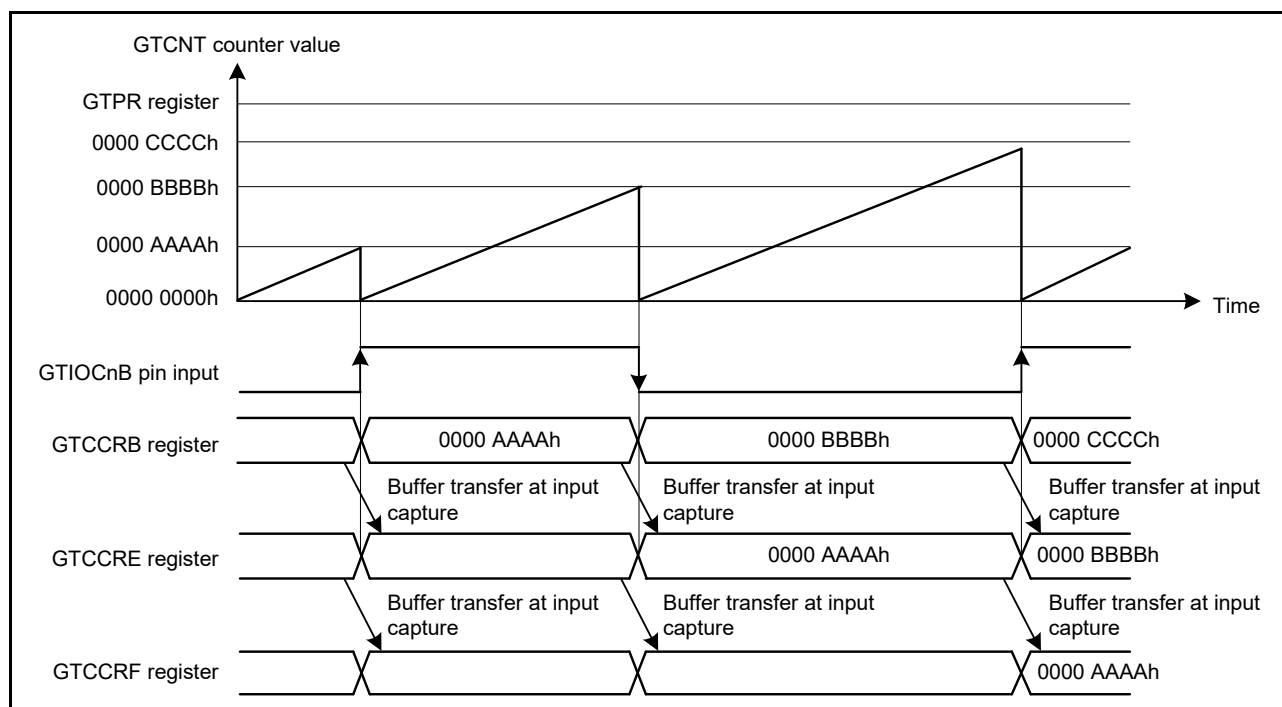


Figure 26.31 Example of Double Buffer Operation of the GTCCRA and GTCCRB Registers (Counting up to Produce a Sawtooth Wave, and Clearing the GTCNT Counter and Input Capture on Both Edges of the Input on the GTIOcNB Pin) (n = 0 to 3)

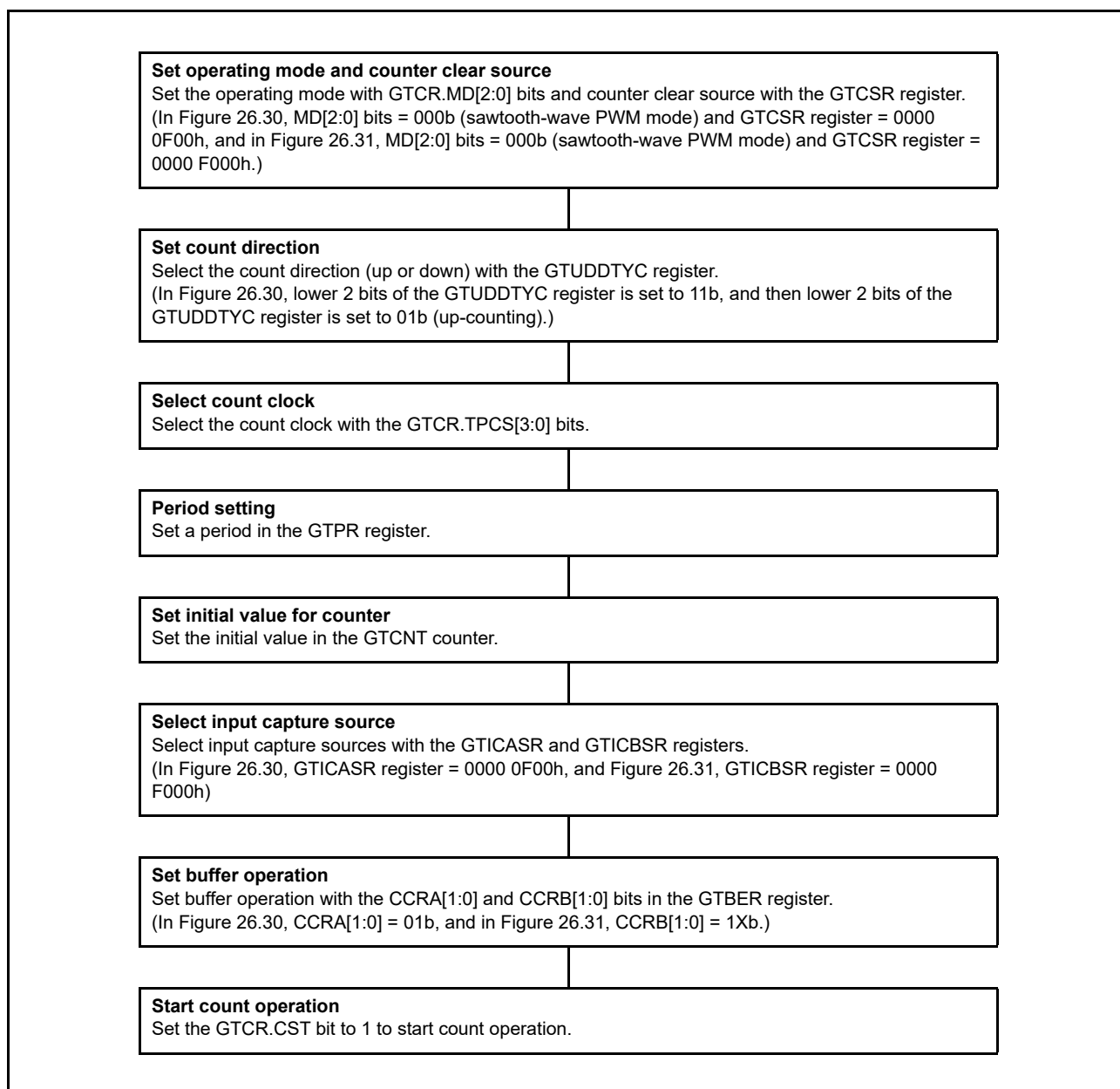


Figure 26.32 Example for Setting Buffer Operation of the GTCCRA and GTCCRB Registers (for Input Capture)

26.3.2.3 Buffer Operation for the GTADTRA and GTADTRB Registers

The GTADTBRA register can function as the GTADTRA buffer register and the GTADTDBRA register can function as the GTADTBRA buffer register (double buffer register for the GTADTRA register). Similarly, the GTADTBRB register can function as the GTADTRB buffer register and the GTADTDBRB register can function as the GTADTBRB buffer register (double buffer register for the GTADTRB register).

To set the GTADTRA or GTADTRB register to function as a double buffer, set the GTBER.ADTDA or ADTDB bit to 1. For single buffer operation, set 0. Not to function as buffer, set the GTBER.ADTTA[1:0] or ADTTB[1:0] bits to 00b. The buffer transfer timing can be set with the ADTTA[1:0] and ADTTB[1:0] bits to an overflow (in up-counting) or an underflow (in down-counting) in sawtooth-wave mode, with ADTTA[1:0] and ADTTB[1:0] bits to 01b for a crest, to 10b for a trough, or to 11b for both crest and trough in triangle-wave mode.

In sawtooth-wave mode, when the ADTTA[1:0] and ADTTB[1:0] bits are set to value other than 00b and in count operation, the buffer transfer, by similar counter clearing sources in section 26.3.2.1, GTPR Register Buffer Operation, is performed in the same way at an overflow (in up-counting) or an underflow (in down-counting).

Figure 26.33 to Figure 26.35 show examples of buffer operation of the GTADTRA and GTADTRB registers, and Figure 26.36 shows an example for setting buffer operation of the GTADTRA and GTADTRB registers.

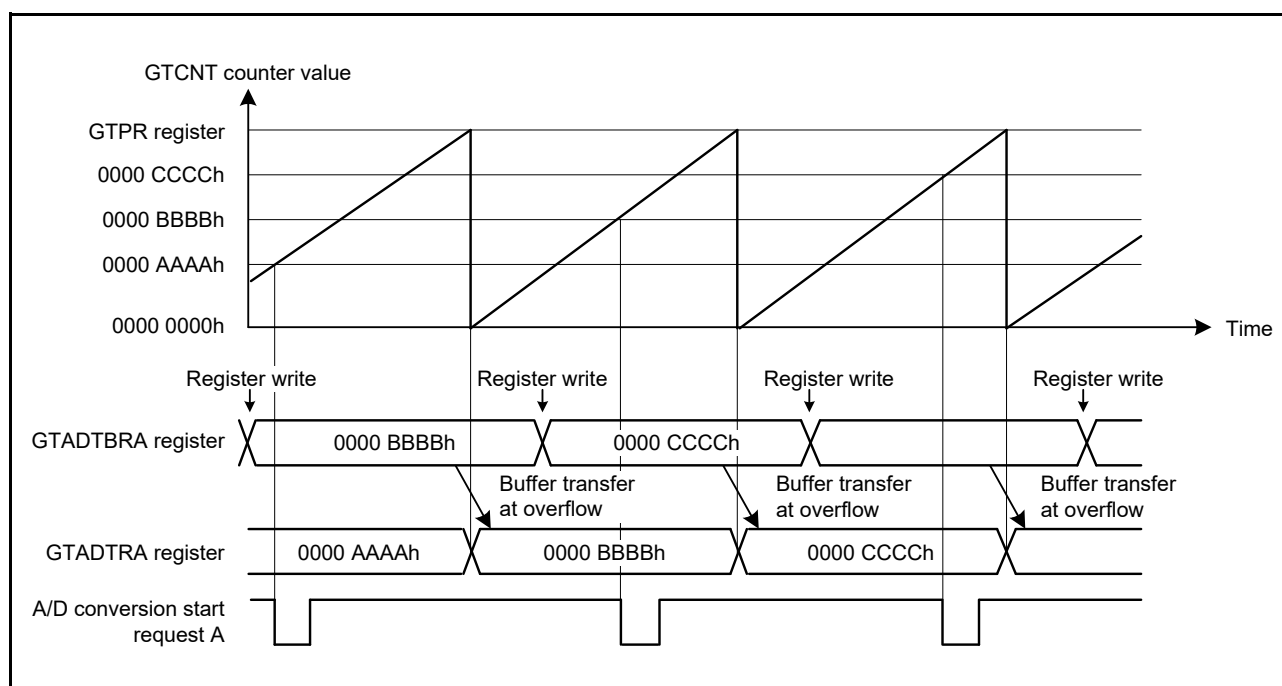


Figure 26.33 Example of Buffer Operation of the GTADTRA and GTADTRB Registers (Sawtooth Waves in Up-Counting, A/D Conversion Start Request Generated by Up-Counting)

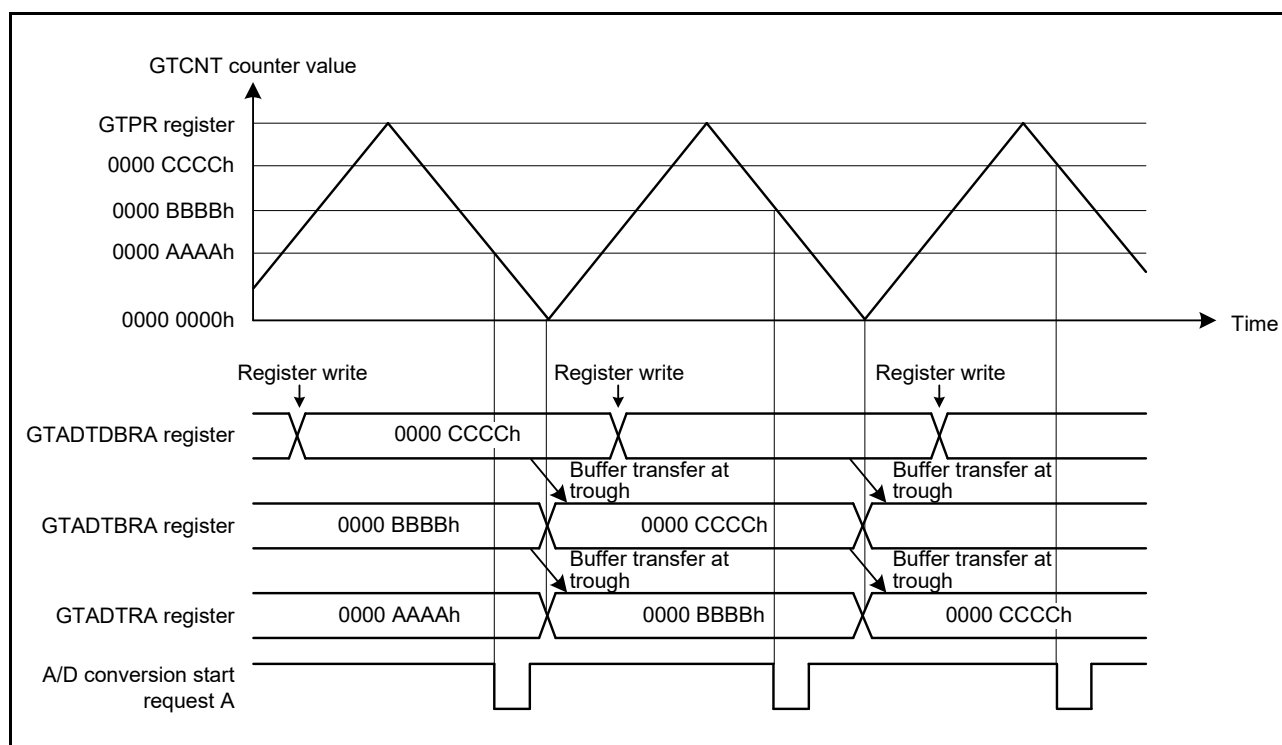


Figure 26.34 Example of Buffer Operation of the GTADTRA and GTADTRB Registers (Sawtooth Waves in Up-Counting, Buffer Transfer at GTADTRA Compare Match, A/D Conversion Start Request Generated by Up-Counting)

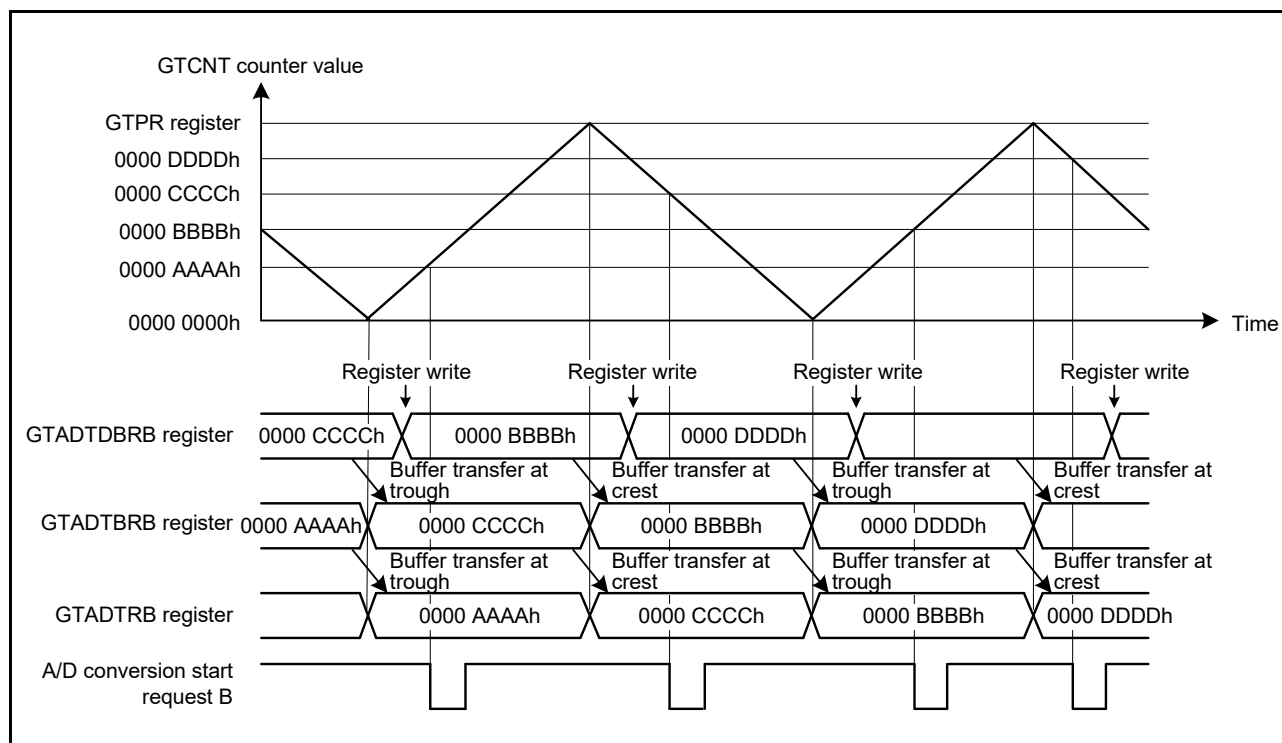


Figure 26.35 Example of Double Buffer Operation of the GTADTRA and GTADTRB Registers (Triangle Waves, Buffer Transfer at Both Troughs and Crests, A/D Conversion Start Request Generated by Both Up- and Down-Counting)

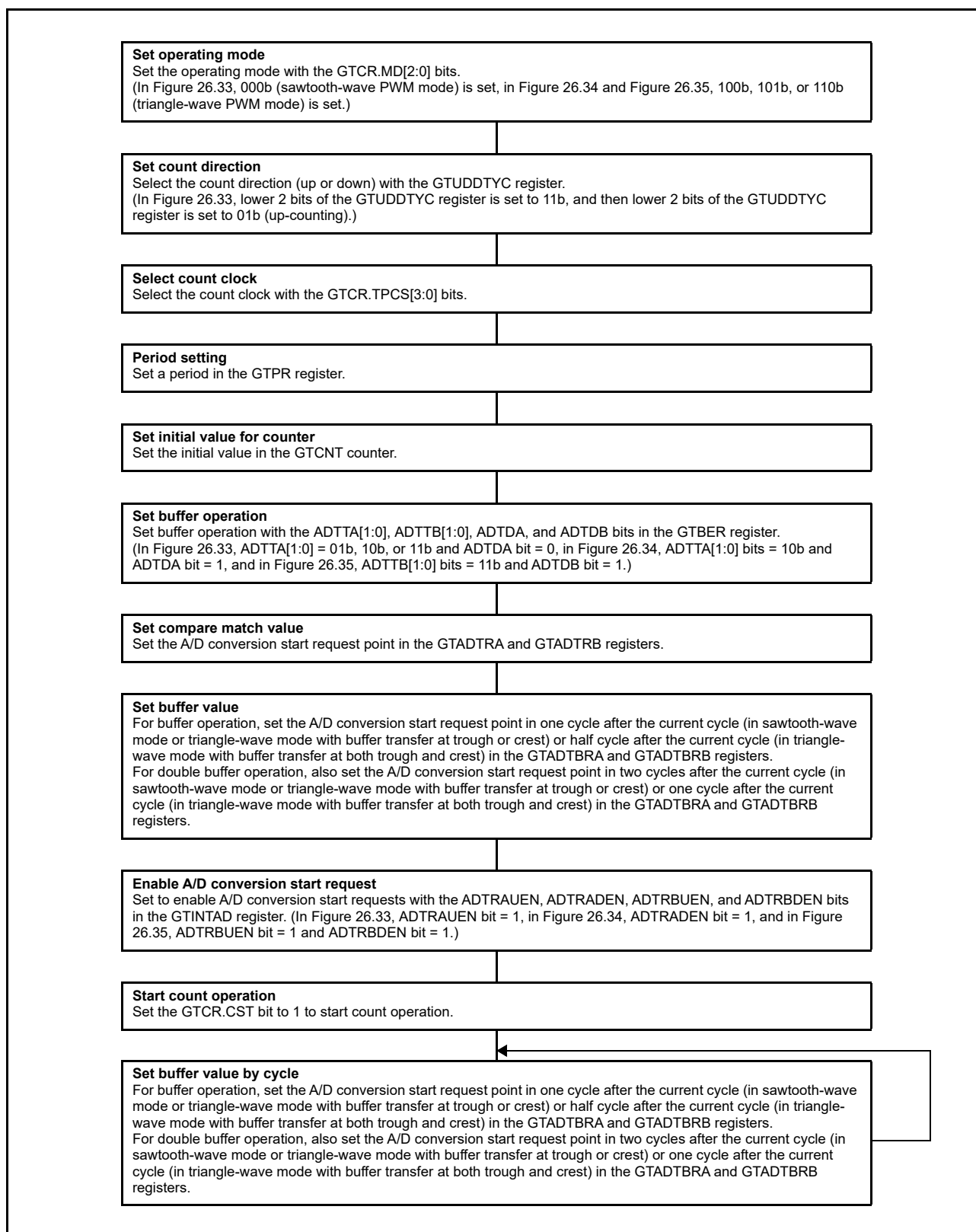


Figure 26.36 Example for Setting Buffer Operation of the GTADTRA and GTADTRB Registers

26.3.3 PWM Output Operating Mode

The GPTW can output PWM waveforms to the GTIOCnA pin or GTIOCnB pin ($n = 0$ to 3) by a compare match between the GTCNT counter and the GTCCRA or GTCCRB register.

By setting the GTDTCR, GTDVU, and GTDVD registers, the compare match value for a negative-phase waveform with dead time can automatically be set to the GTCCRB register.

(1) Sawtooth-Wave PWM Mode

In sawtooth-wave PWM mode, the GTCNT counter performs sawtooth-wave (half-wave) operation by setting the period in the GTPR register and a PWM waveform is output to the GTIOCnA or GTIOCnB pin when a GTCCRA or GTCCRB register compare match occurs. The pin output value can be selected from low output, high output, or toggle output separately for a compare match and for the end of the cycle according to the GTIOR register setting.

Figure 26.37 shows an example of sawtooth-wave PWM mode operation, and Figure 26.38 shows an example for setting sawtooth-wave PWM mode.

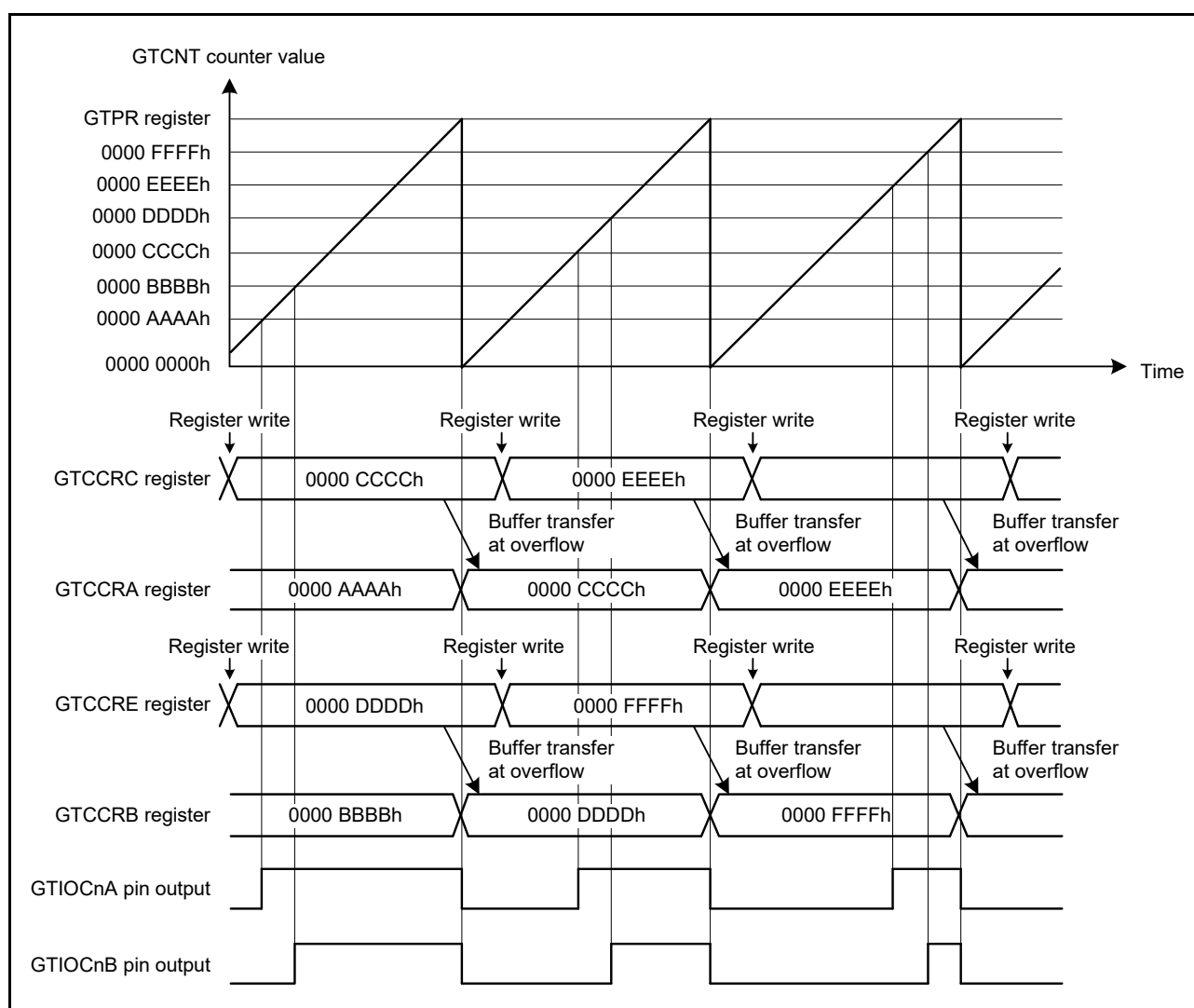


Figure 26.37 Example of Sawtooth-Wave PWM Mode Operation (Up-Counting, Buffer Operation, High Output at GTCCRA/GTCCRB Register Compare Match, Low Output at the End of the Cycle) ($n = 0$ to 3)

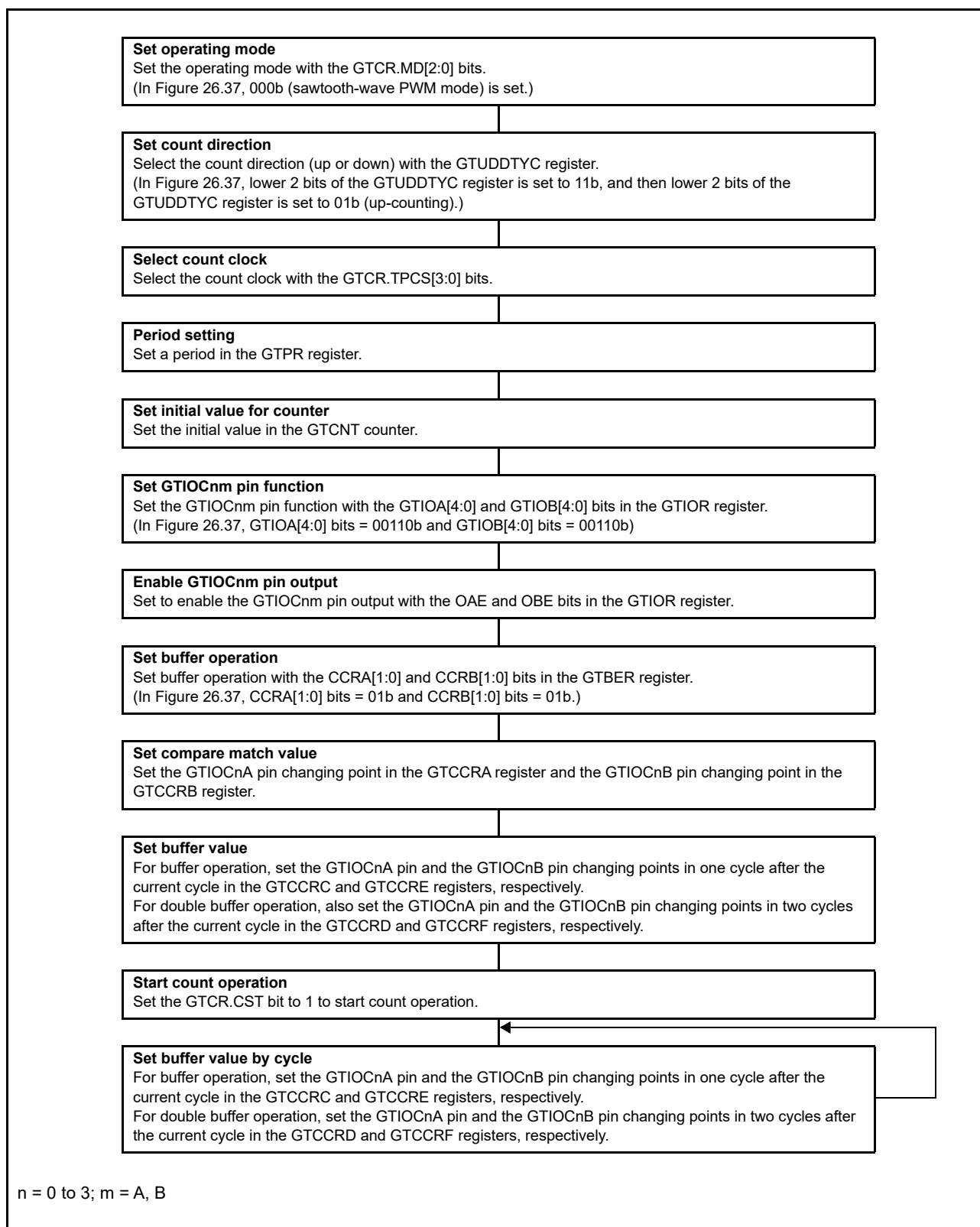


Figure 26.38 Example for Setting Sawtooth-Wave PWM Mode

(2) Sawtooth-Wave One-Shot Pulse Mode

The sawtooth-wave one-shot pulse mode is a mode in which the period is set in the GTPR register, the GTCNT counter performs sawtooth-wave (half-wave) operation and a PWM waveform is output to the GTIOCnA or GTIOCnB pin at a compare match of the GTCCRA or GTCCRB register with buffer operation fixed ($n = 0$ to 3).

Buffer operation in sawtooth-wave one-shot pulse mode is different from the usual buffer operation. Buffer transfer is performed from the GTCCRC register to the GTCCRA register, from the GTCCRE register to the GTCCRB register, from the GTCCRD register to the temporary register A, and from the GTCCRF register to the temporary register B at the end of the cycle, and from the temporary register A to the GTCCRA register at a GTCCRA register compare match and from the temporary register B to the GTCCRB register at a GTCCRB register compare match. The pin output value can be selected from low output, high output, or toggle output separately for a compare match and the end of the cycle according to the GTIOR register setting.

The temporary register A and the temporary register B can perform forcible buffer transfers from the GTCCRD register to the temporary register A and from the GTCCRF register to the temporary register B respectively by writing 1 to the GTBER.CCRSWT bit while the count is stopped.

By setting the GTDTCR, GTDVU, and GTDVD registers, a compare match value for a negative-phase waveform with dead time can automatically be set to the GTCCRB register.

When the GTBER.DBRTCEm ($m = A, B$) bit is set to 1, transfer from an intermediate buffer to the GTCCRm ($m = A, B$) register is repeated on a cyclic basis with using the temporary register x ($x = C, E$) and temporary register m ($m = A, B$) which operate as intermediate buffers for the GTCCRx ($x = C, E$) and GTCCRm ($m = A, B$) registers, respectively, even while buffer transfer is disabled (repeated double buffer operation function during disabling of buffer transfer). For details, refer to section 26.8.2.2, Repeated Double-Buffered Operation When Disabling GTCCR Buffer Transfer.

Figure 26.39 shows an example of sawtooth-wave one-shot pulse mode operation, and Figure 26.40 shows an example for setting sawtooth-wave one-shot pulse mode.

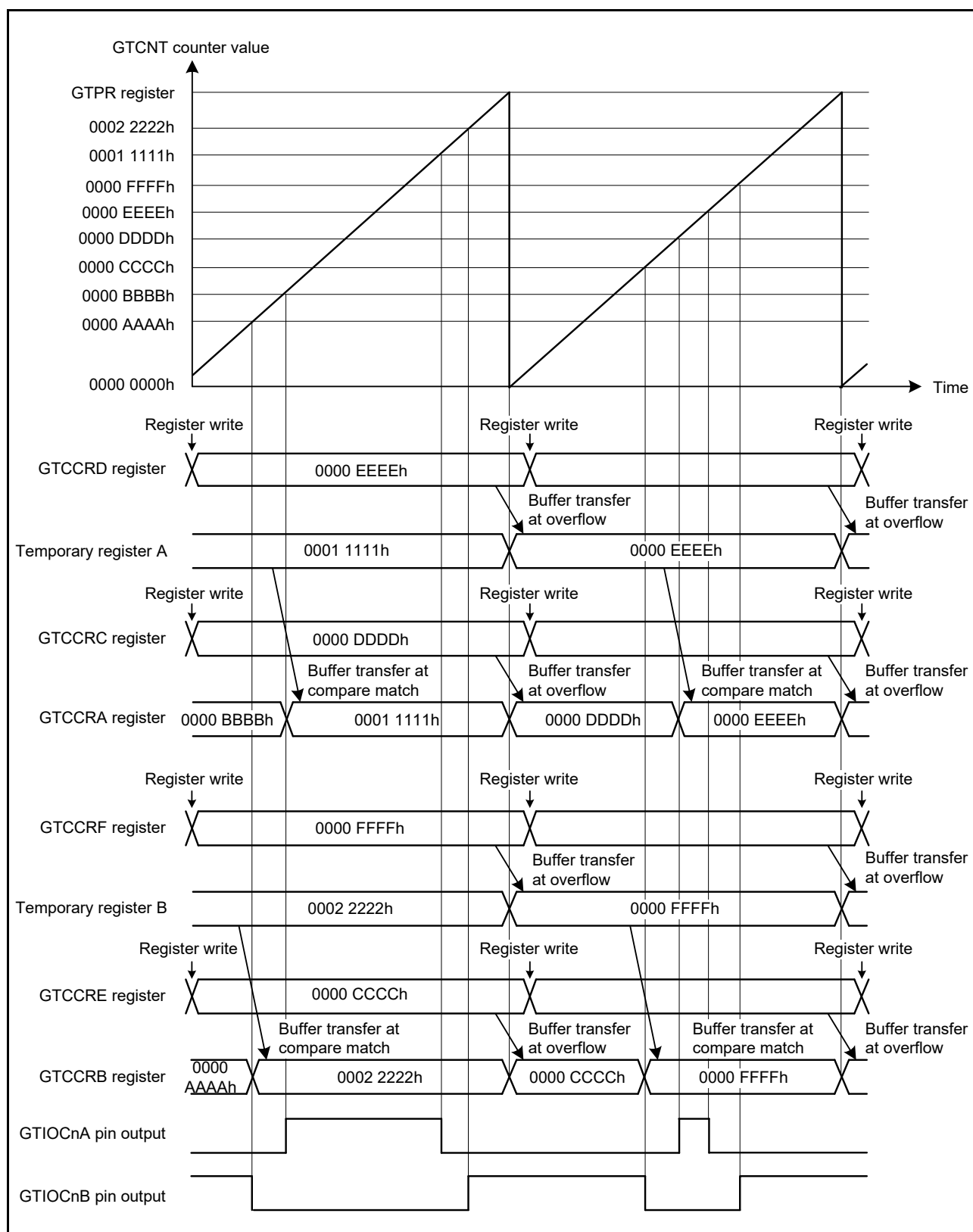


Figure 26.39 Example of Sawtooth-Wave One-Shot Pulse Mode Operation
(Up-Counting, Low Output from the GTIOCnA Pin and High Output from the GTIOCnB Pin at Initial Output, Output Toggled at GTCCRA/GTCCRB Register Compare Match, Output Retained at the End of the Cycle) (n = 0 to 3)

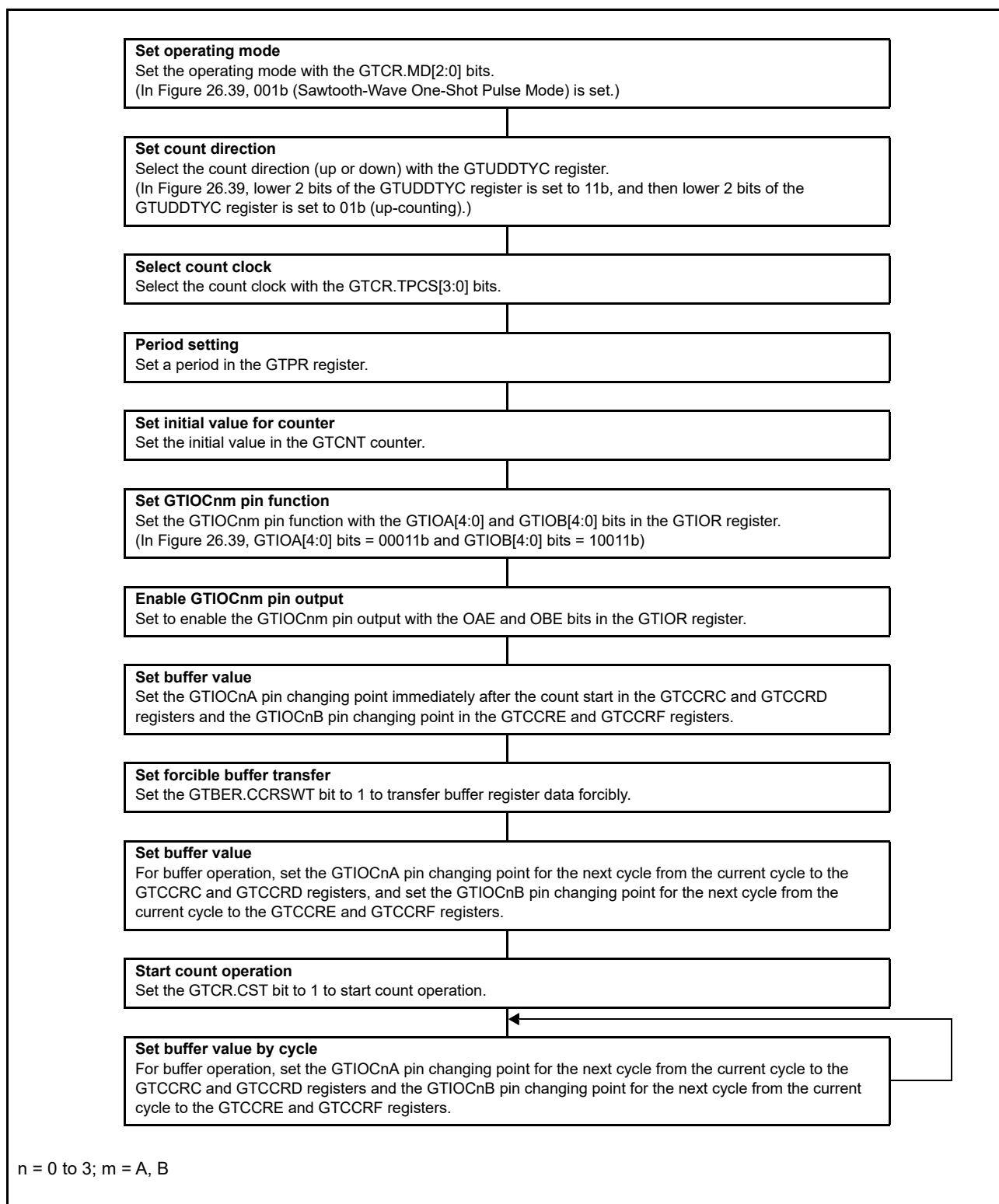


Figure 26.40 Example for Setting Sawtooth-Wave One-Shot Pulse Mode

(3) Triangle-Wave PWM Mode 1 (32-Bit Transfer at Trough)

The triangle-wave PWM mode 1 is a mode in which the period is set in the GTPR register, the GTCNT counter performs triangle-wave (full-wave) operation, and a PWM waveform is output to the GTIOCnA or GTIOCnB pin when a GTCCRA or GTCCRB register compare match occurs ($n = 0$ to 3).

Buffer transfer is performed at the trough. The pin output value can be selected from low output, high output, or toggle output separately for a compare match and for the end of the cycle according to the GTIOR register setting.

By setting the GTDTCR, GTDVU, and GTDVD registers, a compare match value for a negative-phase waveform with dead time can automatically be set to the GTCCRB register.

Figure 26.41 shows an example of triangle-wave PWM mode 1 operation, and Figure 26.42 shows an example for setting triangle-wave PWM mode 1.

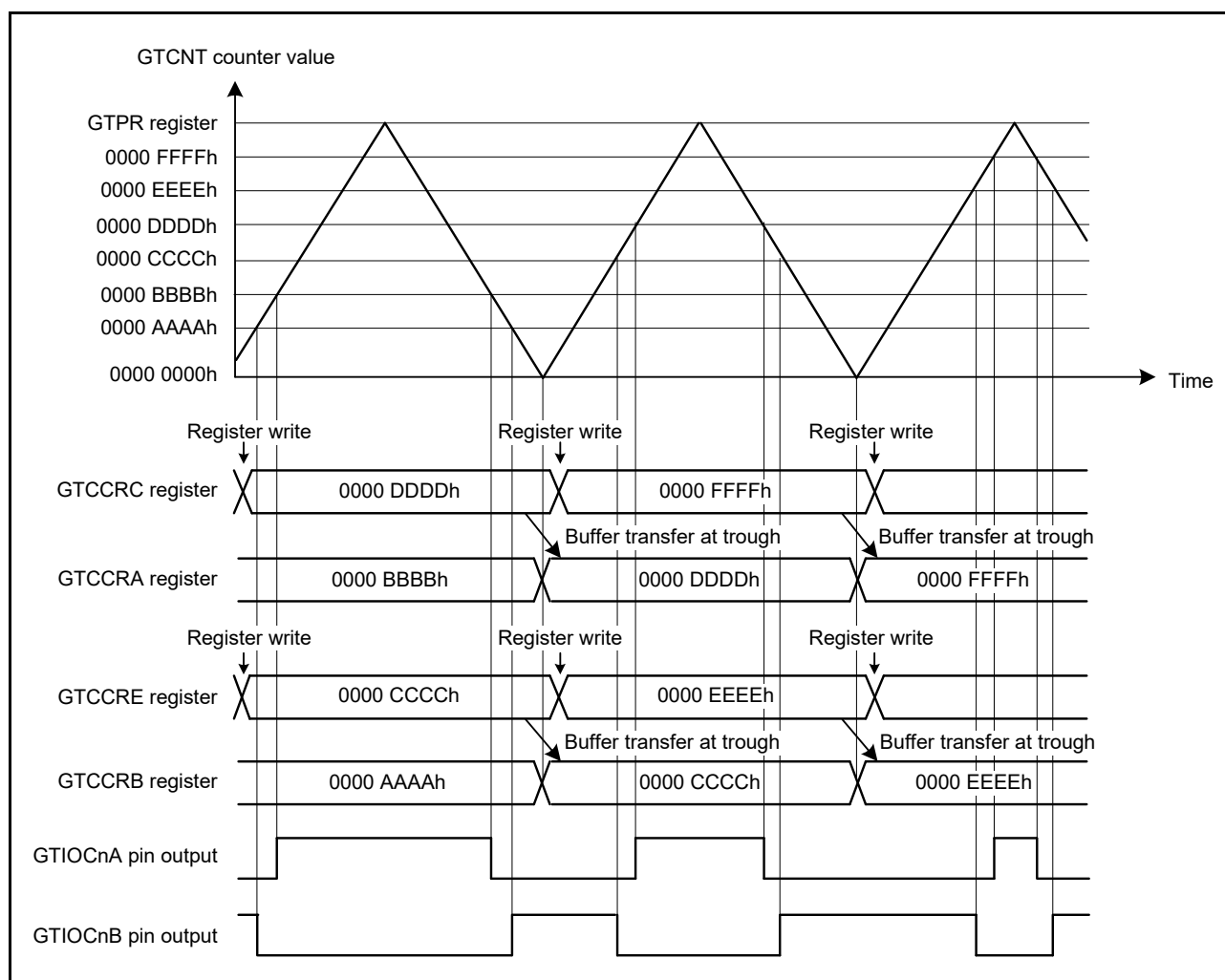


Figure 26.41 Example of Triangle-Wave PWM Mode 1 Operation
(Buffer Operation, Low Output from the GTIOCnA Pin and High Output from the GTIOCnB Pin at Initial Output, Output Toggled at GTCCRA/GTCCRB Register Compare Match, Output Retained at the End of the Cycle) ($n = 0$ to 3)

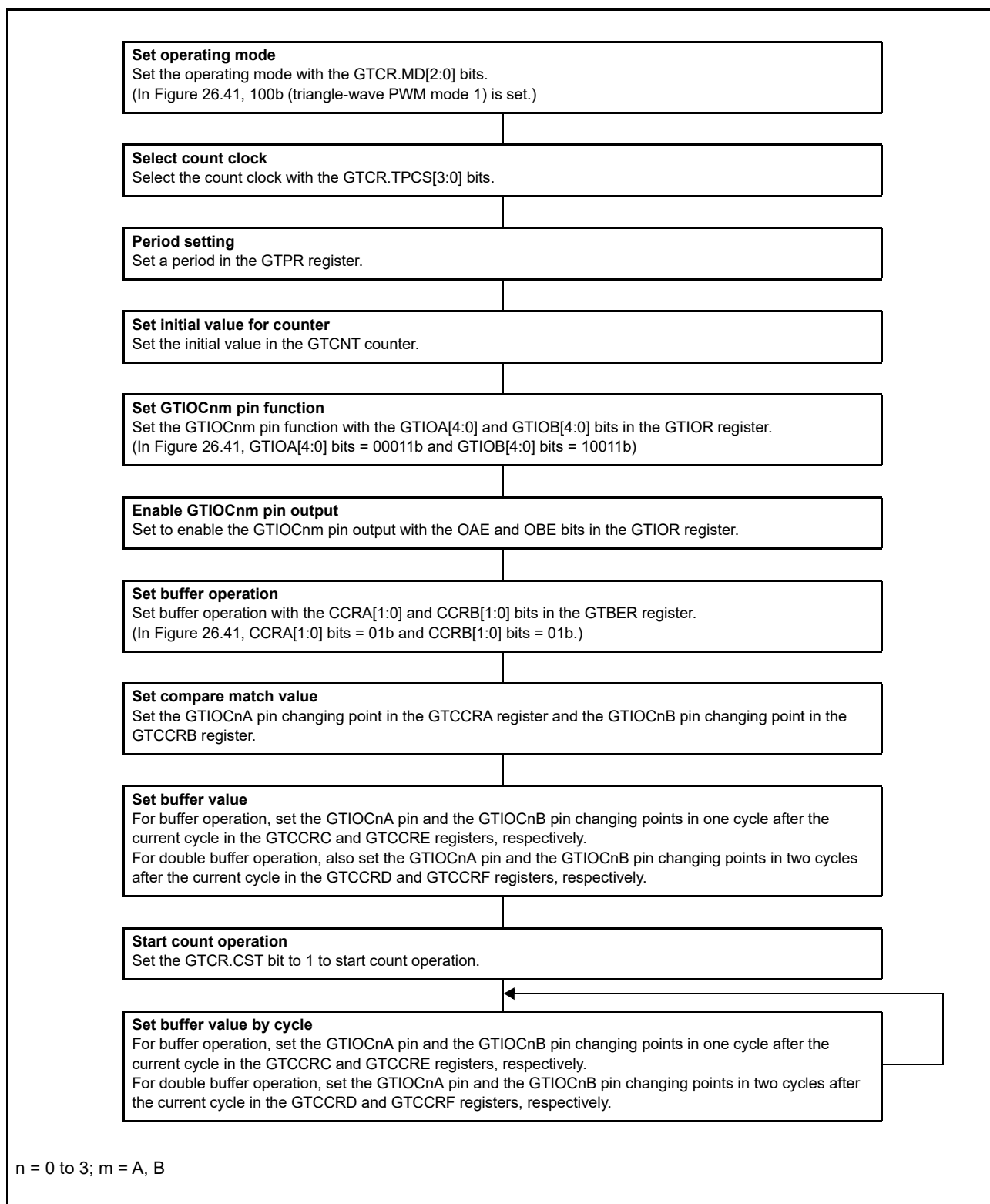


Figure 26.42 Example for Setting Triangle-Wave PWM Mode 1

(4) Triangle-Wave PWM Mode 2 (32-Bit Transfer at Crest and Trough)

Similarly to triangle-wave PWM mode 1, in triangle-wave PWM mode 2 the period is set in the GTPR register, the GTCNT counter performs triangle-wave (full-wave) operation, and a PWM waveform is output to the GTIOCnA or GTIOCnB pin when a GTCCRA or GTCCRB register compare match occurs ($n = 0$ to 3). The buffer transfer is performed at both crests and troughs. The pin output value can be selected from low output, high output, or toggle output separately for a compare match and for the end of the cycle according to the GTIOR register setting.

By setting the GTDTCR, GTDVU, and GTDVD registers, a compare match value for a negative-phase waveform with dead time can automatically be set to the GTCCRB register.

Figure 26.43 shows an example of triangle-wave PWM mode 2 operation, and Figure 26.44 shows an example for setting triangle-wave PWM mode 2.

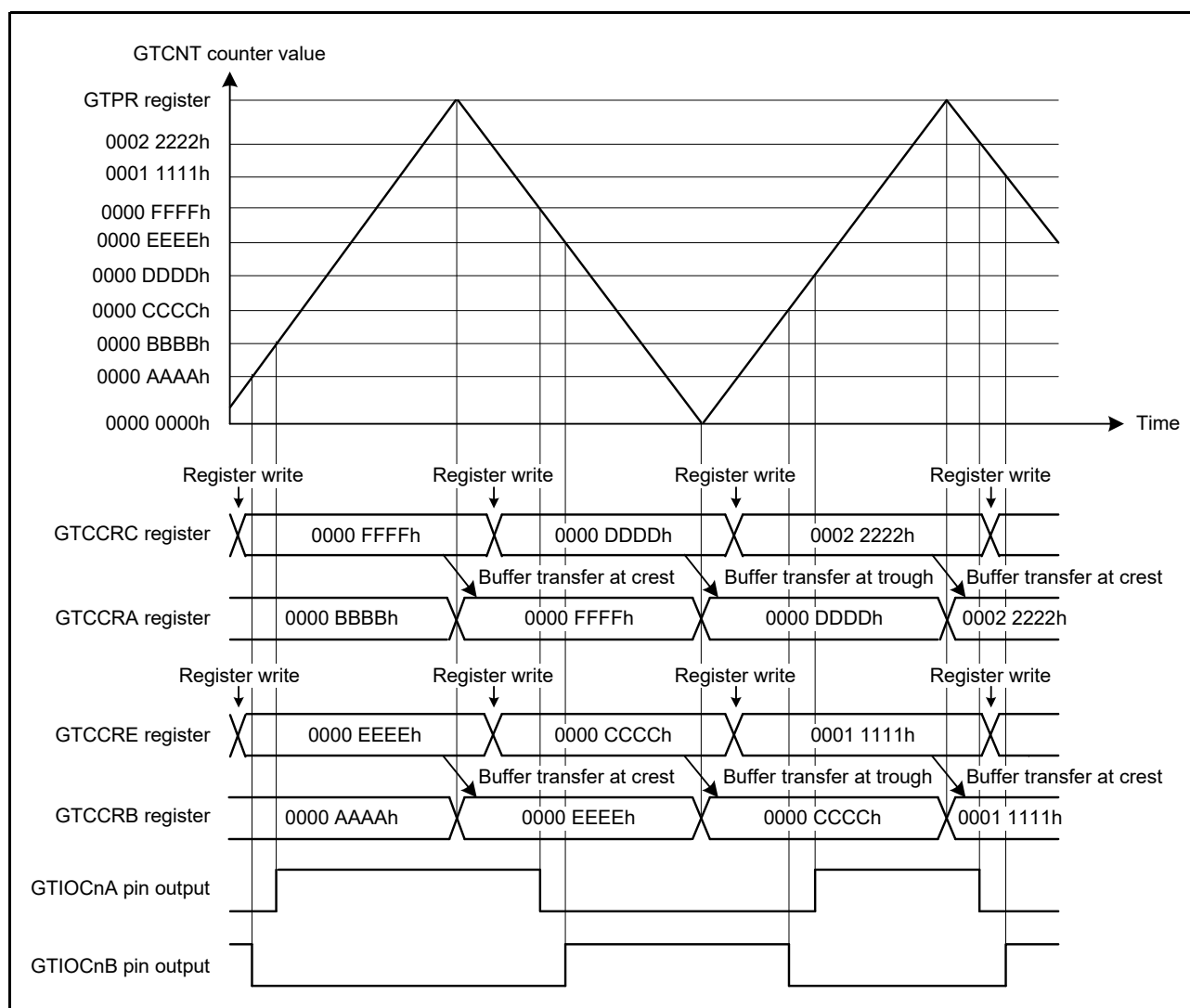


Figure 26.43 Example of Triangle-Wave PWM Mode 2 Operation
 (Buffer Operation, Low Output from the GTIOCnA Pin and High Output from the GTIOCnB Pin at Initial Output, Output Toggled at GTCCRA/GTCCRB Register Compare Match, Output Retained at the End of the Cycle) ($n = 0$ to 3)

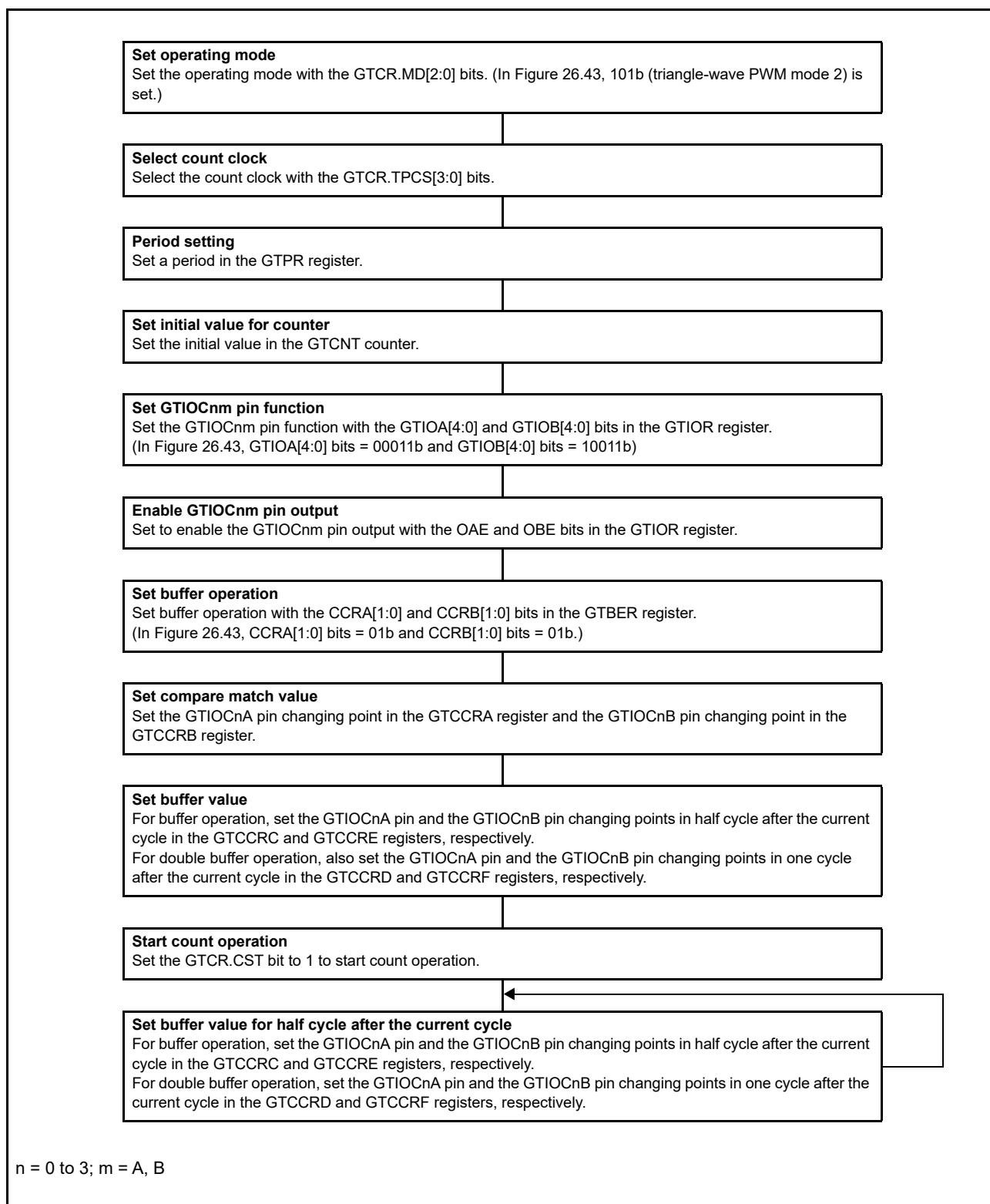


Figure 26.44 Example for Setting Triangle-Wave PWM Mode 2

(5) Triangle-Wave PWM Mode 3 (64-Bit Transfer at Trough)

The triangle-wave PWM mode 3 is a mode in which the period is set in the GTPR register, the GTCNT counter performs triangle-wave (full-wave) operation and a PWM waveform is output to the GTIOCnA or GTIOCnB pin at a compare match of the GTCCRA or GTCCRB register with buffer operation fixed ($n = 0$ to 3). Buffer operation in triangle-wave PWM mode 3 is different from the usual buffer operation. Buffer transfer is performed from the GTCCRC register to the GTCCRA register, from the GTCCRE register to the GTCCRB register, from the GTCCRD register to the temporary register A, and from the GTCCRF register to the temporary register B at the trough, and from the temporary register A to the GTCCRA register and from the temporary register B to the GTCCRB register at the crest. The pin output value can be selected from low output, high output, or toggle output separately for a compare match and for the end of the cycle according to the GTIOR register setting.

By setting the GTDTCR, GTDVU, and GTDVD registers, a compare match value for a negative-phase waveform with dead time can automatically be set to the GTCCRB register.

When the GTBER.DBRTEC m ($m = A, B$) bit is set to 1, transfer from an intermediate buffer to the GTCCR m ($m = A, B$) register is repeated on a cyclic basis with using the temporary register x ($x = C, E$) and temporary register m ($m = A, B$) which operate as intermediate buffers for the GTCCR x ($x = C, E$) and GTCCR m ($m = A, B$) registers, respectively, even while buffer transfer is disabled (repeated double buffer operation function during disabling of buffer transfer). For details, refer to section 26.8.2.2, Repeated Double-Buffered Operation When Disabling GTCCR Buffer Transfer.

Figure 26.45 shows an example of triangle-wave PWM mode 3 operation, and Figure 26.46 shows an example for setting triangle-wave PWM mode 3.

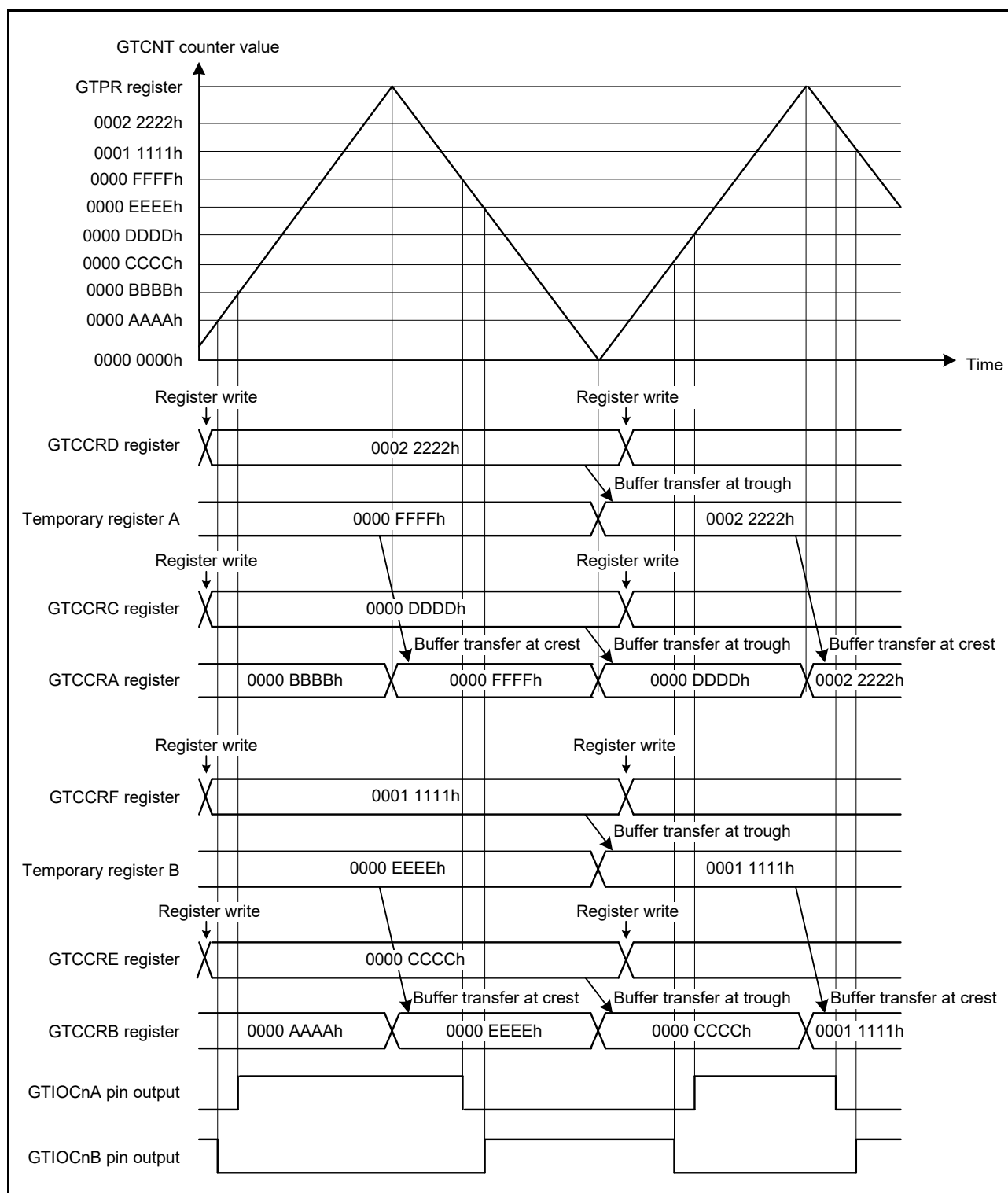


Figure 26.45 Example of Triangle-Wave PWM Mode 3 Operation (Low Output from the GTIOcNA Pin and High Output from the GTIOcNB Pin at Initial Output, Output Toggled at GTCCRA/GTCCRB Register Compare Match, Output Retained at the End of the Cycle) (n = 0 to 3)

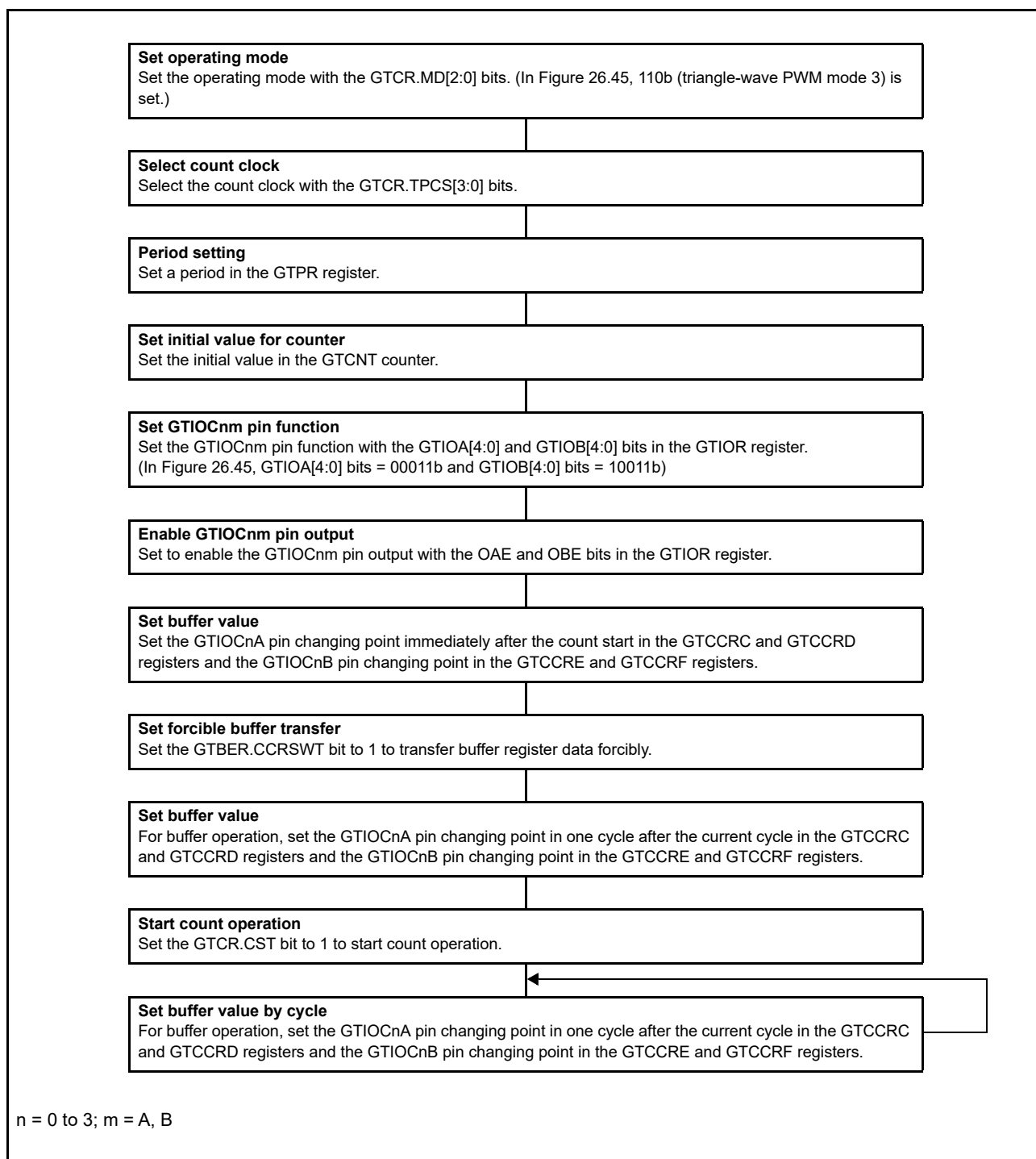


Figure 26.46 Example for Setting Triangle-Wave PWM Mode 3

26.3.4 Automatic Dead Time Setting Function

By setting the GTDTCR register, a compare match value for a negative waveform with dead time obtained by a compare match value for a positive waveform (the GTCCRA register value) and specified dead time values (the GTDVU and GTDVD registers values) can automatically be set to the GTCCRB register.

The automatic dead time setting function can be used in sawtooth-wave one-shot pulse mode and all the triangle PWM modes.

Dead time can be separately set for the first half and second half of a waveform. Dead time for the changing point in the first half of a negative waveform is set in the GTDVU register and that in the second half is set in the GTDVD register. The same dead time can also be set for the first and second halves by setting the GTDTCR.TDFER bit to 1.

The GTDBU register can be used as a buffer register of the GTDVU register, and the GTDBD register can be used as a buffer register of the GTDVD register. Buffer transfer is performed at the end of the cycle (in sawtooth-wave mode: either of an overflow of the GTCNT counter (up-counting), an underflow (down-counting), or the GTCNT counter clearing; in triangle-wave mode: a trough).

The change point of the negative-phase waveform, which is automatically calculated, is obtained by reading the GTCCRB register. Writing to the GTCCRB register is prohibited when the automatic dead time setting function is used.

Do not set the dead-time that makes the change point of the waveform exceeding the count period. When any dead-time setting which would generate a dead-time error is made, adjust the change points of the positive- and negative-phase waveforms to generate waveforms with secured dead-time as shown in Table 26.8. The adjusted change point of the negative-phase waveform is automatically set in the GTCCRB register. An internal signal is used to judge the change point of the positive-phase waveform, thus the value of the GTCCRA register is not updated by the adjusted value.

If the order of the change point becomes inconsistent by adjustment of the waveform change point due to occurrence of dead time errors, or if the change point exceeds the count period even after the adjustment, the complementary relation between the positive- and negative-phases cannot be guaranteed.

If dead-time exceeds the count period by setting 0000 0000h or a value greater than or equal to the setting value of the GTPR register is set in the GTCCRA register in triangle-wave PWM mode, output change is controlled by the output protection function (refer to section 26.8.4, Output Protection Function for GTIOCNm Pin Output ($n = 0$ to 3; $m = A, B$)). When GTCCRA register is greater than or equal to [GTPR register + GTDVm register], [GTPR register – 1] is set in the GTCCRB register as the upper limit. ($m = U, D$)

Automatic setting for a dead time value to the GTCCRB register is performed at the next count clock after the register value for calculating the automatic setting value is updated. In triangle-wave mode, it also can be done at the next count clock from the current crest.

Table 26.8 Adjustment of the Waveform Change Point When a Dead-Time Error Occurs

Mode	Count Direction	Period	Condition for Dead-Time Error	Change Point of the Positive-Phase Waveform after Adjustment	Change Point of the Negative-Phase Waveform after Adjustment
Sawtooth-wave one-shot pulse mode	Up-counting	First half	$GTCCRA - GTDVU < 0$	GTDVU	0
		Second half	$GTCCRA + GTDVD > GTPR$	$GTPR - GTDVD$	GTPR
	Down-counting	First half	$GTCCRA + GTDVU > GTPR$	$GTPR - GTDVU$	GTPR
		Second half	$GTCCRA - GTDVD < 0$	GTDVD	0
Triangle-wave PWM mode 1/2/3	Up-counting	(First half)	$GTCCRA - GTDVU \leq 0$	$GTDVU + 1$	1
	Down-counting	(Second half)	$GTCCRA - GTDVD < 0$	GTDVD	0

Figure 26.47 to Figure 26.50 show examples of automatic dead time setting function operation. Figure 26.51 and Figure 26.52 show the setting examples.

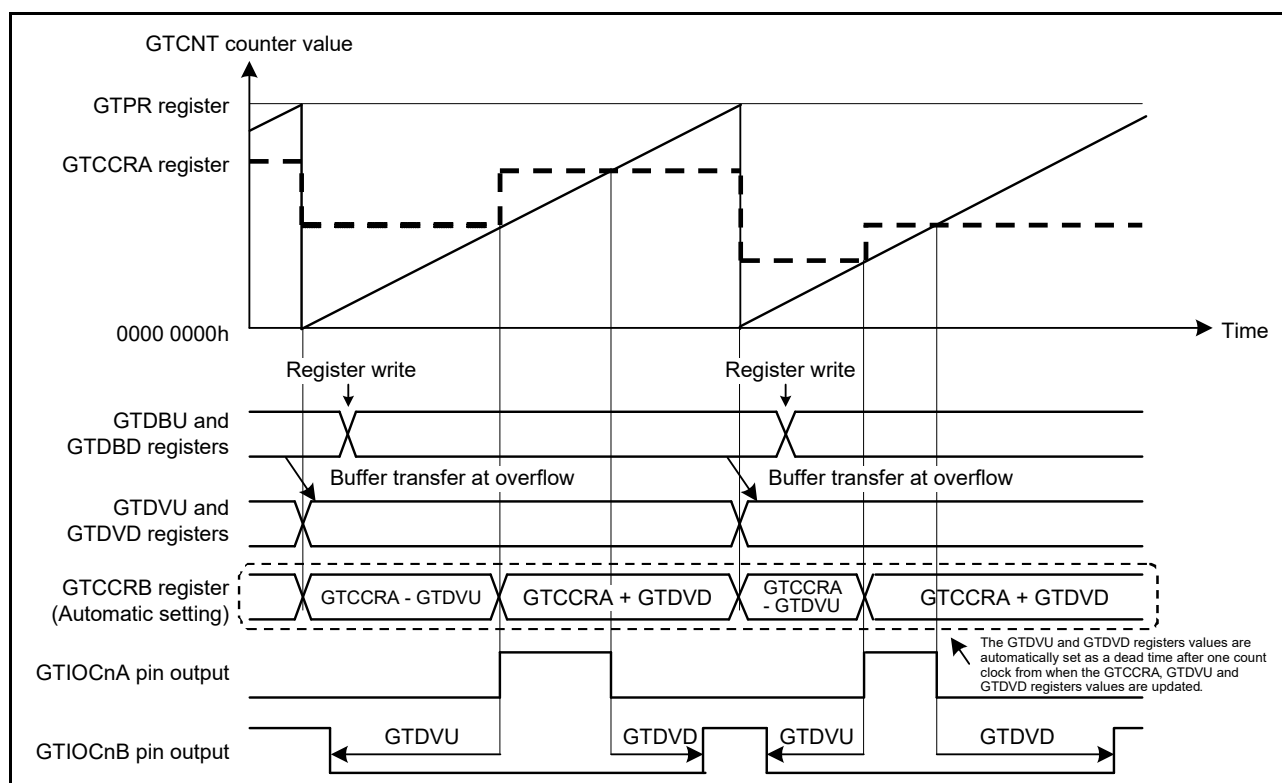


Figure 26.47 Example of Automatic Dead Time Setting Function Operation
(Sawtooth-Wave One-Shot Pulse Mode, Up-Counting, GTDVU and GTDVD Registers Set to Buffer Operation, Active Level: High) (n = 0 to 3)

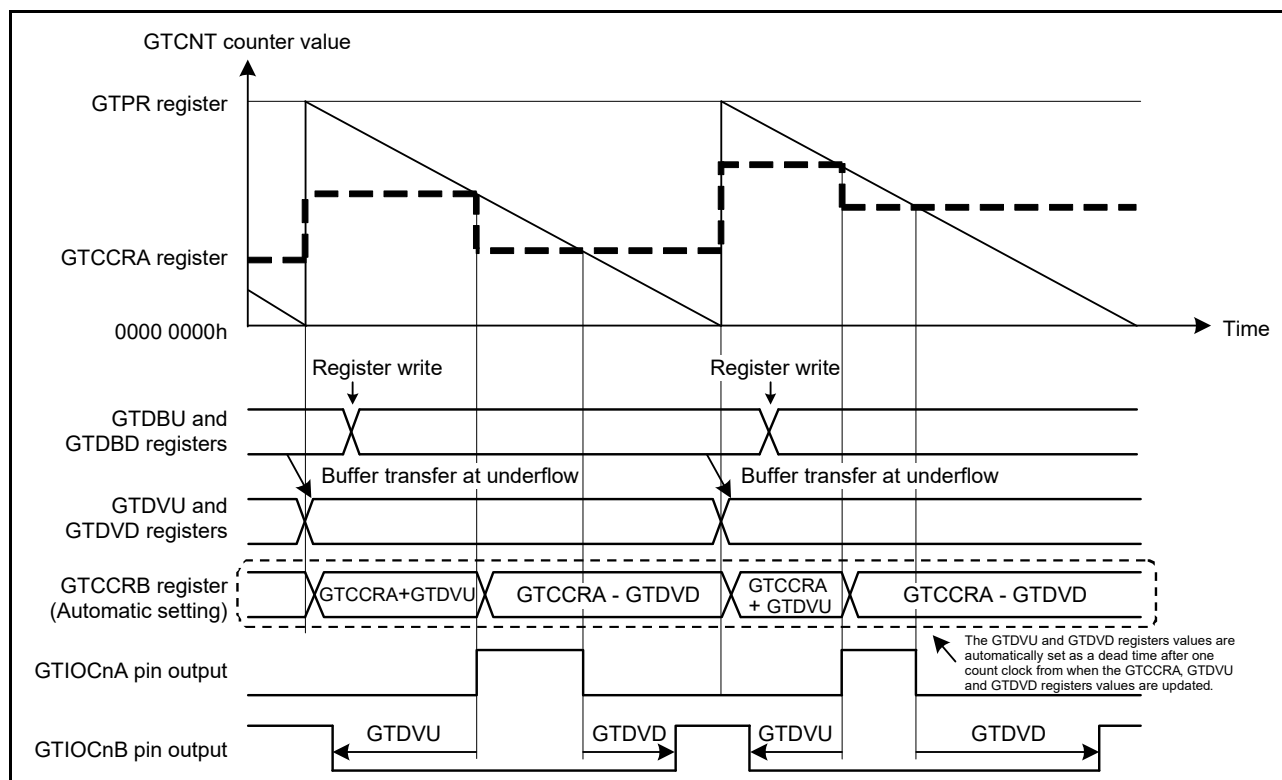


Figure 26.48 Example of Automatic Dead Time Setting Function Operation
(Sawtooth-Wave One-Shot Pulse Mode, Down-Counting, GTDVU and GTDVD Registers Set to Buffer Operation, Active Level: High) (n = 0 to 3)

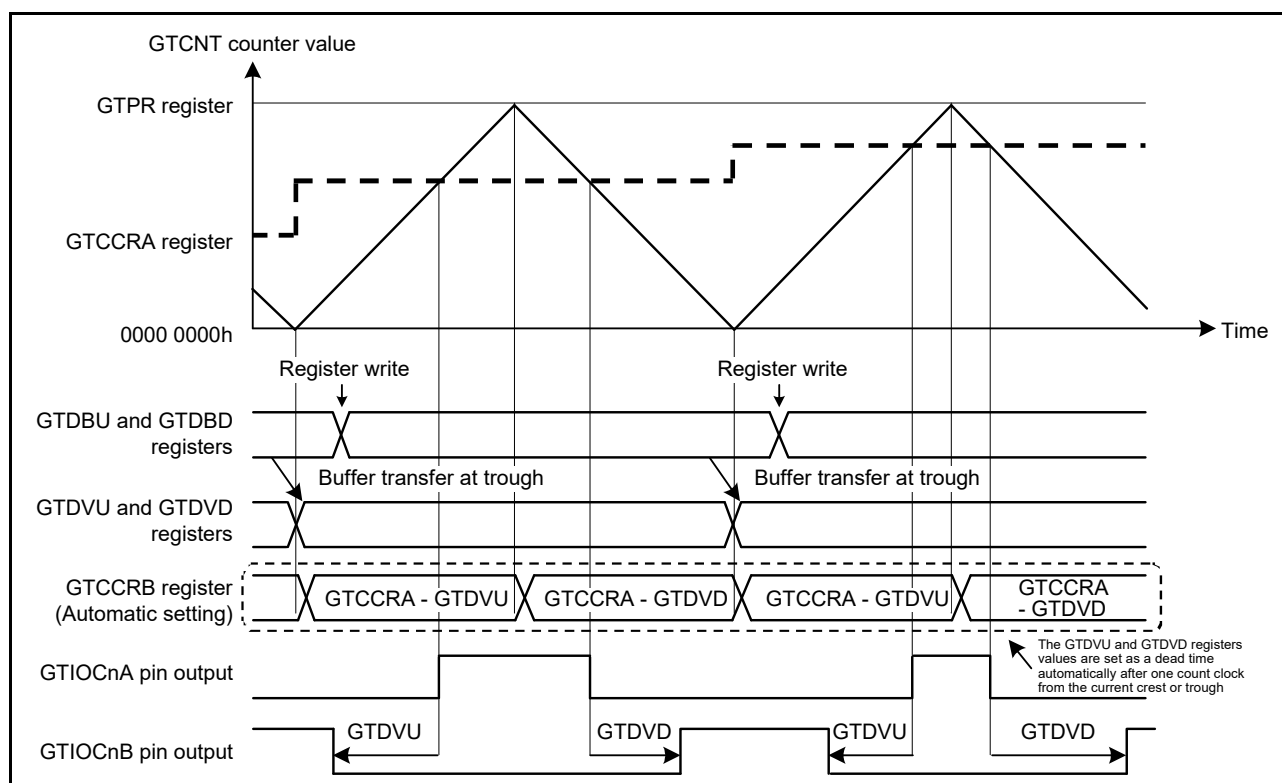


Figure 26.49 Example of Automatic Compare-Match Value Setting Function with Dead Time (Triangle-Wave PWM Mode 1, GTDVU and GTDVD Registers Set to Buffer Operation, Active Level: High) (n = 0 to 3)

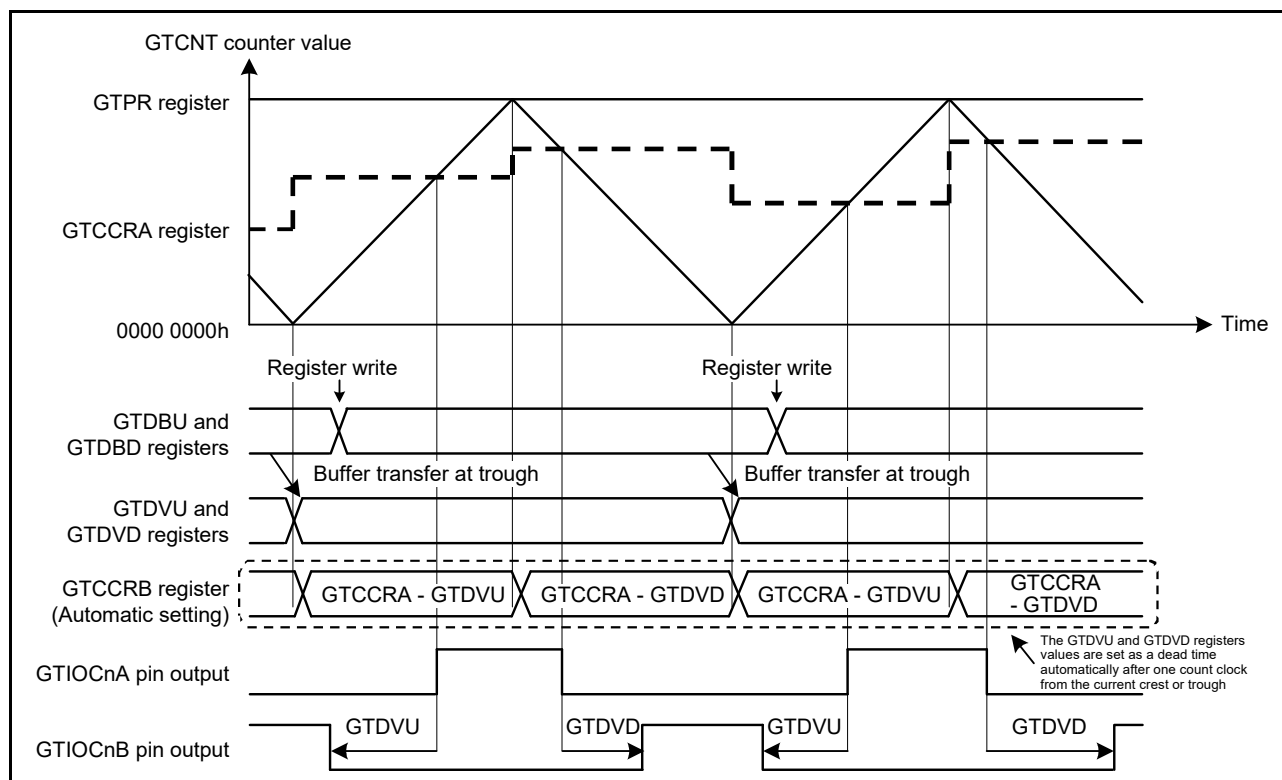


Figure 26.50 Example of Automatic Compare-Match Value Setting Function with Dead Time (Triangle-Wave PWM Mode 2 or 3, GTDVU and GTDVD Registers Set to Buffer Operation, Active Level: High) (n = 0 to 3)

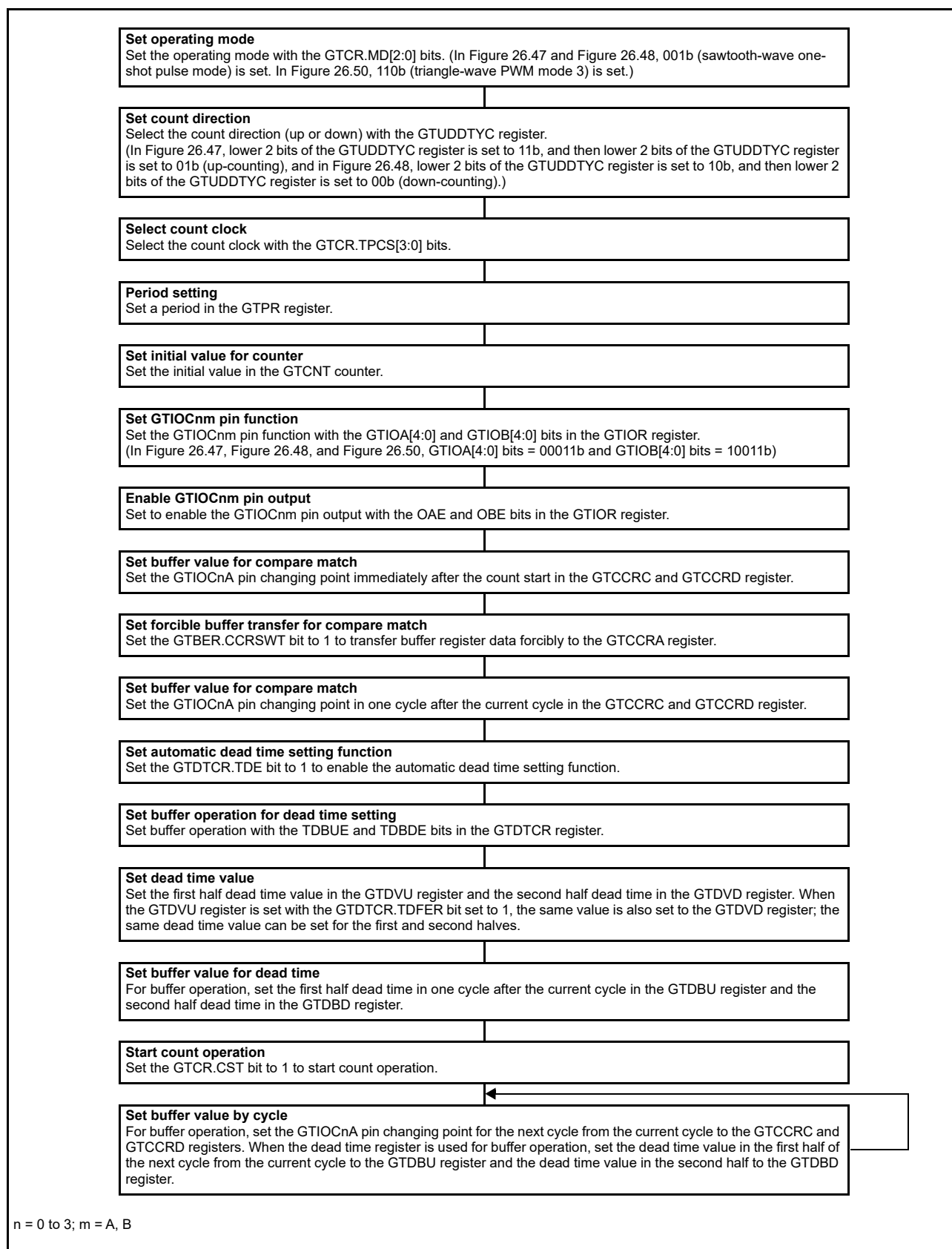


Figure 26.51 Example for Setting Automatic Dead Time Setting Function (Sawtooth-Wave One-Shot Pulse Mode, Triangle-Wave PWM Mode 3)

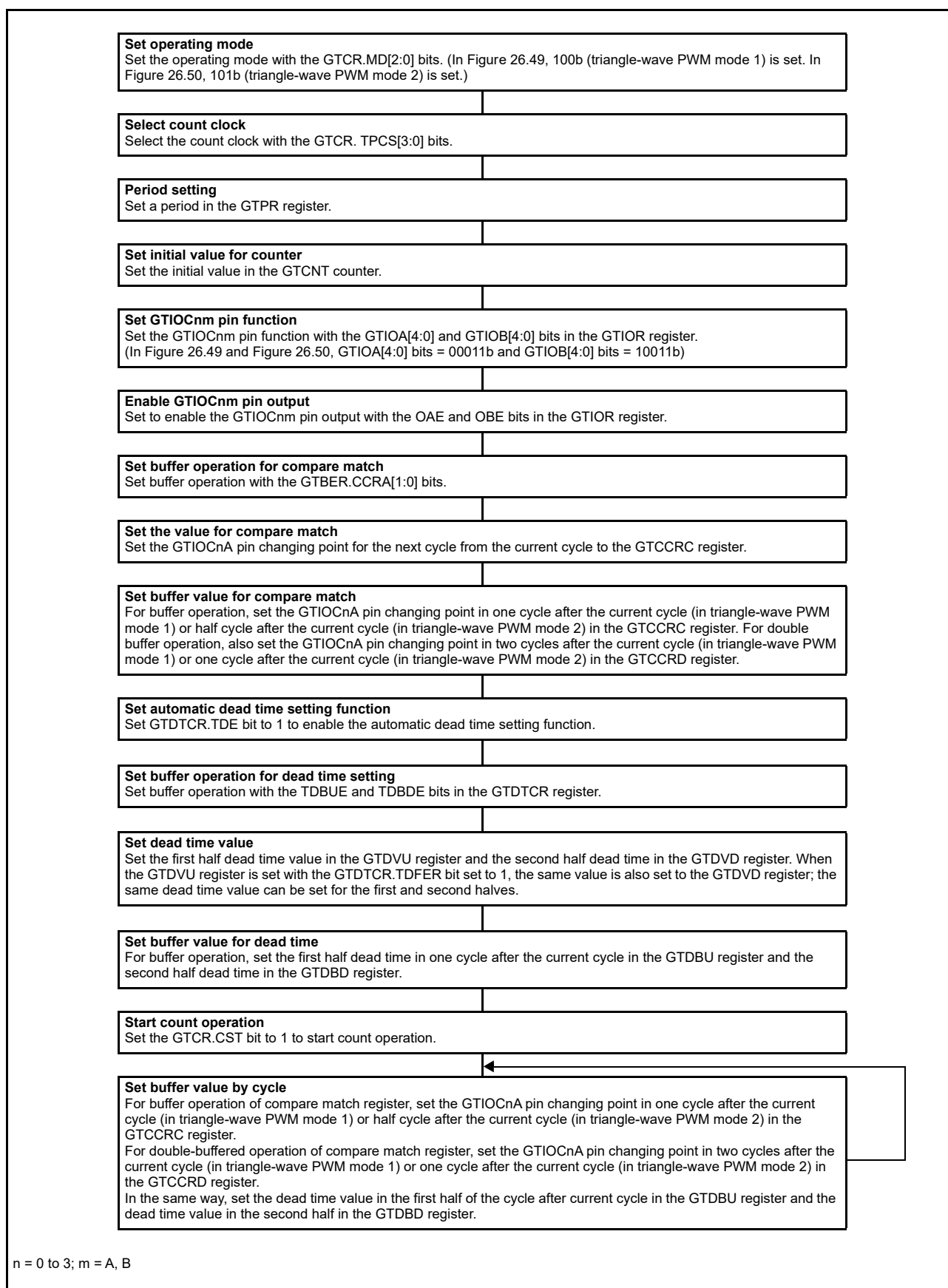


Figure 26.52 Example for Setting Automatic Dead Time Setting Function (Triangle-Wave PWM Mode 1 or 2)

26.3.5 Count Direction Changing Function

The count direction of the GTCNT counter can be changed by modifying the GTUDDTYC.UD bit.

In sawtooth-wave mode, if the UD bit is modified during count operation, the count direction is changed at an overflow (when modified during up-counting) or an underflow (when modified during down-counting). If the UD bit is modified while count operation is stopped and the GTUDDTYC.UDF bit is 0, the UD bit modification is not reflected at the start of counting and the count direction is changed at an overflow or an underflow. If the UDF bit is set to 1 while count operation is stopped, the UD bit value at that time is reflected at the start of counting.

In triangle-wave mode, changing the value of the GTUDDTYC.UD bit during counting does not switch the direction of counting. Likewise, changing the value of the GTUDDTYC.UD bit while the GTUDDTYC.UDF bit is 0 and counting is stopped does not actually update the value of the bit. If 1 is written to the GTUDDTYC.UDF bit while counting is stopped, the value of the GTUDDTYC.UD bit at that time is reflected from the time counting is started.

When counting direction is switched during count operation in sawtooth-wave mode, the value of the GTPR register after start of up-counting is reflected to the count period in up-counting operation, and the value of the GTPR register after the start of down-counting is reflected to the count period in down-counting operation.

Figure 26.53 shows an example of count direction changing function operation.

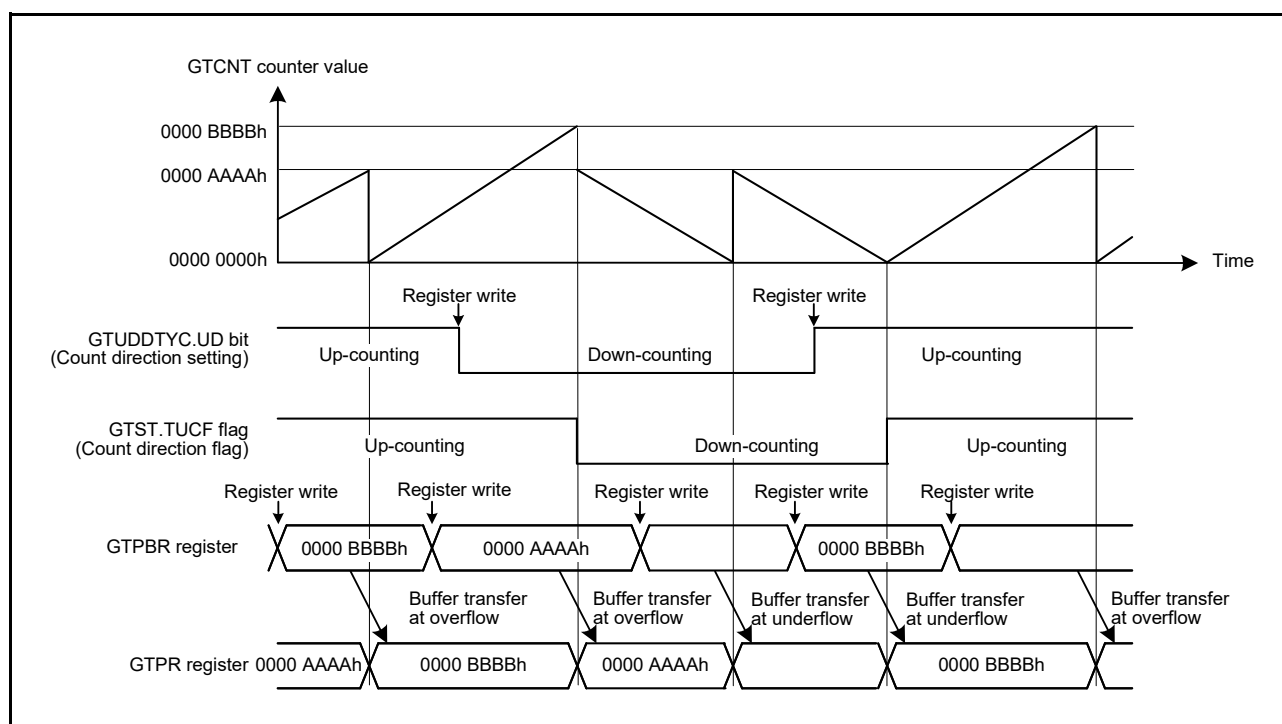


Figure 26.53 Example of Count Direction Changing Function Operation (during Buffer Operation)

26.3.6 Duty Cycle 0%/100% Output Function

Changing the value of the GTUDDTYC.OADTY[1:0] bits and GTUDDTYC.OBDTY[1:0] bits specifies the output duty setting on the GTIOCnA and GTIOCnB pins to 0% or 100%. (n = 0 to 3)

In sawtooth-wave mode, if the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed during the count operation, the output duty-cycle setting is reflected at an overflow (when changed during up-counting) or an underflow (when changed during down-counting). When the GTUDDTYC.OADTYF bit or GTUDDTYC.OBDTYF bit is 0 while count operation is stopped and the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed, the output duty-cycle setting changed at the start of counting is not reflected, but the output duty-cycle setting changed at an overflow or underflow is reflected. When the OADTYF bit or OBDTYF bit is 1 while count operation is stopped and the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed, the value of the OADTY[1:0] bits or OBDTY[1:0] bits at that time is reflected at the start of counting.

In triangle-wave mode, if the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed during the count operation, the output duty-cycle setting changed at an underflow is reflected.

When the OADTYF bit or OBDTYF bit is 0 while count operation is stopped and the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed, the output duty-cycle setting changed at the start of counting is not reflected, but the output duty-cycle setting changed at an underflow is reflected. When the OADTYF bit or OBDTYF bit is 1 while count operation is stopped and the value of the OADTY[1:0] bits or OBDTY[1:0] bits is changed, the value of the OADTY[1:0] bits or OBDTY[1:0] bits at that time is reflected at the start of counting.

During the operation under the settings of duty cycle for 0%/100%, compare match operation inside the GPTW continues to perform interrupt outputs and buffer operations.

When the output duty setting is changed from 0% or 100% due to a compare match, the GTIOR.GTIOA[3:2] bits and the OADTYR bits determine the level output on the GTIOCnA pin and GTIOR.GTIOB[3:2] bits and OBDTYR bits determine the level output on the GTIOCnB pin at the end of the cycle. When the GTIOA[3:2], GTIOB[3:2] bits are set as 01b, output is driven Low at the end of the cycle, and when set as 10b, output is driven High at the end of the cycle. The output is retained at the end of the cycles when the GTIOA[3:2] or GTIOB[3:2] bits are 00b, and toggled at the end of the cycles when the GTIOA[3:2] or GTIOB[3:2] bits are 11b. A value to be output is selectable in the OADTYR or OBDTYR bit when the corresponding GTIOA[3:2] or GTIOB[3:2] bits are either 00b or 11b. Table 26.9 lists the output values at the end of the cycle when the output setting is changed from duty 0% or 100% to compare match.

Table 26.9 Output Value after Release of Duty 0%/100%

GTIOR.GTIOm[3:2]	Value at Compare Match Output at the End of the Cycle in the Case of Masking by a Duty Cycle 0% or 100%	The GTUDDTYC.OmDTYR Bit at Duty 0% Setting		The GTUDDTYC.OmDTYR Bit at Duty 100% Setting	
		0	1	0	1
00b (Output retained at the end of the cycle)	0	0	0	1	0
	1	0	1	1	1
01b (Low output at the end of the cycle)	—	0	0	0	0
10b (High output at the end of the cycle)	—	1	1	1	1
11b (Toggle output at the end of the cycle)	0	1	1	0	1
	1	1	0	0	0

m = A, B

Figure 26.54 shows an example of operation of the output of the duty cycle 0% or 100% ($n = 0$ to 3).

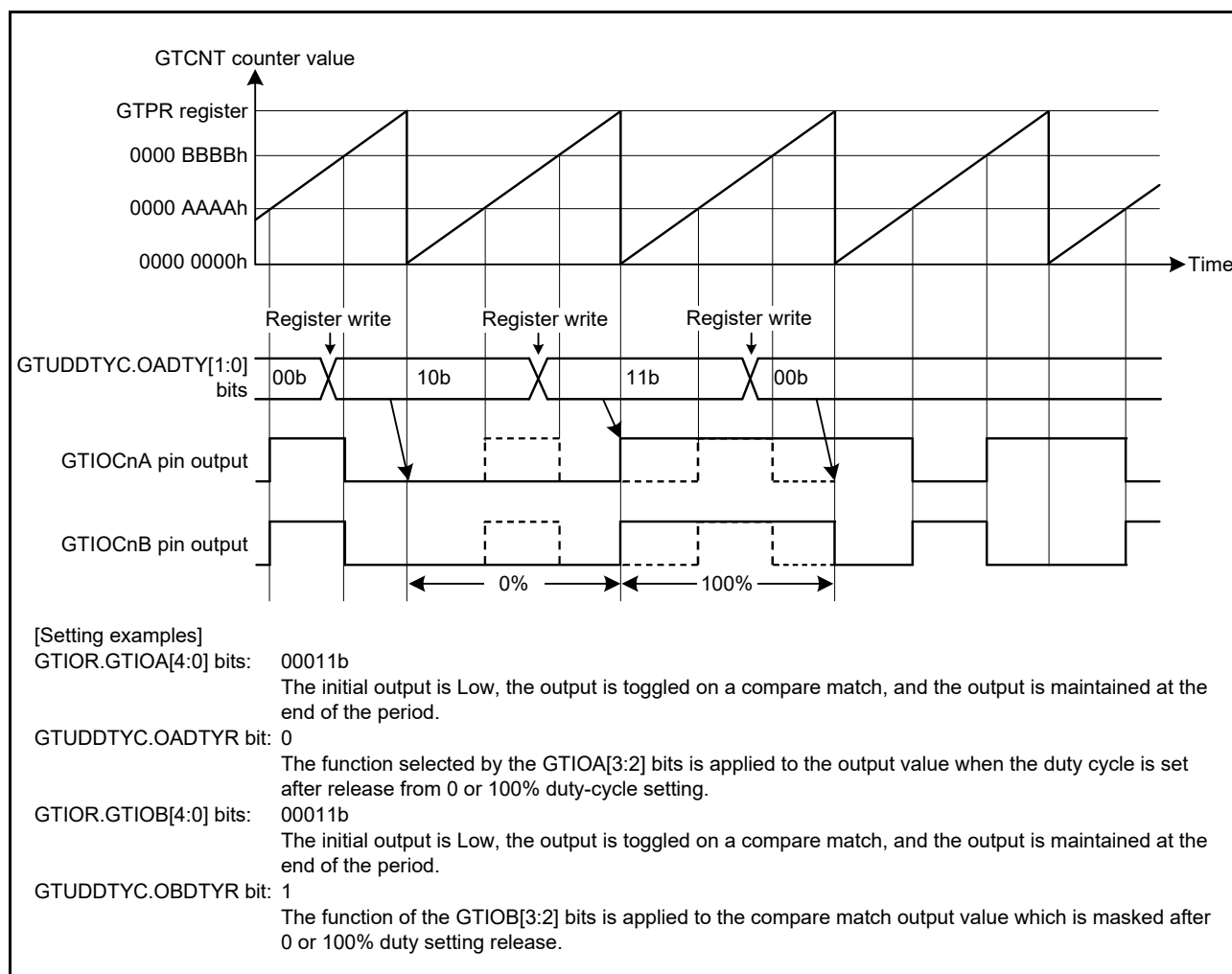


Figure 26.54 Example of Operation of Output of Duty Cycle 0% or 100% ($n = 0$ to 3)

26.3.7 Hardware Count Start/Count Stop and Clear Operation

The GTCNT counter can be started, stopped, or cleared by hardware sources in this MCU. There are 3 types of hardware sources, including external trigger input, ELC event input, and GTIOCNm pin input (n = 0 to 3; m = A, B)

26.3.7.1 Hardware Start Operation

The GTCNT counter can be started by a hardware source. Select a hardware source to start counting using the GTSSR register, and enable to start counting.

Figure 26.55 shows an example of count start operation by a hardware source. Figure 26.56 shows the setting example.

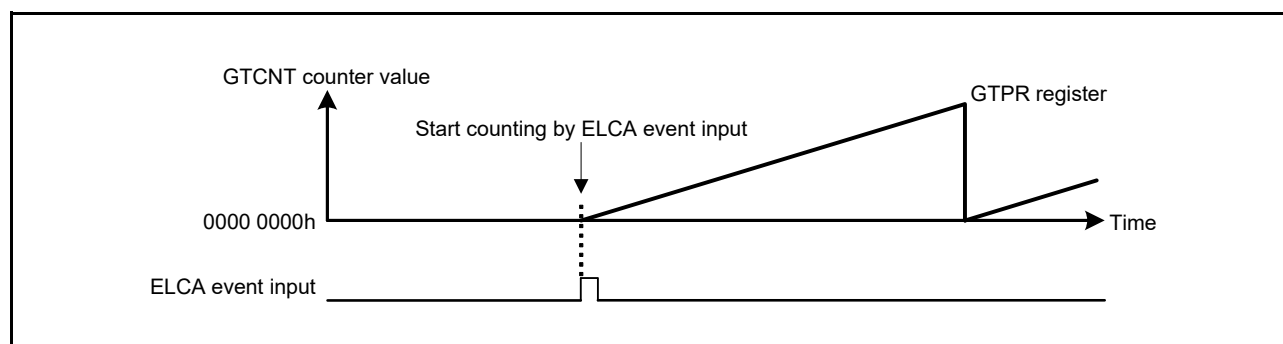


Figure 26.55 Example of Count Start Operation by Hardware Source (At starting by ELCA event)

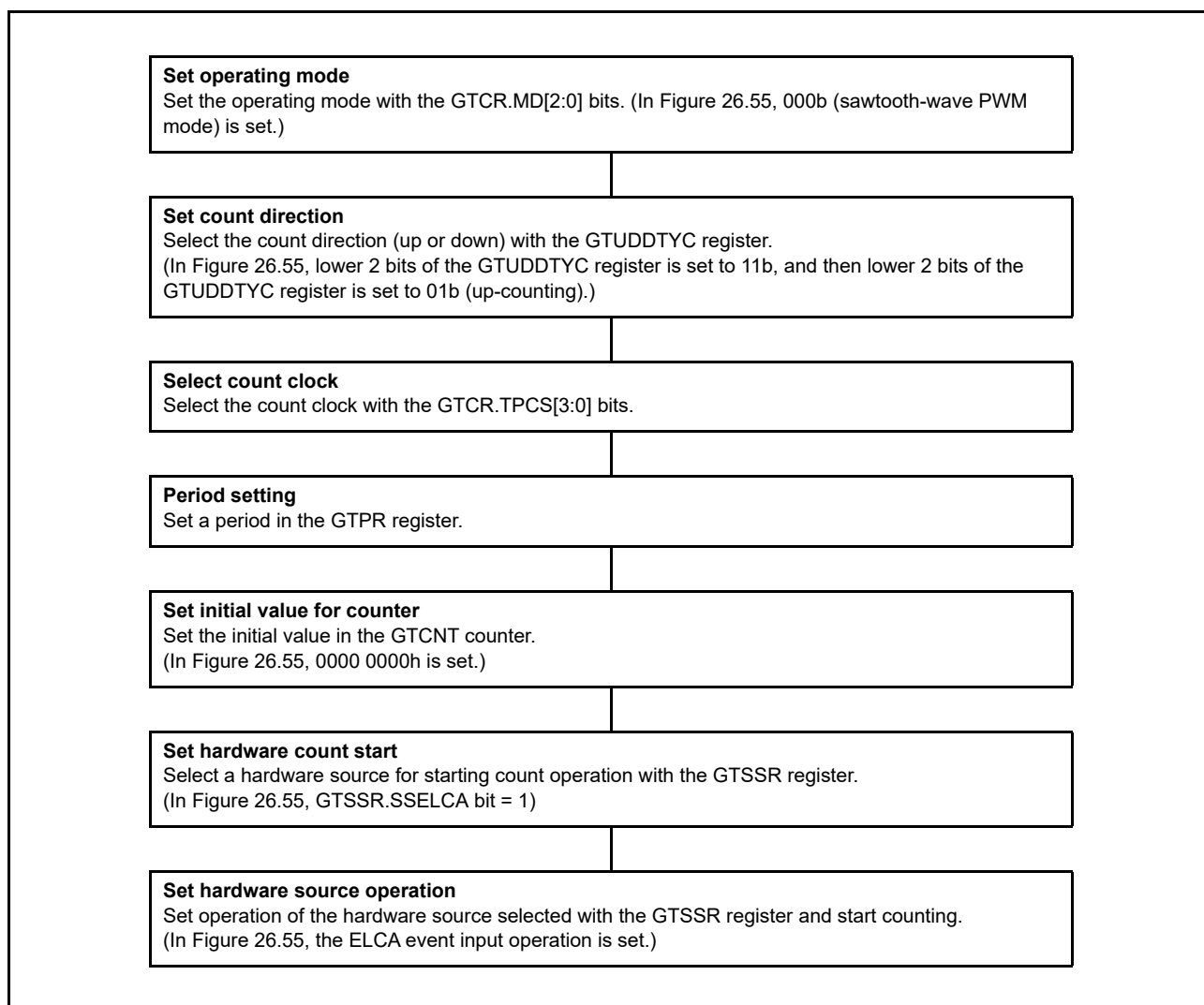


Figure 26.56 Example for Setting Count Start Operation by Hardware Source

Figure 26.57 shows an example of timing of operations to start counting in response to a rising edge of the input on the GTETRGA pin.

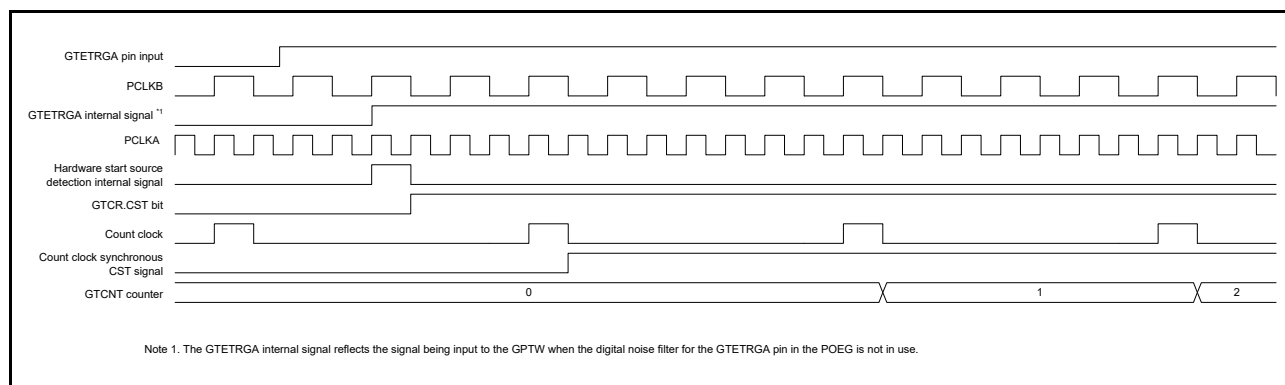


Figure 26.57 Example of Timing of Operations to Start Counting in Response to a Rising Edge of the Input on the GTETRGA Pin

Figure 26.58 shows an example of timing of operations to start counting in response to a rising edge of the input on the GTIOCnA pin.

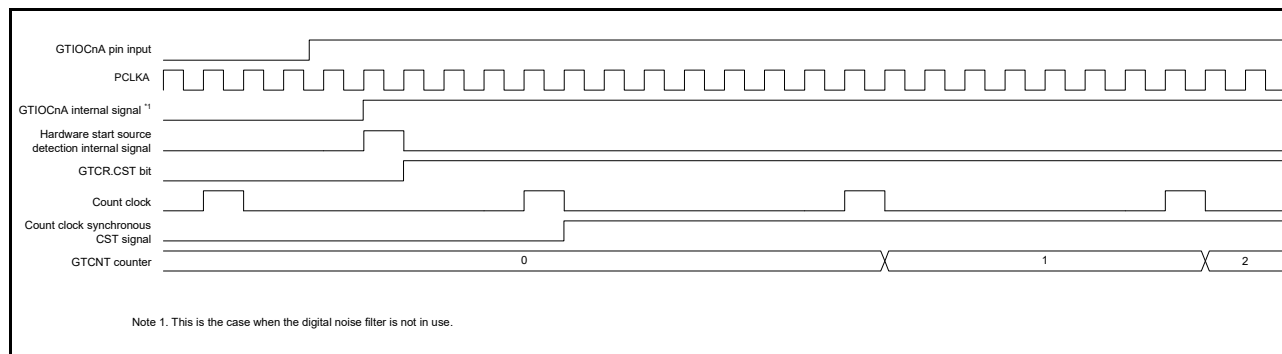


Figure 26.58 Example of Timing of Operations to Start Counting in Response to a Rising Edge of the Input on the GTIOCnA Pin

Figure 26.59 shows an example of timing of operations to start counting in response to event input from the ELCA. This is an example of operations to start counting by the GPTW1.GTCNT counter in response to a signal. An event signal is output to the ELC after matches in comparison with the GPTW0.GTCCRA register. This is selected as a trigger for output to the GPTW by the ELC as event source A. The GPTW0 compare match A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and outputs the GPTW event source A signal after 1 cycle of PCLKB has elapsed.

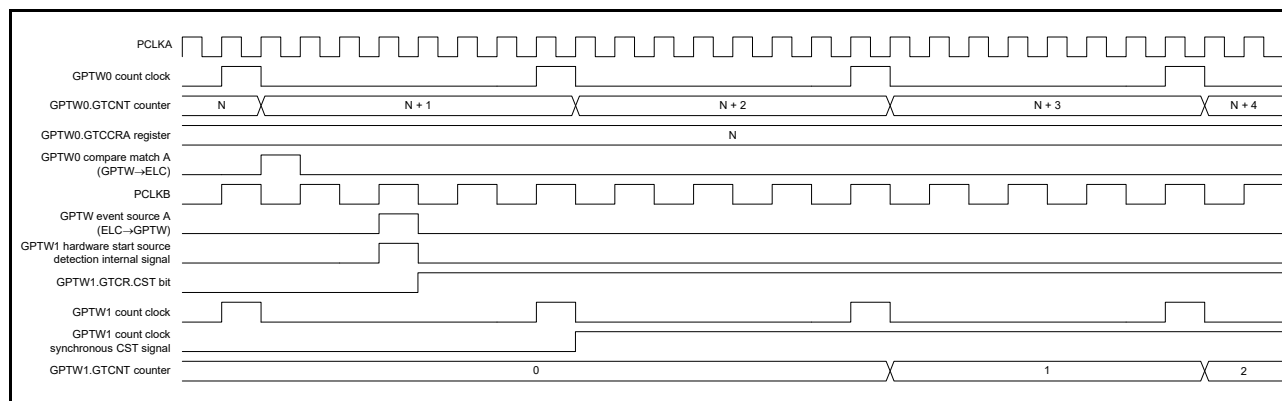


Figure 26.59 Example of Timing of Operations to Start Counting in Response to Event Input from the ELCA

26.3.7.2 Hardware Stop Operation

The GTCNT counter can be stopped by a hardware source. Select a hardware source to stop counting with the GTPSR register, and enable to stop counting.

Figure 26.60 shows an example of count stop operation by a hardware source. Figure 26.61 shows the setting example. In this example, the count operation is stopped by the ELCA event input, and is restarted by the ELCB event input.

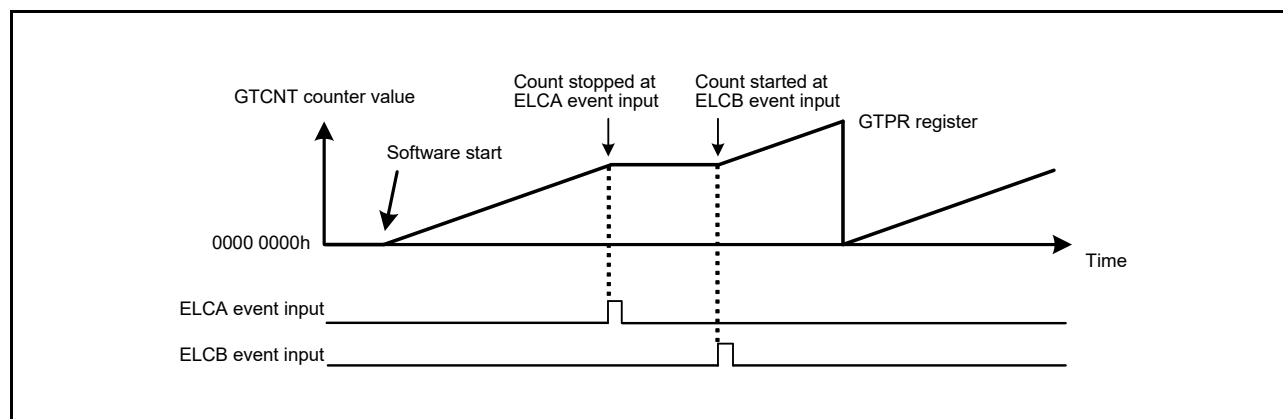


Figure 26.60 Example of Count Stop Operation by Hardware Source
(Started by Software, Stopped at ELCA event input, and Restarted at ELCB event input)

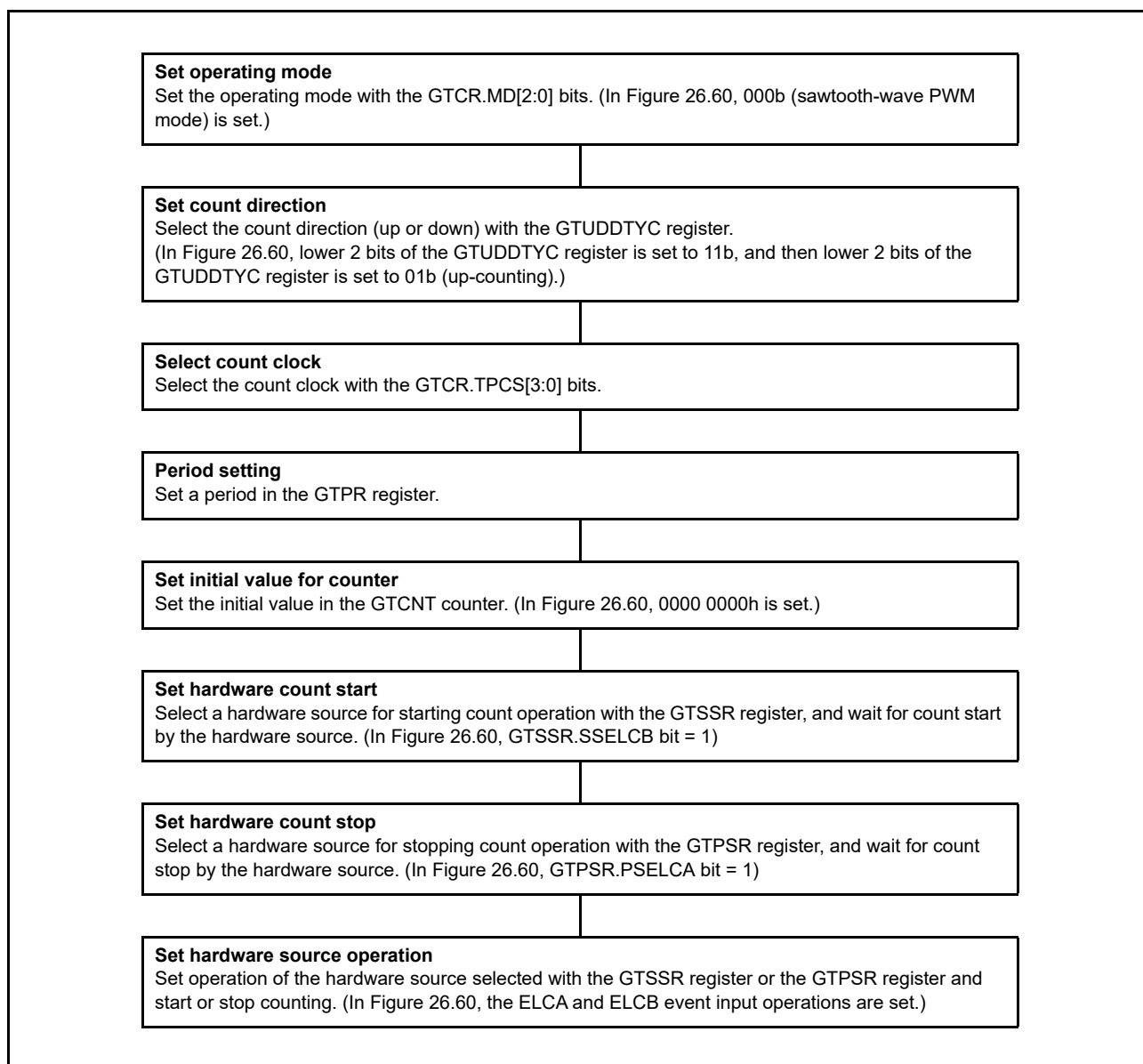


Figure 26.61 Example for Setting Count Stop Operation by Hardware Source

Figure 26.62 shows an example of count start/stop operation by a hardware source. Figure 26.63 shows the setting example. In this example, the counter operates during the high-level periods of the external trigger input GTETRGA pin.

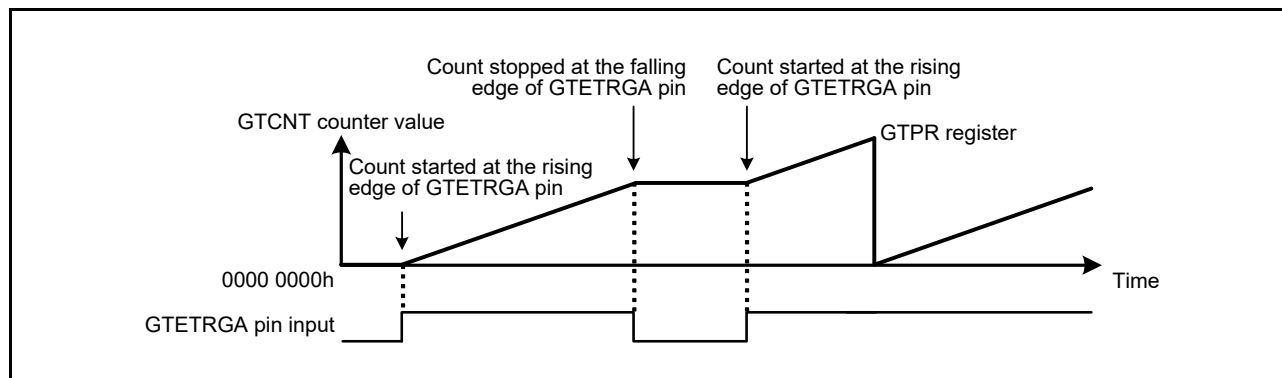


Figure 26.62 Example of Count Start/Stop Operation by Hardware Source (Started at Rising Edge of GTETRGA Pin Input, Stopped at Falling Edge of GTETRGA Pin Input)

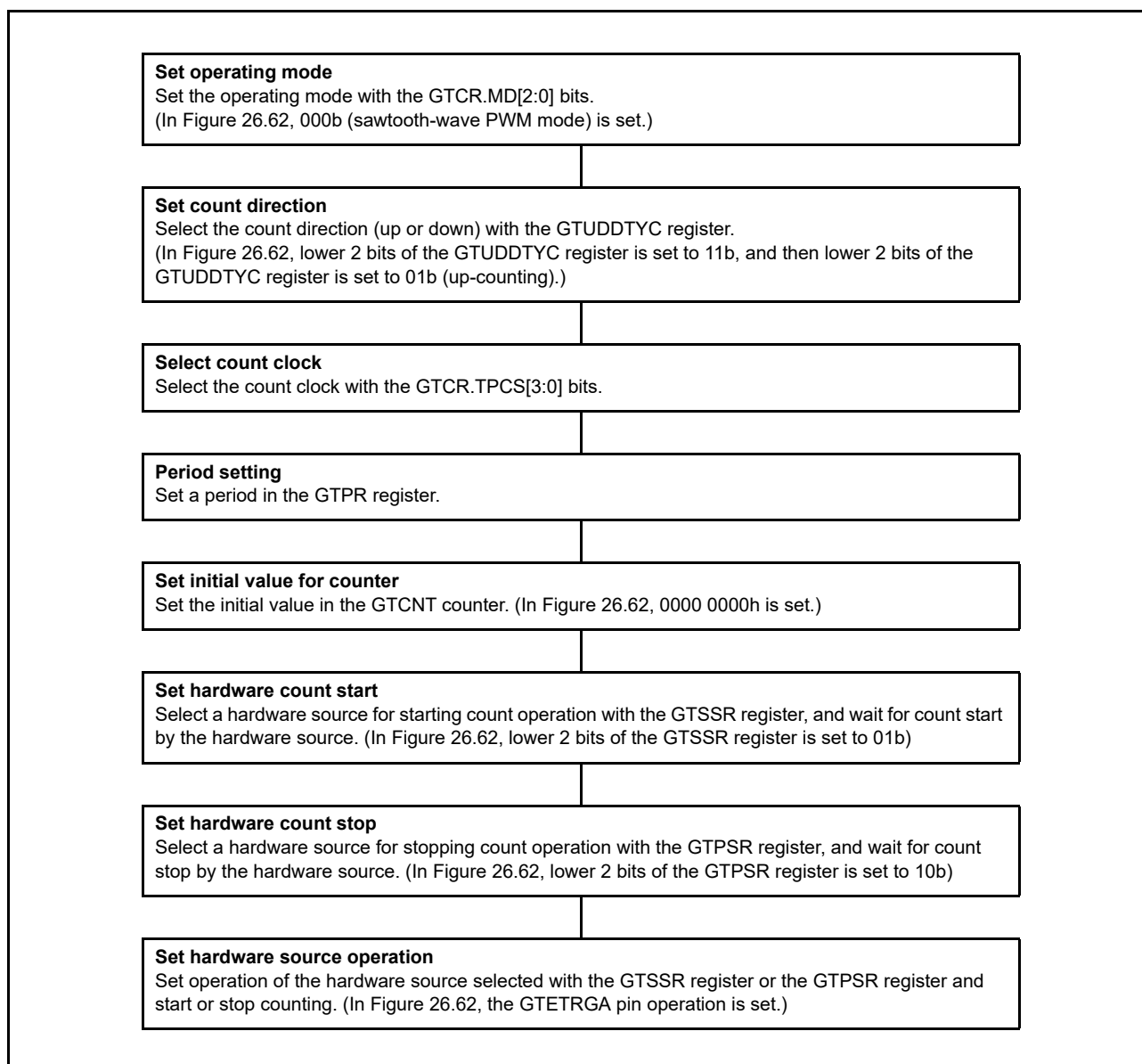


Figure 26.63 Example for Setting Count Start/Stop Operation by Hardware Source

Figure 26.64 shows an example of timing of operations to stop counting in response to a rising edge of the input on the GTETRGA pin.

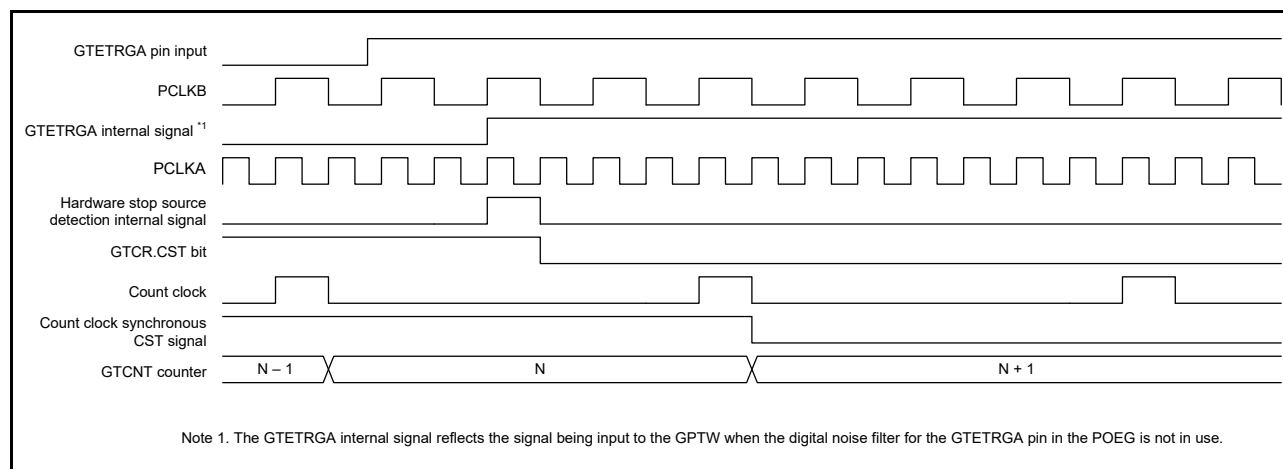


Figure 26.64 Example of Timing of Operations to Stop Counting in Response to a Rising Edge of the Input on the GTETRGA Pin

Figure 26.65 shows an example of timing of operations to stop counting in response to a rising edge of the input on the GTIOCnA pin.

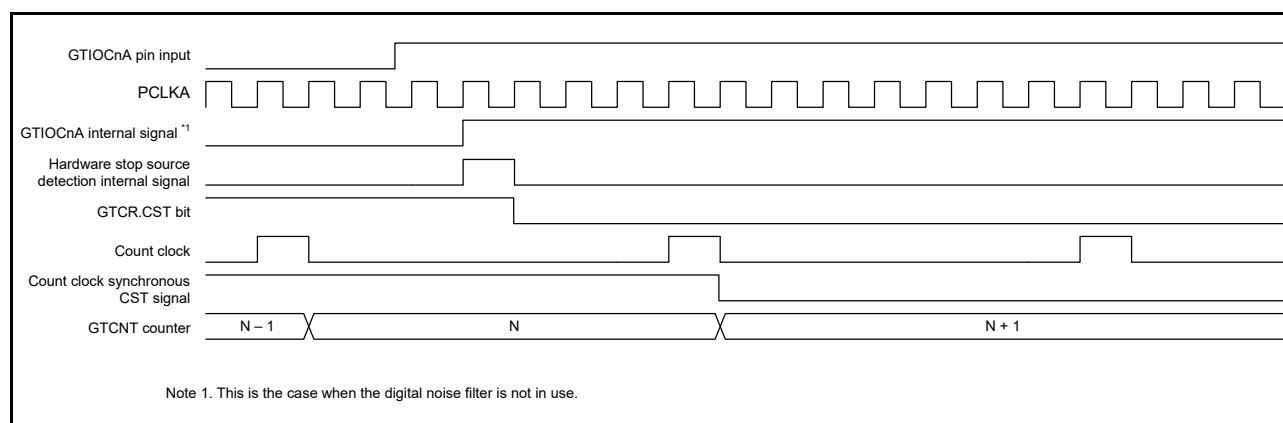


Figure 26.65 Example of Timing of Operations to Stop Counting in Response to a Rising Edge of the Input on the GTIOCnA Pin

Figure 26.66 shows an example of timing of operations to stop counting in response to event input from the ELCA. This is an example of operations to stop counting by the GPTW1.GTCNT counter in response to a signal. An event signal is output to the ELC after matches in comparison with the GPTW0.GTCCRA register. This is selected as a trigger for output to the GPTW by the ELC as event source A. The GPTW0 compare match A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and outputs the GPTW event source A signal after 1 cycle of PCLKB has elapsed.

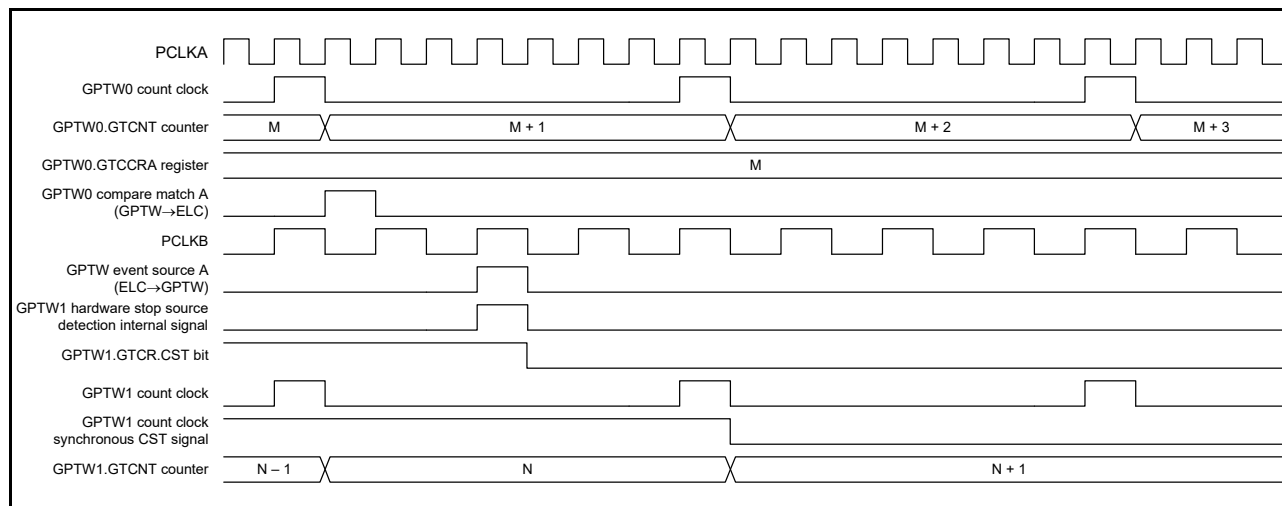


Figure 26.66 Example of Timing of Operations to Stop Counting in Response to Event Input from the ELCA

26.3.7.3 Hardware Clear Operation

The GTCNT counter can be cleared by a hardware source. Select a hardware source to clear the counter with the GTCSCR register, and enable clearing the counter.

Note that the GTCIV/GTCIU interrupt (overflow/underflow interrupt) is not generated when the GTCNT counter is cleared by a hardware source or by software.

Figure 26.67 and Figure 26.68 show examples of the GTCNT counter clearing operation by a hardware source. Figure 26.69 shows the setting example. In this example, the GTCNT counter is started at the ELCA event input, and is stopped/cleared at the ELCB event input.

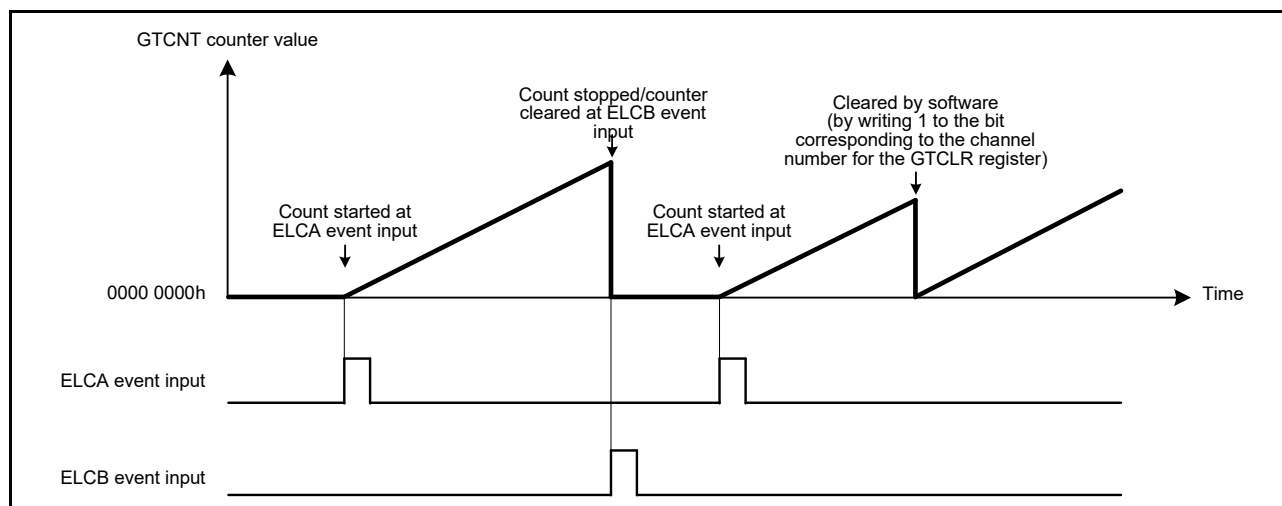


Figure 26.67 Examples of Counter Clearing Operation by Hardware Source (Sawtooth-Wave Up-Counting, Started at ELCA event input, and Count Stopped/Counter Cleared at ELCB event input)

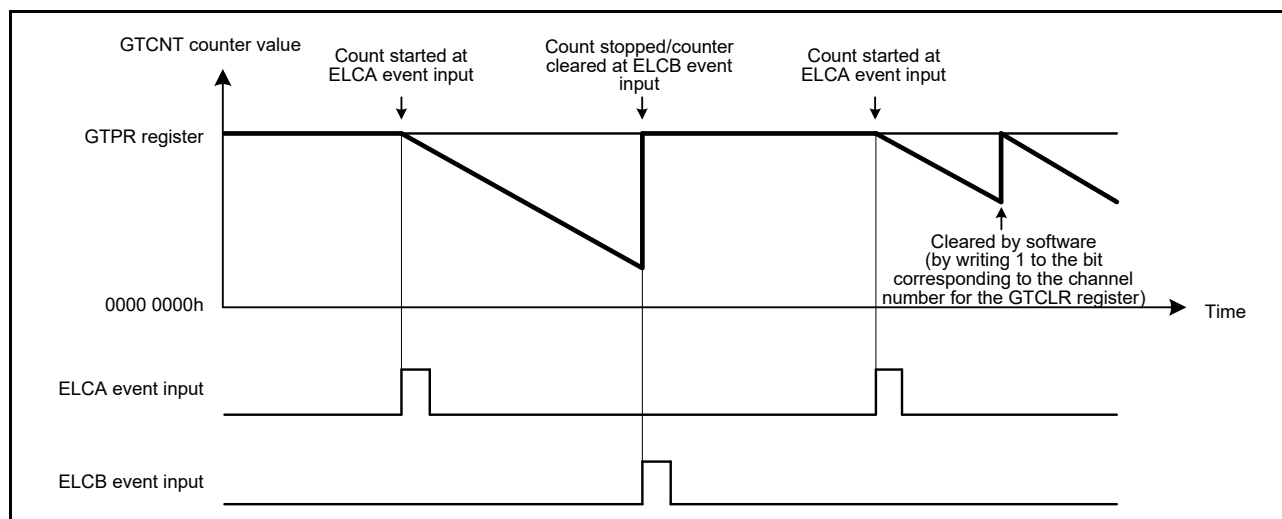


Figure 26.68 Examples of Counter Clearing Operation by Hardware Source (Sawtooth-Wave Down-Counting, Started at ELCA event input, and Count Stopped/Counter Cleared at ELCB event input)

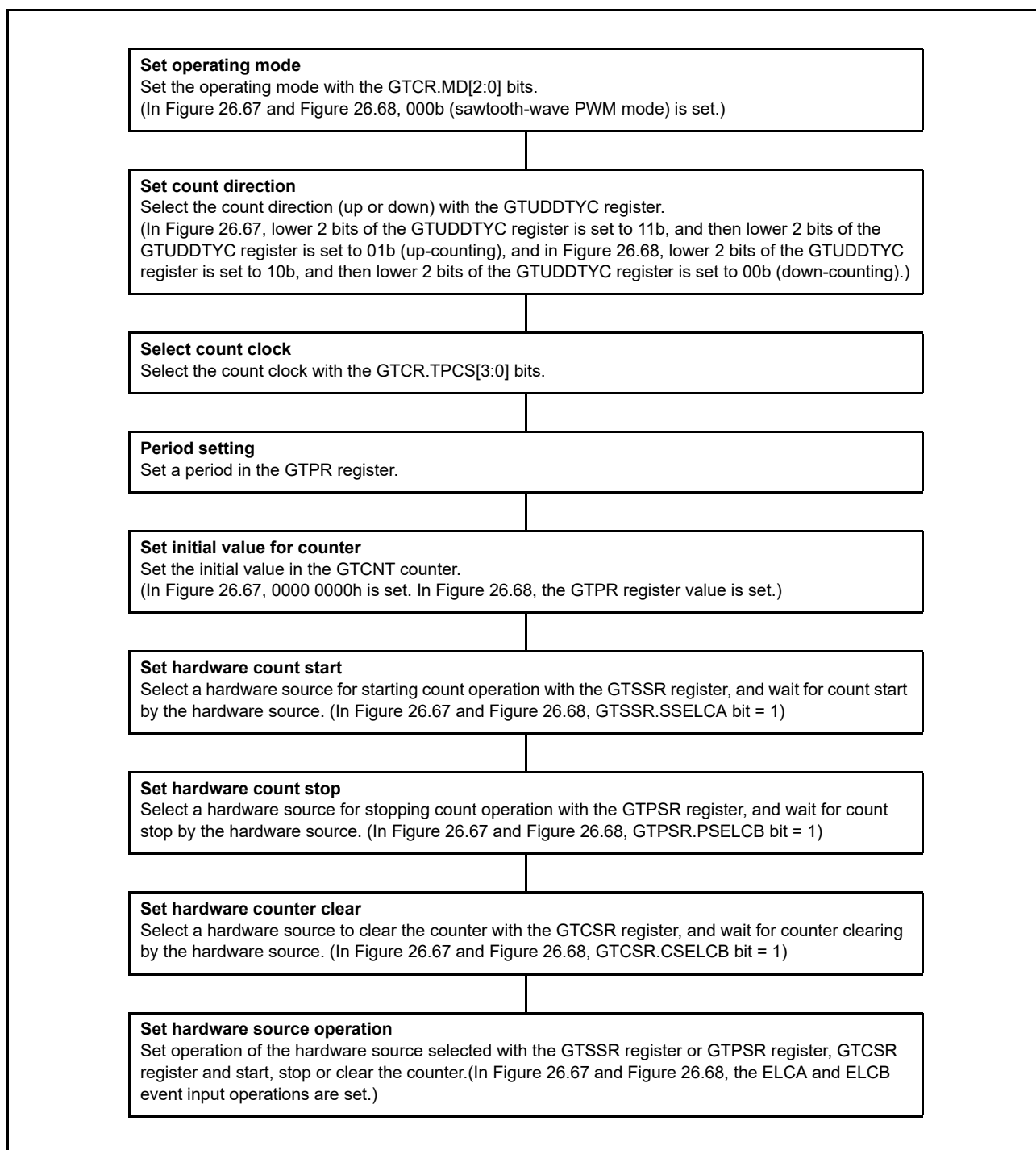


Figure 26.69 Example for Setting Counter Clearing Operation by Hardware Source

The GTCIV/GTCIU interrupt (overflow/underflow interrupt) does not occur when the counter is cleared by a hardware source. In the same way, the GTCIV/GTCIU interrupt does not occur when the counter is cleared by software. Figure 26.70 shows the relationship between counter clearing by a hardware source and the GTCIV interrupt.

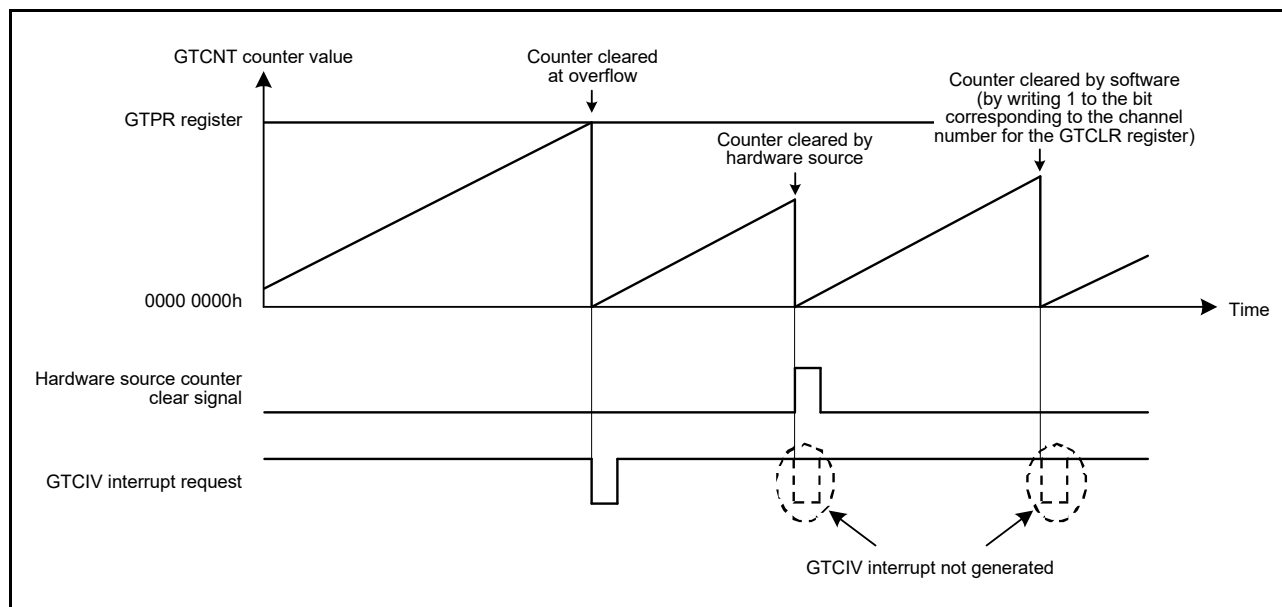


Figure 26.70 Relationship between Counter Clearing by Hardware Source and GTCIV Interrupt

Figure 26.71 shows an example of the timing of operations to clear the counter in response to a rising edge of the input on the GTETRGA pin when a clock signal produced by frequency-dividing the PCLKA signal is used as the counter clock for the GTCNT counter.

The GTCNT counter is cleared when counting is in progress after the GPTW has detected the internal clearing signal.

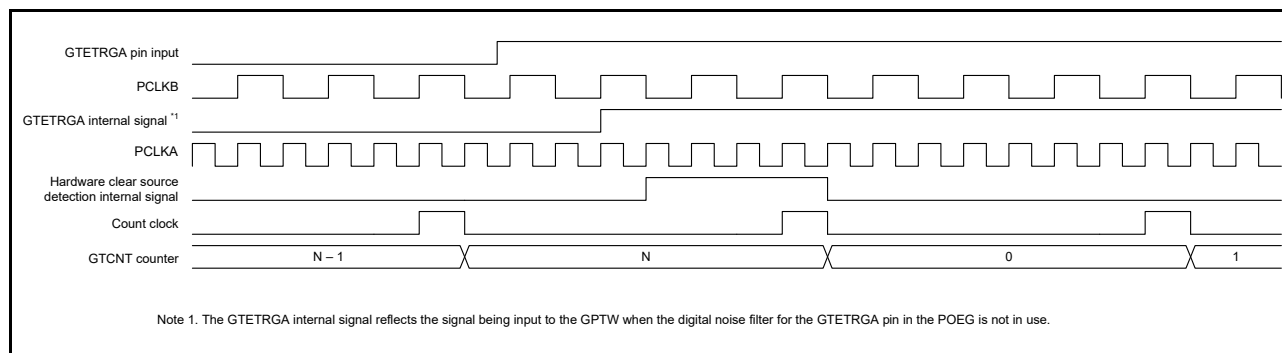


Figure 26.71 Example of the Timing of Operations for Counter Clearing in Response to a Rising Edge of the Input on the GTETRGA Pin (During the Counting of Cycles of Clock Signal Produced by Dividing the PCLKA Frequency)

Figure 26.72 shows an example of the timing of operations to clear the counter in response to a rising edge of the input on the GTIOCnA pin when a clock signal produced by frequency-dividing the PCLKA signal is used as the counter clock for the GTCNT counter.

The GTCNT counter is cleared when counting is in progress after the GPTW has detected the internal clearing signal.

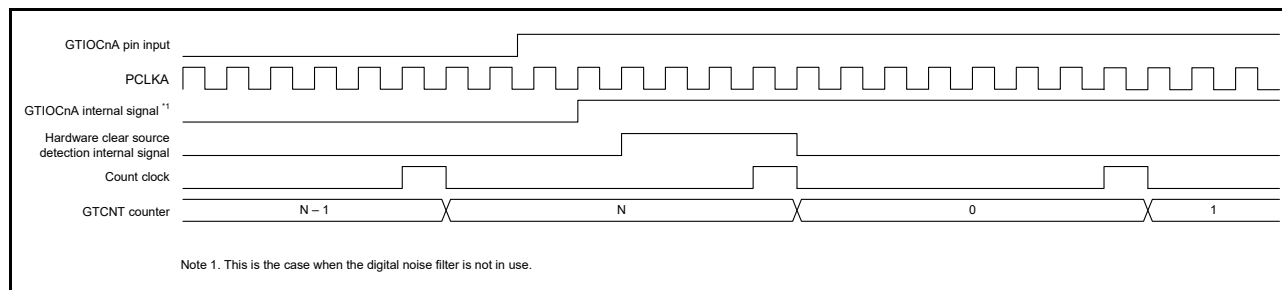


Figure 26.72 Example of the Timing of Operations for Counter Clearing in Response to a Rising Edge of the Input on the GTIOCnA Pin (During the Counting of Cycles of Clock Signal Produced by Dividing the PCLKA Frequency)

Figure 26.73 shows an example of the timing of operations to clear the counter in response to event input from the ELCA when a clock signal produced by frequency-dividing the PCLKA signal is used as the counter clock for the GTCNT counter.

This is an example of operations to clear the GPTW1.GTCNT counter in response to a signal. An event signal is output to the ELC after matches in comparison with the GPTW0.GTCCRA register. This is selected as a trigger for output to the GPTW by the ELC as event source A.

The GPTW0 compare match A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and outputs the GPTW event source A signal after 1 cycle of PCLKB has elapsed. The GTCNT counter is cleared when counting is in progress after the GPTW has detected the internal clearing signal.

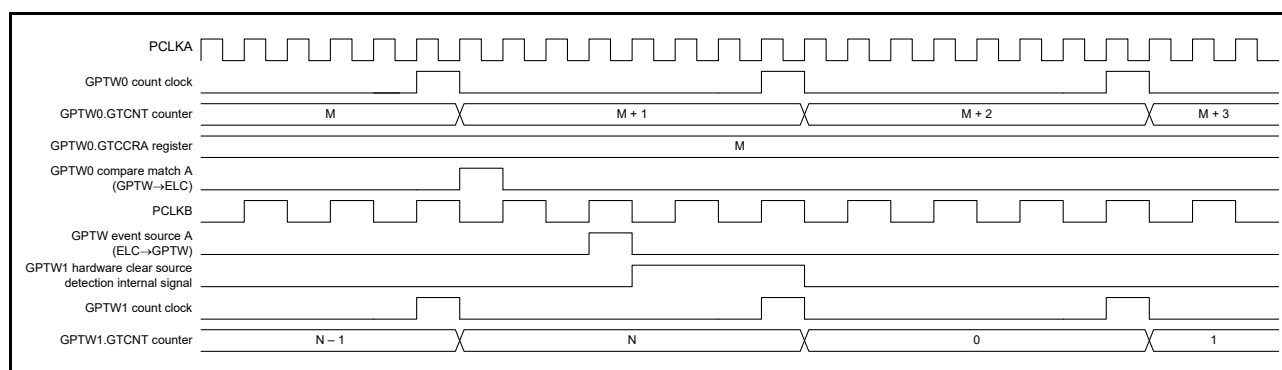


Figure 26.73 Example of the Timing of Operations for Counter Clearing in Response to Event Input from the ELCA (During the Counting of Cycles of Clock Signal Produced by Dividing the PCLKA Frequency)

Figure 26.74 shows an example of the timing of operations to clear the counter in response to a rising edge of the input on the GTETRGA pin input when counting is triggered by a hardware source.

The GTCNT counter is cleared in synchronization with PCLKA after the GPTW has detected the internal clearing signal.

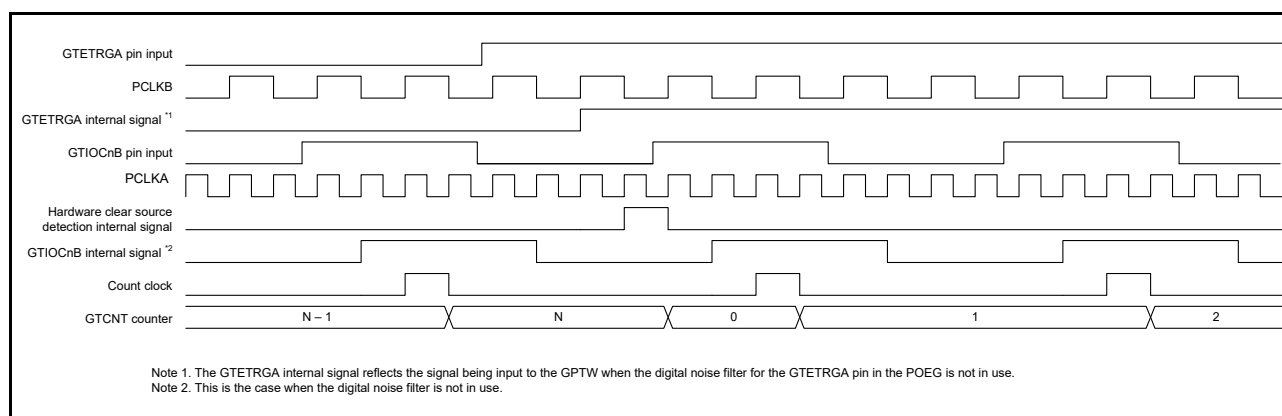


Figure 26.74 Example of the Timing of Operations for Counter Clearing in Response to a Rising Edge of the Input on the GTETRGA Pin (During Counting Triggered by Hardware Source)

Figure 26.75 shows an example of the timing of operations to clear the counter in response to a rising edge of the input on the GTIOCnA pin input when counting is triggered by a hardware source.

The GTCNT counter is cleared in synchronization with PCLKA after the GPTW has detected the internal clearing signal.

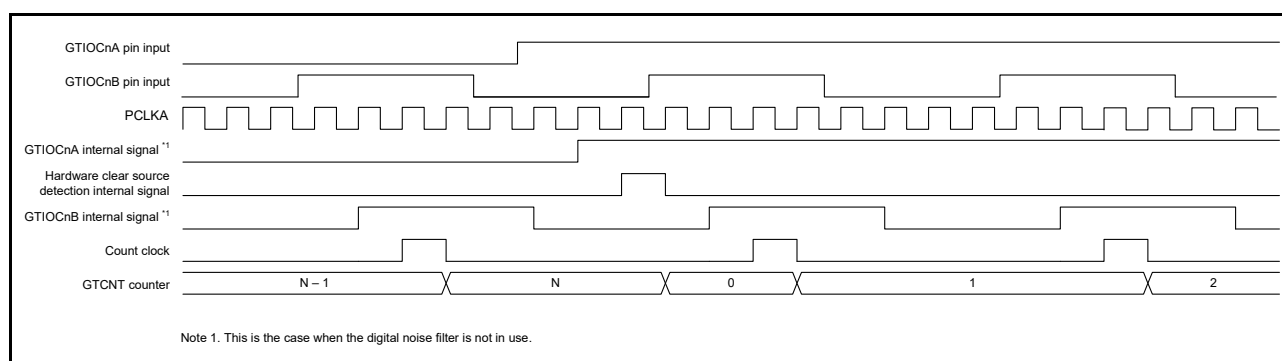


Figure 26.75 Example of the Timing of Operations for Counter Clearing in Response to a Rising Edge of the Input on the GTIOCnA Pin (During Counting Triggered by Hardware Source)

Figure 26.76 shows an example of the timing of operations for clearing the counter in response to the input of an event signal from the ELCA when counting is triggered by a hardware source.

This is an example of operations to clear the GPTW1.GTCNT counter in response to a signal. An event signal is output to the ELC after matches in comparison with the GPTW0.GTCCRA register. This is selected as a trigger for output to the GPTW by the ELC as event source A.

The GPTW0 compare match A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and outputs the GPTW event source A signal after 1 cycle of PCLKB has elapsed. The GTCNT counter is cleared in synchronization with PCLKA after the GPTW has detected the internal clearing signal.

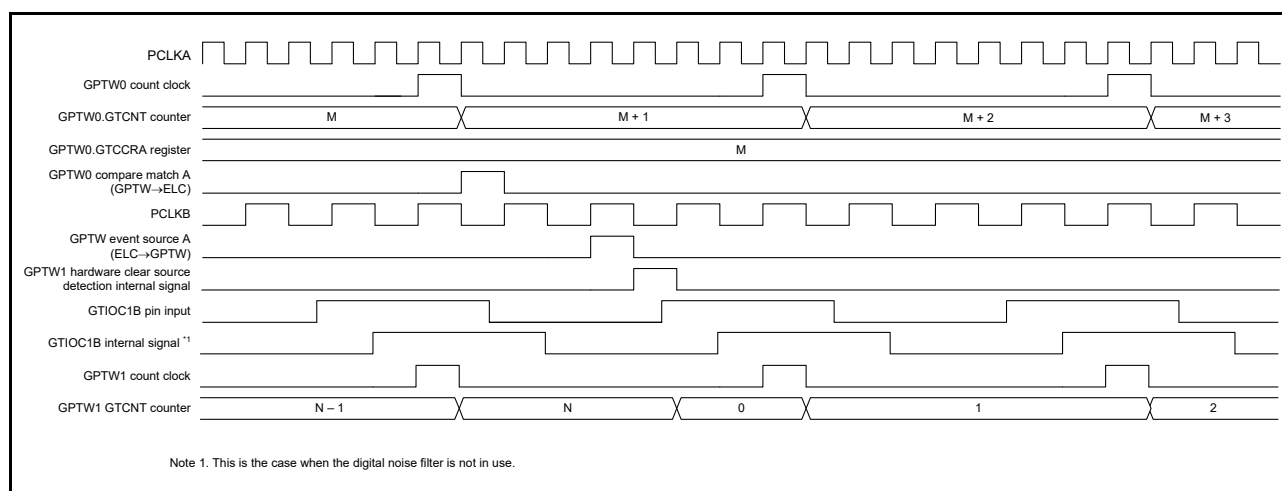


Figure 26.76 Example of the Timing of Operations for Counter Clearing in Response to Event Input from the ELCA (During Counting Triggered by Hardware Source)

26.3.8 Synchronous Operation

Synchronous operation on channels (synchronous start/stop/clear operation) can be performed.

26.3.8.1 Synchronous Operation by Software

The count operations for respective channels can be started, stopped and cleared simultaneously by setting multiple bits of the GTSTR, GTSTP, and GTCLR registers to 1 at the same time.

Count start with phase differences among channels is possible firstly by setting the GTCNT counter value before counting starts, and then, by setting simultaneously multiple bits in the GTSTR register to 1.

The Figure 26.77 shows an example of simultaneous start/stop/clearing of four channels by software, and Figure 26.78 shows an example of phase shift start among four channels by software.

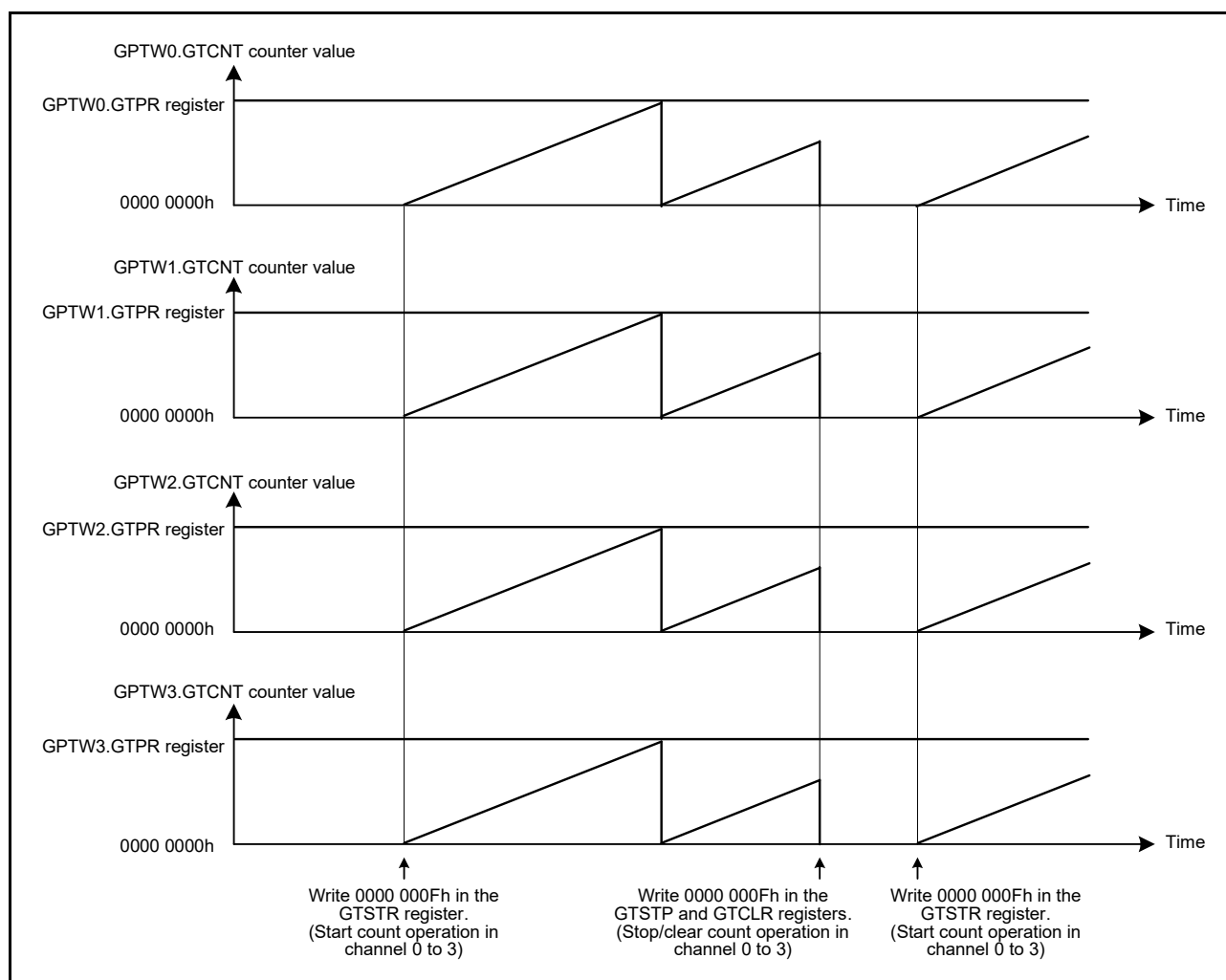


Figure 26.77 Example of Simultaneous Start/Stop/Clearing Operation by Software (with the Same Count Period (GTPR Register Value))

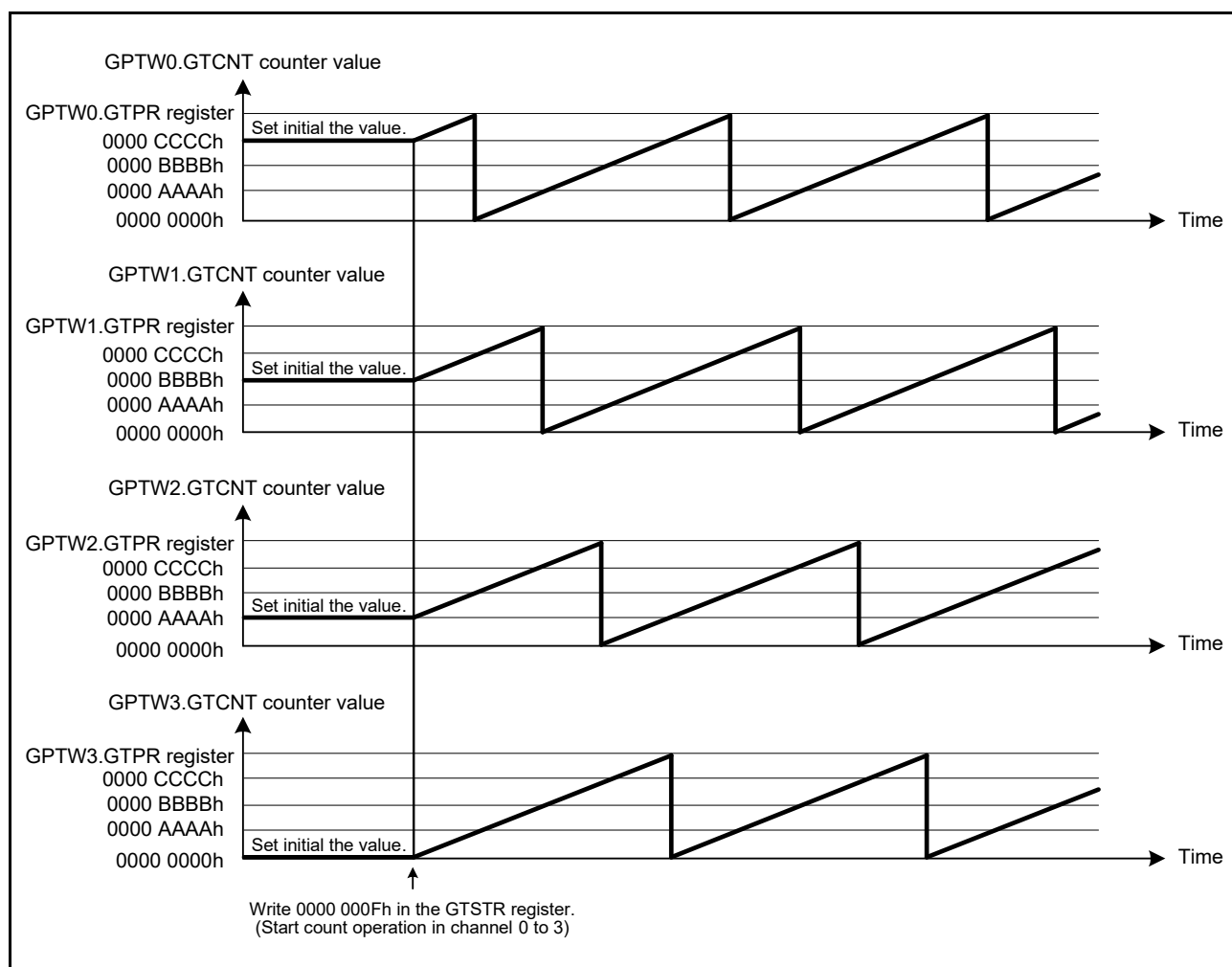


Figure 26.78 Example of Software Phase Shift Start (with Same Count Period (GTPR Register Value))

26.3.8.2 Synchronous Operation by Hardware Source

The count operations for respective channels can be started/stopped/cleared simultaneously by following hardware sources: external trigger input, and ELC event input.

Synchronous operation by the GTIOCnA and GTIOCnB pin inputs ($n = 0$ to 3) can be performed by setting the ELC event by input capture as a hardware source.

Figure 26.79 shows an example of simultaneous start/stop/clearing operation of four channels by hardware source.

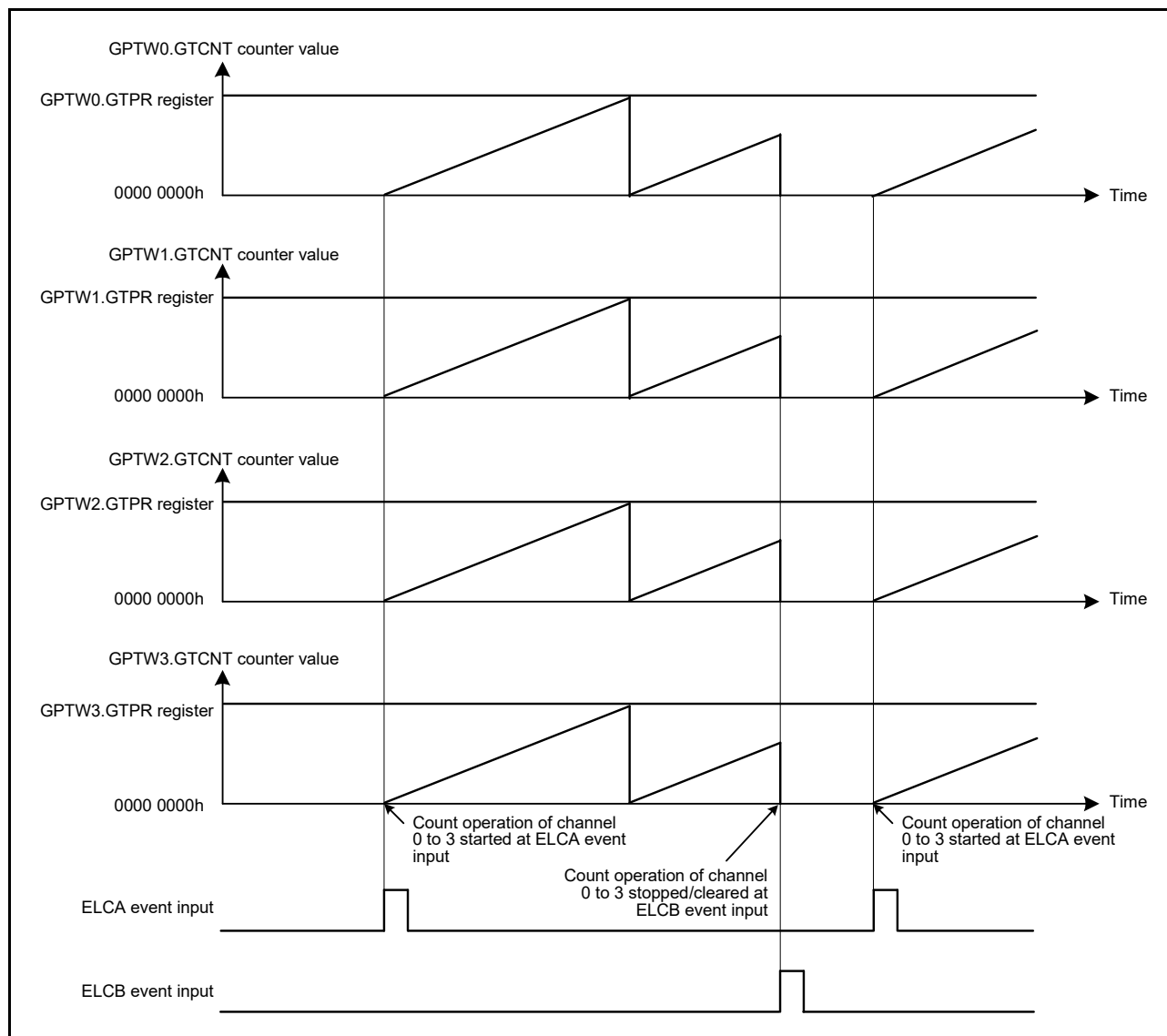


Figure 26.79 Example of Simultaneous Start/Stop/Clearing Operation by Hardware Source (with Same Count Period (GTPR Register Value))

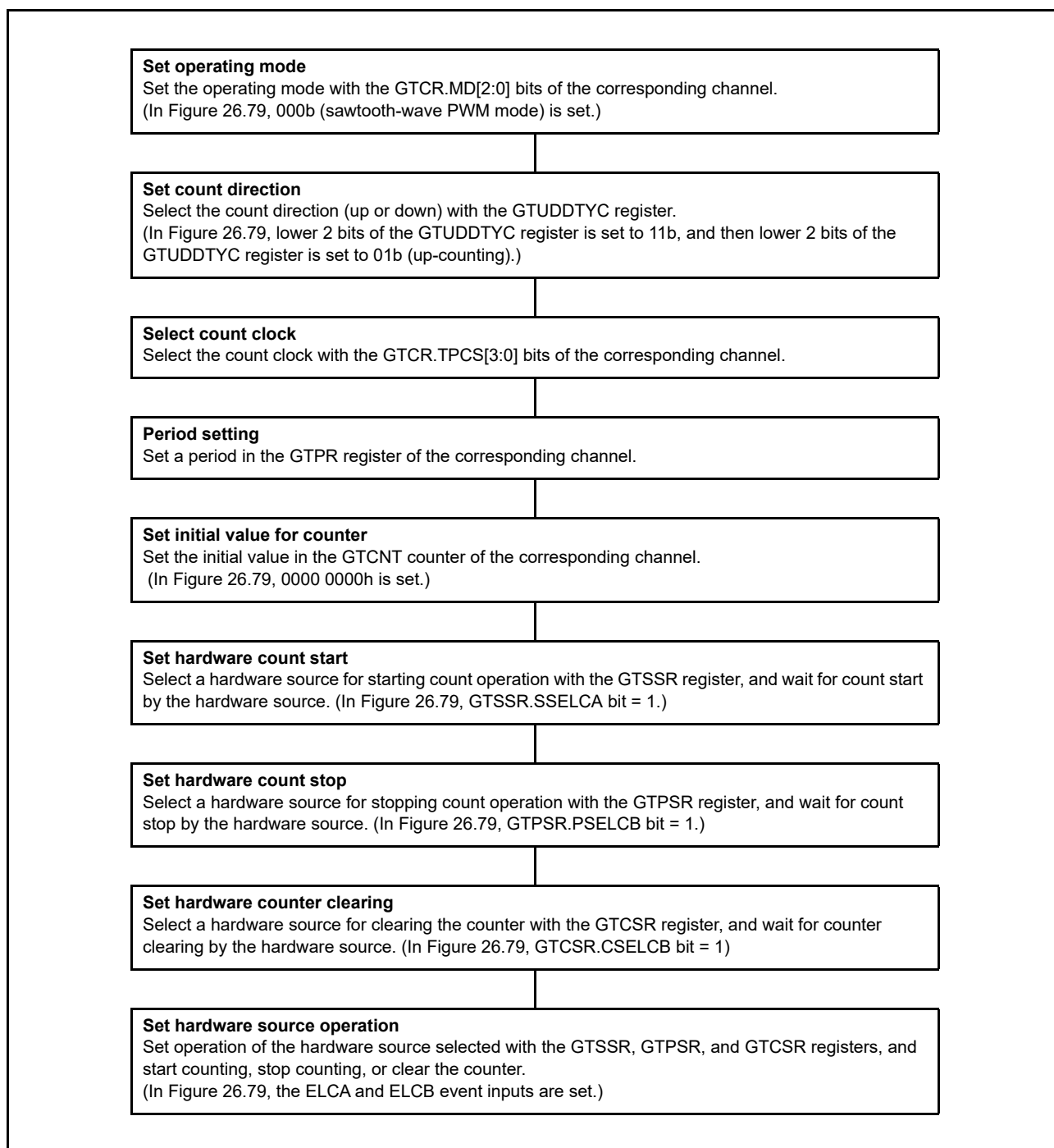


Figure 26.80 Example for Setting Simultaneous Start by Hardware Source

26.3.9 PWM Output Operation Examples

(1) Synchronous PWM Output

The GPTW can output two phases of linked PWM waveforms for one channel, or 8 phases of linked PWM waveforms for a maximum of 4 channels by synchronizing operation of the channels.

Figure 26.81 shows an example in which four channels perform synchronous operation in sawtooth-wave PWM mode and eight phases of PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 3) is set so that it will output low as the initial output, high at a GTCCRA register compare match, and low at the end of the cycle. The GTIOCnB pin is set so that it will output low as the initial output, high at a GTCCRB register compare match, and low at the end of the cycle.

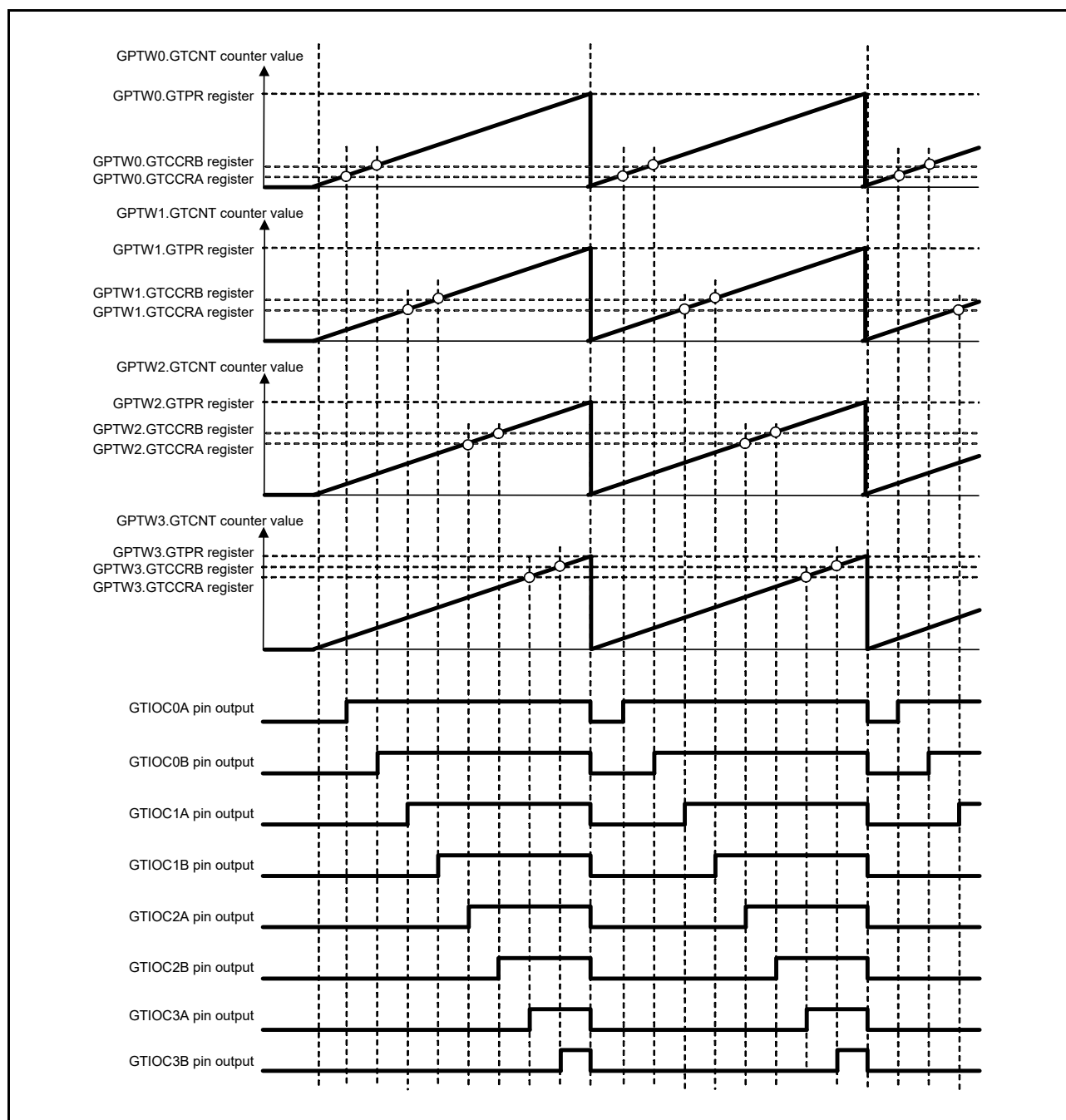


Figure 26.81 Example of Synchronous PWM Output

(2) Three-Phase Sawtooth-Wave Complementary PWM Output

Figure 26.82 shows an example in which three channels perform synchronous operation in sawtooth-wave PWM mode and three-phase complementary PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 2) is set so that it will output low as the initial output, high at a GTCCRA register compare match, and low at the end of the cycle. The GTIOCnB pin is set so that it will output high as the initial output, low at a GTCCRB register compare match, and high at the end of the cycle.

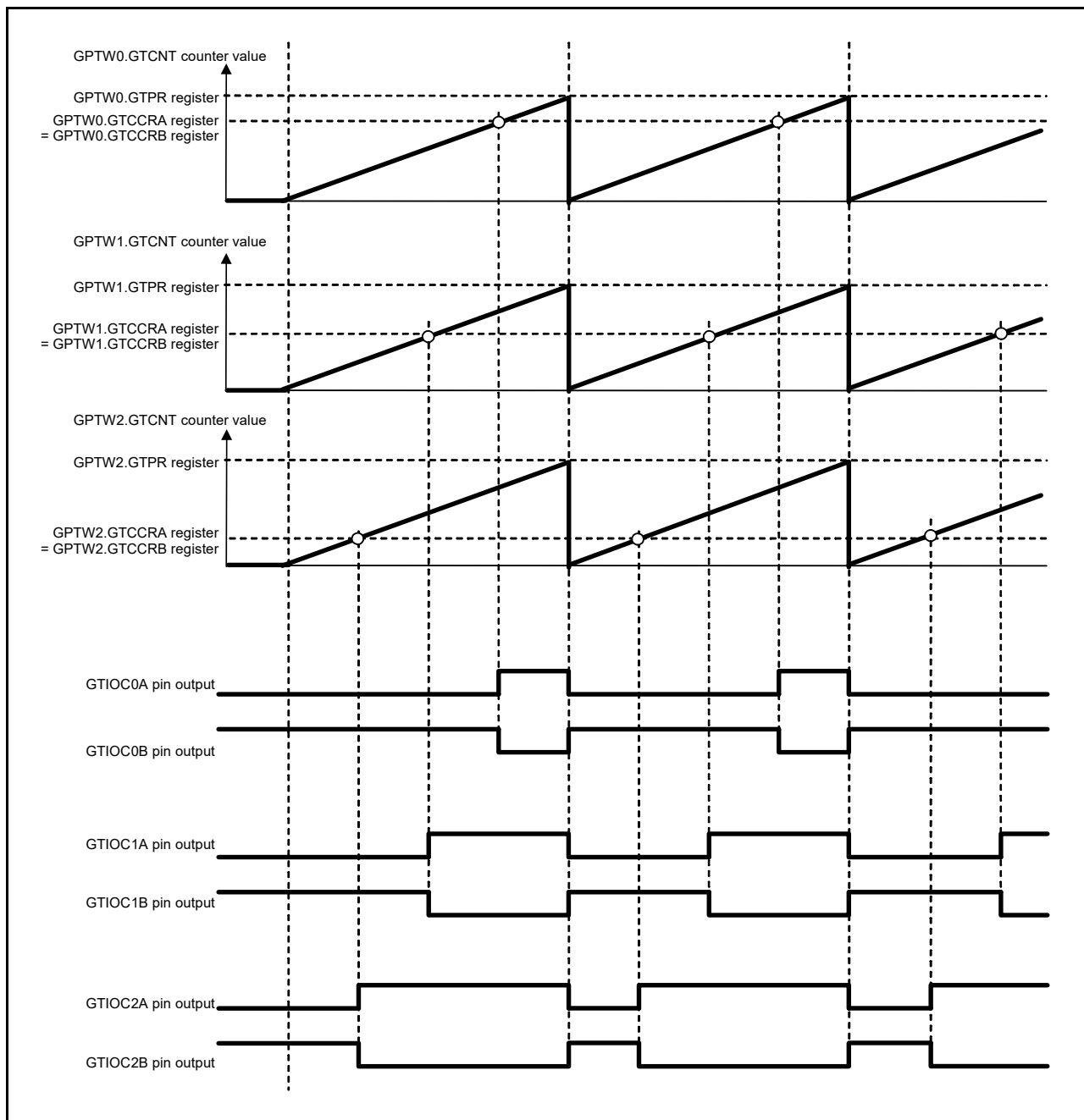


Figure 26.82 Example of Three-Phase Sawtooth-Wave Complementary PWM Output

(3) Three-Phase Sawtooth-Wave Complementary PWM Output with Automatic Dead Time Setting

Figure 26.83 shows an example in which three channels perform synchronous operation in sawtooth-wave one-shot pulse mode with automatic dead time setting and three-phase complementary PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 2) is set so that it will output low as the initial output, toggle the output at a GTCCRA register compare match, and retain the output at the end of the cycle. The GTIOCnB pin is set so that it will output high as the initial output, toggle the output at a GTCCRB register compare match, and retain the output at the end of the cycle.

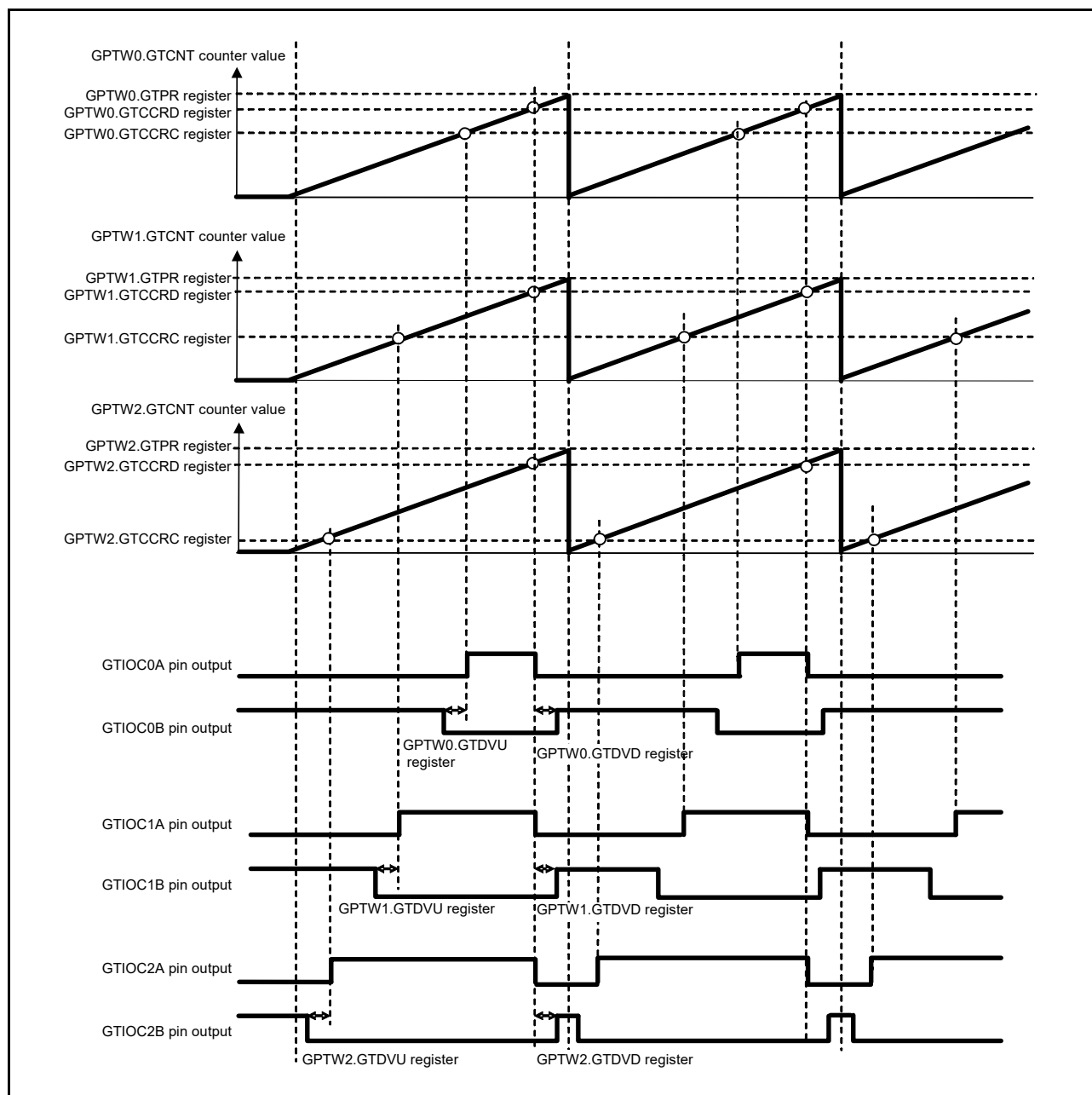


Figure 26.83 Example of Three-Phase Sawtooth-Wave Complementary PWM Output with Automatic Dead Time Setting

(4) Three-Phase Triangle-Wave Complementary PWM Output

Figure 26.84 shows an example in which three channels perform synchronous operation in triangle-wave PWM mode 1 and three-phase complementary PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 2) is set so that it will output low as the initial output, toggle the output at a GTCCRA register compare match, and retain the output at the end of the cycle. The GTIOCnB pin is set so that it will output high as the initial output, toggle the output at a GTCCRB register compare match, and retain the output at the end of the cycle.

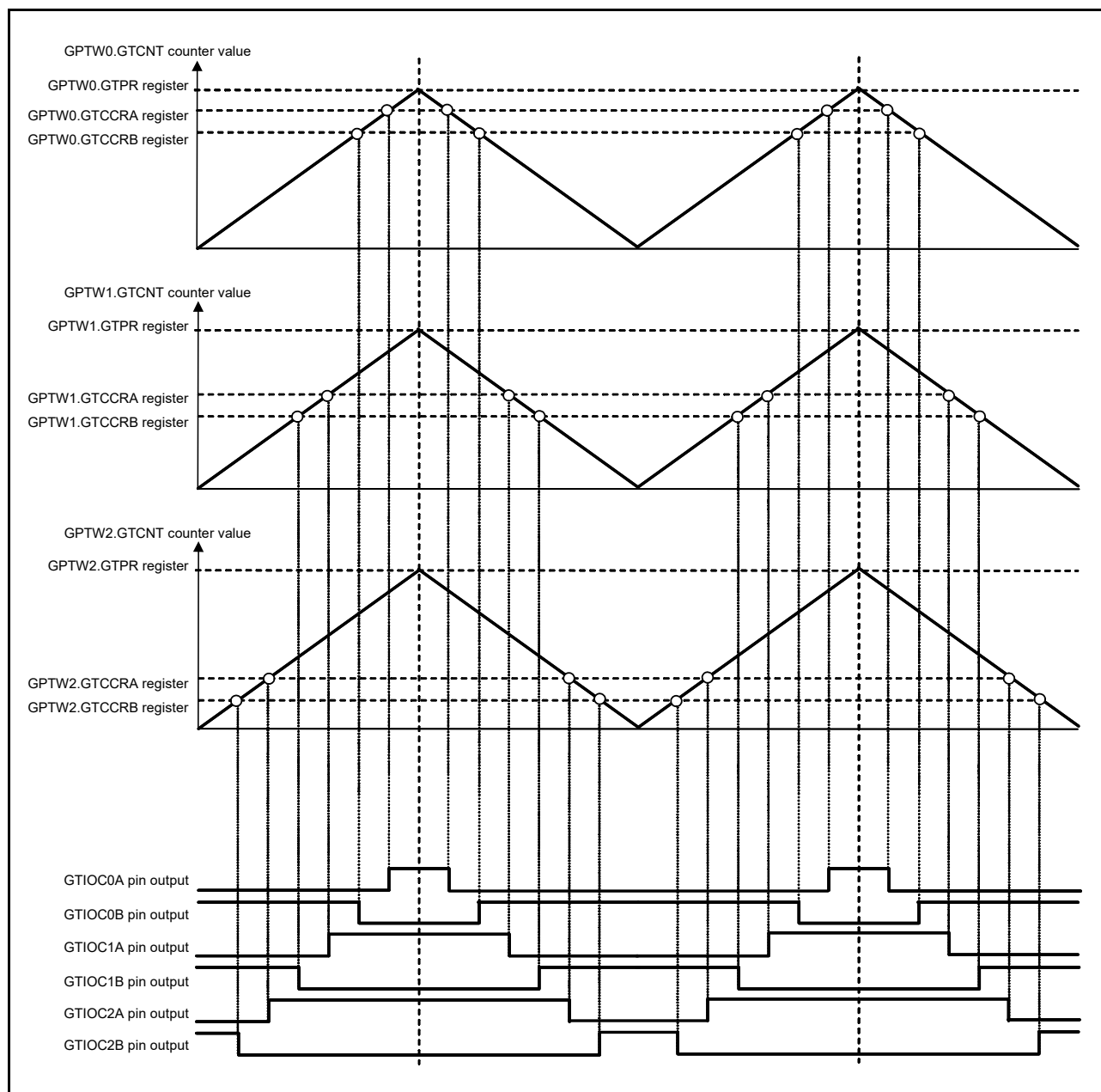


Figure 26.84 Example of Three-Phase Triangle-Wave Complementary PWM Output

(5) Three-Phase Triangle-Wave Complementary PWM Output with Automatic Dead Time Setting

Figure 26.85 shows an example in which three channels perform synchronous operation in triangle-wave PWM mode 1 with automatic dead time setting and three-phase complementary PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 2) is set so that it will output low as the initial output, toggle the output at a GTCCRA register compare match, and retain the output at the end of the cycle. The GTIOCnB pin is set so that it will output high as the initial output, toggle the output at a GTCCRB register compare match, and retain the output at the end of the cycle.

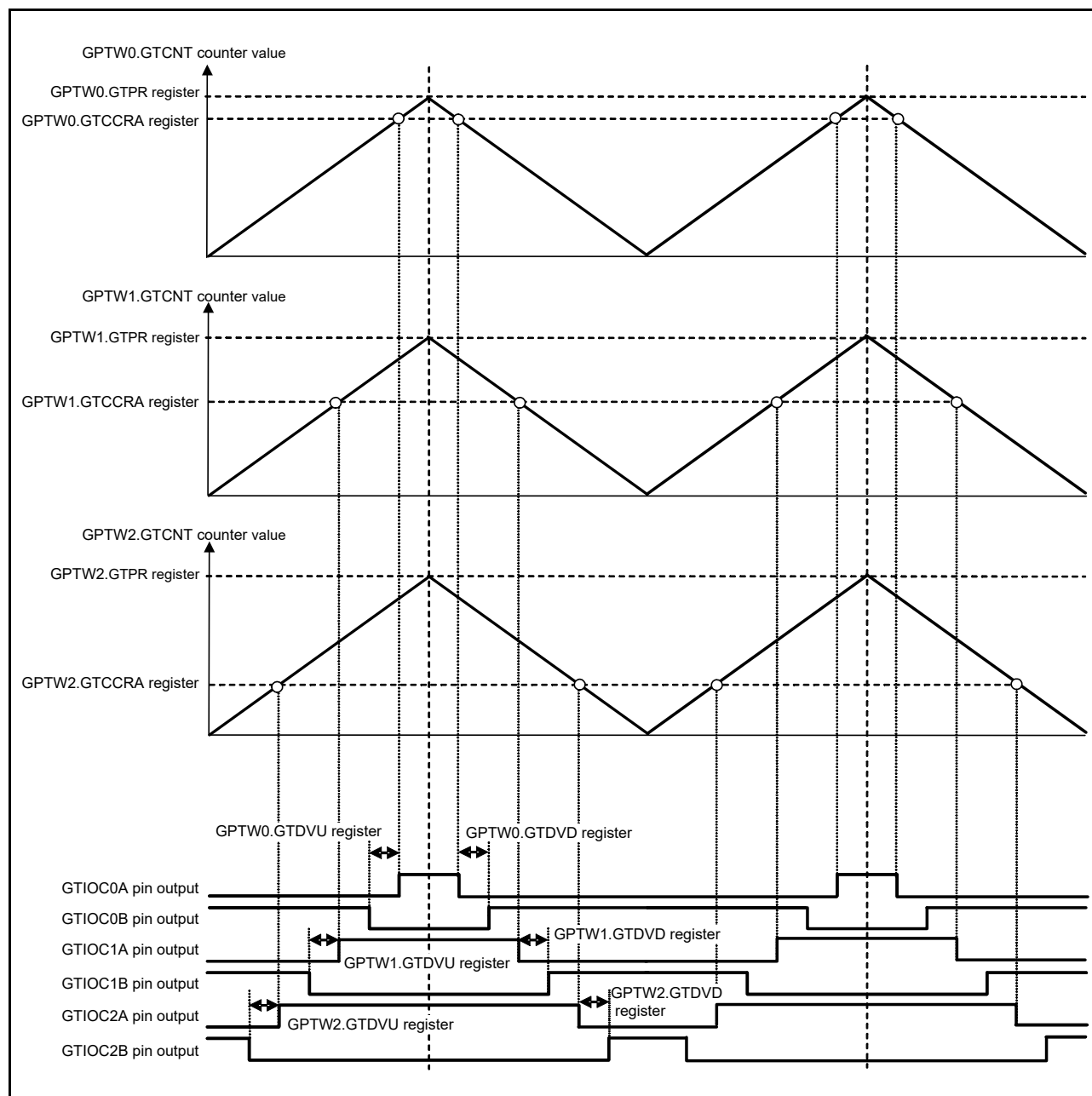


Figure 26.85 Example of Three-Phase Triangle-Wave Complementary PWM Output with Automatic Dead Time Setting

(6) Three-Phase Asymmetric Triangle-Wave Complementary PWM Output with Automatic Dead Time Setting

Figure 26.86 shows an example in which three channels perform synchronous operation in triangle-wave PWM mode 3 with automatic dead time setting and three-phase complementary PWM waveforms are output. The GTIOCnA pin ($n = 0$ to 2) is set so that it will output low as the initial output, toggle the output at a GTCCRA register compare match, and retain the output at the end of the cycle. The GTIOCnB pin is set so that it will output high as the initial output, toggle the output at a GTCCRB register compare match, and retain the output at the end of the cycle.

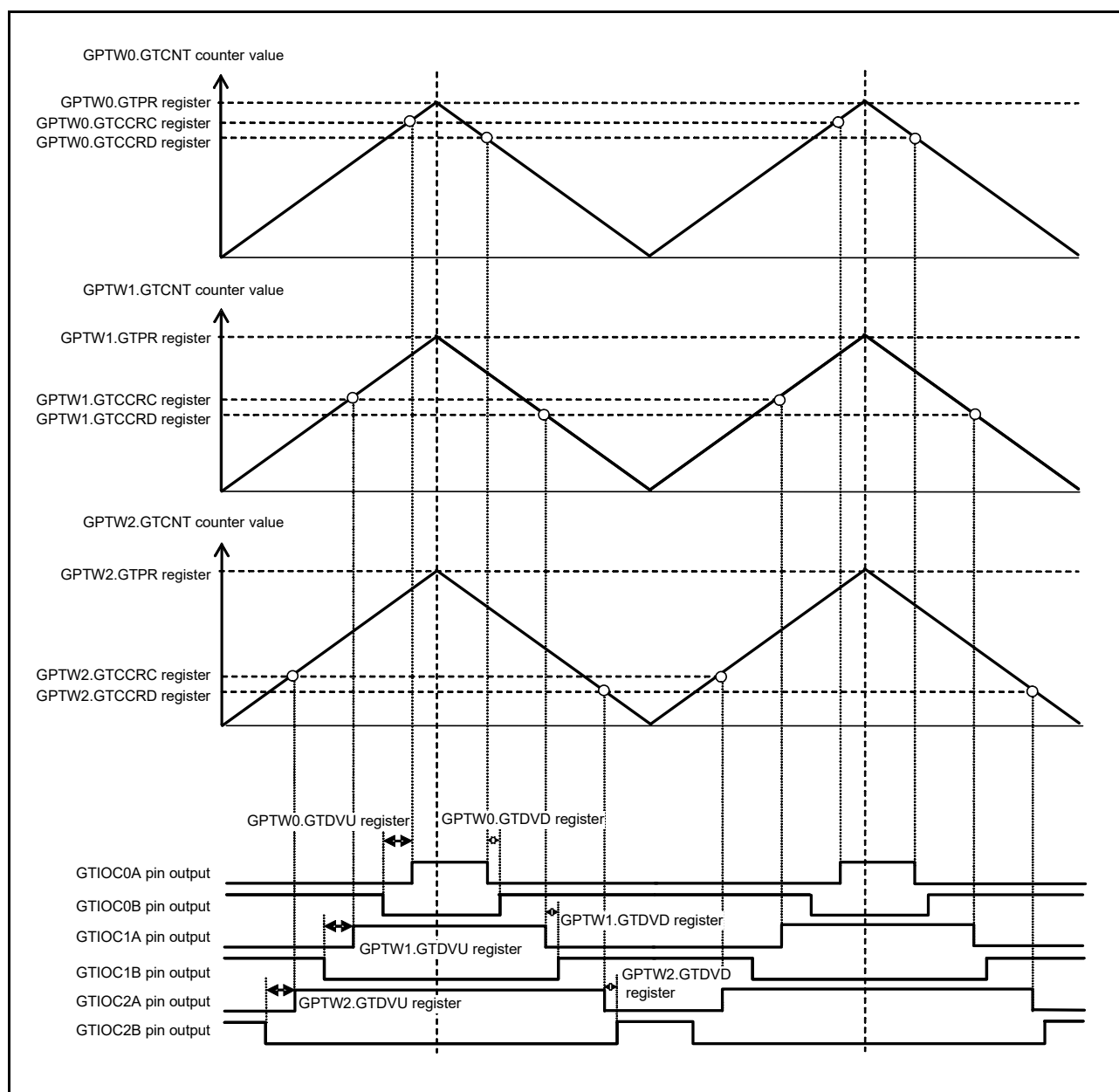


Figure 26.86 Example of Three-Phase Asymmetric Triangle-Wave Complementary PWM Output with Automatic Dead Time Setting

26.3.10 Phase Counting Mode

The GTCNT counter can be up-counting or down-counting by detecting the phase difference between the GTIOCnA pin input (n = 0 to 3) and the GTIOCnB pin input.

Detection of the phase difference can be performed by setting the relationship between the edge and the level of the GTIOCnA pin input and the GTIOCnB pin input using the GTUPSR and GTDNSR registers for input combination. Refer to section 26.3.1.1, Counter Operation for count operation.

Figure 26.87 to Figure 26.96 show the examples of setting procedure for the phase counting mode 1 to 5, and the Table 26.10 to Table 26.19 show the setting of the GTUPSR and GTDNSR registers and up-counting and down-counting conditions.

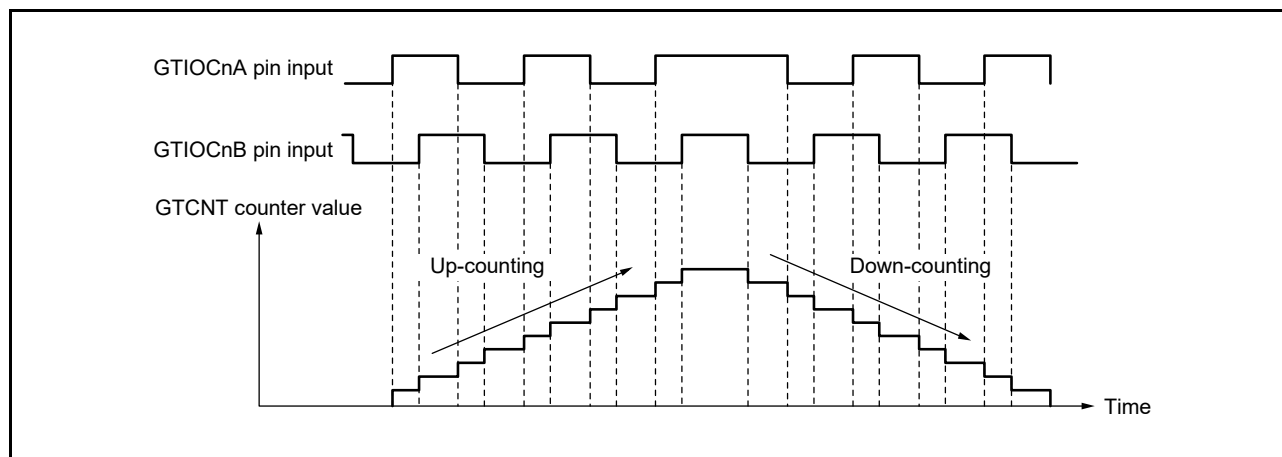


Figure 26.87 Example of setting procedure for the phase counting mode 1 (n = 0 to 3)

Table 26.10 Up-counting and down-counting conditions for the phase counting mode 1 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Up-counting	GTUPSR = 0000 6900h GTDNSR = 0000 9600h
Low			
	Low		
	High		
High		Down-counting	
Low			
	High		
	Low		

: Rising edge

: Falling edge

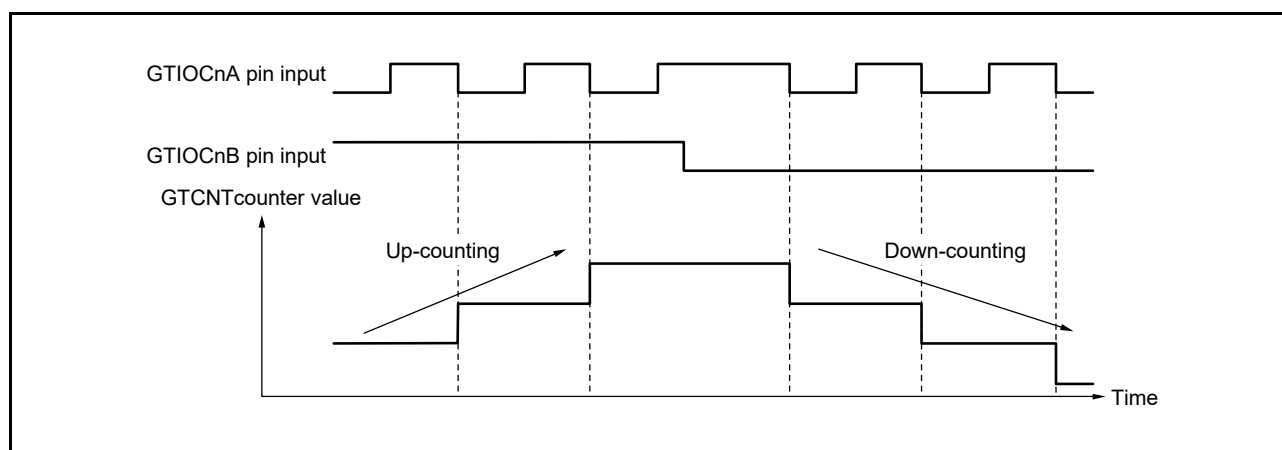


Figure 26.88 Example of setting procedure for the phase counting mode 2 (n = 0 to 3)

Table 26.11 Up-counting and down-counting conditions for the phase counting mode 2 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 0800h GTDNSR = 0000 0400h
Low			
	Low		
	High	Up-counting	
High		Don't care	
Low			
	High		
	Low	Down-counting	

: Rising edge

: Falling edge

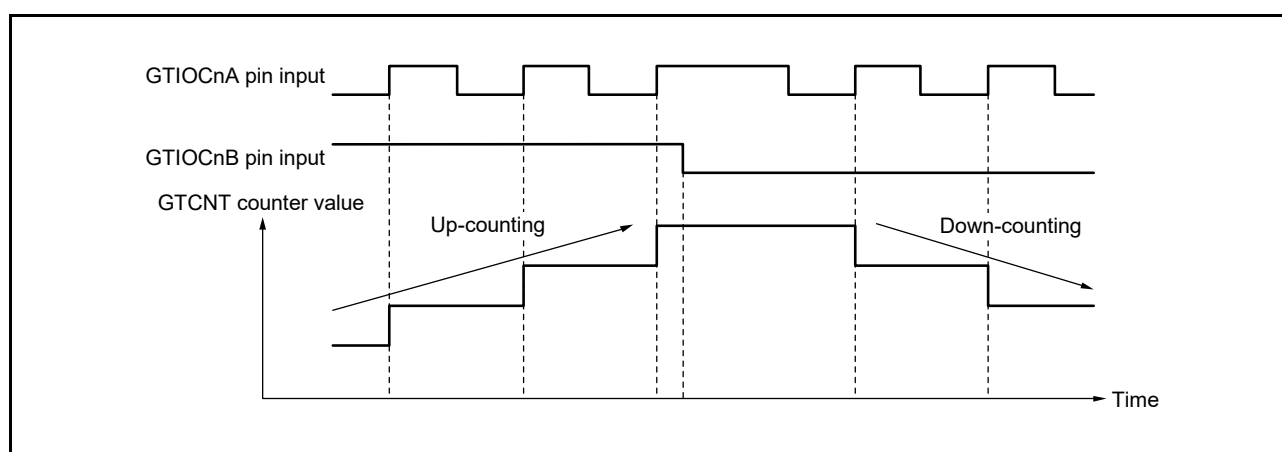
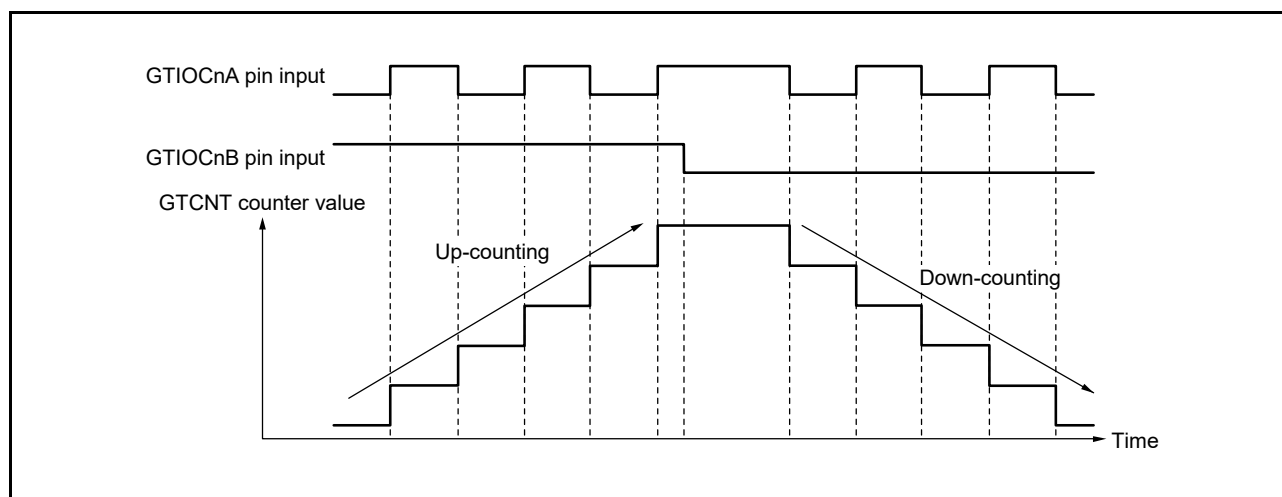


Figure 26.89 Example of setting procedure for the phase counting mode 2 (n = 0 to 3)

Table 26.12 Up-counting and down-counting conditions for the phase counting mode 2 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 0200h GTDNSR = 0000 0100h
Low			
	Low	Down-counting	
	High	Don't care	
High			
Low		Up-counting	
	High		
	Low	Don't care	

 : Rising edge

 : Falling edge
**Figure 26.90 Example of setting procedure for the phase counting mode 2 (n = 0 to 3)****Table 26.13 Up-counting and down-counting conditions for the phase counting mode 2 (n = 0 to 3)**

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 0A00h GTDNSR = 0000 0500h
Low			
	Low	Down-counting	
	High	Up-counting	
High		Don't care	
Low			
	High	Up-counting	
	Low	Down-counting	

 : Rising edge

 : Falling edge

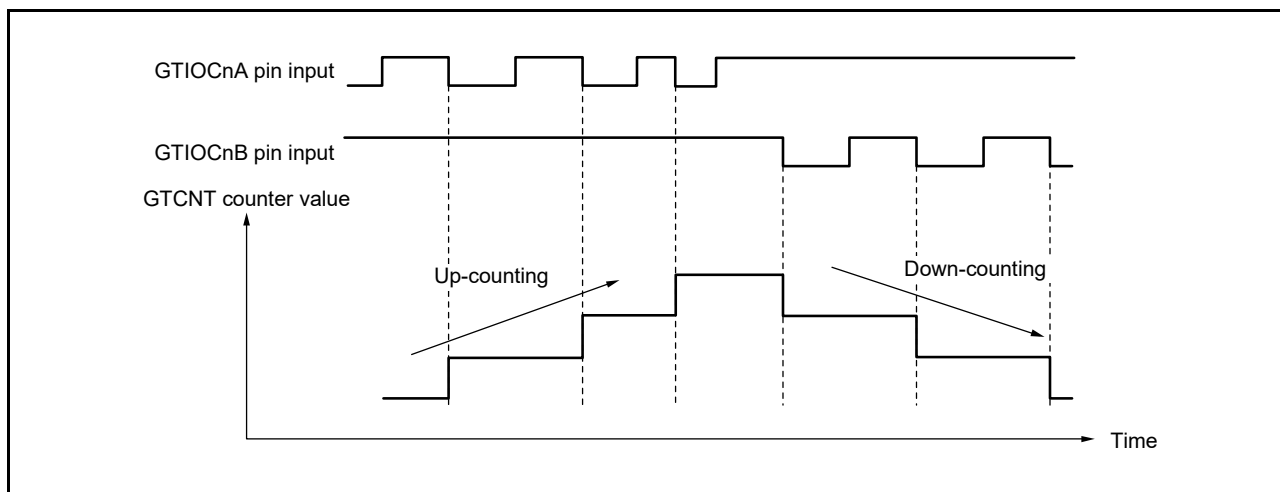


Figure 26.91 Example of setting procedure for the phase counting mode 3 (n = 0 to 3)

Table 26.14 Up-counting and down-counting conditions for the phase counting mode 3 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 0800h GTDNSR = 0000 8000h
Low			
	Low		
	High	Up-counting	
High		Down-counting	
Low		Don't care	
	High		
	Low		

: Rising edge

: Falling edge

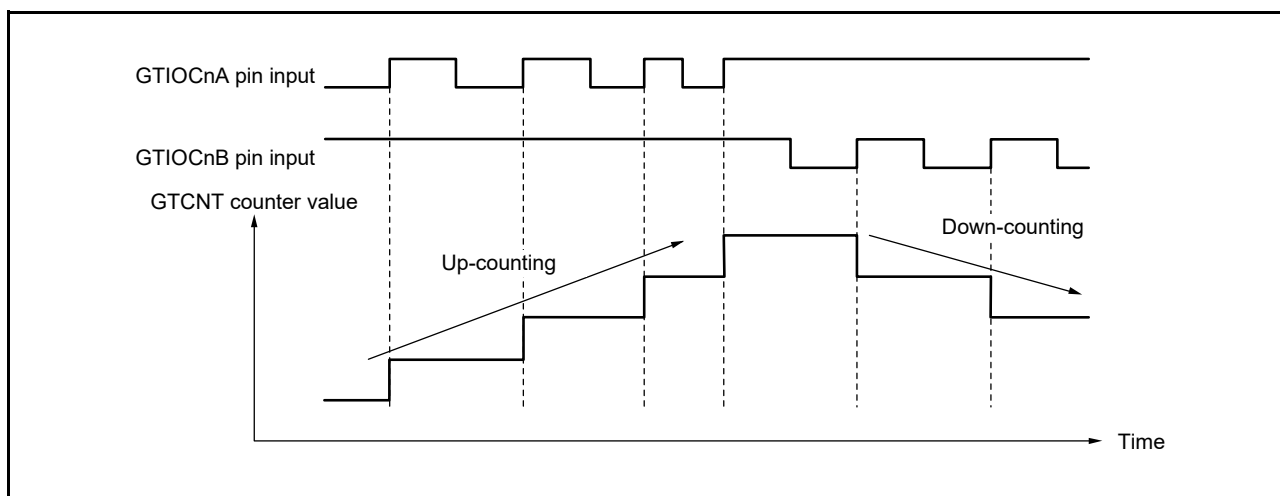


Figure 26.92 Example of setting procedure for the phase counting mode 3 (n = 0 to 3)

Table 26.15 Up-counting and down-counting conditions for the phase counting mode 3 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Down-counting	GTUPSR = 0000 0200h GTDNSR = 0000 2000h
Low		Don't care	
	Low		
	High		
High			
Low			
	High	Up-counting	
	Low	Don't care	

 : Rising edge

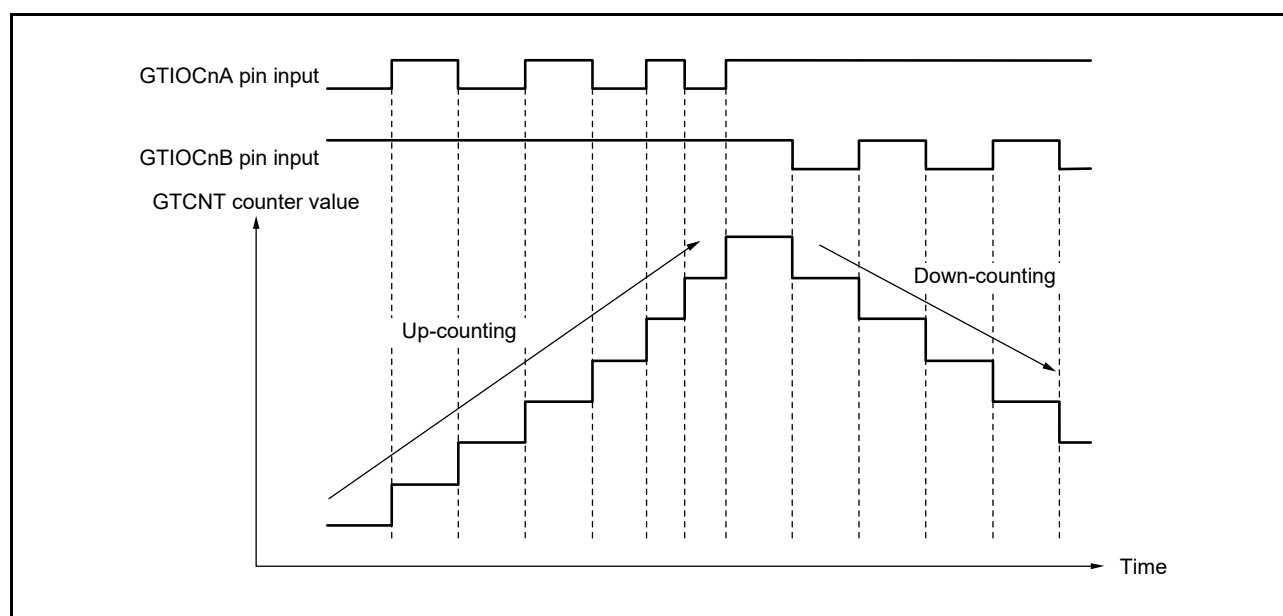
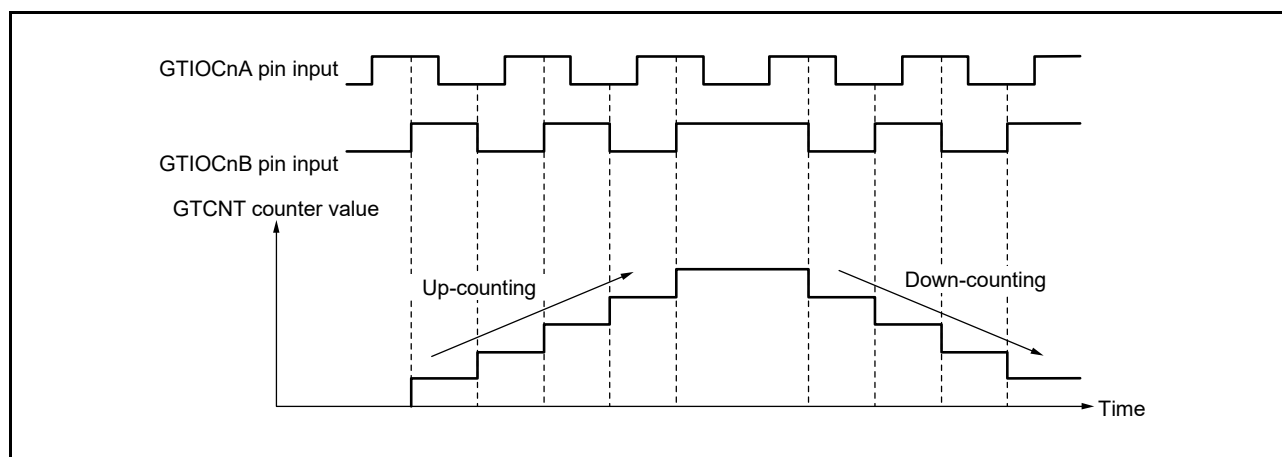
 : Falling edge
**Figure 26.93 Example of setting procedure for the phase counting mode 3 (n = 0 to 3)**

Table 26.16 Up-counting and down-counting conditions for the phase counting mode 3 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Down-counting	GTUPSR = 0000 0A00h GTDNSR = 0000 A000h
Low		Don't care	
	Low		
	High	Up-counting	
High		Down-counting	
Low		Don't care	
	High	Up-counting	
	Low	Don't care	

 : Rising edge

 : Falling edge
**Figure 26.94 Example of setting procedure for the phase counting mode 4 (n = 0 to 3)****Table 26.17 Up-counting and down-counting conditions for the phase counting mode 4 (n = 0 to 3)**

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Up-counting	GTUPSR = 0000 6000h GTDNSR = 0000 9000h
Low			
	Low	Don't care	
	High		
High		Down-counting	
Low			
	High	Don't care	
	Low		

 : Rising edge

 : Falling edge

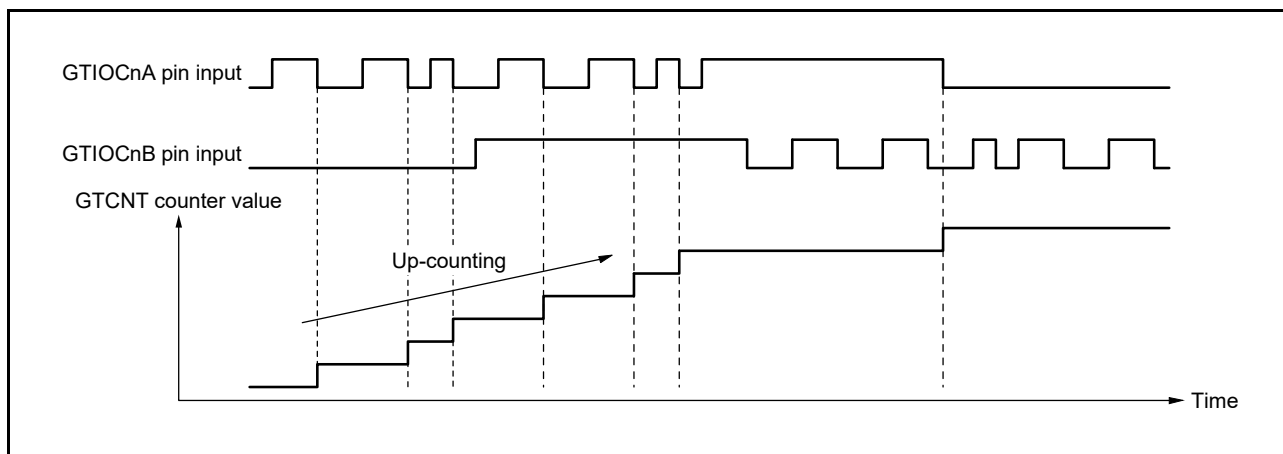


Figure 26.95 Example of setting procedure for the phase counting mode 5 ($n = 0$ to 3)

Table 26.18 Up-counting and down-counting conditions for the phase counting mode 5 ($n = 0$ to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 0C00h GTDNSR = 0000 0000h
Low			
	Low		
	High	Up-counting	
High		Don't care	
Low			
	High		
	Low	Up-counting	

: Rising edge

: Falling edge

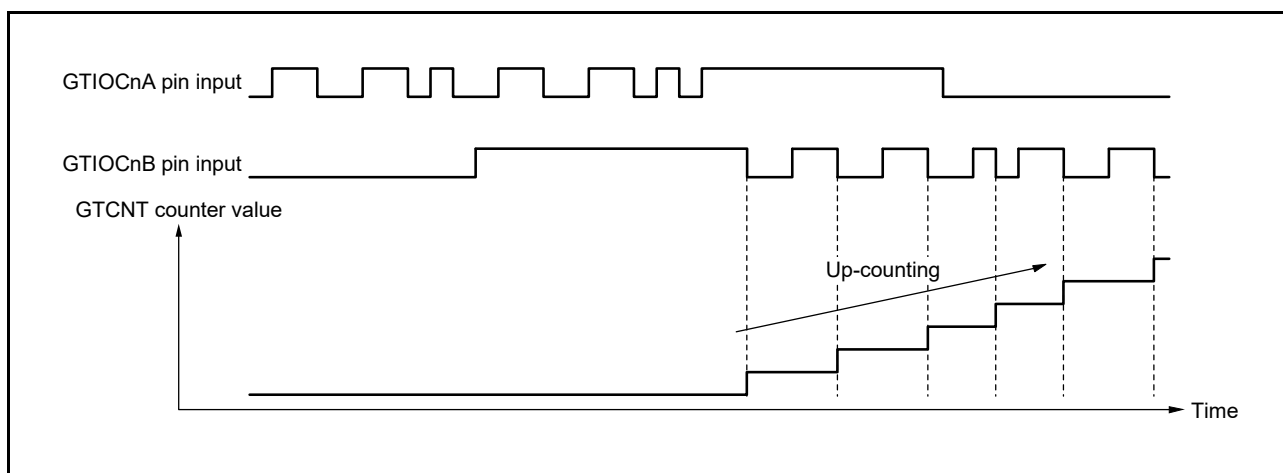


Figure 26.96 Example of setting procedure for the phase counting mode 5 ($n = 0$ to 3)

Table 26.19 Up-counting and down-counting conditions for the phase counting mode 5 (n = 0 to 3)

GTIOCnA pin input	GTIOCnB pin input	Operation	Setting of register
High		Don't care	GTUPSR = 0000 C000h GTDNSR = 0000 0000h
Low		Up-counting	
	Low	Don't care	
	High		
High		Up-counting	
Low		Don't care	
	High		
	Low		

: Rising edge

: Falling edge

26.4 Interrupt Sources

26.4.1 Interrupt Sources and Priorities

There are three kinds of interrupt sources; GTCCRm register (m = A to F) input capture/compare match, GTCNT counter overflow (GTPR register compare match)/underflow, and dead time error.

Each interrupt source has its own status flag and control bit for generating interrupt request signals, where the generation of interrupt request signals can be enabled or disabled individually.

When an interrupt source condition is satisfied, an interrupt request is generated if the corresponding interrupt request enable or disable bit is 1.

For details, refer to section 15, Interrupt Controller (ICUD). Table 26.20 shows a list of GPTW interrupt sources.

Table 26.20 GPTW Interrupt Sources (n = 0 to 3)

Channel	Name	Interrupt Source
GPTWn	GTCIA _n	GTCCRA register input capture/compare match
	GTCIB _n	GTCCRB register input capture/compare match
	GTCIC _n	GTCCRC register compare match
	GTCID _n	GTCCRD register compare match
	GDTEn	Dead time error
	GTCIE _n	GTCCRE register compare match
	GTCIF _n	GTCCRF register compare match
	GTCIV _n	GTCNT counter overflow (GTPR register compare match)
	GTCIU _n	GTCNT counter underflow

(1) GTCIA_n Interrupt (n = 0 to 3)

When the GTINTAD.GTINTA bit is 1, an interrupt request is generated under the following conditions.

- When the GTCCRA register functions as a compare match register, the GTCNT counter value matches with the GTCCRA register.
- When the GTCCRA register functions as an input capture register, the input-capture signal has caused transfer of the GTCNT counter value to the GTCCRA register.

(2) GTCIB_n Interrupt (n = 0 to 3)

When the GTINTAD.GTINTB bit is 1, an interrupt request is generated under the following conditions.

- When the GTCCRB register functions as a compare match register, the GTCNT counter value matches with the GTCCRB register.
- When the GTCCRB register functions as an input capture register, the input-capture signal has caused transfer of the GTCNT counter value to the GTCCRB register.

(3) GTCIC_n Interrupt (n = 0 to 3)

When the GTINTAD.GTINTC bit is 1, an interrupt request is generated under the following condition.

- When the GTCCRC register functions as a compare match register, the GTCNT counter value matches with the GTCCRC register.

A compare match is not performed and thus interrupt is not requested under the following conditions.

- GTCR.MD[2:0] bits = 001b (sawtooth-wave one-shot pulse mode)
- GTCR.MD[2:0] bits = 110b (triangle-wave PWM mode 3)
- GTBER.CCRA[1:0] bits = 01b, 10b, 11b (buffer operation with the GTCCRC register)

(4) GTCIDn Interrupt (n = 0 to 3)

When the GTINTAD.GTINTD bit is 1, an interrupt request is generated under the following condition.

- When the GTCCRD register functions as a compare match register, the GTCNT counter value matches with the GTCCRD register.

A compare match is not performed and thus interrupt is not requested under the following conditions.

- GTCR.MD[2:0] bits = 001b (sawtooth-wave one-shot pulse mode)
- GTCR.MD[2:0] bits = 110b (triangle-wave PWM mode 3)
- GTBER.CCRA[1:0] bits = 10b, 11b (buffer operation with the GTCCRD register)

(5) GTCIE n Interrupt (n = 0 to 3)

When the GTINTAD.GTINTE bit is 1, an interrupt request is generated under the following condition.

- When the GTCCRE register functions as a compare match register, the GTCNT counter value matches with the GTCCRE register.

A compare match is not performed and thus interrupt is not requested under the following conditions.

- GTCR.MD[2:0] bits = 001b (sawtooth-wave one-shot pulse mode)
- GTCR.MD[2:0] bits = 110b (triangle-wave PWM mode 3)
- GTBER.CCRB[1:0] bits = 01b, 10b, 11b (buffer operation with the GTCCRE register)

(6) GTCIFn Interrupt (n = 0 to 3)

When the GTINTAD.GTINTF bit is 1, an interrupt request is generated under the following condition.

- When the GTCCRF register functions as a compare match register, the GTCNT counter value matches with the GTCCRF register.

A compare match is not performed and thus interrupt is not requested under the following conditions.

- GTCR.MD[2:0] bits = 001b (sawtooth-wave one-shot pulse mode)
- GTCR.MD[2:0] bits = 110b (triangle-wave PWM mode 3)
- GTBER.CCRB[1:0] bits = 10b, 11b (buffer operation with the GTCCRF register)

(7) GTCIVn Interrupt (n = 0 to 3)

When the GTINTAD.GTINTPR[0] bit is 1, an interrupt request is generated under the following conditions.

- In sawtooth-wave mode, interrupt requests are enabled at overflows (the GTCNT counter value changes from the GTPR register value to 0 during up-counting).
- In triangle-wave mode, interrupt requests are enabled at crests (the GTCNT counter value changes from the GTPR register value to the GTPR register value minus 1).
- In counting driven by a hardware source, interrupt requests are enabled on overflows (that is, when the GTCNT counter value changes from the GTPR register value to 0 during up-counting).

(8) GTCIU n Interrupt (n = 0 to 3)

When the GTINTAD.GTINTPR[1] bit is 1, an interrupt request is generated under the following conditions.

- In sawtooth-wave mode, interrupt requests are enabled at underflows (the GTCNT counter value changes from 0 to the GTPR register value during down-counting).
- In triangle-wave mode, interrupt requests are enabled at troughs (the GTCNT counter value changes from 0 to 1).
- In count operation by a hardware source, interrupt requests are enabled at underflows (the GTCNT counter value

changes from 0 to the GTPR register value during down-counting).

(9) GDTE_n Interrupt (n = 0 to 3)

When automatic dead time setting has been made, the GTST.DTEF flag becomes 1 when the timer output toggle point with dead time added exceeds the count period. If the GTINTAD.GRPDTE bit is 1 at this time, a dead time error interrupt request (GDTE) is generated.

In addition, when the timer output toggle point with dead time added is back within the count period, the GTST.DTEF flag changes from 1 to 0.

Table 26.21 Relationship between Interrupt Signals and Interrupt Enable Bits (n = 0 to 3)

Interrupt Signal	Interrupt Enable Bit
GTCIA _n	GTINTAD.GTINTA bit
GTCIB _n	GTINTAD.GTINTB bit
GTCIC _n	GTINTAD.GTINTC bit
GTCID _n	GTINTAD.GTINTD bit
GDTE _n	GTINTAD.GRPDTE bit
GTCIE _n	GTINTAD.GTINTE bit
GTCIF _n	GTINTAD.GTINTF bit
GTCIV _n	GTINTAD.GTINTPR[1:0] bits
GTCIU _n	

26.4.2 DMAC/DTC Activation

The DMAC and DTC can be triggered by the interrupt request in each channel. For details, see section 15, Interrupt Controller (ICUD), section 18, DMA Controller (DMACa), section 20, Data Transfer Controller (DTCb).

26.4.3 Interrupt and A/D Conversion Start Request Skipping Function

26.4.3.1 Interrupt Skipping Function by GTITC Register

By setting the GTITC register, the GTCNT counter overflow (GTPR register compare match) interrupt (GTCIV) and underflow interrupt (GTCIU) can be skipped. Other interrupts and A/D conversion start request signals can be skipped in coordination with the GTCIV/GTCIU skipping function. However, the dead time error interrupts cannot be linked with the GTCIV/GTCIU skipping function.

The interrupt skipping function is related only to the GTITC register setting, and is not related to setting of the GTINTAD register interrupt enable bit.

When both troughs and crests are counted and skipped in triangle-wave mode, if the number of times of skipping is odd, GTCIV/GTCIU interrupt requests cannot be generated at troughs only or at crests only depending on the skipping counter start timing. Therefore, in order to count both troughs and crests and generate the GTCIV/GTCIU interrupts at troughs only or crests only in triangle-wave mode, the number of times of skipping should be even.

Similarly, in sawtooth-wave mode, when both overflows and underflows are counted and skipped with the count direction changed, GTCIV/GTCIU interrupt requests cannot be generated at overflows only or at underflows only. Therefore, in order to count both overflows and underflows with the count direction changed and generate the GTCIV/GTCIU interrupts at overflows only or underflows only in sawtooth-wave mode, the skipping state should be carefully checked before using.

When changing the skipping count, be sure to release the skipping count setting (GTITC.IVTC[1:0] bits = 00b).

Figure 26.97 to Figure 26.102 show examples of skipping function operation.

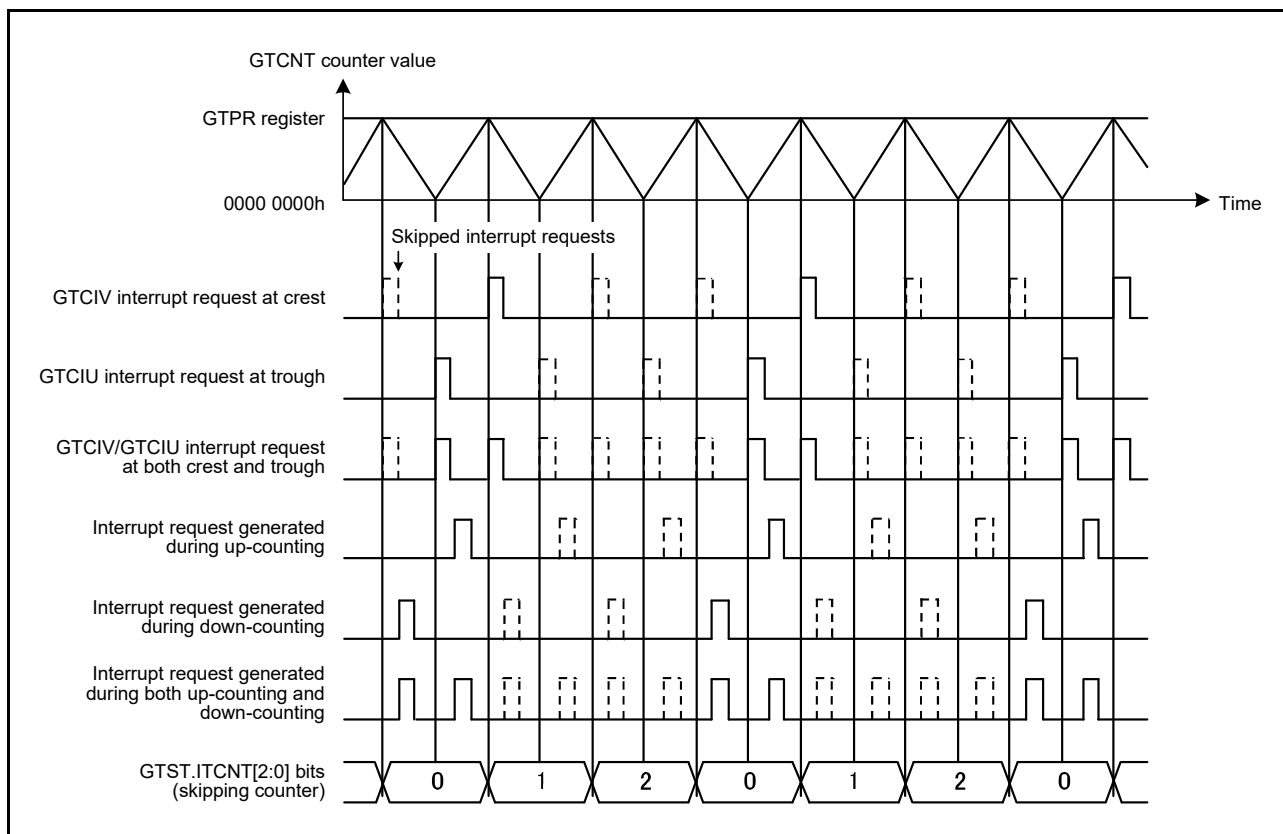


Figure 26.97 Example of Interrupt Skipping Function Operation (Triangle Waves, Counting and Skipping Crests, Skipping Count: 2)

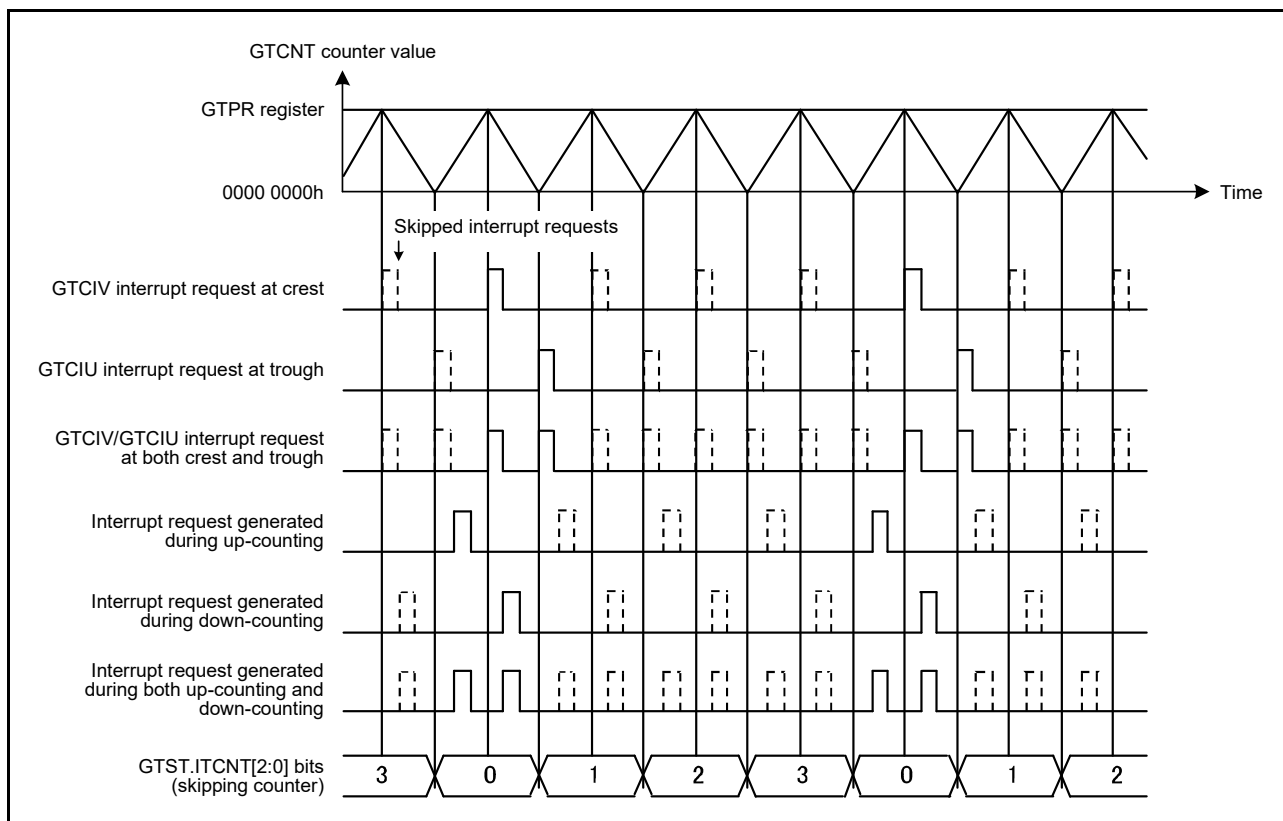


Figure 26.98 Example of Interrupt Skipping Function Operation (Triangle Waves, Counting and Skipping Troughs, Skipping Count: 3)

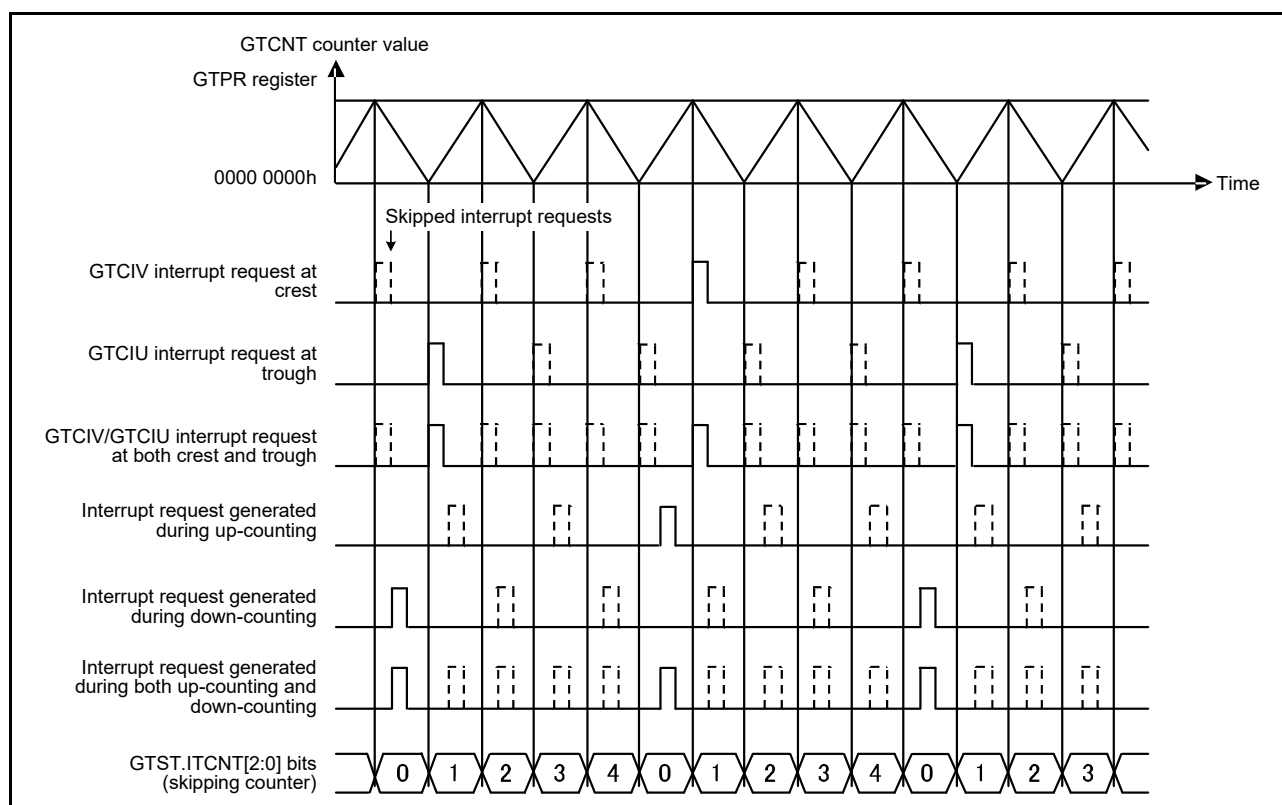


Figure 26.99 Example of Interrupt Skipping Function Operation (Triangle Waves, Counting and Skipping Both Troughs and Crests, Skipping Count: 4)

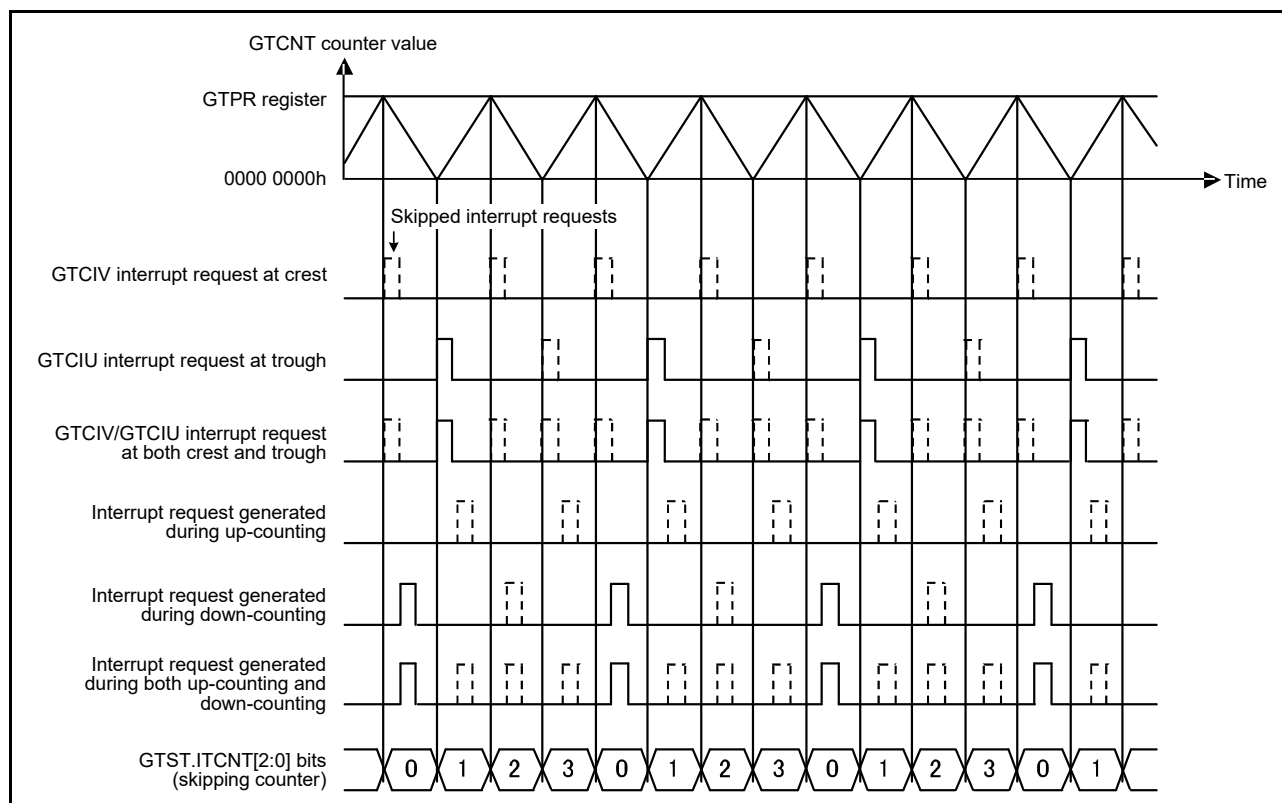


Figure 26.100 Example of Interrupt Skipping Function Operation (Triangle Waves, Counting and Skipping Both Troughs and Crests, Skipping Count: 3, Skipping Started at Up-Counting)

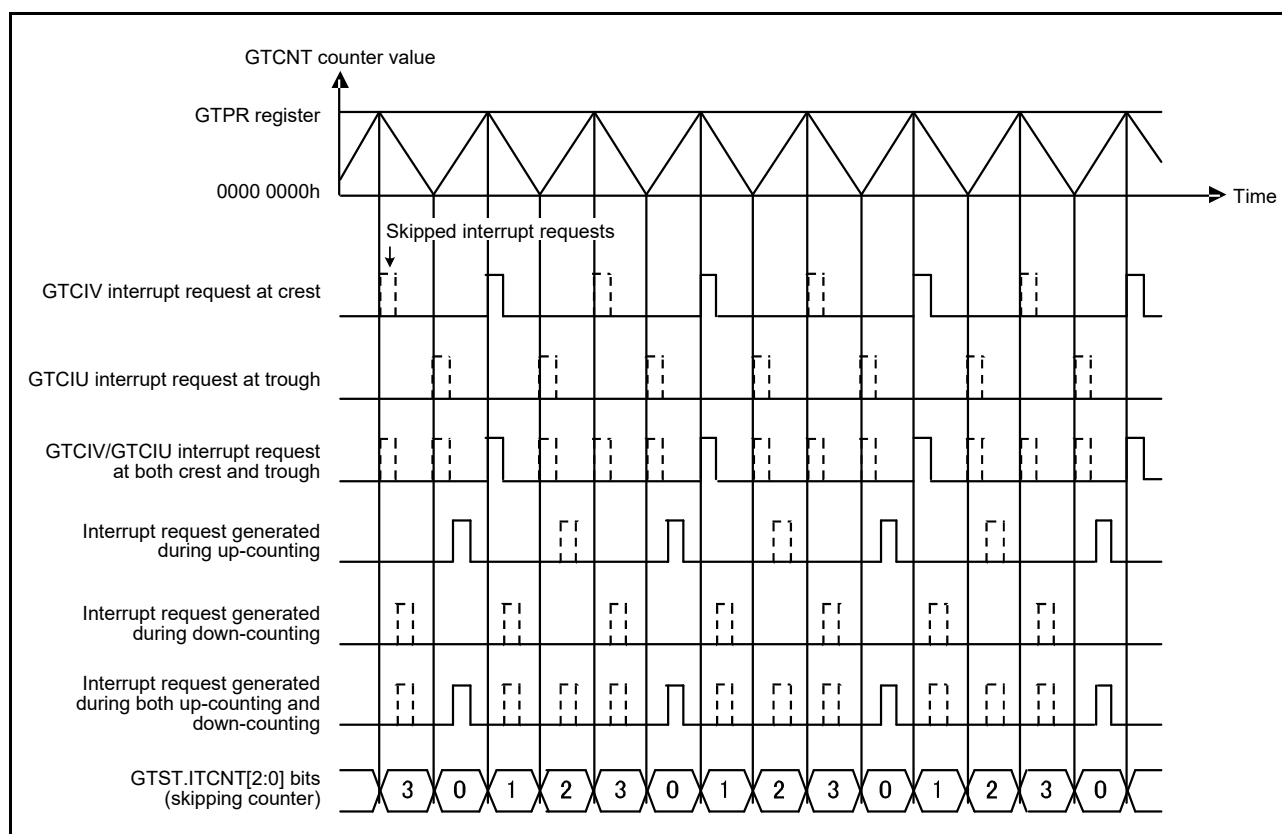


Figure 26.101 Example of Interrupt Skipping Function Operation (Triangle Waves, Counting and Skipping Both Troughs and Crests, Skipping Count: 3, Skipping Started at Down-Counting)

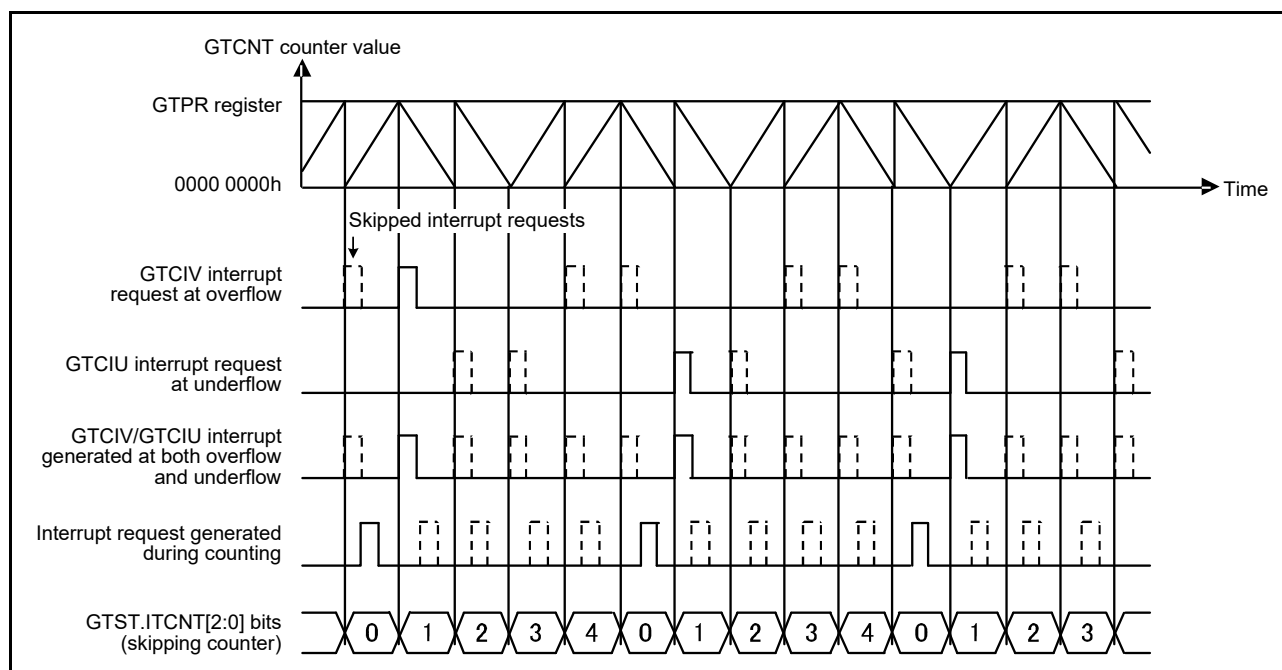


Figure 26.102 Example of Interrupt Skipping Function Operation (Sawtooth Waves, Operation with Count Direction Changed, Counting and Skipping Both Overflows and Underflows, Skipping Count: 4)

26.4.3.2 Extended Interrupt Skipping Function

Overflow/underflow interrupt, compare match/input capture interrupt, A/D conversion start request, and buffer transfer can be skipped by counting the GTCNT counter overflow or underflow based on settings of the GTEITC, GTEITLI1, GTEITLI2 and GTEITLB registers. The dead time error interrupt cannot be skipped.

Overflow/underflow interrupt, compare match/input capture interrupt, A/D conversion start request, and buffer transfer can individually be set for skipping and the skipping period using the GTEITLI1, GTEITLI2, and GTEITLB registers. The skipping period is set, related to the operation of the two independent extended interrupt skipping counters (EITCNT1[3:0] and EITCNT2[3:0] bits in the GTEITC registers), as a period that either of the counter value for such skipping counters is other than 0 or other than the skipping count. The skipping period can also be set as a period that the both skipping counters are set as other than 0 or other than the skipping count for the counter value.

Figure 26.103 shows the counter operation for interrupt skipping by the GTITC register and extended interrupt skipping.

The counter operation for extended interrupt skipping is set by the GTEITC register.

The EITCNT1[3:0] bits count the count source (in Figure 26.103, a crest is selected) selected by the extended interrupt counter 1 count source select bit (EIVTC1[1:0] bits) when the initial value is set for 0, and repeat the count operation which returns to 0 when the skipping count (in Figure 26.103, count is 2) set by the extended interrupt skipping 1 skipping count setting bit (EIVTT1[3:0] bits) is achieved.

The EITCNT2[3:0] bits can set the initial value, count the count source (in **Figure 26.103**, a trough is selected) selected by the extended interrupt counter 2 count source select bit (EIVTC2[1:0] bits), and repeat the count operation which return to 0 when the skipping count (in **Figure 26.103**, count is 2) set by the EIVTT2[3:0] bits is achieved. The initial value is set when the GTEITC register is written by the access of upper 16 bits or 32 bits and when the written value to the EIVTC2[1:0] bits is other than 00b, while the setting for the extended interrupt skipping counter 2 is not in counting (EIVTC2[1:0] bits are 00b).

When the initial value is set, the written value to the extended interrupt skipping counter 2 initial value bit (EITCNT2IV[3:0] bits) is set as the initial value for the EITCNT2[3:0] bits.

The extended interrupt skipping counter starts up-counting at the first count clock after the setting is modified from not counting to counting.

The interrupt skipping count counter (GTST.ITCNT[2:0] bits) by the GTITC register is reset to 000b when the GTCNT counter operation is stopped; however, the EITCNT1[3:0] and EITCNT2[3:0] bits which are an extended interrupt skipping function retain the value even after the GTCNT counter operation is stopped, and the counting can be resumed from the value before the counter is stopped. When values for the EITCNT1[3:0] and EITCNT2[3:0] bits are to be reset (0000b), set the EIVTC1[1:0] and EIVTC2[1:0] bits to the setting for not counting (not skipping) (00b).

When the skipping count is to be changed, change the count after stopping the skipping counter operation (set either of the EIVTC1[1:0] bit or the EIVTC2[1:0] bit to 00b).

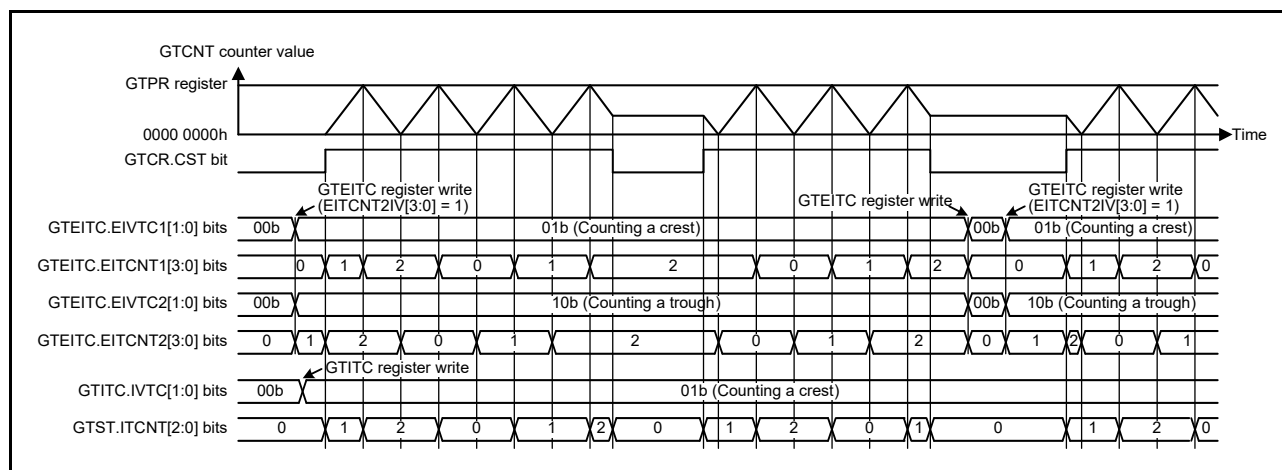


Figure 26.103 Counter Operation for Interrupt Skipping

Interrupt skipping by the GTEITLI1 register and A/D conversion start request skipping by the GTEITLI2 register can be performed simultaneously with interrupt skipping by the GTITC register. A skipping period at this time is represented by ORed skipping periods of respective registers.

Figure 26.104 shows corresponding interrupt skipping operations by different registers are performed simultaneously.

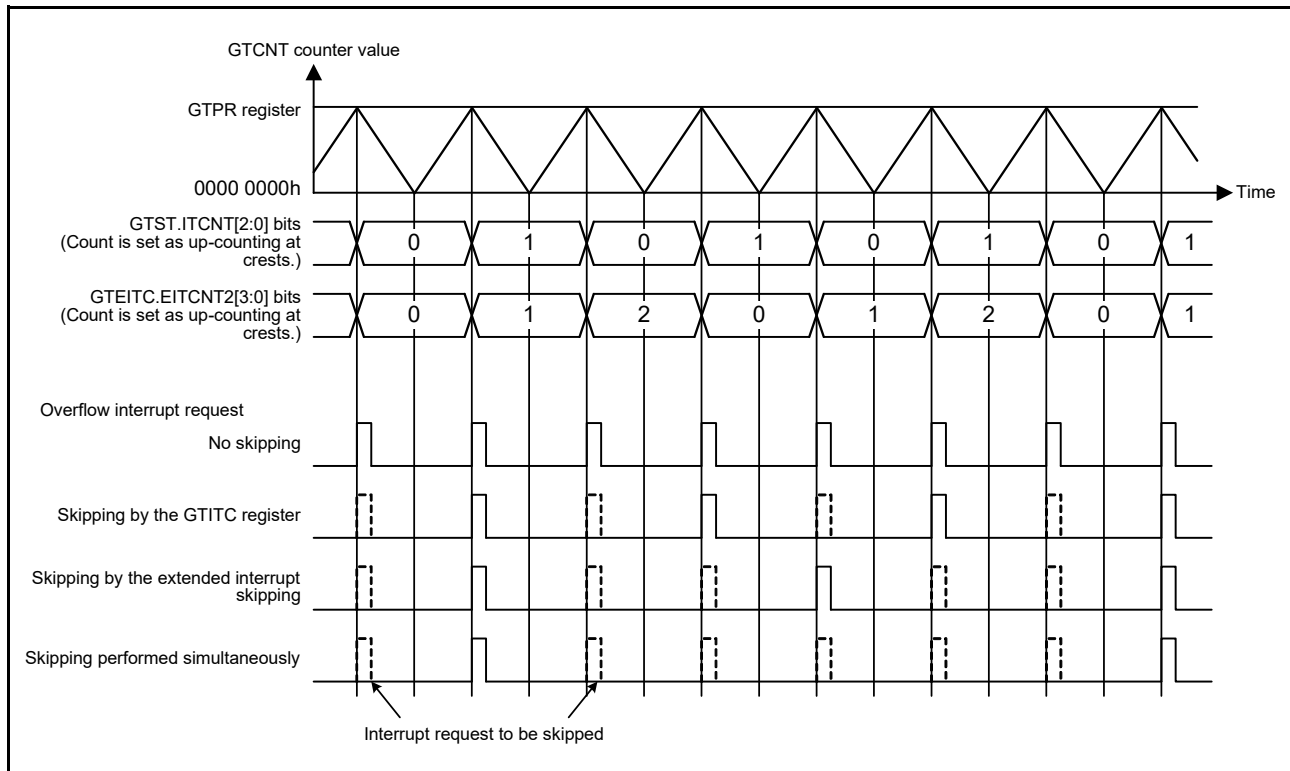


Figure 26.104 Example of Interrupt Skipping Operation
(Skipping by the GTITC Register: Counting Crests, Extended Interrupt Skipping: EIVTC1[1:0] bits = 00b, EIVTC2[1:0] bits = 01b, EITLV[2:0] bits = 010b)

Skipping of ELC event source outputs corresponding to interrupts for which skipping can be set up in the GTEITLI1 register is only based on the settings of the GTITC register and the extended interrupt skipping register, and has no connection with the setting of the interrupt enable bit in the GTINTAD register. The interrupt enable bit is used only for the interrupt signal output after skipping.

When A/D conversion start request skipping which can be set by the GTEITLI2 register is performed, the ELC event output depend on the A/D conversion start request enable bit in the GTINTAD register. All the operations by A/D conversion start request which are set as disable by the GTINTAD register is not performed.

A buffer transfer skipping by the GTEITLB register is performed in all of buffer operation which is enabled in the GTBER and GTDTCR registers, or all buffer operations performed by sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3.

An interrupt skipping and a buffer transfer skipping are operated individually. An interrupt output without performing a buffer transferred performing a buffer transfer without an interrupt output can also be operated.

Figure 26.105 to Figure 26.112 show examples of skipping function operation.

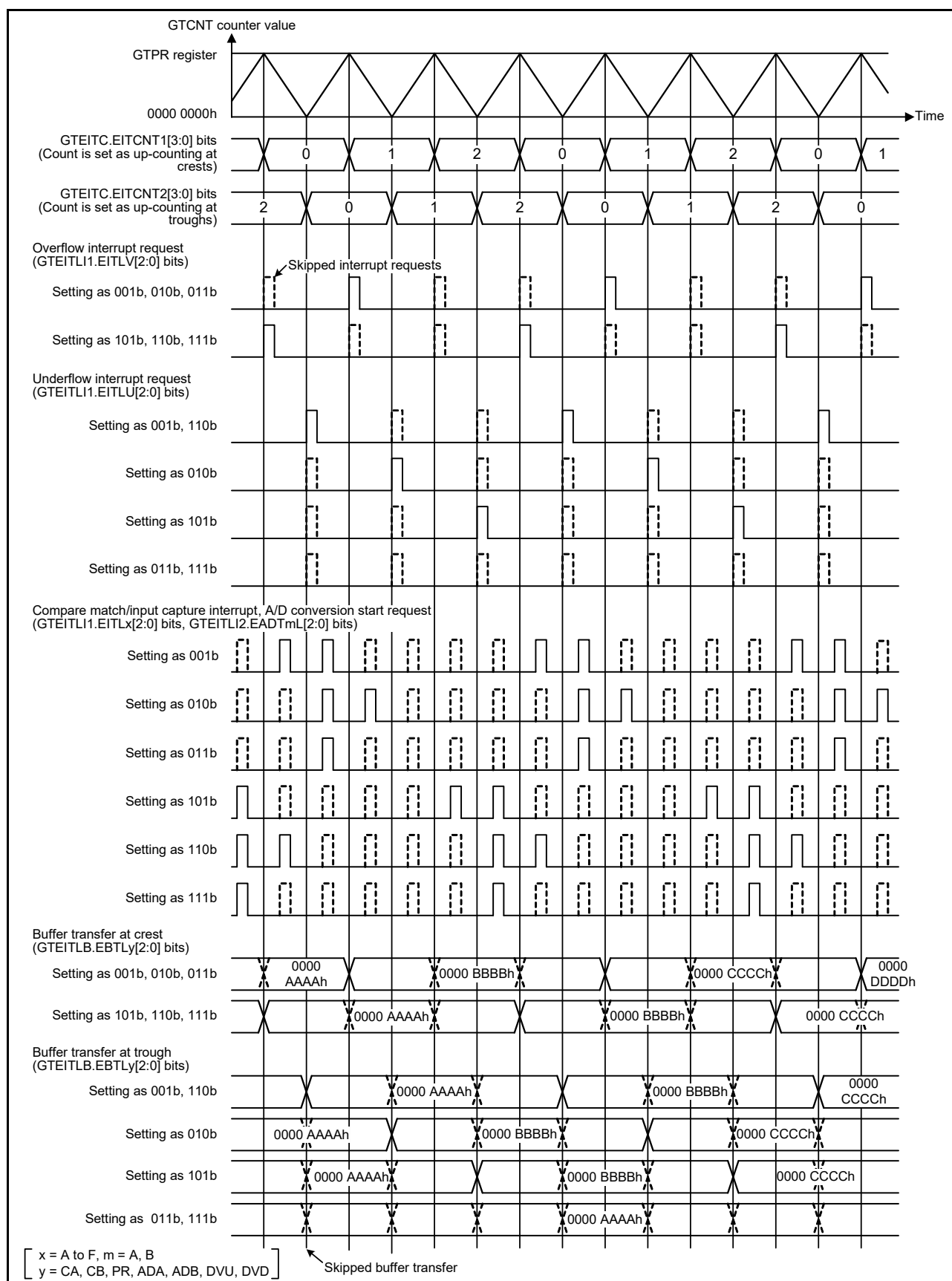


Figure 26.105 Example of Extended Interrupt Skipping Function (Triangle Waves, Counting Crests, Extended Interrupt Skipping 1 Skipping Count: 2, Counting Troughs, Extended Interrupt Skipping 2 Skipping Count: 2, Extended Interrupt Skipping Counter 2 Initial Value: 0)

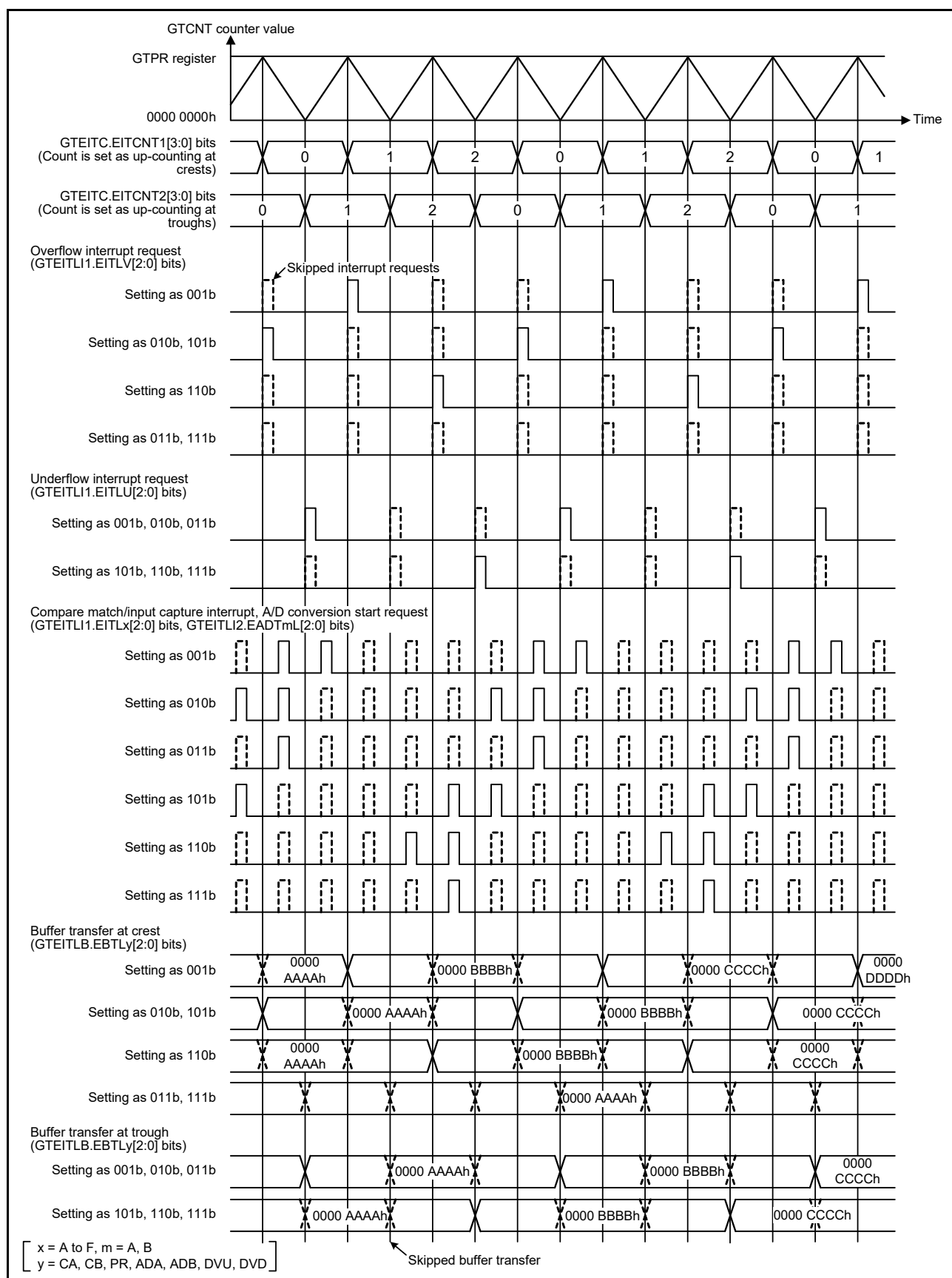


Figure 26.106 Example of Extended Interrupt Skipping Function (Triangle Waves, Count Crests by the Extended Interrupt Skipping 1 with Skipping Count: 2, Count Troughs by the Extended Interrupt Skipping 2 with Skipping Count: 2, Extended Interrupt Skipping Counter 2 Initial Value: 1)

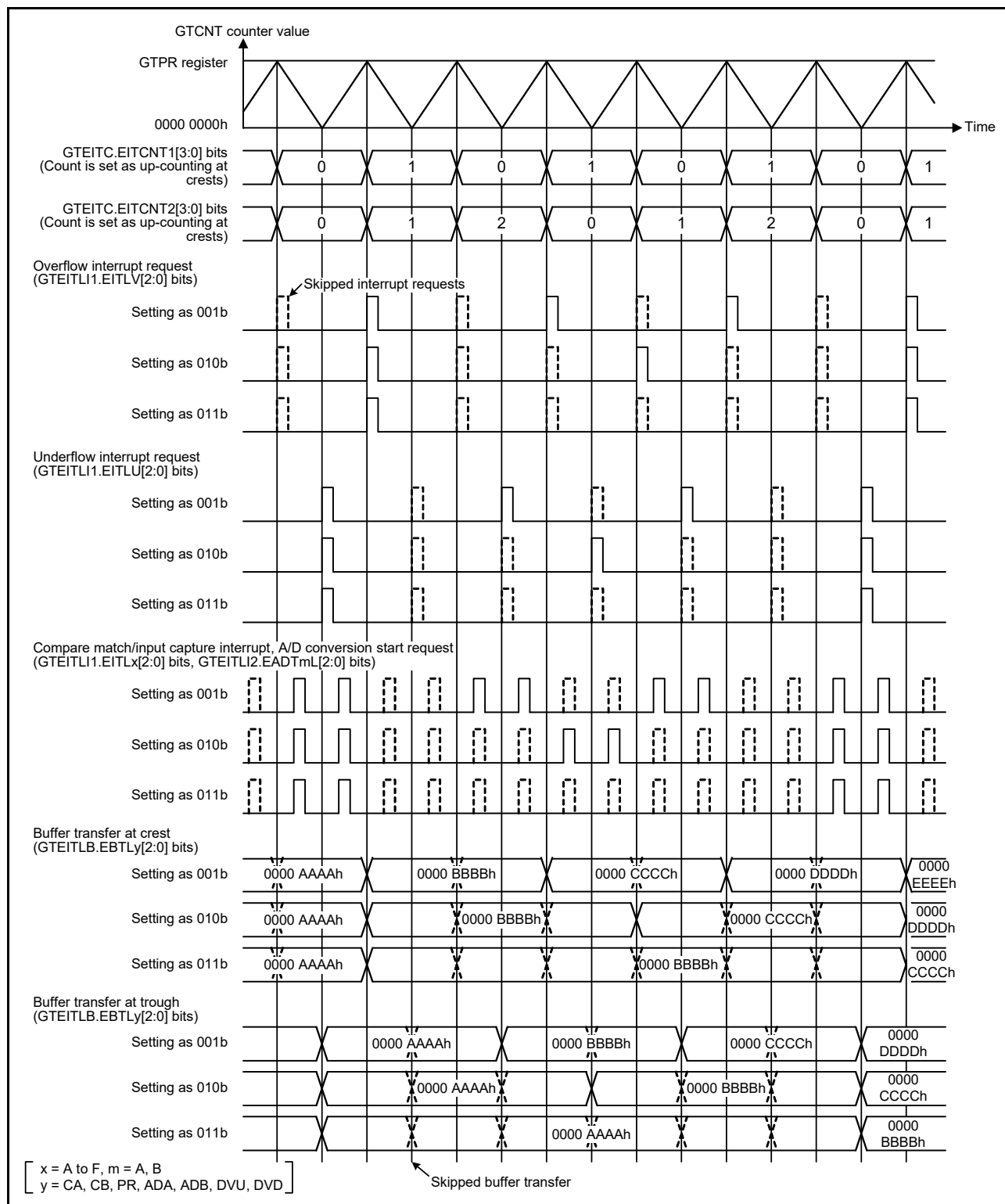


Figure 26.107 Example of Extended Interrupt Skipping Function Operation
 (Triangle Waves, Counting Crests, Extended Interrupt Skipping 1 Skipping Count: 1, Counting Crests, Extended Interrupt Skipping 2 Skipping Count: 2, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as 0)

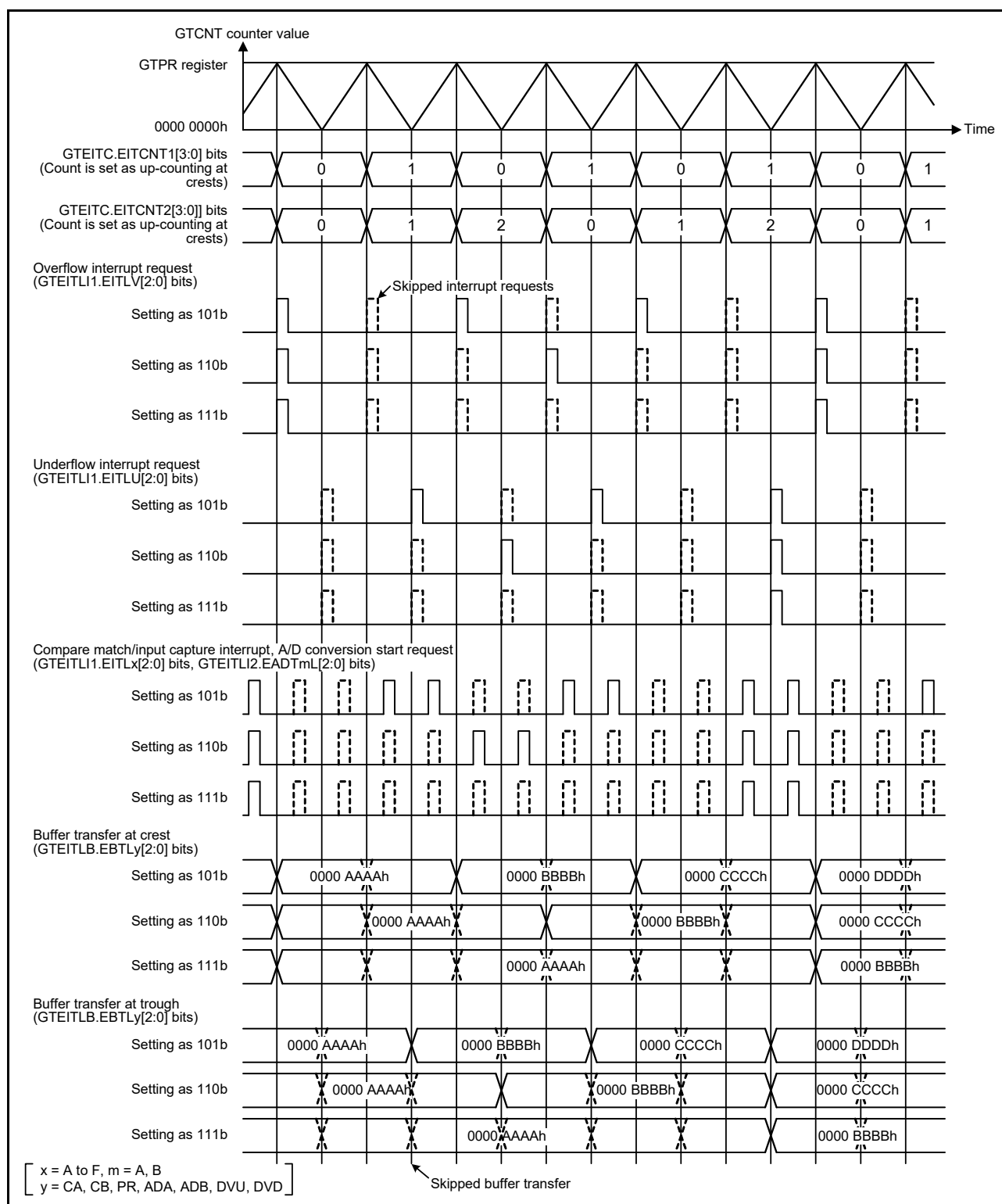


Figure 26.108 Example of Extended Interrupt Skipping Function Operation
 (Triangle Waves, Counting Crests, Extended Interrupt Skipping 1 Skipping Count: 1, Counting Crests, Extended Interrupt Skipping 2 Skipping Count: 2, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as GTEITC.EIVTTk bit.)

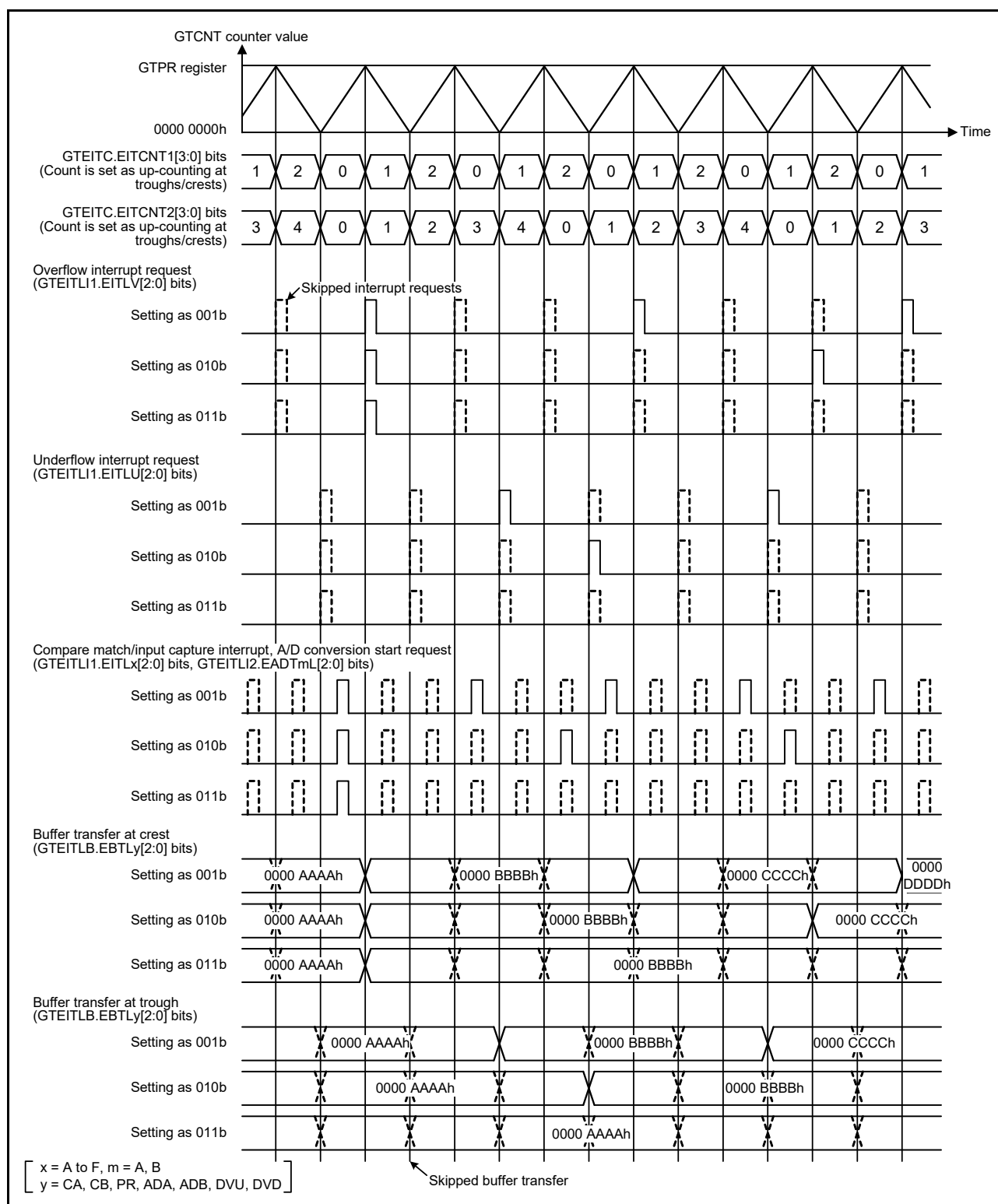


Figure 26.109 Example of Extended Interrupt Skipping Function Operation
 (Triangle Waves, Counting both Crests and Troughs, Extended Interrupt Skipping 1 Skipping Count: 2, Counting both Crests and Troughs, Extended Interrupt Skipping 2 Skipping Count: 4, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as 0)

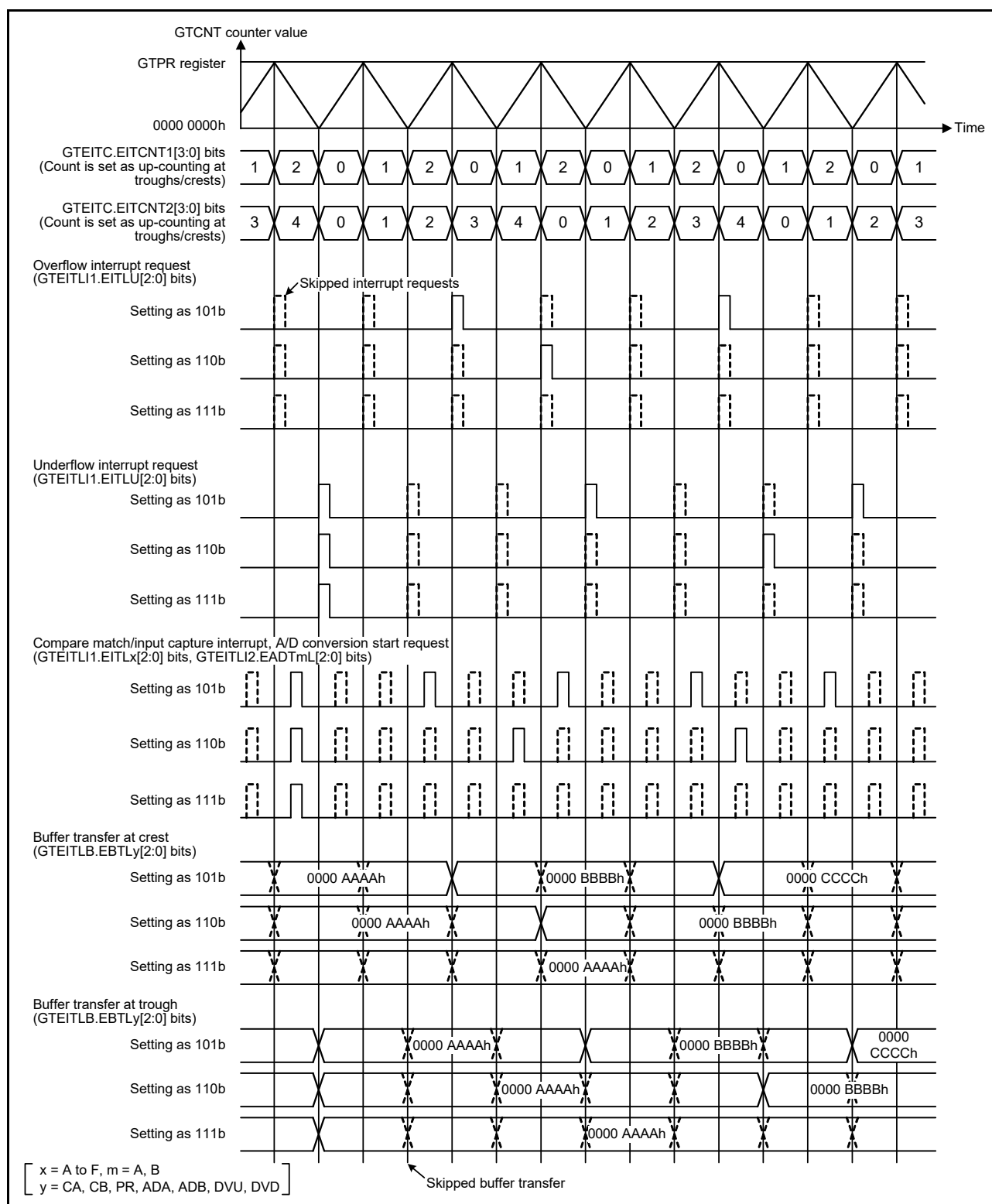


Figure 26.110 Example of Extended Interrupt Skipping Function Operation
 (Triangle Waves, Counting both Crests and Troughs, Extended Interrupt Skipping 1 Skipping Count: 2, Counting both Crests and Troughs, Extended Interrupt Skipping 2 Skipping Count: 4, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as GTEITC.EIVTTk bit.)

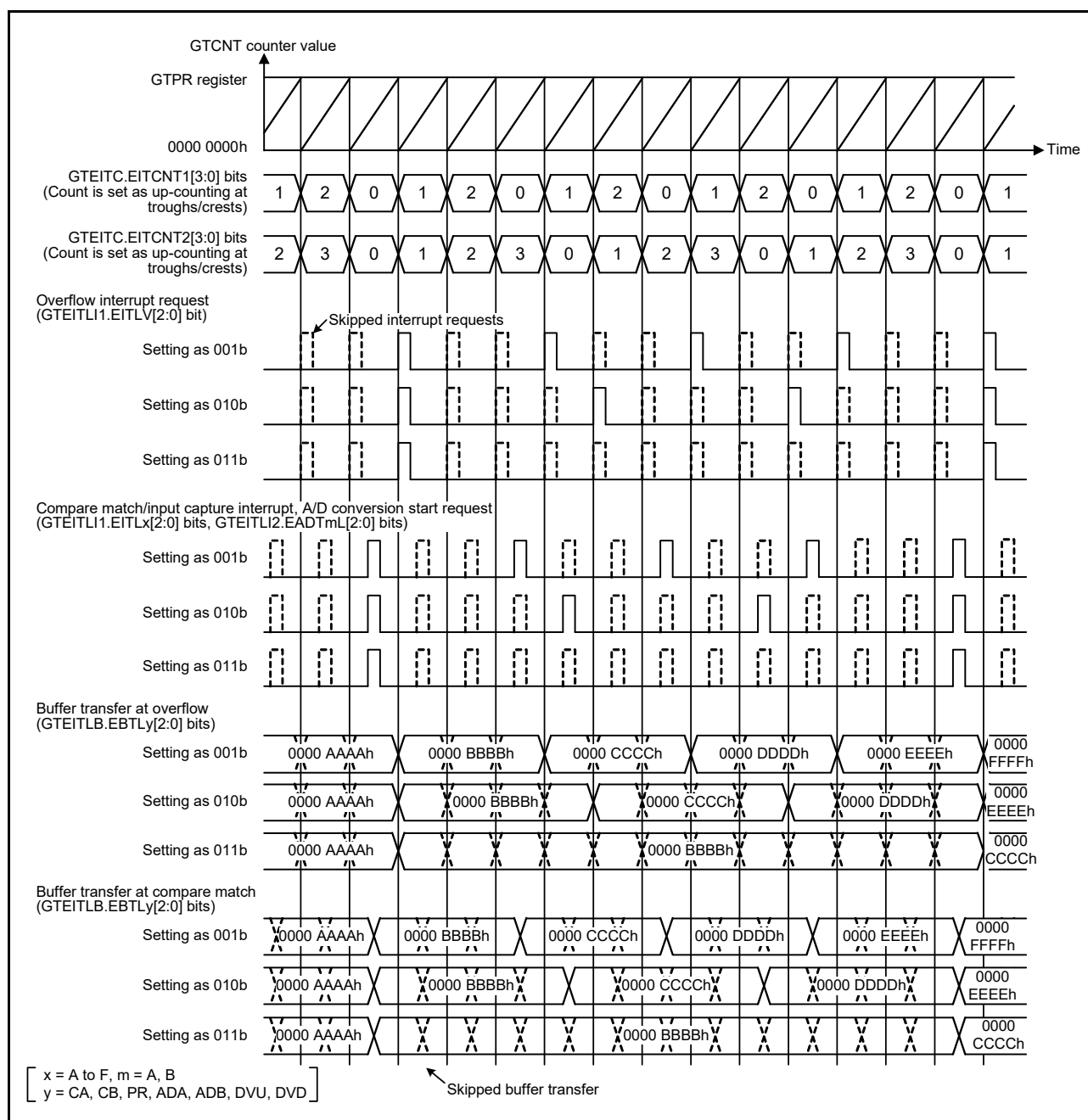


Figure 26.111 Example of Extended Interrupt Skipping Function Operation
 (Up-Counting in Sawtooth Waves, Counting Overflow, Extended Interrupt Skipping 1 Skipping Count: 2, Counting Overflow, Extended Interrupt Skipping 2 Skipping Count: 3, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as 0)

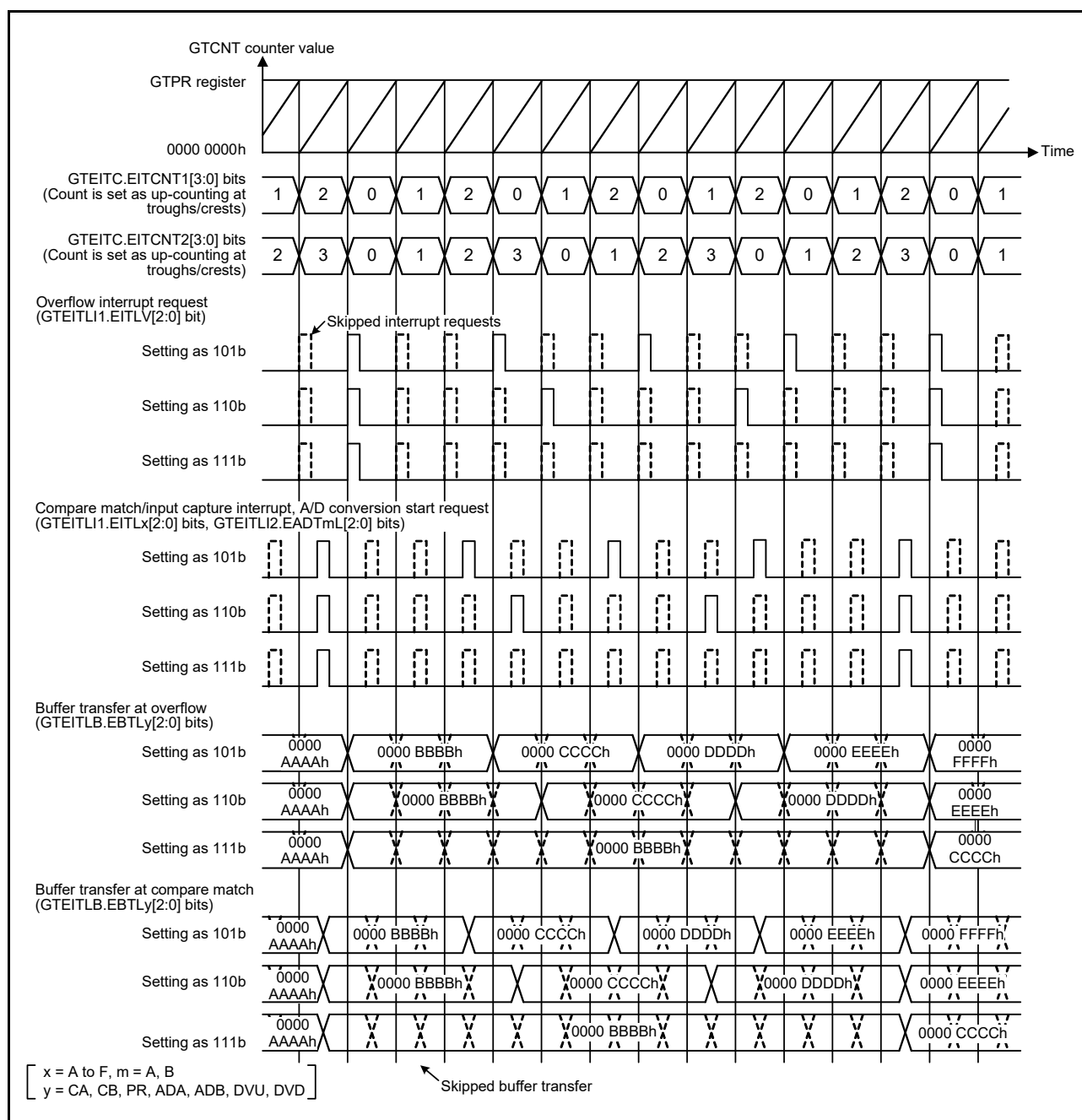


Figure 26.112 Example of Extended Interrupt Skipping Function Operation
 (Up-Counting Sawtooth Waves, Counting Overflow, Extended Interrupt Skipping 1 Skipping Count: 2, Counting Overflow, Extended Interrupt Skipping 2 Skipping Count: 3, Extended Interrupt Skipping Counter 2 Initial Value: 0, Skipped in the Period other than GTEITC.EITCNTk bits (k = 1, 2) as GTEITC.EIVTTk bit.)

Figure 26.113 shows an example of the extended interrupt skipping operation on the input capture. When the setting is for the input capture operation setting (GTCR.ICDS bit = 0) at count stop of the GTCNT counter, the interrupt by the input capture and the extended skipping on buffer transfer is enabled even at count stop of the GTCNT counter. When the input capture is generated at count stop of the GTCNT counter by setting the ICDS bit as 0, an interrupt and a buffer transfer are skipped if the skipping counter value is the same with the skipping period set in a corresponding interrupt function select bit.

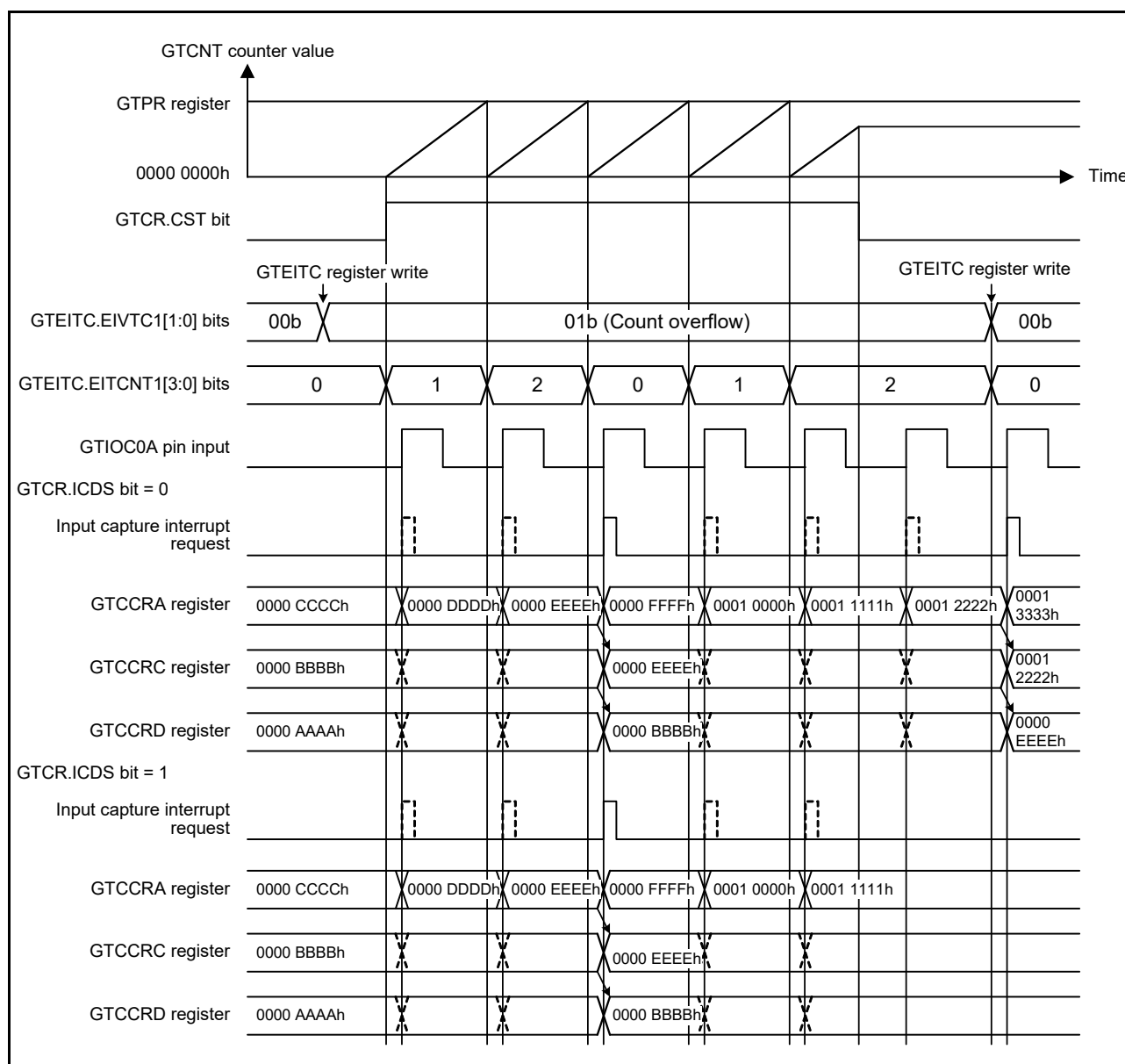


Figure 26.113 Example of Input Capture Operation in the Extended Interrupt Skipping Function Operation (Up-Counting Sawtooth Waves, Counting Overflow, Extended Interrupt Skipping 1 Skipping Count: 2, Skipped in the Period other than the EITCNT1 as 0, Input Capture at the Input Rising)

Figure 26.114 shows an example for setting the extended interrupt skipping.

The extended interrupt skipping counter 2 initial value is set by the written value to the ETICNT2IV[3:0] bits which is applied to change the setting from not counting the extended interrupt skipping counter 2 count source (GTEITC.EIVTC2[1:0] bits = 00b) to counting (EIVTC2[1:0] bits = other than 00b). Writing to the extended interrupt skipping counter 2 initial value bit (ETICNT2IV[3:0] bits) is performed only when the setting of the above mentioned extended interrupt skipping counter 2 initial value is written.

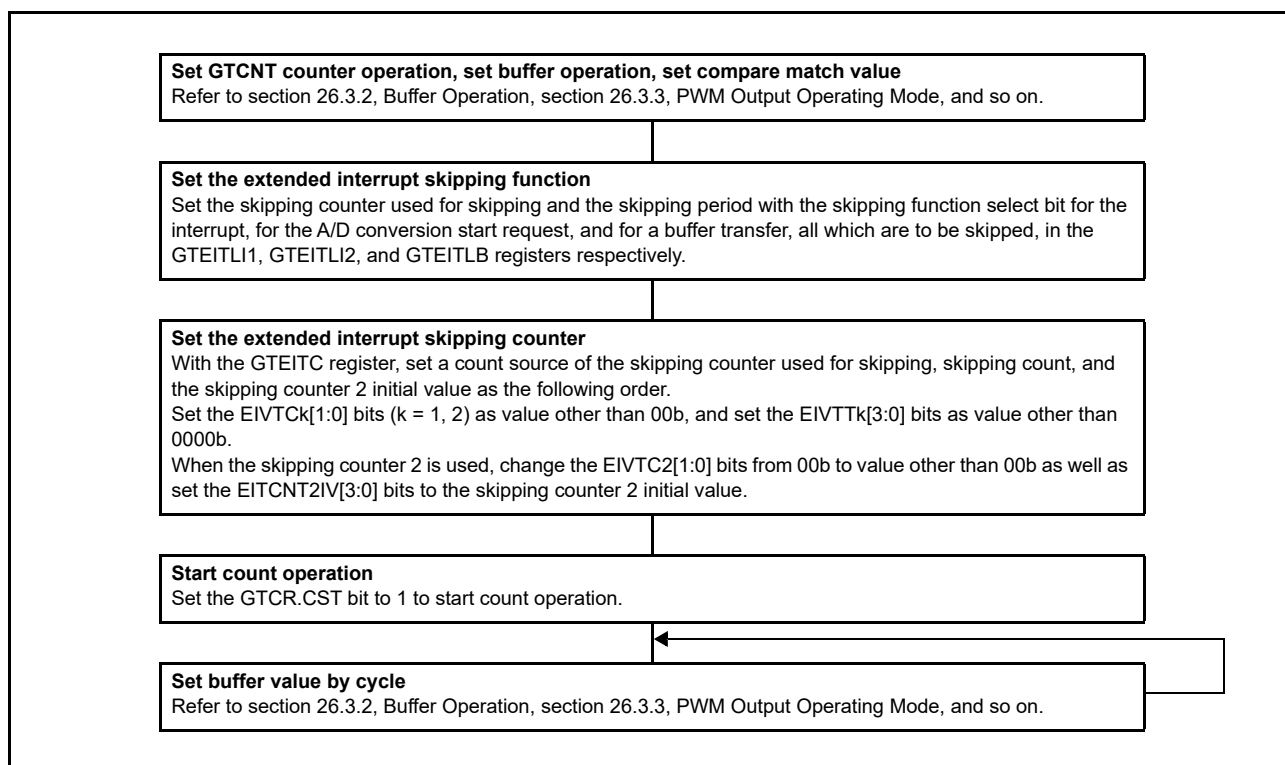


Figure 26.114 Example for Setting the Extended Interrupt Skipping

26.5 A/D Conversion Start Request

An A/D conversion start request can be issued at a compare match between the GTCNT counter and the GTADTRA or GTADTRB register, up-counting only, down-counting only, or both up-counting and down-counting can be specified by setting the GTINTAD register.

During event count operation, the A/D conversion start request cannot be generated.

The A/D conversion start request is not directly output to the A/D converter, but an event signal to the ELC are output. The GTADTRA and GTADTRB registers each has two buffer registers. Buffer operation with the GTADTRA register used together with the GTADTBRA and GTADTDBRA registers, and buffer operation with the GTADTRB register used together with the GTADTBRB and GTADTDBRB registers can be performed.

The timing of the generation of requests to start A/D conversion can be monitored by an external pin. When the A/D conversion start request signal to be monitored is selected in the GTADSMR.ADSMSk bit ($k = 0, 1$) and when the output is enabled in the ADSMENk bit, a signal is output synchronized with a cycle frame of the timer used to generate the A/D conversion start request signal, of which the output is driven high at the generation of the A/D conversion start request signal by the GTADSMk pin, or at the end of the cycle of which the output is driven low. When a signal to request the start of A/D conversion is generated at the end of the cycle, the generation of this signal has priority in terms of monitoring output and the output remains at the high level till the end of the next cycle. The registers (GTADTRA and GTADTRB) that are sources of generating the A/D conversion start request signals and their counting directions can be checked by the A/D conversion start request flags (ADTRAUF, ADTRADF, ADTRBUF, and ADTRBDF) in the GTST register. When the output of the same A/D conversion start request signal monitoring output is enabled for multiple channels, ORed signals will be output from the GPTW.

Figure 26.115 shows an example of A/D conversion start request operation, Figure 26.116 shows an example for setting A/D conversion start request operation, and Figure 26.117 shows an example for A/D conversion start request timing operation.

Figure 26.117 shows an example of the output of A/D conversion start request A by the ELC as start source 0 for the A/D converter. The A/D conversion start request A signal is output by the ELC in response to a match in comparison with the GTADTRA register. The A/D conversion start request A signal is synchronized with PCLKA. The ELC receives the signal in synchronization with PCLKB, and then outputs the A/D conversion start source 0 signal after 1 cycle of the PCLKB has elapsed.

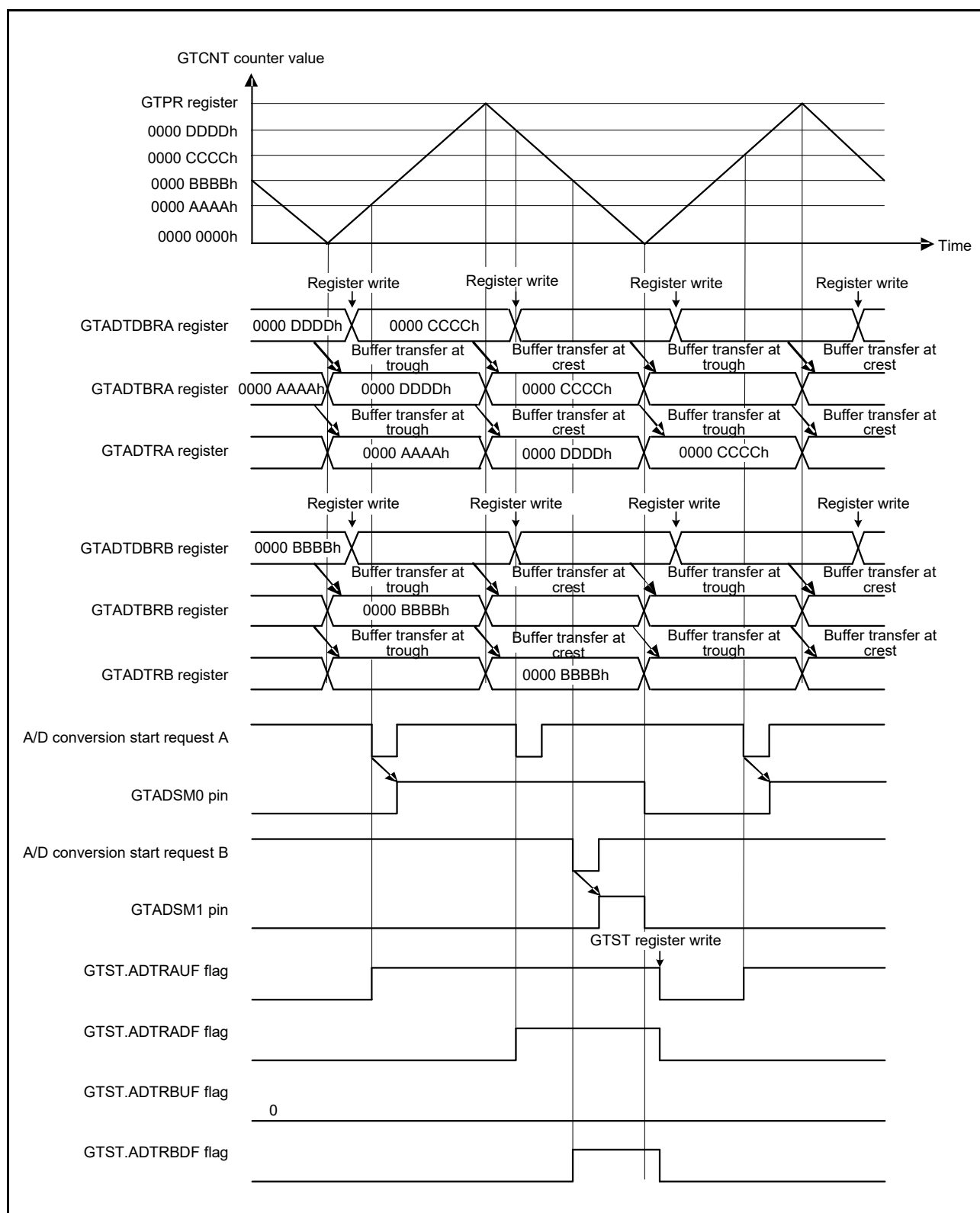


Figure 26.115 Example of A/D Conversion Start Request Timing Operation
 (Triangle Waves, Double Buffer Operation, Buffer Transfer at Both Troughs and Crests, A/D Conversion Start Requested by GTADTRA Register at Both Up-Counting and Down-Counting, A/D Conversion Start Requested by GTADTRB Register at Down-Counting, Monitoring of the GTADTRA Register Up-Counting by the GTADSM0 Pin, Monitoring of the GTADTRB Register Down-Counting by the GTADSM1 Pin)

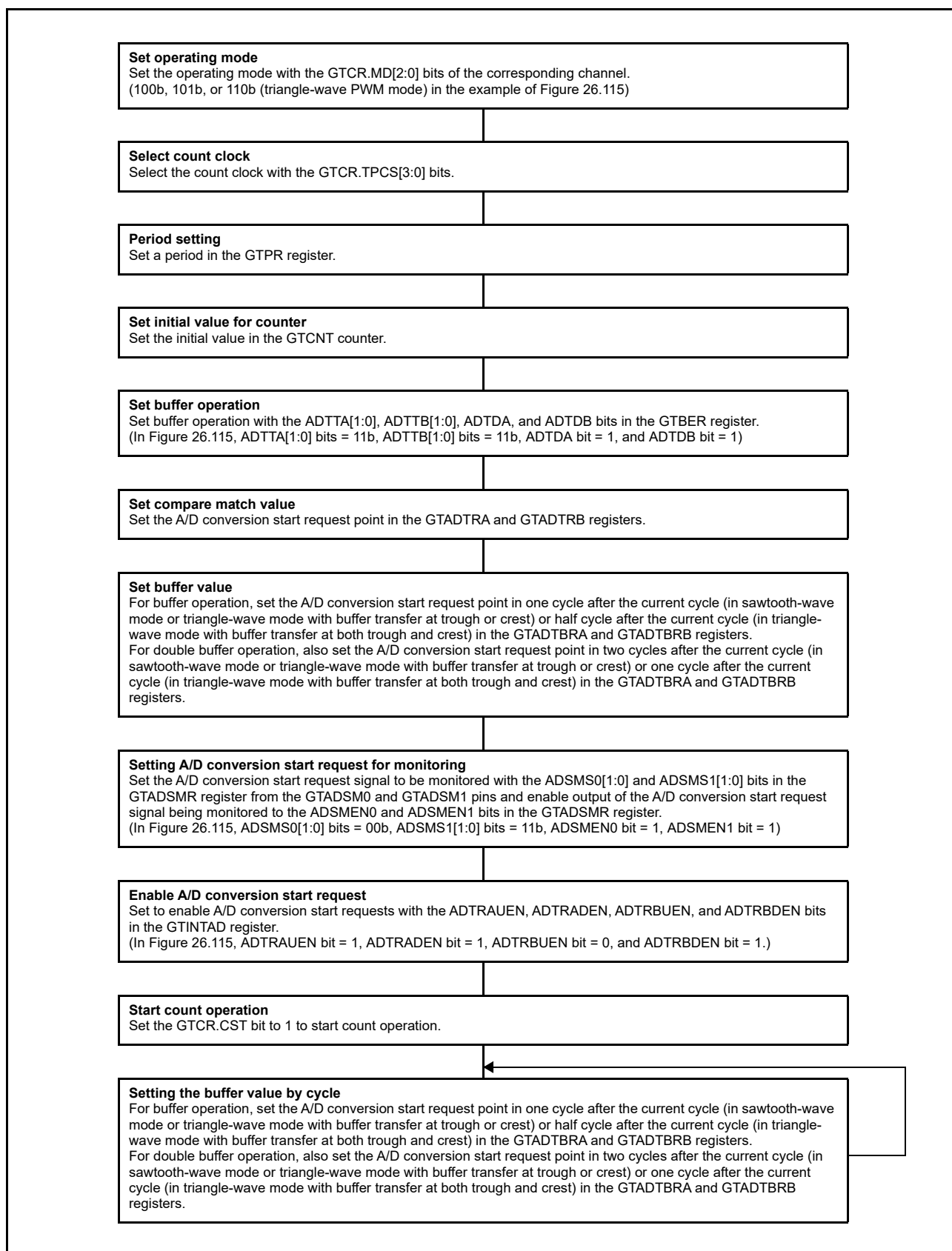


Figure 26.116 Example for Setting A/D Conversion Start Request Operation

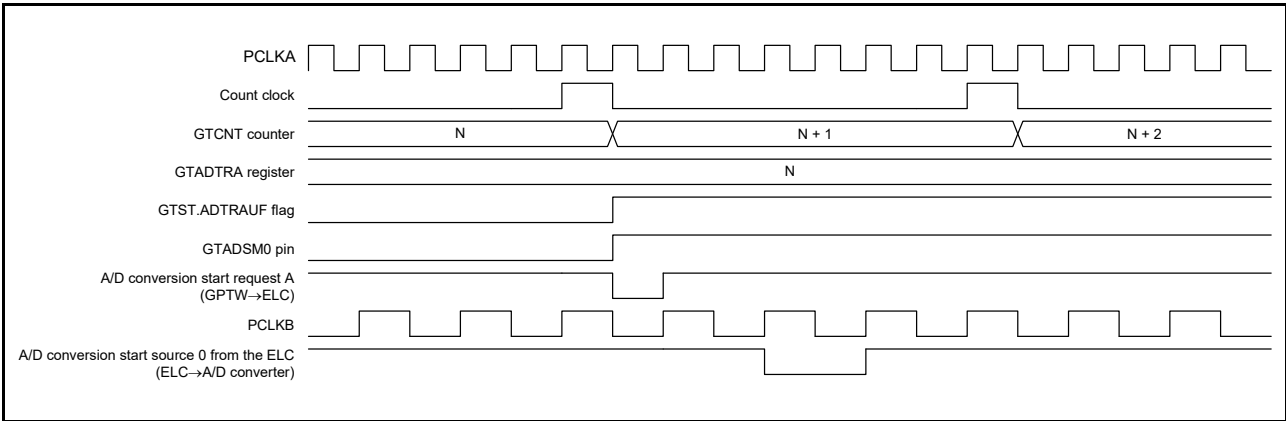


Figure 26.117 Example of A/D Conversion Start Request Timing Operation

26.6 Operations Linked by the ELC

26.6.1 Event Signal Output to the ELC

GPTW is capable of operation linked with another module set in advance when its interrupt request signal is used as an event signal by the event link controller (ELC).

The event signal can be output regardless of the setting of the corresponding interrupt request enable bits except the A/D conversion start request.

The A/D conversion start request during up-counting/down-counting can be enabled/disabled individually with the A/D conversion start request enable bit to output to the ELC.

Each GPTW channel has the following 10 event signals for output to the ELC. A dead time error interrupt does not have a corresponding event signal.

- Generation of compare match A interrupt
- Generation of compare match B interrupt
- Generation of compare match C interrupt
- Generation of compare match D interrupt
- Generation of compare match E interrupt
- Generation of compare match F interrupt
- Generation of overflow interrupt
- Generation of underflow interrupt
- Generation of A/D conversion start request A
- Generation of A/D conversion start request B

26.6.2 GPTW Operations in Response to Receiving Event Signals from the ELC

The GPTW can perform the following operations by the signals for sources A to H which are output from the ELC. Each signal is provided to all channels. Select an event source for a desired operation by a relevant source select register in a channel.

- Operation of count start/count stop/counter clearing
- Operation of up-counting/down-counting
- Operation of input capture A and B

Refer to operation by hardware sources in section 26.3.1.1, Counter Operation for respective operations.

26.7 Noise Filter Function

Each pin for use in input capture to the GPTW is equipped with a noise filter. The noise filter samples input signals at the sampling clock and removes the pulses of which length is less than three sampling periods.

The noise filter functionality includes enabling and disabling of the noise filter for each pin and setting of the sampling clock for each channel.

Figure 26.118 shows the timing of noise filtering.

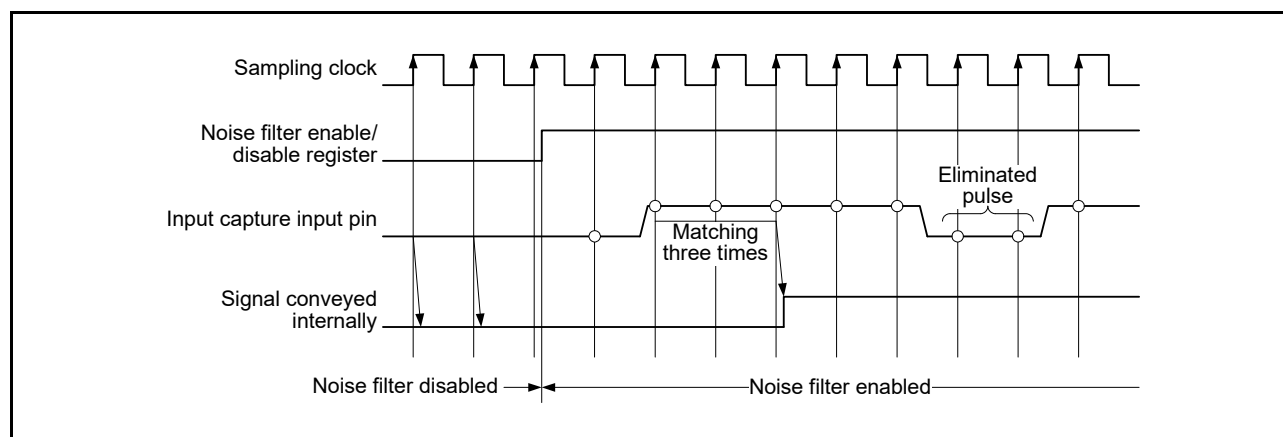


Figure 26.118 Timing of Noise Filtering

If noise filtering is enabled, input capture operation is performed on the edges of noise-filtered signal after a delay of $(\text{minimum sampling interval} \times 2 + \text{PCLKA})$ due to noise filtering for the input capture input.

26.8 Protection Function

26.8.1 Write-Protection for Registers

In order to avoid incorrect writing to registers, write access to registers can be enabled or disabled per channel by setting the GTWP.WP bit for the given channel.

The WP bits enable or disable writing to the registers listed below.

GTSSR, GTPSR, GTCSR, GTUPSR, GTDNSR, GTICASR, GTICBSR, GTCR, GTUDDTYC, GTIOR, GTINTAD, GTST, GTBER, GTITC, GTCNT, GTCCRA, GTCCRB, GTCCRC, GTCCRD, GTCCRE, GTCCRF, GTPR, GTPBR, GTPDBR, GTADTRA, GTADTBRA, GTADTDBRA, GTADTRB, GTADTBRB, GTADTDBRB, GTDTCR, GTDVU, GTDVD, GTDBU, GTDBD, GTSOS, GTSOTR, GTADSMR, GTEITC, GTEITLI1, GTEITLI2, GTEITLB

Every bit in registers GTSTR, GTSTP, and GTCLR, which can update the corresponding registers in other channels and can be updated by any of the corresponding registers in other channels conversely, can be protected by setting the GTWP.STRWP, STPWP, and CLRWP bits, respectively, per channel.

Likewise, writing to the GTSECSR and GTSECR registers, which can control all channels by writing to the GTSECSR and GTSECR registers of a given channel, can be enabled or disabled by the setting of the GTWP.CMNWP bit.

Protection by the GTWP register only targets writing operation by the CPU. Updating the register generated in related to the CPU writing is not protected.

26.8.2 Disabling of Buffer Operation

If the timing of buffer register write is too late for the buffer transfer timing, buffer operation can be suspended with setting the BD[0], BD[1], BD[2], and BD[3] bits in the GTBER register.

Specifically, buffer transfer can be temporarily disabled, even though a buffer transfer condition is generated during buffer register write, by setting the BD[0], BD[1], BD[2], and BD[3] bits to 1 (buffer operation disabled) before buffer register write, and setting the bits to 0 (buffer operation enabled) after completion of writing to all the buffer registers.

The BD[0], BD[1], BD[2], and BD[3] bits can be set on channel basis by writing directly to the GTBER register or can be set simultaneously by setting the GTSECR register for multiple channels which were set by the GTSECSR register.

Figure 26.119 shows an example of operation for disabling buffer operation.

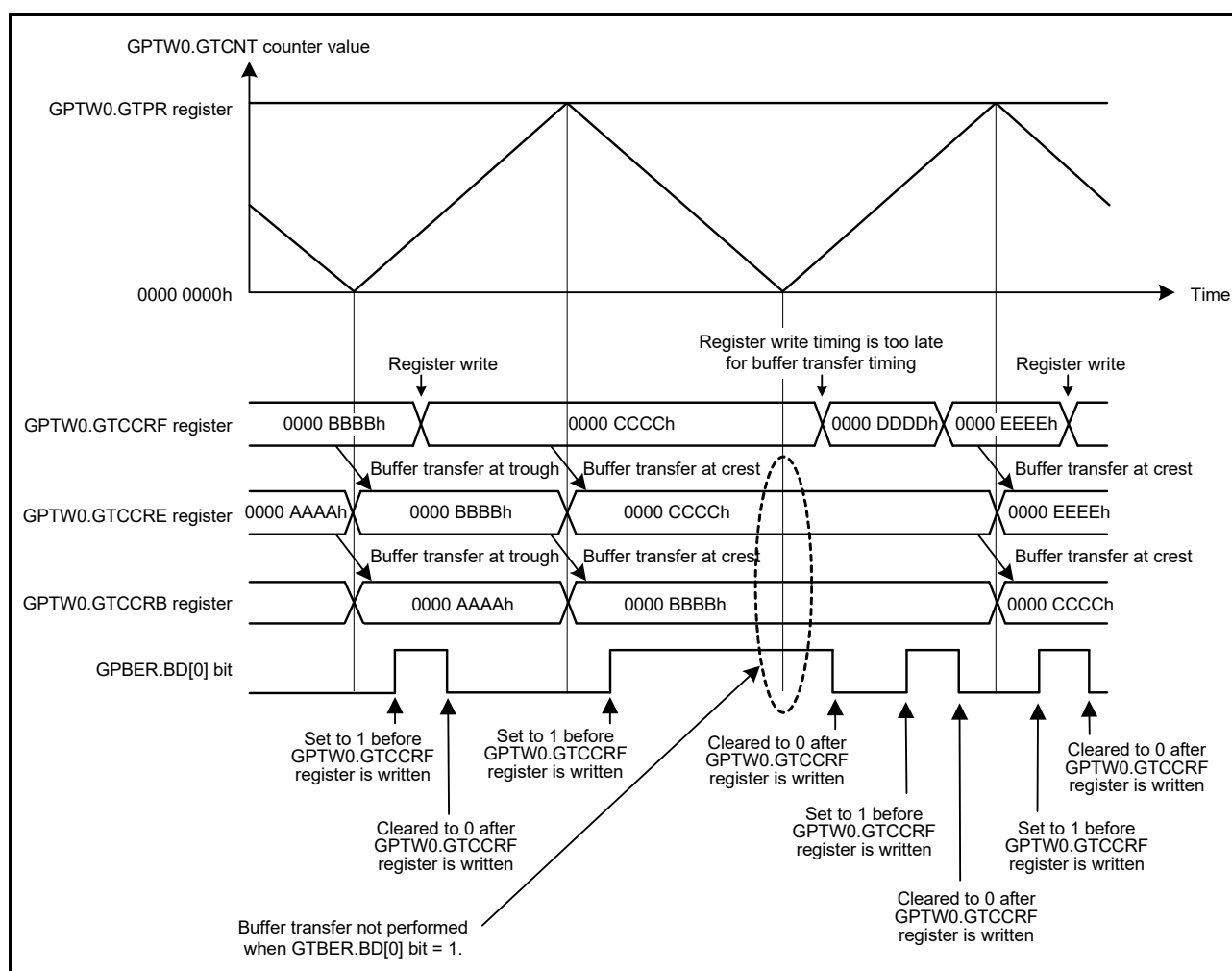


Figure 26.119 Example of Operation for Disabling Buffer Operation (Triangle Waves, Double Buffer Operation, Buffer Transfer at Both Troughs and Crests)

26.8.2.1 Simultaneous Control of Buffer Operations of Multiple Channels

The GTBER.BD bit can be set by writing directly to the GTBER register per channel or by making settings in the GTSECR register for multiple channels that have already set in the GTSECSR register.

Follow the procedure below to simultaneously set the GTBER.BD bits of multiple channels.

(1) Select the channels for simultaneously setting of the GTSECSR register

Set the GTSECSR register so that the values at the bit positions for the corresponding channels for simultaneously setting of the GTBER.BD bits become 1. All GTSECSR registers can be updated by writing to the GTSECSR register of any channel.

(2) Simultaneously set the GTBER.BD bits by updating the GTSECR register

In the GTSECR register, set the operation of the GTBER.BD bits (enabling or disabling of buffer operation) which are to be simultaneously set. Writing to a GTSECR register from any channel updates the GTBER.BD bits in all channels corresponding to the bits set as 1 in the GTSECSR register, in accordance with the value of the GTSECR register. Figure 26.120 shows an example of simultaneously controlling the enabling or disabling of buffer operation for multiple channels.

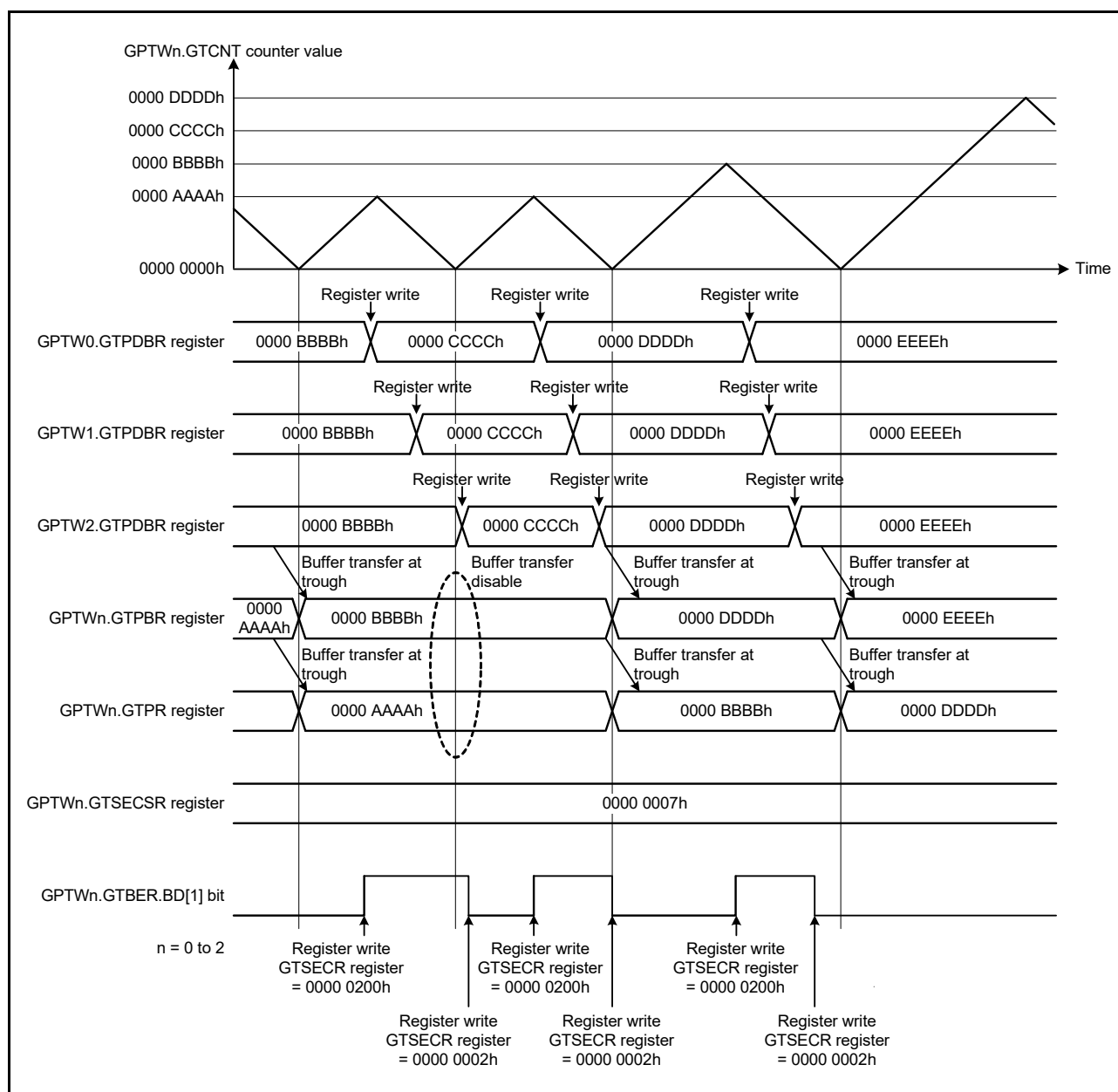


Figure 26.120 Example of Multiple Channel Operation for Disabling Buffer Operation (Triangle Waves, Double Buffer Operation)

26.8.2.2 Repeated Double-Buffered Operation When Disabling GTCCR Buffer Transfer

When a GTBER.DBRTEC_m ($m = A, B$) bit is set to 1 in sawtooth-wave one-shot pulse mode or triangle-wave PWM mode 3, transfer from the intermediate buffer to the GTCCR_m ($m = A, B$) register is repeated on a cyclic basis even while buffer transfer is disabled by the setting of the GTBER.BD[0] bit or by the buffer transfer extended skipping function.

(1) In sawtooth-wave one-shot pulse mode

In sawtooth-wave one-shot pulse mode, the compare match value for the first half of a waveform is stored in temporary register x ($x = C, E$) as the intermediate buffer for the GTCCR_x ($x = C, E$) register and the compare match value for the second half of a waveform is stored in temporary register m ($m = A, B$) as the intermediate buffer for the GTCCR_y ($y = D, F$) register, respectively, for compare match values during repeated buffer operation, and the given values are alternately transferred to the GTCCR_m ($m = A, B$) registers.

Table 26.22 lists the types of buffer transfer of the GTCCR register during counting in sawtooth-wave one-shot pulse mode. While counting is stopped, the setting of the value in the temporary register is transferred through forcible buffer transfer.

In forcible buffer transfer, the values of the GTCCR_y ($y = D, F$) registers are transferred to temporary registers m ($m = A, B$), and the values of the GTCCR_x ($x = C, E$) are transferred to temporary registers x ($x = C, E$), when the setting of the corresponding GTBER.DBRTEC_m ($m = A, B$) bit is 1, respectively.

When the setting of the GTBER.DBRTEC_m ($m = A, B$) bit is 1, values written by the CPU to the GTCCR_m ($m = A, B$) registers are reflected as the values of temporary registers x ($x = C, E$).

Table 26.22 GTCCR Buffer Operation in Sawtooth-Wave One-Shot Pulse Mode

GTBER.DBRTEC _m	Buffer transfer	Timing of transfer				
		GTCCR _x ↓ GTCCR _m	GTCCR _x ↓ Temporary register x	Temporary register x ↓ GTCCR _m	GTCCR _y ↓ Temporary register m	Temporary register m ↓ GTCCR _m
0	Transfer-enabled period	Overflow or underflow	No transfer	No transfer	Overflow or underflow	GTCCR _m compare match
	Transfer-disabled period	No transfer	No transfer	No transfer	No transfer	No transfer
1	Transfer-enabled period	Overflow or underflow	Overflow or underflow	No transfer	Overflow or underflow	GTCCR _m compare match
	Transfer-disabled period	No transfer	No transfer	Overflow or underflow	No transfer	GTCCR _m compare match

Figure 26.121 shows the operation of generating transfer-disabled period by extended buffer transfer skipping as an example of repeated double buffer operations when the GTCCR buffer transfer disabled in sawtooth-wave one-shot pulse mode.

Figure 26.122 shows the operation of generating transfer-disabled period by updating the GTBER.BD[0] bit as an example of repeated double-buffer operations when the GTCCR buffer transfer is disabled in sawtooth-wave one-shot pulse mode.

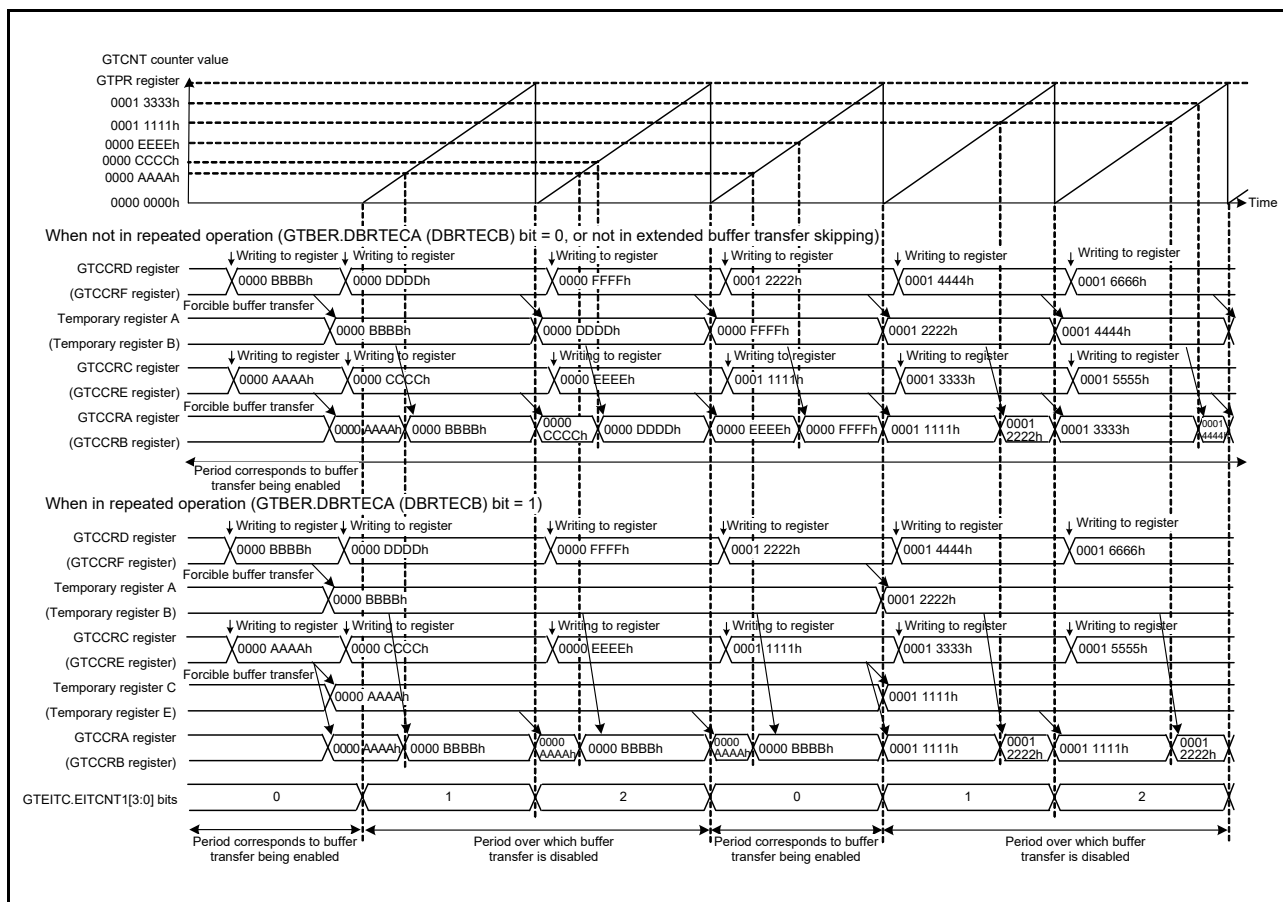


Figure 26.121 Example of Repeated Double-Buffer Operation When GTCCR Buffer Transfer is Disabled (Sawtooth-Wave One-Shot Pulse Mode, Using Extended Buffer Transfer Skipping, GTBER.BD[0] is Constantly 0)

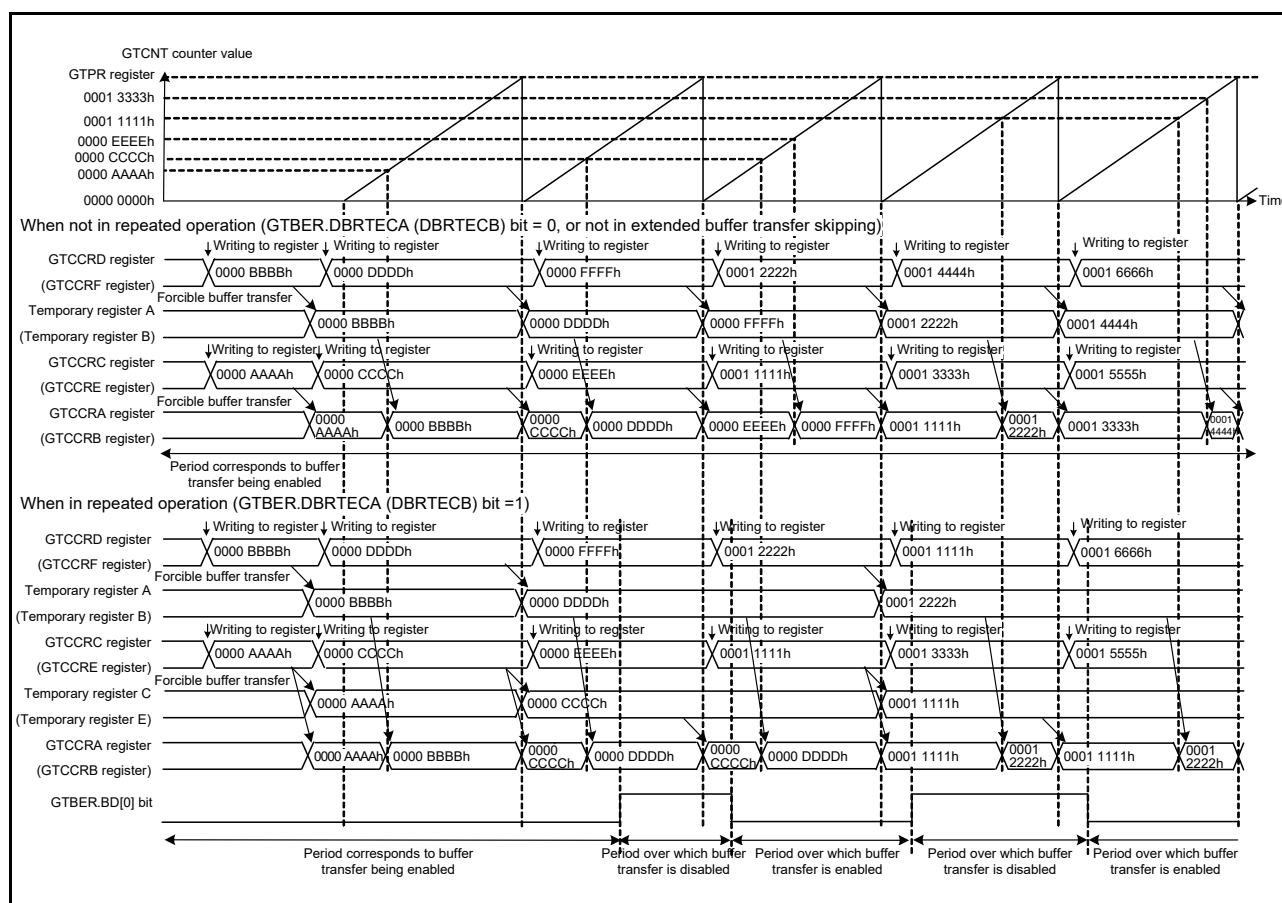


Figure 26.122 Example of Repeated Double-Buffer Operation When GTCCR Buffer Transfer is Disabled (Sawtooth-Wave One-Shot Pulse Mode, Updating the GTBER.BD[0] Bit)

(2) In triangle-wave PWM mode 3

In triangle-wave PWM mode 3, the compare match value for the first half of a waveform is stored in the temporary register x (x = C, E) as the intermediate buffer for the GTCCR_x (x = C, E) register and the compare match value for the second half of a waveform is stored in temporary register m (m = A, B) as the intermediate buffer for the GTCCR_y (y = D, F) register, respectively, for compare match values during repeated buffer operation, and the given values are alternately transferred to the GTCCR_m (m = A, B) register.

Table 26.23 lists the types of buffer transfer of the GTCCR register during counting operation in triangle-wave PWM mode 3.

While counting is stopped, the setting of the value in the temporary register is transferred through forcible buffer transfer. In forcible buffer transfer, the values of the GTCCR_y (y = D, F) registers are transferred to temporary registers m (m = A, B), and the values of the GTCCR_x (x = C, E) are transferred to temporary registers x (x = C, E), when the setting of the corresponding GTBER.DBRTEC_m (m = A, B) bit is 1, respectively.

When the setting of the GTBER.DBRTEC_m (m = A, B) bit is set to 1, values written by the CPU to the GTCCR_m (m = A, B) registers are reflected as the values of temporary registers x (x = C, E).

Table 26.23 GTCCR Buffer Operation in Triangle-Wave PWM Mode 3

GTBER.DBRTEC _m	Buffer transfer	Timing of transfer				
		GTCCR _x ↓ GTCCR _m	GTCCR _x ↓ Temporary register x	Temporary register x ↓ GTCCR _m	GTCCR _y ↓ Temporary register m	Temporary register m ↓ GTCCR _m
0	Transfer-enabled period	Trough	No transfer	No transfer	Trough	Crest
	Transfer-disabled period	No transfer	No transfer	No transfer	No transfer	No transfer
1	Transfer-enabled period	Trough	Trough	No transfer	Trough	Crest
	Transfer-disabled period	No transfer	No transfer	Trough	No transfer	Crest

Figure 26.123 shows the operation of generating transfer-disabled period by extended buffer transfer skipping as an example of repeated double buffer operations when the GTCCR buffer transfer disabled in triangle-wave PWM mode 3. Figure 26.124 shows the operation of generating transfer-disabled period by updating the GTBER.BD[0] bit as an example of repeated double-buffer operations when the GTCCR buffer transfer is disabled in triangle-wave PWM mode 3.

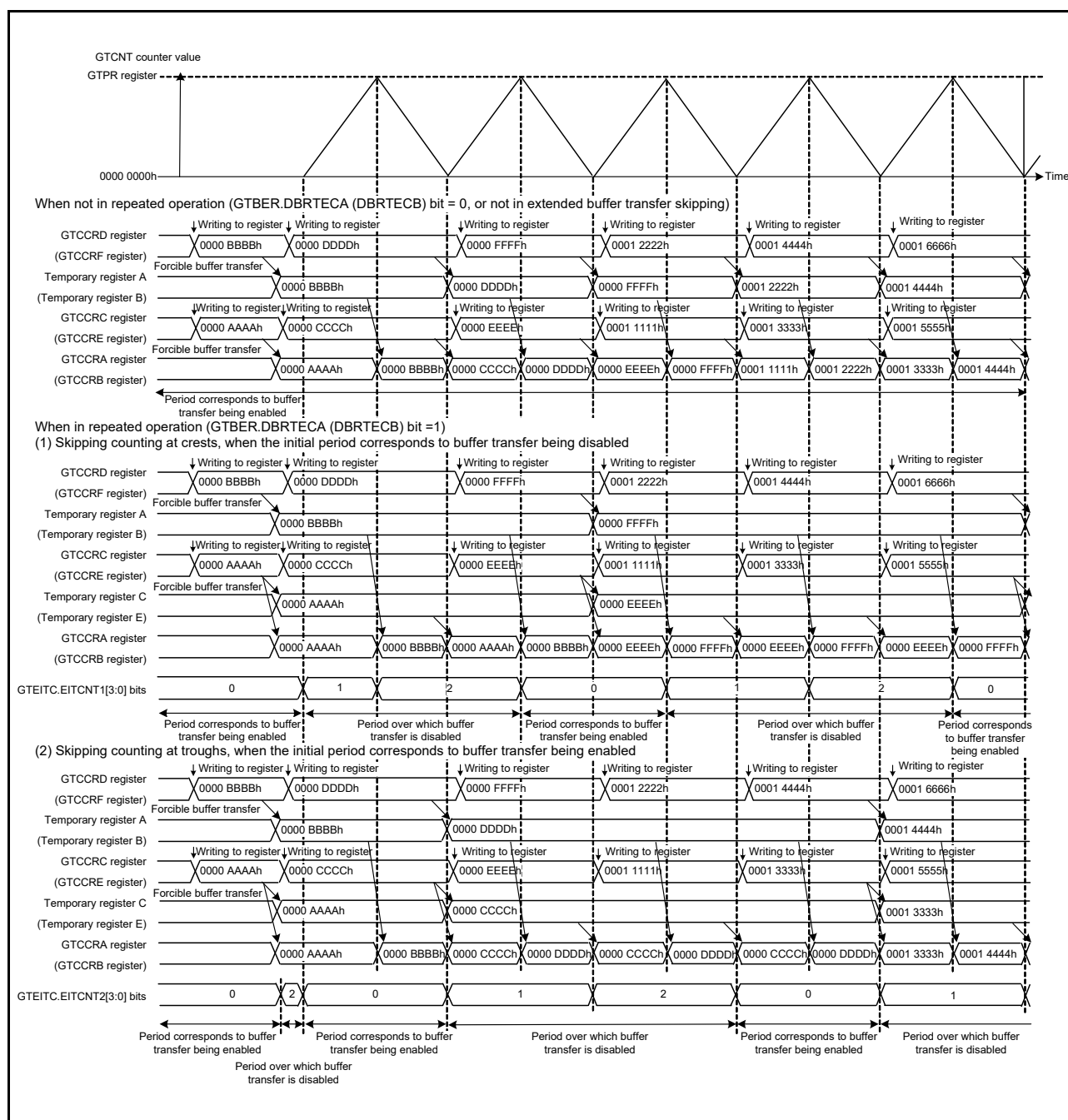


Figure 26.123 Example of Repeated Double-Buffer Operation When GTCCR Buffer Transfer is Disabled (Triangle-Wave PWM Mode 3, Using Extended Buffer Transfer Skipping, GTBER.BD[0] is Constantly 0)

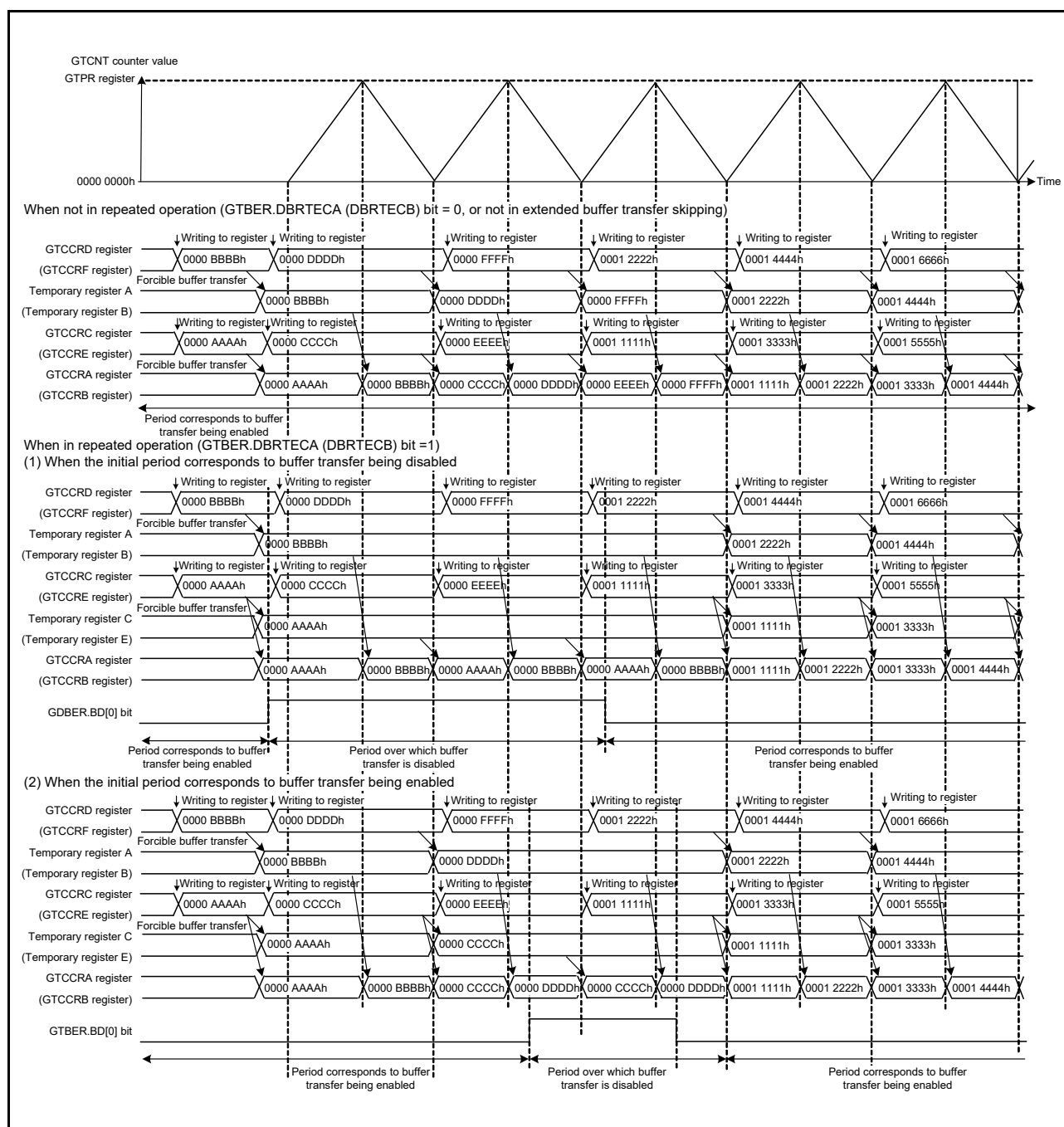


Figure 26.124 Example of Repeated Double-Buffer Operation When GTCCR Buffer Transfer is Disabled (Triangle-Wave PWM Mode 3, Updating the GTBER.BD[0] Bit)

26.8.3 GTIOCnm Pin Output Negate Control (n = 0 to 3; m = A, B)

For protection from system failure, the output negate control which changes the GTIOCnm pin output forcibly by the output stop request from the POEG is provided.

Output protection is required when a dead-time error or the same output level being on the GTIOCnA and GTIOCnB pins is detected. At that time, the detection of disabling of output is output to the group of the POEG set in the GTINTAD.GRP[1:0] bits based on the setting of the output disable detection enable bits, i.e. the GRPDTE, GRPABH, and GRPABL bits in the GTINTAD register. The POEG takes the logical OR of detection of the types described above and other forms of detection of the disabling of output, and outputs requests for the disabling of output to the GPTW. By setting the GTINTAD.GRP[1:0] bits, one output stop request can be selected out of four output stop requests which are input from the POEG as output stop request signal common in the GTIOCnA and GTIOCnB pins. Selected output stop request can be checked by reading the GTST.ODF flag.

The states of outputs at the time of control for their negation can be set with the GTIOR.OADF[1:0] bits for the GTIOCnA pin and with the GTIOR.OBDF[1:0] bits for the GTIOCnB pin.

Transition to the output negate condition by generating the output stop request from the POEG is performed asynchronously, while a release from the output negate condition by making the output stop request no longer satisfied is performed at the end of the cycle. It is after 3 PCLKA at shortest when the output negate condition is released after the output stop request becomes no longer satisfied. To reliably control output negation, clear the flag of POEG for which the condition for the request to disable the output is no longer satisfied after 4 cycles of PCLKA.

To release the output stop condition during event count operation or without waiting the end of the cycle, set the OADF[1:0] bits to 00b for the GTIOCnA pin, and set the OBDF[1:0] bits to 00b for the GTIOCnB pin.

Figure 26.125 shows the output negate control operation for the GTIOCnm pin output.

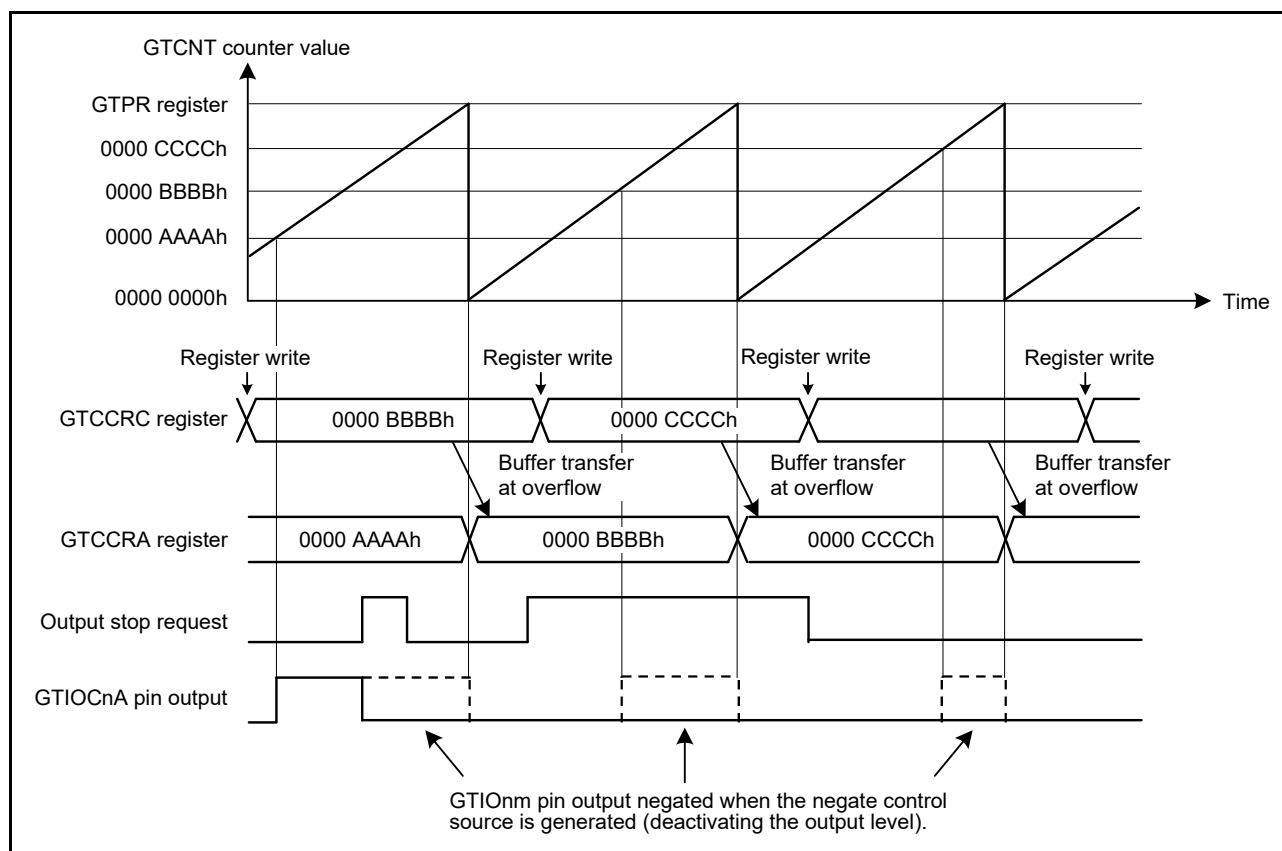


Figure 26.125 Example of Operation for GTIOCnm Pin Output Negate Control
(Sawtooth-Wave Up-Counting, Buffer Operation, Active Level: High Output at GTCCRA Register Compare Match, Low Output at the End of the Cycle) and Low Output at Output Negate)
(n = 0 to 3; m = A, B)

26.8.4 Output Protection Function for GTIOCnm Pin Output (n = 0 to 3; m = A, B)

To prepare for a case when an incorrect value (0000 0000h or a value greater than or equal to the GTPR register value) is set in the GTCCRA register, the output protection function for the GTIOCnm pin output (disabling function) is activated when the automatic dead time is set (GTDTCR.TDE bit = 1) in triangle-wave PWM mode.

The status of the output protection function can be read from the GTSOS.SOS[1:0] bits.

Figure 26.126 shows the output protection function state transition.

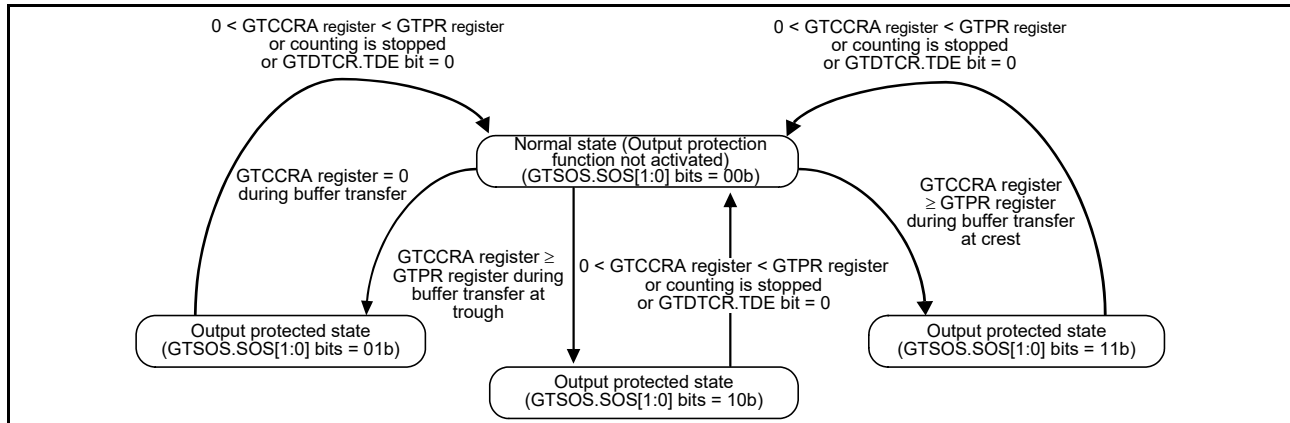


Figure 26.126 Output Protection Function

(1) Output Protection Function when the GTCCRA Register is Set to 0000 0000h during Buffer Transfer

Figure 26.127 and Figure 26.128 show examples of output protection function operation when the GTCCRA register is set to 0000 0000h during buffer transfer at troughs, and Figure 26.129 and Figure 26.130 show examples when the GTCCRA register is set to 0000 0000h during buffer transfer at crests.

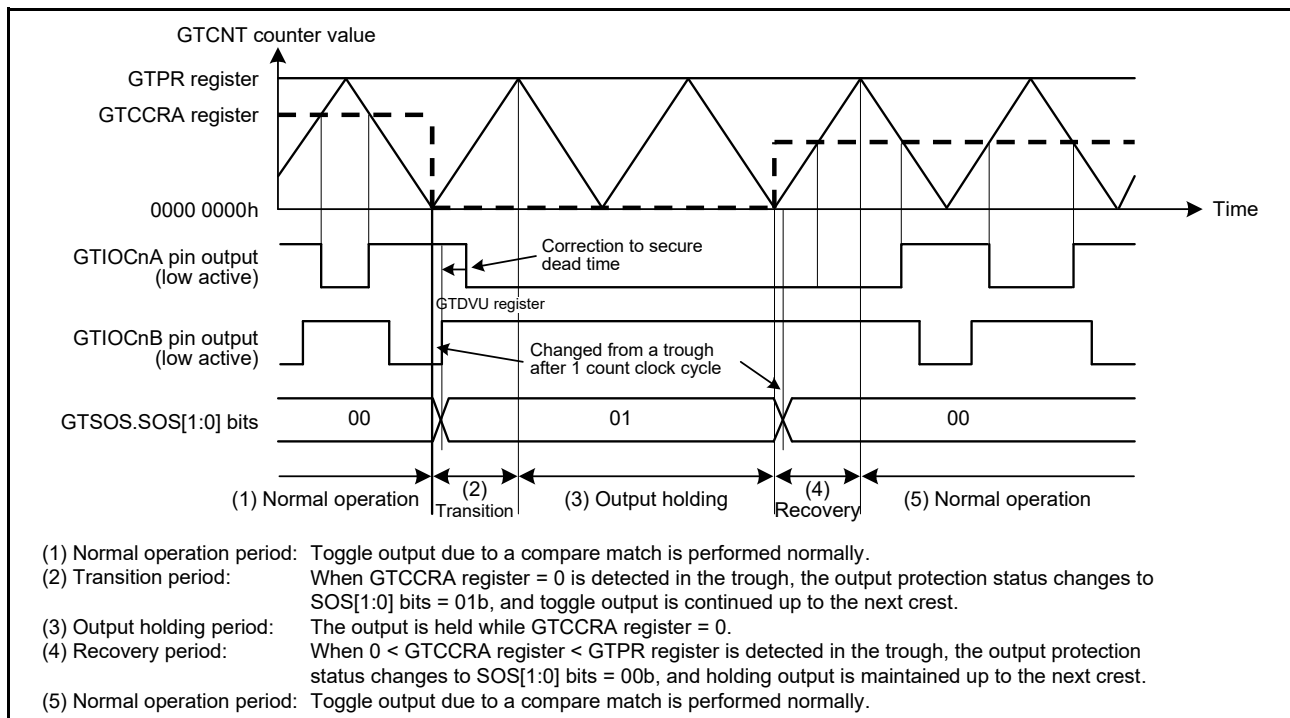


Figure 26.127 Example of Output Protection Function Operation When the GTCCRA Register is Set to 0000 0000h during Buffer Transfer at Troughs (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Troughs, Active Level: Low) (n = 0 to 3)

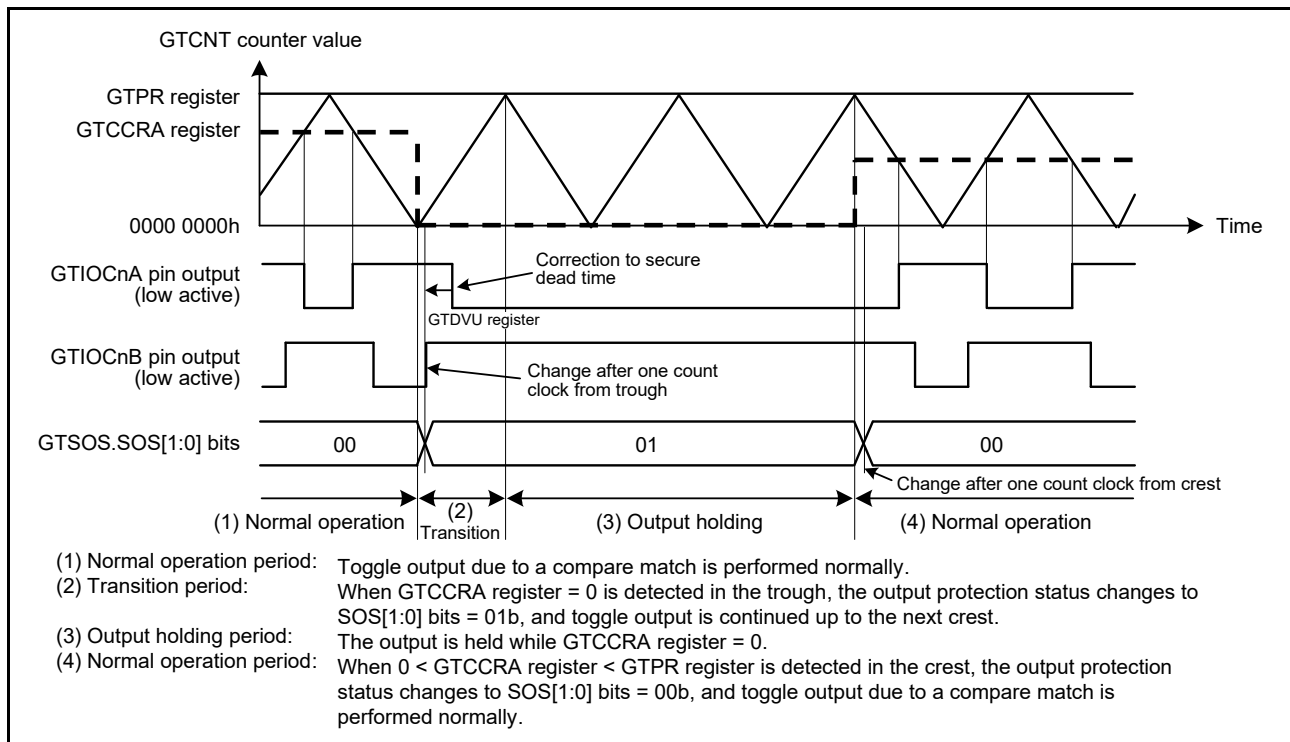


Figure 26.128 Example of Output Protection Function Operation When the GTCCRA Register is Set to 0000 0000h during Buffer Transfer at Troughs (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Crests, Active Level: Low) ($n = 0$ to 3)

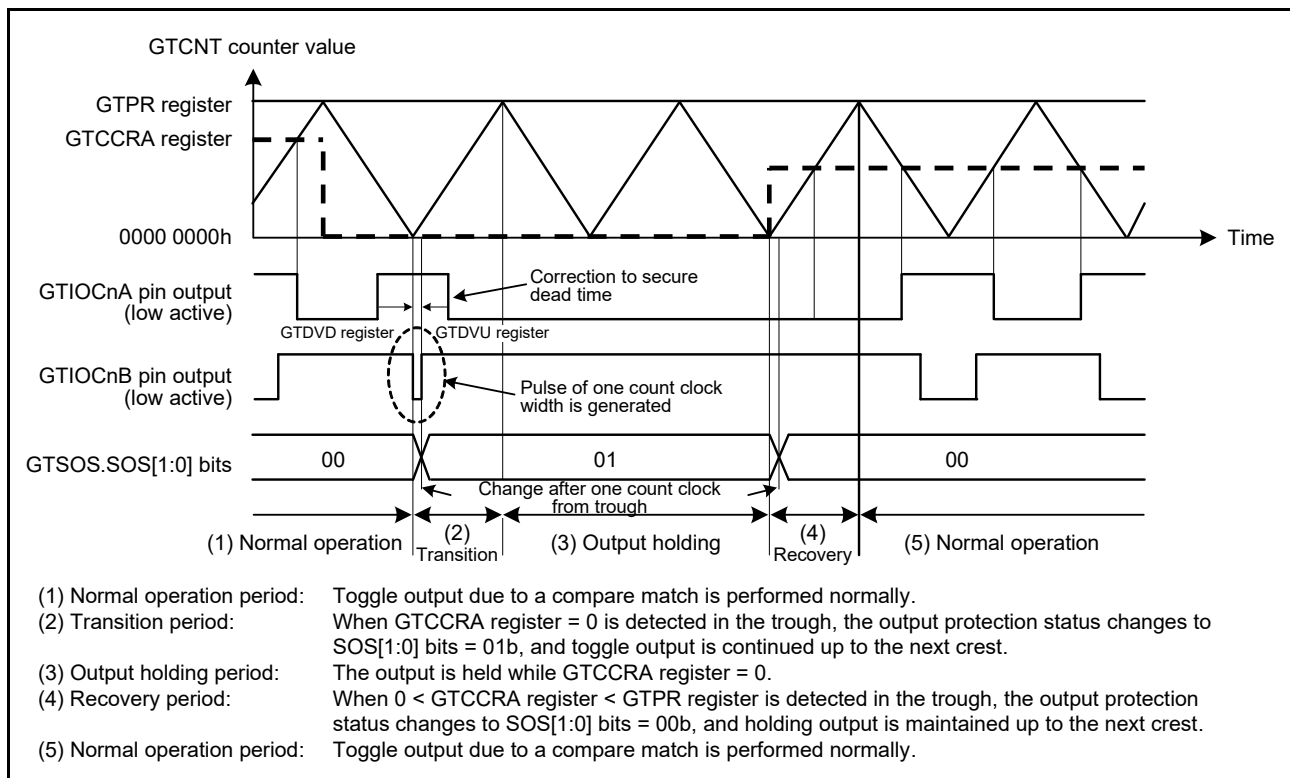


Figure 26.129 Example of Output Protection Function Operation When the GTCCRA Register is Set to 0000 0000h during Buffer Transfer at Crests (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Troughs, Active Level: Low) ($n = 0$ to 3)

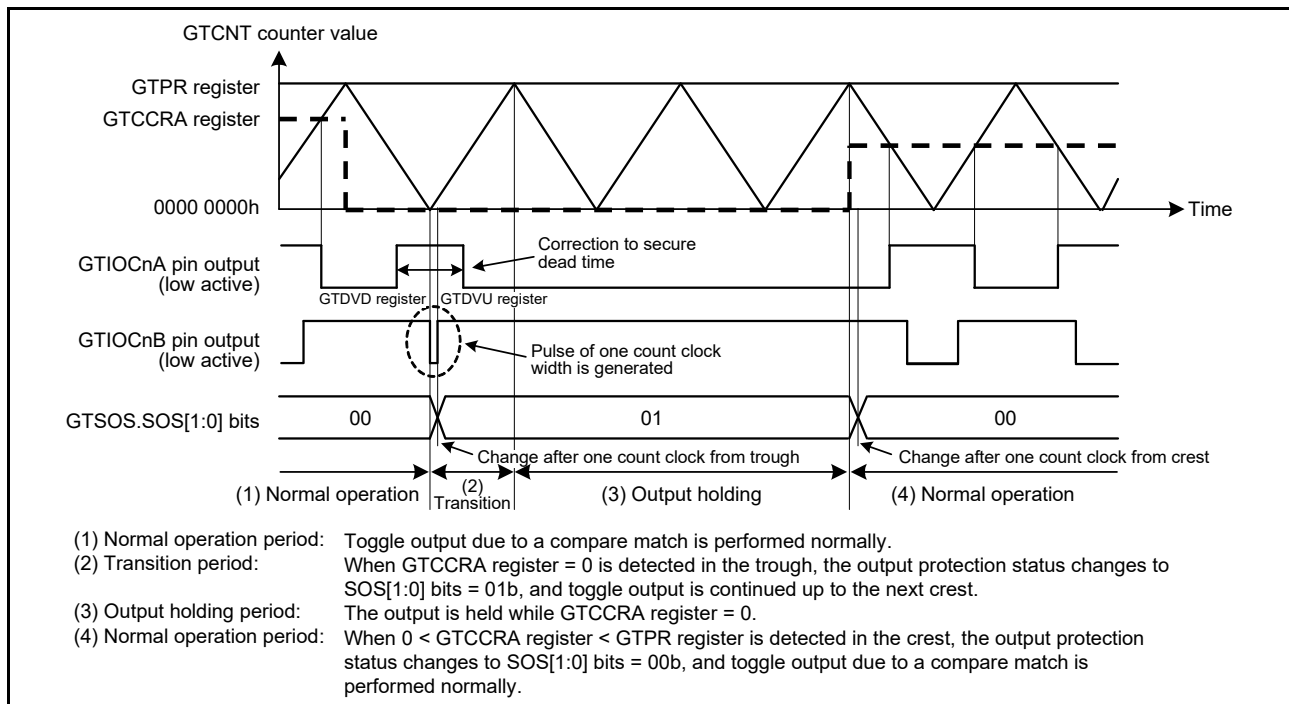


Figure 26.130 Example of Output Protection Function Operation When the GTCCRA Register is Set to 0000 0000h during Buffer Transfer at Crests (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Crests, Active Level: Low) ($n = 0$ to 3)

(2) Output Protection Function when GTCCRA Register $\geq$ GTPR Register is Set during Buffer Transfer at Troughs

Figure 26.131 and Figure 26.132 show examples of output protection function operation when GTCCRA register $\geq$ GTPR register is set during buffer transfer at troughs.

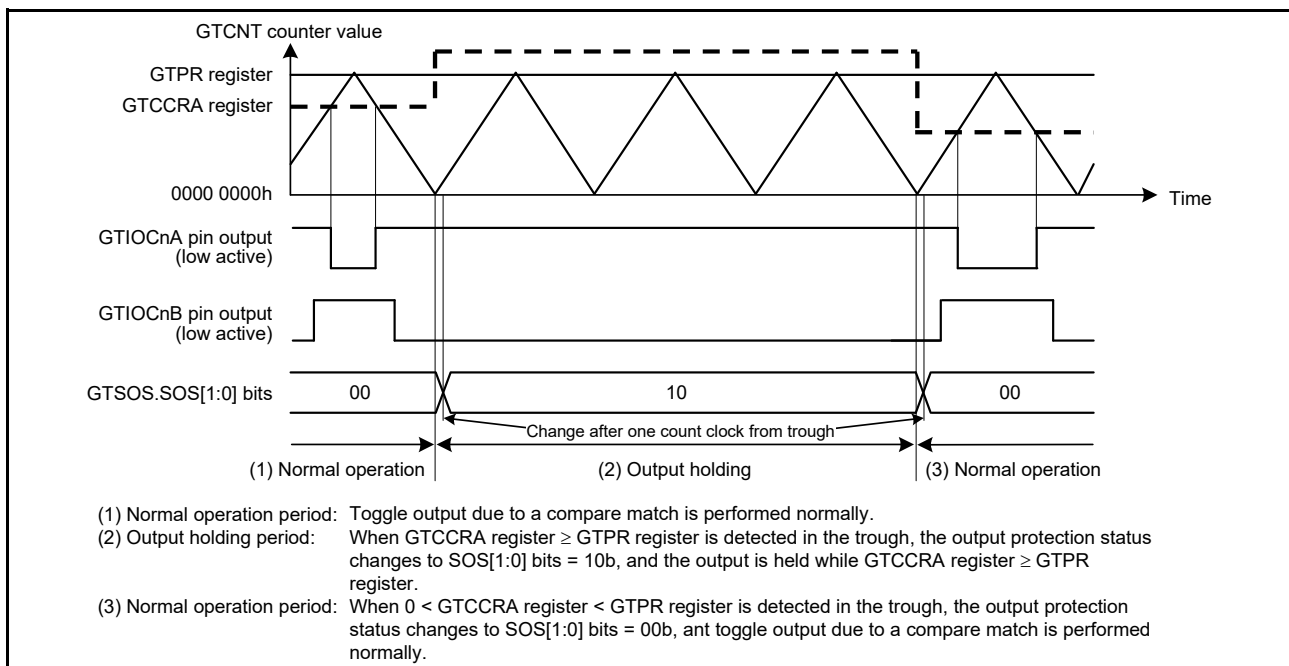


Figure 26.131 Example of Output Protection Function Operation When GTCCRA Register $\geq$ GTPR Register is set during Buffer Transfer at Troughs (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Troughs, Active Level: Low) ($n = 0$ to 3)

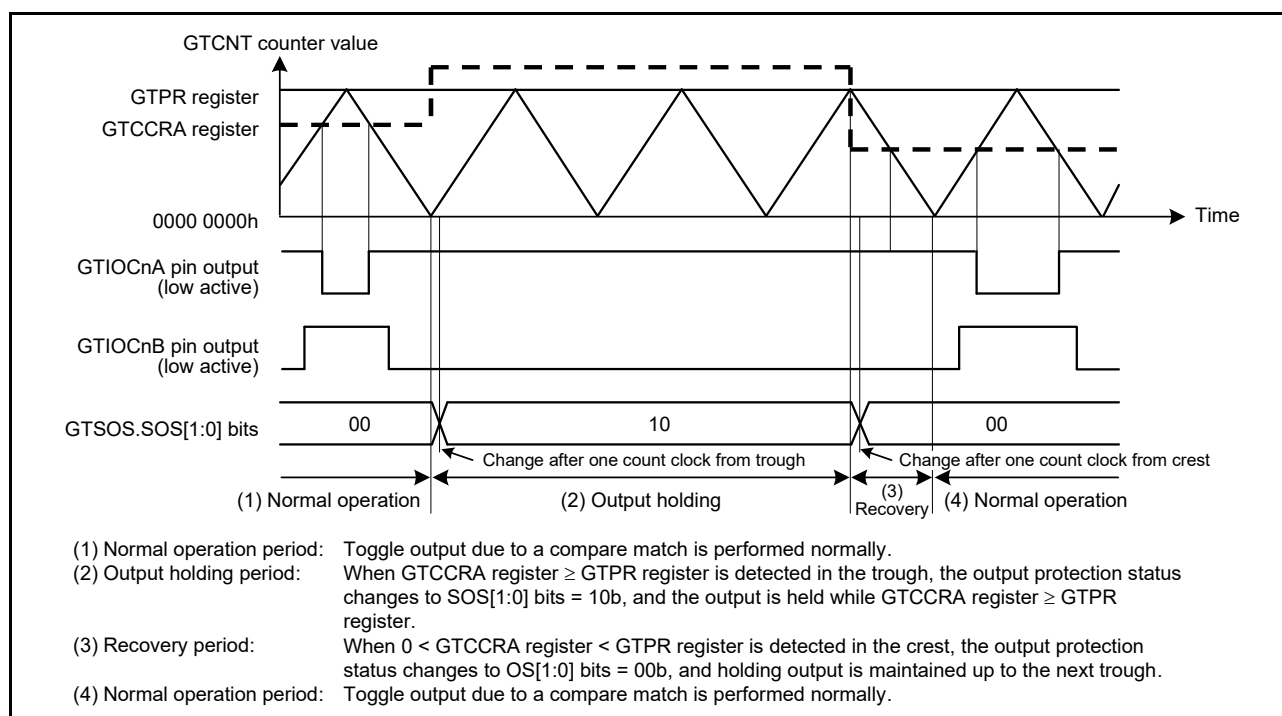


Figure 26.132 Example of Output Protection Function Operation When GTCCRA Register $\geq$ GTPR Register is Set during Buffer Transfer at Troughs (Restored to $0 < GTCCRA < GTPR$ Register during Buffer Transfer at Crests, Active Level: Low) (n = 0 to 3)

(3) Output Protection Function when GTCCRA Register $\geq$ GTPR Register is Set during Buffer Transfer at Crests

Figure 26.133 and Figure 26.134 show examples of output protection function operation when GTCCRA register $\geq$ GTPR register is set during buffer transfer at crests.

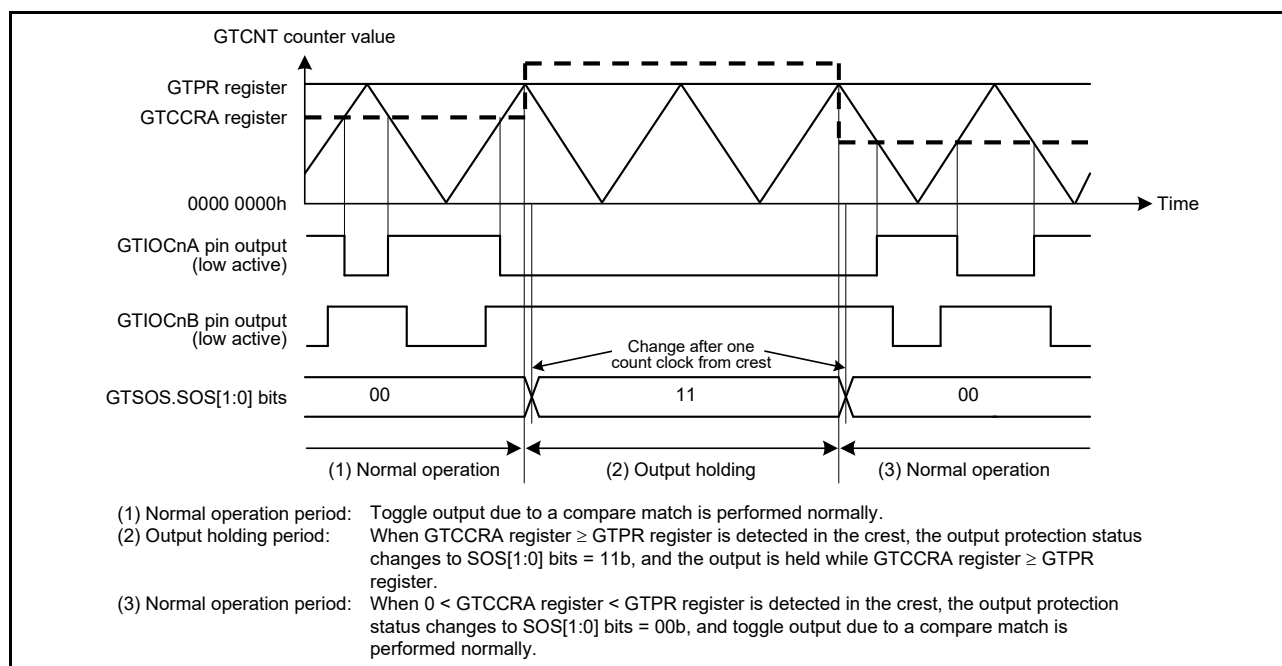


Figure 26.133 Example of Output Protection Function Operation When GTCCRA Register $\geq$ GTPR Register is Set during Buffer Transfer at Crests (Restored to $0 < GTCCRA < GTPR$ Register during Buffer Transfer at Crests, Active Level: Low) (n = 0 to 3)

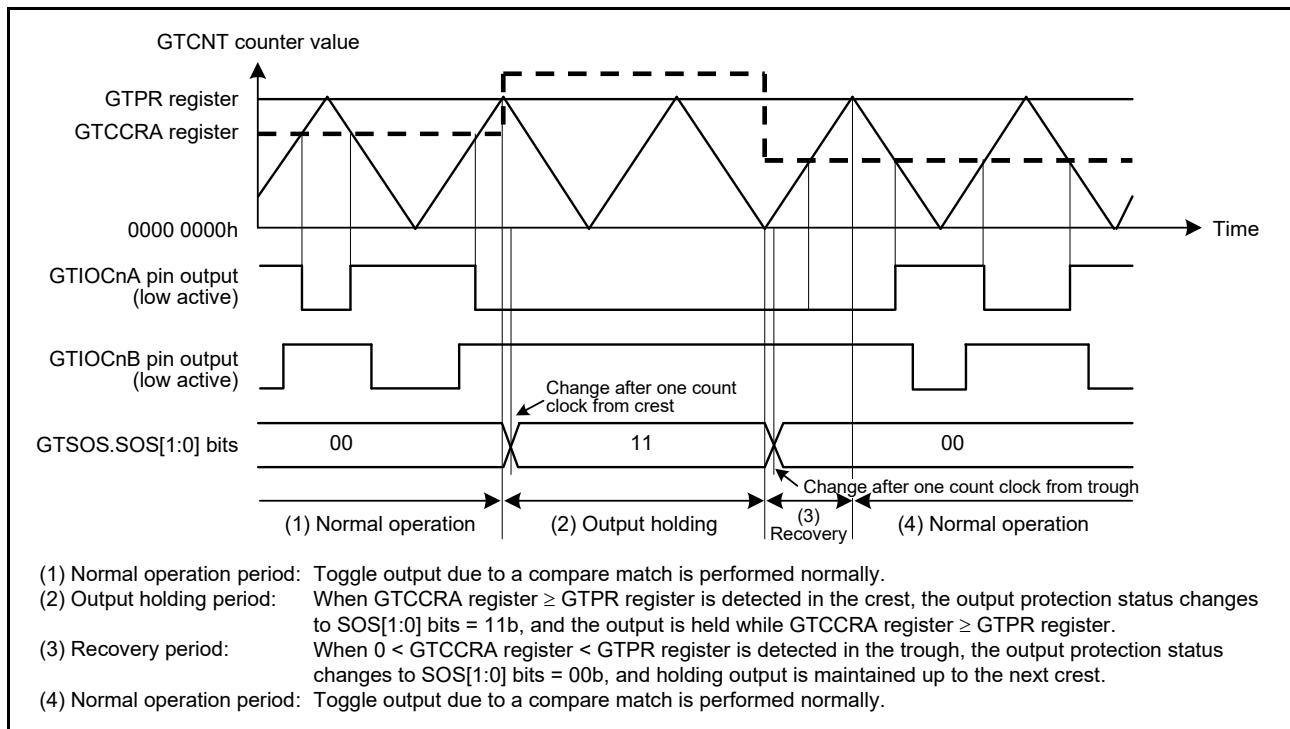


Figure 26.134 Example of Output Protection Function Operation When GTCCRA Register $\geq$ GTPR Register is Set during Buffer Transfer at Crests (Restored to $0 <$ GTCCRA Register $<$ GTPR Register during Buffer Transfer at Troughs, Active Level: Low) (n = 0 to 3)

(4) Restricted Specification of Output Protection Function

Even if an incorrect value (0000 0000h or a value greater than or equal to the GTPR register value) is set in the GTCCRA register during count operation, the output protection functions in a specific way such that one of the positive- and negative-phase outputs becomes non-active. However, the output protection does not operate normally.

- When the GTCCRA register value at the start of count operation is greater than 0000 0000h, and less than the setting value of the GTPR register

(5) Temporary Release of Output Protection Function

When the GTSOS.SOS[1:0] bits = 10b (protected state in which GTCCRA register $\geq$ GTPR register has occurred during transfer at trough), the protected state of the GTIOcNB pin output can be temporarily released by setting the GTSOTR.SOTR bit to 1. The SOS[1:0] bits retain 10b even if the output protection function is released.

When the SOTR bit is set to 0, the GTIOcNB pin output protection can be restarted.

Figure 26.135 shows an example of the operation of temporary release of output protection when the setting of the GTCCRA register $\geq$ GTPR register during buffer transfer at troughs.

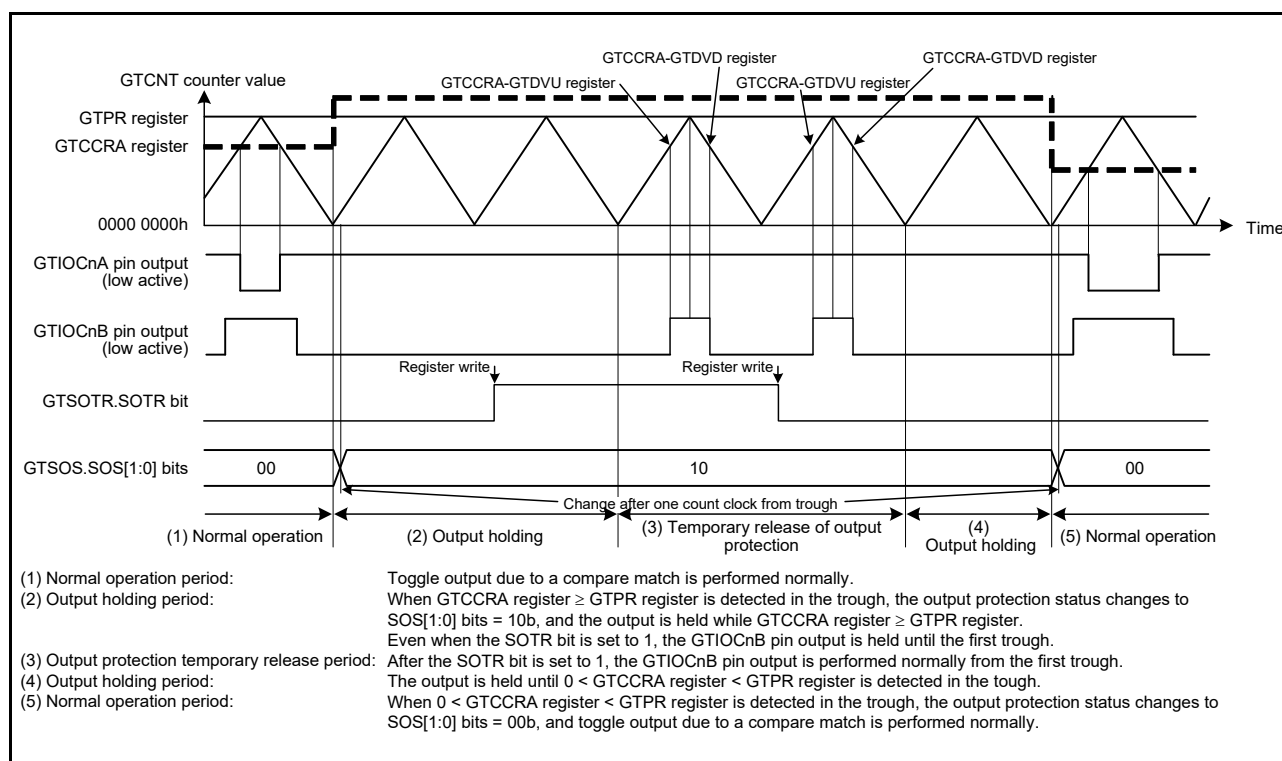


Figure 26.135 Example of the Operation of Temporary Release of Output Protection When the Setting of the GTCCRA Register $\geq$ GTPR Register during Buffer Transfer at Troughs (Restored to $0 < \text{GTCCRA Register} < \text{GTPR Register}$ during Buffer Transfer at Troughs, Active Level: Low) ($n = 0$ to 3)

26.9 Initialization Method of Output Pins

26.9.1 Pin Settings after Reset

The GPTW registers are initialized at a reset. Initialize the GPTW for output to the external pins by making the port mode settings and setting the OAE and OBE bits in the GTIOR register, and then start the counter counting.

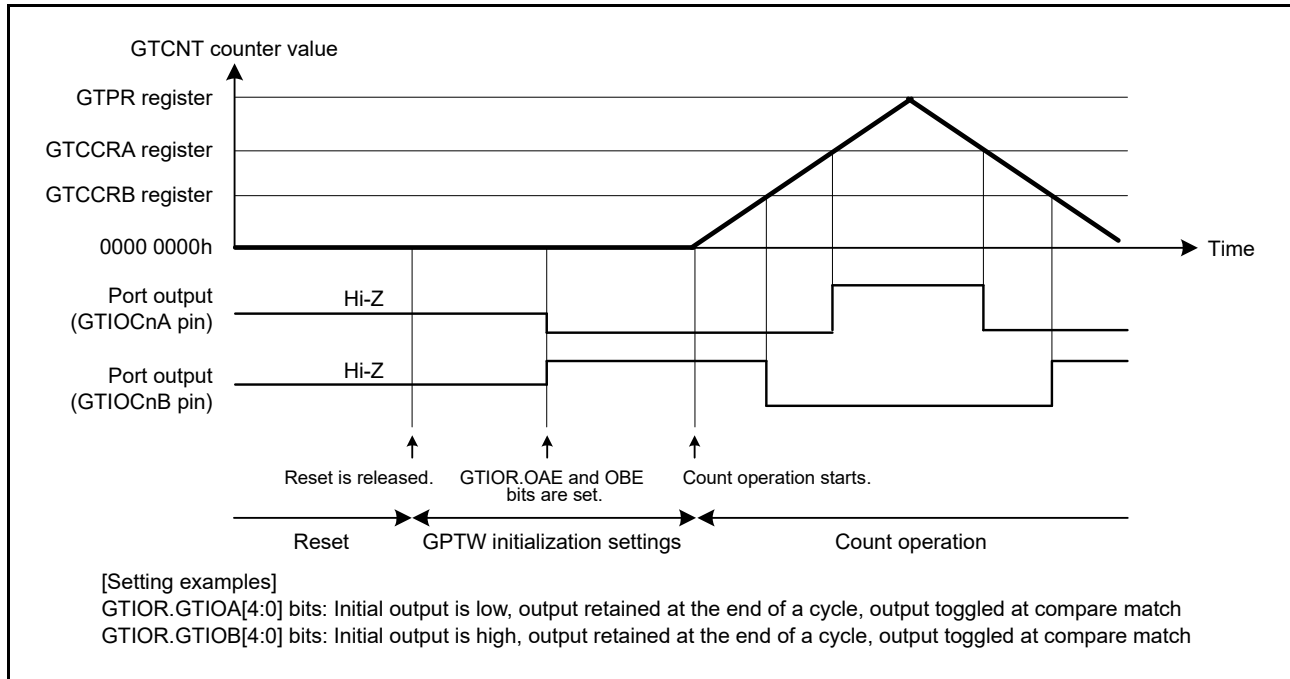


Figure 26.136 Example of Pin Settings after Reset (n = 0 to 3)

26.9.2 Pin Initialization Due to Error during Operation

If an error occurs during GPTW operation, the following four types of pin processing can be performed before pin initialization.

- (1) Set OAHLD and OBHLD bits in the GTIOR register to 1 and retain the outputs at count stop.
- (2) Set OAHLD and OBHLD bits to 0, specify arbitrary output values at OADFLT and OBDFLT bits in the GTIOR register, and output the arbitrary values at count stop.
- (3) Set the pin to output an arbitrary value as a general output port by setting the PDR, PODR, and PMR registers of the I/O port in advance. Set the OAE and OBE bits in the GTIOR register to 0 and the control bit in PMR register that corresponds to the pin to 0 to allow the arbitrary values to be output from the pin set as a general output port when an error occurs.
- (4) Drive the output to a high impedance state using the POEG function.

When automatic dead time setting has been made, set the GTDTCR.TDE bit to 0 once after counting is stopped.

When counting is stopped, only the values of registers that are changed by a GPTW external source will change. If counting is resumed, operation will carry on from where it was stopped.

If counting was stopped, registers should be initialized before counting is started.

26.10 Usage Notes

26.10.1 Module Stop Function Setting

Operation of the GPTW can be disabled or enabled by the module stop control register. The initial setting is for operation of the GPTW to be halted. Register access is enabled by clearing module stop state. For details, see section 11, Low Power Consumption.

26.10.2 Settings of the GTCCRm Register during Compare Match Operation (m = A to F)

(1) When Automatic Dead Time Setting has been Made in Triangle-Wave PWM Mode

The GTCCRA register should satisfy the following conditions:

- GTCCRA register > GTDVU register,
- GTCCRA register > GTDVD register, and
- GTCCRA register < GTPR register.

When the GTCCRA register is set to 0000 0000h or a value greater than or equal to the GTPR register value during count operation, the output protection function is activated.

However, if the following conditions are not satisfied, the output protection does not operate normally.

- When the GTCCRA register value at the start of count operation is 0000 0001h or greater, and less than the setting value of the GTPR register

For details, refer to section 26.8.4, Output Protection Function for GTIOcnm Pin Output (n = 0 to 3; m = A, B).

(2) When Automatic Dead Time Setting has not been Made in Triangle-Wave PWM Mode

Set a value greater than 0000 0000h, and less than the setting value of the GTPR register in the GTCCRA register. When 0000 0000h or the same value as that of the GTPR register is set in the GTCCRA register, compare match is generated in one cycle only when [GTCCRA register = 0000 0000h] or [GTCCRA register = GTPR register] is met. Furthermore, a value exceeding the setting value of the GTPR register is set in the GTCCRA register, compare match does not occur.

Similarly, set a value greater than 0000 0000h, and less than the setting value of the GTPR register in the GTCCRB register. When 0000 0000h or the same value as that of the GTPR register is set in the GTCCRB register, compare match is generated in one cycle only when [GTCCRB register = 0000 0000h] or [GTCCRB register = GTPR register] is met. Furthermore, a value exceeding the setting value of the GTPR register is set in the GTCCRB register, compare match does not occur.

(3) When Automatic Dead Time Setting has been Made in Sawtooth-Wave One-Shot Pulse Mode

The GTCCRC and GTCCRD registers should be set to satisfy the following restrictions. If the restrictions are not satisfied, correct output waveforms with secured dead time may not be obtained.

- In up-counting:
 - GTCCRC register < GTCCRD register
 - GTCCRC register > GTDVU register
 - GTCCRD register < GTPR register – GTDVD register
- In down-counting:
 - GTCCRC register > GTCCRD register
 - GTCCRC register < GTPR register – GTDVU register
 - GTCCRD register > GTDVD register

(4) When Automatic Dead Time Setting has not been Made in Sawtooth-Wave One-Shot Pulse Mode

The GTCCRC and GTCCRD registers should be set to satisfy the following restrictions. If the restrictions are not satisfied, two compare matches do not occur and pulse output cannot be performed.

- In up-counting: $0 < \text{GTCCRC register} < \text{GTCCRD register} < \text{GTPR register}$
- In down-counting: $\text{GTPR register} > \text{GTCCRC register} > \text{GTCCRD register} > 0$

Similarly, GTCCRE and GTCCRF registers should be set to satisfy the following restrictions. If the restrictions are not satisfied, two compare matches do not occur and pulse output cannot be performed.

- In up-counting: $0 < \text{GTCCRE register} < \text{GTCCRF register} < \text{GTPR register}$
- In down-counting: $\text{GTPR register} > \text{GTCCRE register} > \text{GTCCRF register} > 0$

(5) In Sawtooth-Wave PWM Mode

The GTCCRA register should be set with the range of $0000\ 0000\text{h} < \text{GTCCRA register} < \text{GTPR register}$. If $\text{GTCCRA register} = 0000\ 0000\text{h}$ or $\text{GTCCRA register} = \text{GTPR register}$ is set, a compare match occurs within the cycle only when $\text{GTCCRA register} = 0000\ 0000\text{h}$ or $\text{GTCCRA register} = \text{GTPR register}$ is satisfied. If $\text{GTCCRA register} > \text{GTPR register}$ is set, no compare match occurs.

Similarly, GTCCRB register should be set with the range of $0000\ 0000\text{h} < \text{GTCCRB register} < \text{GTPR register}$. If $\text{GTCCRB register} = 0000\ 0000\text{h}$ or $\text{GTCCRB register} = \text{GTPR register}$ is set, a compare match occurs within the cycle only when $\text{GTCCRB register} = 0000\ 0000\text{h}$ or $\text{GTCCRB register} = \text{GTPR register}$ is satisfied. If $\text{GTCCRB register} > \text{GTPR register}$ is set, no compare match occurs.

26.10.3 Setting Range of the GTCNT Counter

Set the range of the GTCNT counter within the range of $0 \leq \text{GTCNT counter} \leq \text{GTPR register}$.

26.10.4 The GTCNT Counter Start/Stop

The GTCNT counter start/stop control by the GTCR.CST bit is synchronized with the count clock selected by the GTCR.TPCS[3:0] bits. Since the GTCNT counter starts or stops after one count clock cycle selected by the TPCS[3:0] bits following the CST bit updating, events until the GTCNT counter actually starts are ignored, where events may be accepted and the interrupt may be generated after the CST bit becomes 0.

26.10.5 Order of Priority in Events

(1) GTCNT Counter

Order of priority in events to update the GTCNT counter is shown below.

Table 26.24 Order of Priority in Updating GTCNT Counter

The GTCNT counter updating source	Order of priority
CPU writing (GTCNT counter writing/GTCLR register writing)	High
Clearing by a hardware source set by the GTCSR register	
Up-counting/down-counting by a hardware source set by the GTUPSR and GTDNSR registers	
Count operation	
	Low

In case of contention between up-counting by the GTUPSR register and down-counting by the GTDNSR register, the counter value is not updated.

When updating of the GTCNT counter conflicts with reading by the CPU, the value before updating is reflected.

(2) The GTCR.CST Bit

In case that start/stop by a hardware source set by the GTSSR and GTPSR registers conflicts with the CPU writing (GTCR register writing/GTSTR register writing /GTSTP register writing), the CPU writing takes priority.

In case of contention between start by a hardware source set by the GTSSR register and stop by a hardware source set by the GTPSR register, the state of the CST bit does not change.

In case of contention between the CST bit updating and the CPU reading (GTCR register reading/GTSTR register reading /GTSTP register reading), data before updating is read.

(3) GTCCRm Register (m = A to F)

In case of contention between the GTCCRm register writing and an input capture/buffer transfer, an input capture/buffer transfer takes priority over the GTCCRm register writing.

In case that an input capture conflicts with the CPU writing or the counter updating by a hardware source, the counter value before updating is captured.

In case of contention between the GTCCRm register updating and the CPU reading, data before updating is read.

(4) GTPR Register

In case of contention between a buffer transfer and the GTPR register writing, the GTPR register writing takes priority over a buffer transfer.

In case of contention between the GTPR register updating and the CPU reading, data before updating is read.

(5) GTADTRm Register (m = A, B)

In case of contention between a buffer transfer and the GTADTRm register writing, the GTADTRm register writing takes priority over a buffer transfer.

In case of contention between the GTADTRm register updating and the CPU reading, data before updating is read.

(6) GTDVM Register (m = U, D)

In case of contention between a buffer transfer and the GTDVM register writing, the GTDVM register writing takes priority over a buffer transfer.

In case of contention between the GTDVM register updating and the CPU reading, data before updating is read.